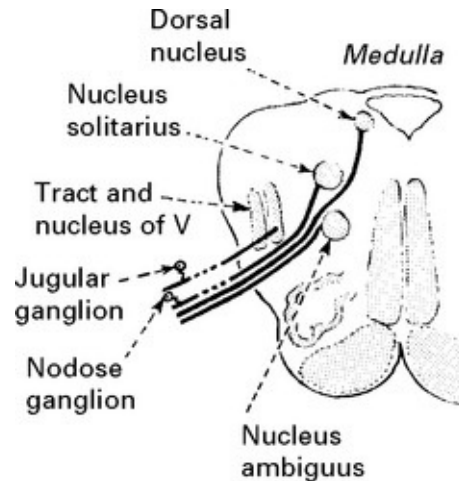
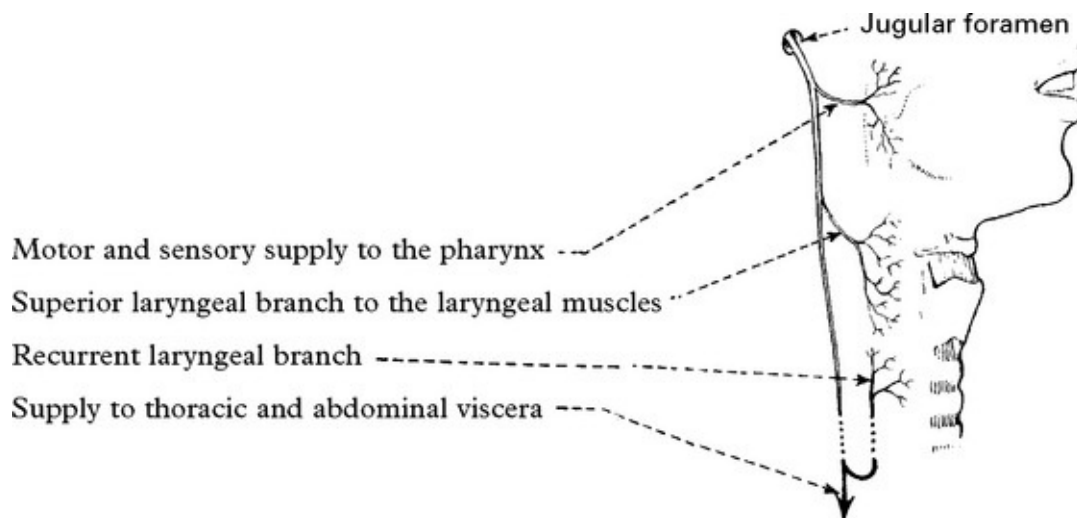




The nerve emerges from the brain stem as a series of converging rootlets. It exits from the cranial cavity by the jugular foramen where both ganglia lie.

Extracranial branches:



Disorders of the vagus nerve cause:

Palatal weakness

Unilateral – minimal symptoms.

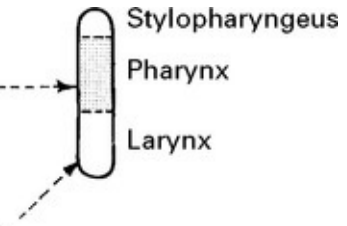
Bilateral – nasal regurgitation of fluid, nasal quality of speech.

Pharyngeal weakness

Pharyngeal muscles are represented by the middle part of the nucleus ambiguus.
Unilateral – pharyngeal wall droops on the affected side.
Bilateral – marked dysphagia.

Laryngeal weakness

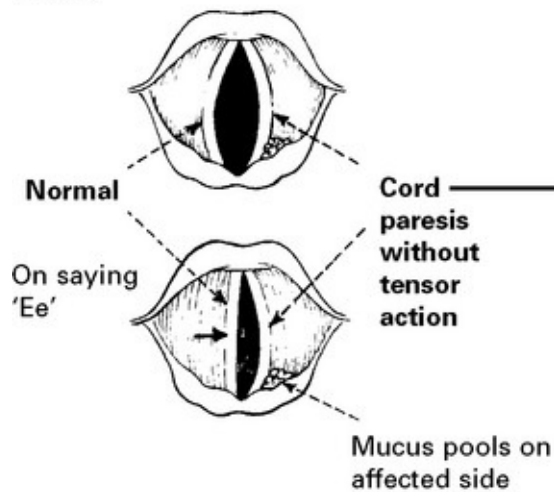
Motor fibres arise in the lowest part of the nucleus ambiguus.
Fibres to *tensors of the vocal cords* pass in *superior laryngeal nerves*.
Fibres to *adductors and abductors of the vocal cords* are supplied by the *recurrent laryngeal nerves*.



Clinical examination (see [page 17](#))

Direct examination of the vocal cords helps identification of the lesion site.

At rest



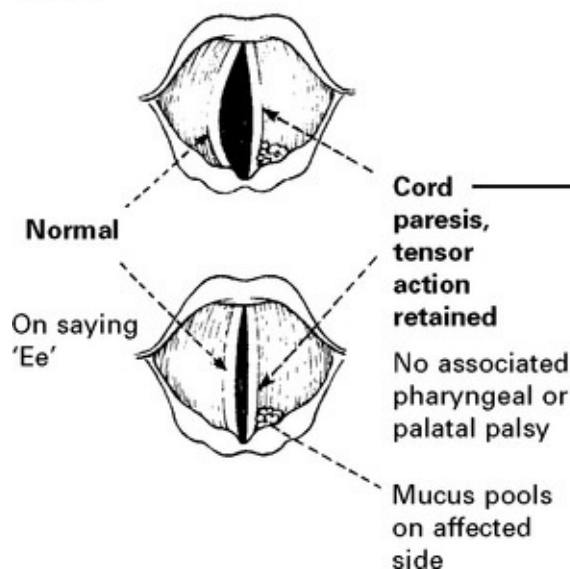
**Cord
paresis
without
tensor
action**

→ *Vagus nerve lesion above the origin of the superior and recurrent laryngeal nerves.*

Unilateral damage produces mild dysphagia, hoarseness and reduced vocal strength.

Bilateral damage at this level causes bilateral cord paresis. The cough is weak. Pharyngeal and palatal involvement cause marked dysphagia and nasal regurgitation. Breathlessness and stridor do not occur.

At rest



**Cord
paresis,
tensor
action
retained**

→ *Lesion of recurrent laryngeal nerve.*

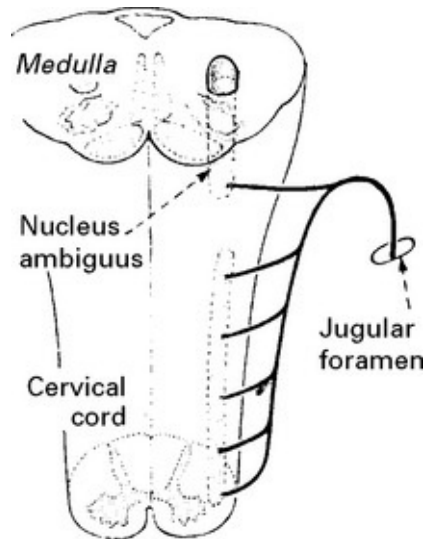
Unilateral damage produces hoarseness with breathless speech and stridor.

Bilateral recurrent laryngeal nerve lesions cause stridor and breathlessness on exertion. Approximation of the vocal cords may necessitate tracheostomy.

Causes (see [page 179](#))

ELEVENTH (ACCESSORY) CRANIAL NERVE

This is a purely motor nerve supplying the sternomastoid and trapezius muscles.



The cranial portion of the accessory nerve arises from the lowest part of the nucleus ambiguus in the medulla.

The spinal part arises in the ventral grey matter of the upper five cervical segments, ascends alongside the spinal cord and passes through the foramen magnum. After joining with the cranial portion it exits as the accessory nerve through the jugular foramen.

The *supranuclear connections* act on the ipsilateral sternomastoid (turning the head to the contralateral side) and on the contralateral trapezius. This results in:

- head turning away from the relevant hemisphere during the seizure
- head turning towards the relevant hemisphere with cerebral infarction.

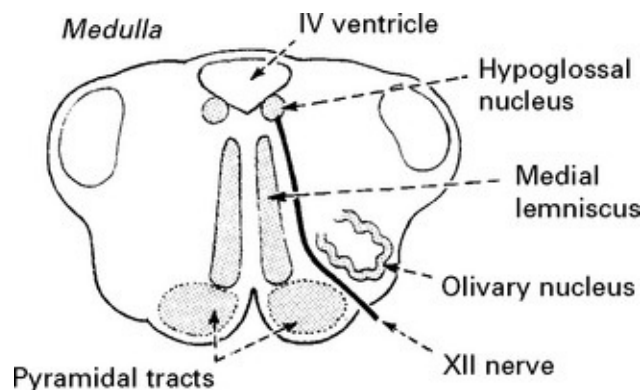
Unilateral lower motor neuron weakness produces a lower shoulder on the affected side (trapezius) and weakness in turning the head to the opposite side (sternomastoid).

Clinical examination (see [page 17](#)) *Causes* (see [page 179](#))

TWELFTH (HYPOGLOSSAL) CRANIAL NERVE

This is a purely motor nerve which supplies the intrinsic muscles of the tongue.

The nucleus lies in the floor of the IV ventricle and fibres pass ventrally to leave the brain stem lateral to the pyramidal tract.



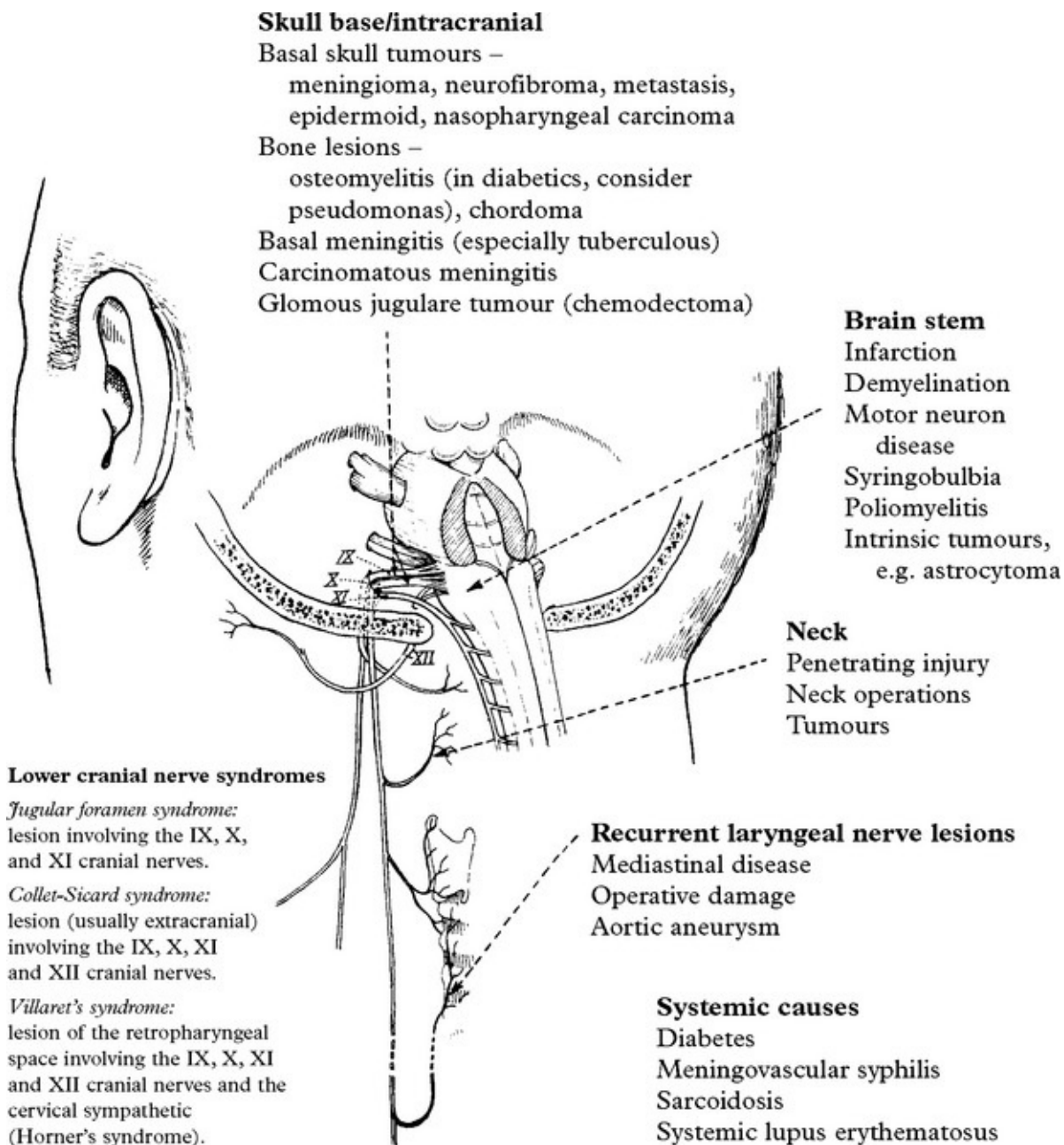
Since each nucleus is bilaterally innervated, a unilateral supranuclear lesion will not produce signs or symptoms. A bilateral supranuclear lesion results in a thin pointed (spastic) tongue which cannot be protruded.

A lesion of the hypoglossal nerve results in atrophy and deviation of the tongue to the weak side

Clinical examination (see [page 18](#)) **Causes** (see [page 179](#))

CAUSES OF LOWER CRANIAL NERVE PALSIES

Lower cranial nerve palsies seldom occur in isolation. Investigations include CT or MR imaging of the skull base. If negative, specific tests for systemic causes and EMG (for nerve and muscle disease) may be required.



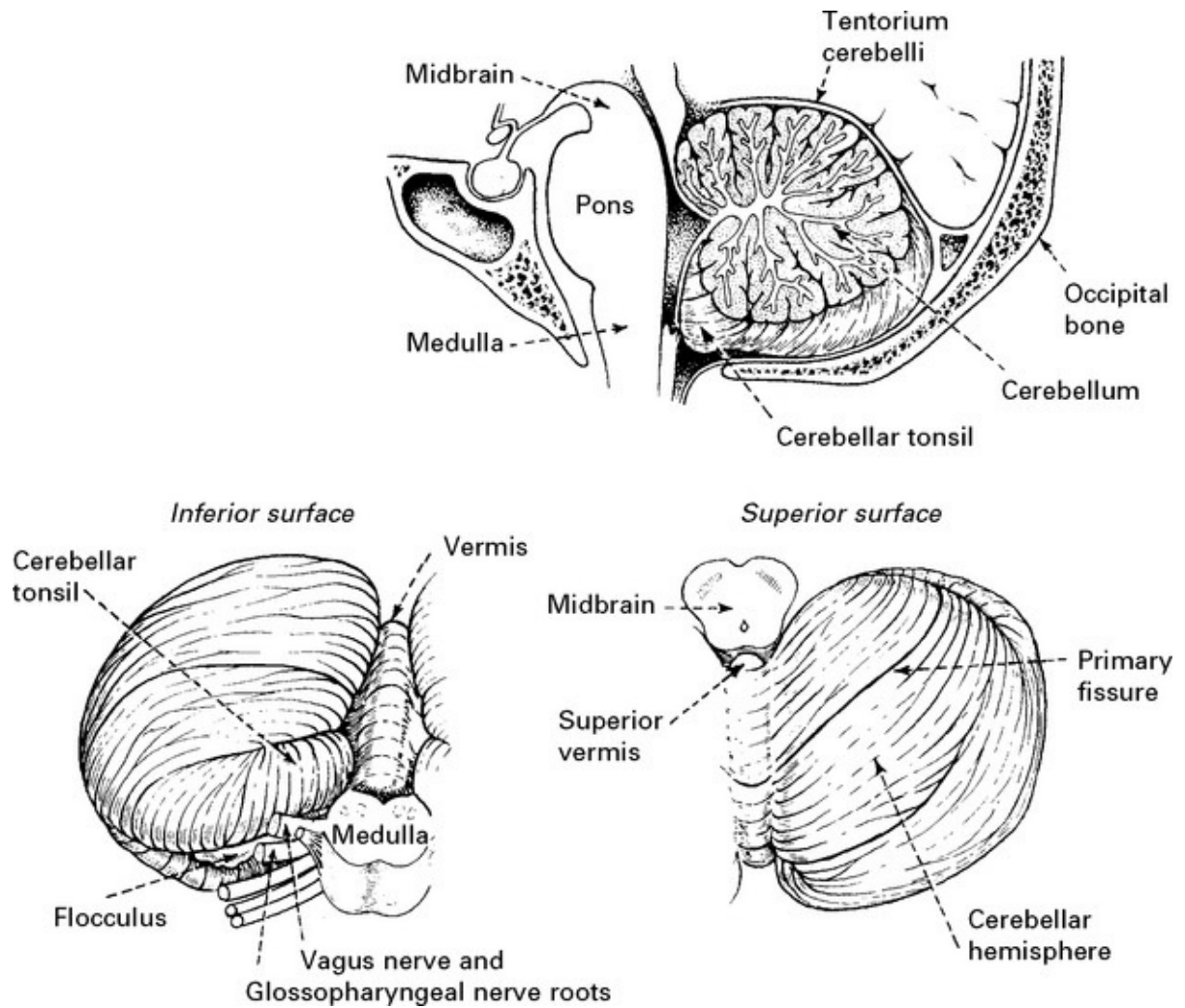
CEREBELLAR DYSFUNCTION

Anatomy

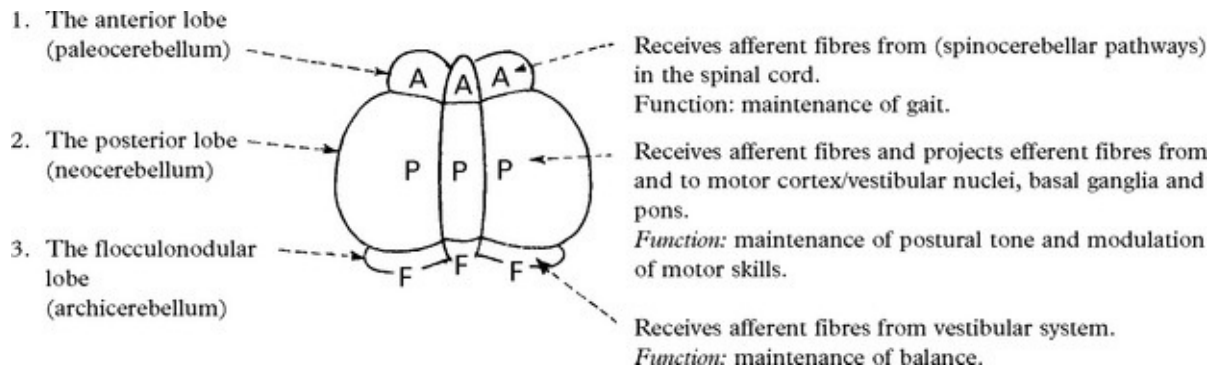
The cerebellum lies in the posterior fossa, posterior to the brain stem,

separated from the cerebrum above by the tentorium cerebelli.

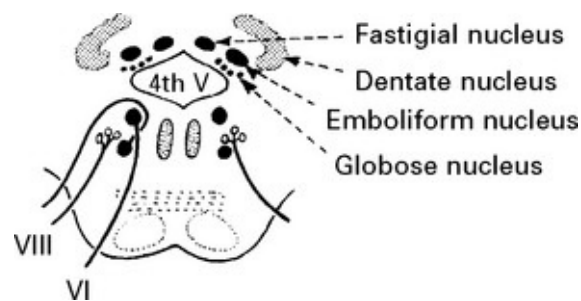
The cerebellum consists of two laterally placed hemispheres and the midline structure – the vermis.



Three major phylogenetic subdivisions of the cerebellum are recognised.

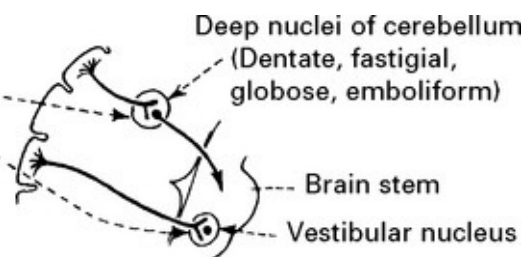


The cerebellar cortex is made up of three cell layers. The middle or Purkinje layer contains Purkinje cells. These are the only neurons capable of transmitting efferent impulses. Deep within the cerebellar hemispheres in the roof of the 4th ventricle, lie four paired nuclei separated by white matter from the cortex.



The efferent system

The Purkinje cells give rise to all efferent axons. These pass either to the deep nuclei of the cerebellum and thence to the brain stem, or to the vestibular nuclei of the brain stem. From there fibres relay back to the cerebral cortex and thalamus, or project into the spinal cord, influencing motor control.



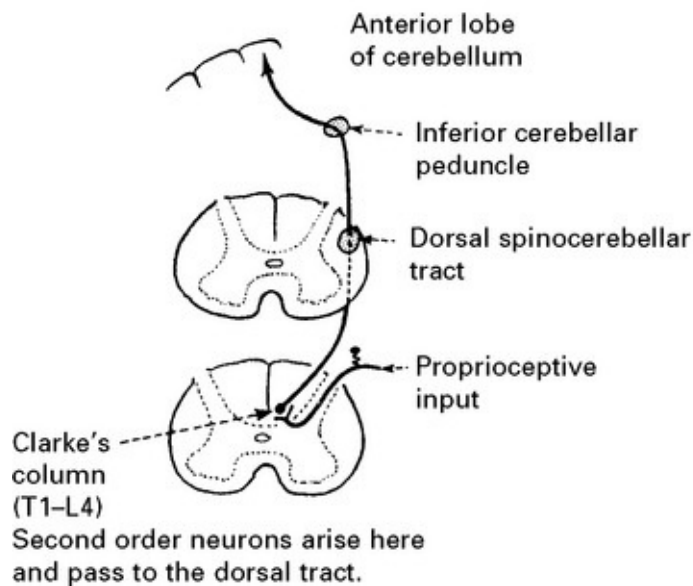
The afferent system

Connections between the vestibular system and the cerebellum are

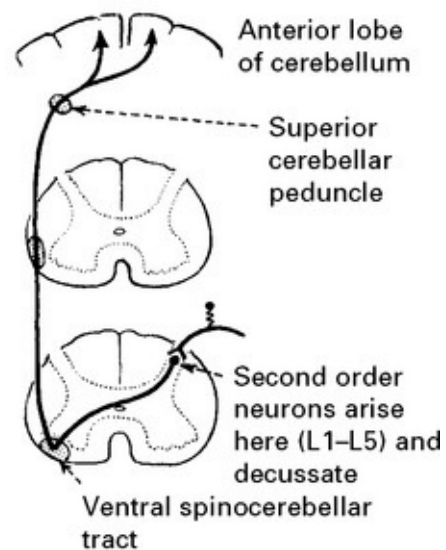
described on [page 173](#).

The spinocerebellar pathways form a major afferent input. These transmit 'subconscious' proprioception from muscles, joints and skin – especially of the lower limbs.

THE DORSAL SPINOCEREBELLAR TRACT



THE VENTRAL SPINOCEREBELLAR TRACT



The cerebellar peduncles: Three peduncles connect the cerebellum to the brain stem:

Superior peduncle – afferent and efferent fibres.

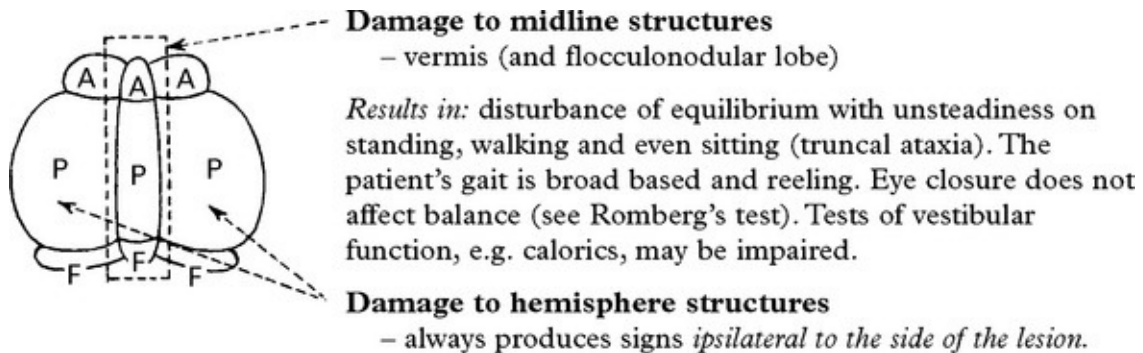
Middle peduncle – afferent fibres only.

Inferior peduncle – afferent and efferent fibres.

SYMPTOMS AND SIGNS OF CEREBELLAR DYSFUNCTION

The close relationship of structures within the posterior fossa makes the identification of exclusively cerebellar symptoms and signs difficult. Disease of the brain stem and its connections may produce identical

results.



Results in: a loss of the normal capacity to modulate fine voluntary movements. Errors or inaccuracies cannot be corrected. The patient complains of impaired limb coordination and certain signs are recognised:

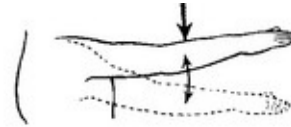
Ataxia of extremities with unsteadiness of gait towards the side of the lesion.

Dysmetria: a breakdown of movement with the patient 'overshooting' the target when performing a specific motor task, e.g. finger-to-nose test.

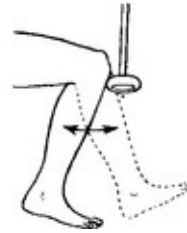
Dysdiadochokinesia: a failure to perform a rapid alternating movement.

Intention tremor: a tremor which increases as the limb approaches its target.

Rebound phenomenon: the outstretched arm swings excessively when displaced.



'Pendular' reflexes: the leg swings backwards and forwards when the knee jerk is elicited.



Eye movements

Nystagmus results from disease affecting cerebellar connections to the vestibular nuclei.

In unilateral disease, amplitude and rate increase when looking towards the diseased side.

Other ocular signs may occur, e.g. ocular dysmetria – an 'overshoot' when the eyes voluntarily fixate.

Disturbance of speech

Scanning dysarthria (where the same emphasis is put on each syllable like scanning a poem) may occur with speech occasionally delivered with sudden unexpected force – *explosive speech*. Whether dysarthria results from hemisphere or midline vermis disease remains debatable.

Dysarthria, like nystagmus, is an inconsistent finding in cerebellar disease.

Titubation

Titubation is a rhythmic 'nodding' tremor of the head from side to side or to and fro, usually associated with distal limb tremor. It appears to be

of little localising value.

Head tilt

Abnormal head tilt suggests a lesion of the anterior vermis. Note that a IV (trochlear) cranial nerve palsy and tonsillar herniation also produce this abnormal posture.

Involuntary movements

Myoclonic jerks and choreiform involuntary movements occur with extensive cerebellar disease involving the deep nuclei.

NOTE: Cerebellar lesions may cause symptoms and signs relating to

- obstructive hydrocephalus
- cranial nerve involvement
- brain stem involvement.

(Extensor spasms from brain stem damage may be wrongly described as ‘cerebellar fits’.)

CLASSIFICATION OF CEREBELLAR DYSFUNCTION

The following disorders are dealt with in their specific sections.

Developmental

- agenesis
- Dandy-Walker malformation
- Arnold-Chiari malformations
- Von Hippel Lindau disease.

Demyelinative

- multiple sclerosis.

– acute disseminated encephalomyelitis (ADEM)
<i>Degenerative/Hereditary</i> – cerebellar degeneration – multi-system atrophy (MSA) – spinocerebellar ataxias (SCA)
<i>Neoplastic</i> – astrocytoma, medulloblastoma, haemangioblastoma, metastasis
<i>Paraneoplastic</i> – subacute cerebellar degeneration
<i>Infectious</i> – abscess formation – acute cerebellitis (viral) – Creutzfeldt–Jakob disease
<i>Metabolic</i> – myxoedema – hypoxia, hypoglycaemia. – alcohol (vitamin B1 deficiency) – inborn disorders of metabolism. (lipid or amino acid metabolism)
<i>Vascular</i> – cerebellar haemorrhage – cerebellar infarction.
<i>Drugs/toxins</i> – alcohol – phenytoin. – carbamazepine.

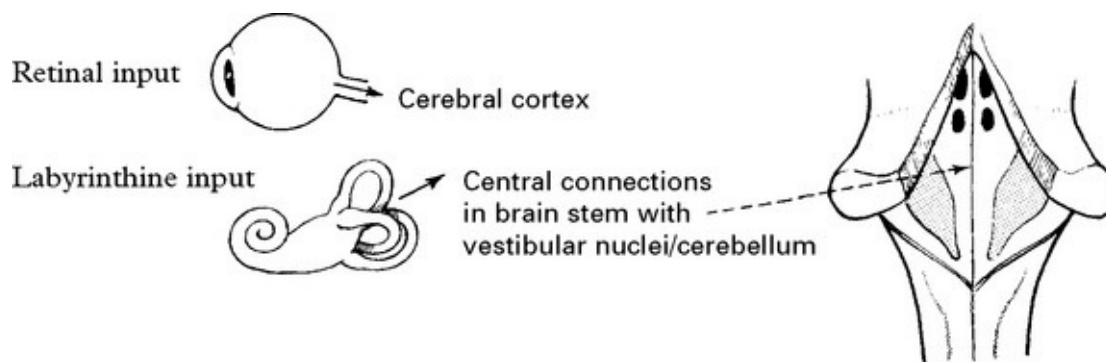
NYSTAGMUS

Nystagmus is defined as an involuntary ‘to and fro’ movement of the

eyes in a horizontal, vertical, rotatory or mixed direction. The presence and characteristics of such movements help localise to the site of neurological disease.

Nystagmus may be *pendular* – equal velocity and amplitude in all directions, or *jerk* – with a fast phase (specifying the direction) and a slow phase.

The normal maintenance of ocular posture and alignment of the eyes with the environment depends upon:



Nystagmus may result from:

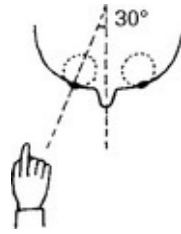
- retinal disease
- labyrinthine disease, or
- disorders affecting the cerebellum or a substantial portion of the brain stem.

Examination for nystagmus

‘Nystagmoid’ movements of the eyes are present in many people at extremes of gaze.

Nystagmus present with the eyes deviated less than 30° from the

midline is abnormal.



When nystagmus is present only with the eyes deviated to one side – *1st degree nystagmus*.

With eyes deviated to one side and in the midline position also – *2nd degree nystagmus*.

When present in all directions of gaze – *3rd degree nystagmus*.

If nystagmus is detected, note the type (jerk or pendular), direction (of fast phase) and degree.

Nystagmus suppressed by visual fixation may appear in darkness, but this requires specialised techniques (electronystagmography – see [page 65](#)) to demonstrate.

RETINAL OR OCULAR nystagmus

Physiological: following moving objects beyond the limits of gaze – optokinetic nystagmus. **Pathological:** occurs when vision is defective. Fixation is impaired and the eyes vainly search.

Nystagmus is:

Rapid Pendular (lacks slow and fast phase) Increased when looking to sides Persistent throughout lifetime
--

Occurs in *congenital cataract, congenital macula defect, albinism*.

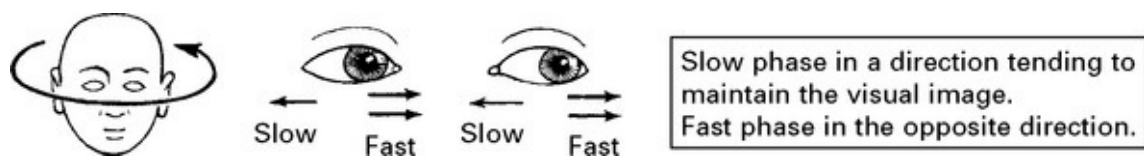
VESTIBULAR nystagmus

Nystagmus arises from:

- natural stimulation of the vestibular apparatus – rotational or linear acceleration.
 - artificially removing or increasing the stimulus from one labyrinth (e.g. caloric testing).
 - damage to vestibular apparatus or the vestibular nerve.
- Creates an imbalance between each side resulting in a slow drift of the eyes towards the damaged side (or side with the reduction in stimulus) followed by a fast compensatory movement to the opposite side.

Physiological

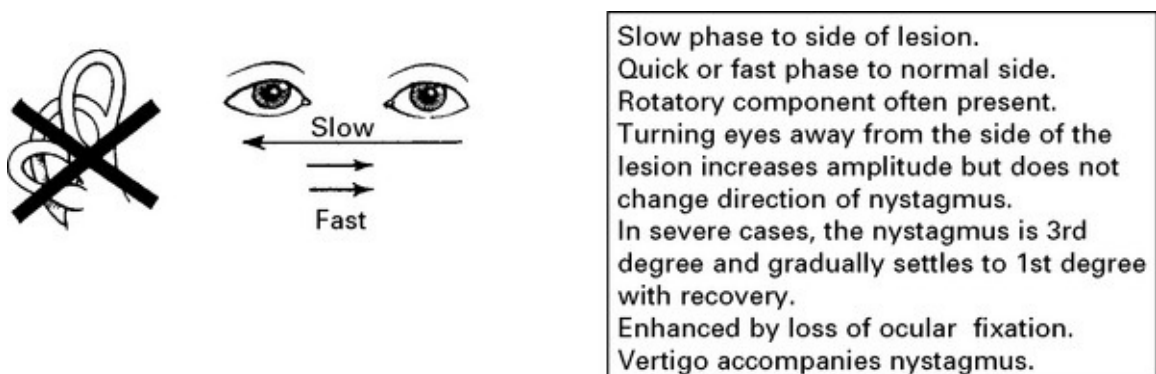
(i) Rotational acceleration produces nystagmus in the plane of rotation.



(ii) Caloric testing sets up convection currents in the lateral semicircular canal producing a horizontal nystagmus (see [page 65](#)).

Pathological

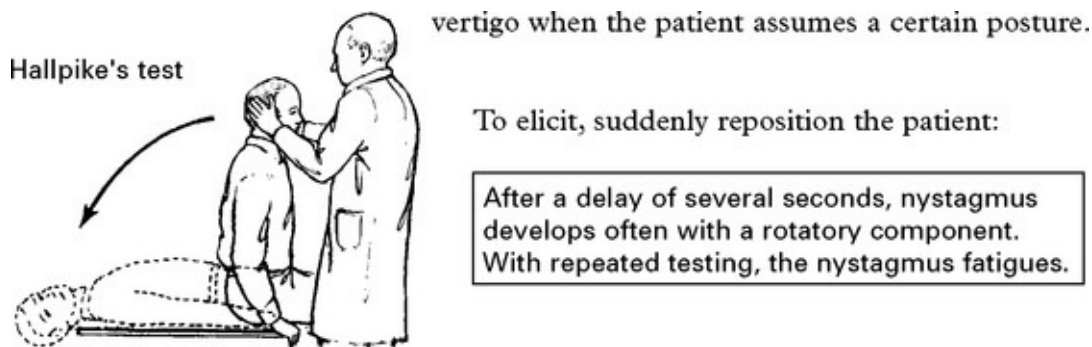
Damage to labyrinth or vestibular nerve.



Often associated with tinnitus and hearing loss. Vertigo and nystagmus settle simultaneously.

Occurs in acute labyrinthine disease – *Menière's disease, vestibular neuronitis, vascular disease.*

POSITIONAL nystagmus: this may occur in labyrinthine disease in association with vertigo when the patient assumes a certain posture.



NYSTAGMUS

CENTRAL NERVOUS SYSTEM nystagmus

Central nystagmus arises from damage to the central vestibular connections in the vestibular nuclei and brain stem. The nystagmus may be horizontal, vertical, rotatory or dissociated (present in one eye only).

The direction (fast phase) is determined by direction of gaze (multidirectional). Vertigo is seldom present.

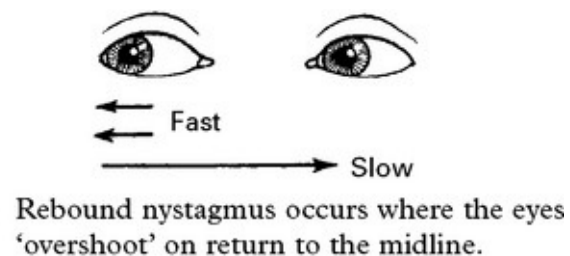
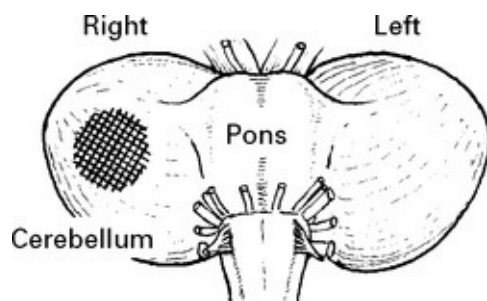
Signs of other nuclear or tract involvement in brain stem should be evident.

Central nystagmus occurs in *vascular disease, demyelination, neoplasms, nutritional disease (Wernicke's encephalopathy), alcohol intoxication and drug toxicity, e.g. phenytoin.*

Posterior fossa lesions may produce positional nystagmus. This may be distinguished from labyrinthine disease by:

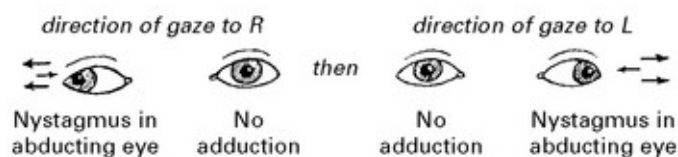
Absence of delay before onset, lack of fatiguing with repetitive testing, and a tendency to occur with any rather than one specific head movement.

Although nystagmus often occurs in cerebellar disease, the role of the cerebellum in its production remains unclear. *The fast phase tends to occur to the side of the cerebellar damage (i.e. the opposite of labyrinthine disease).*

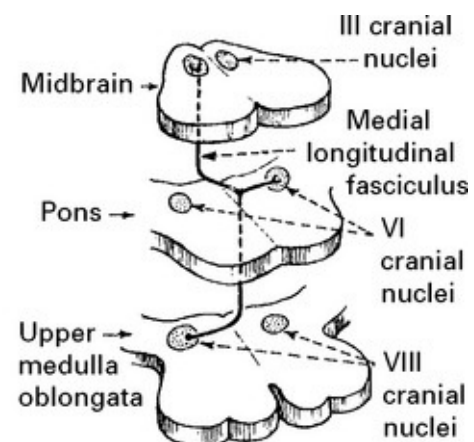


INTERNUCLEAR OPHTHALMOPLEGIA (Ataxic nystagmus)

The median longitudinal fasciculus links, among other structures, the innervation of the lateral rectus with the contralateral medial rectus muscle in order to coordinate horizontal gaze. A lesion of this fasciculus will cause *dissociate nystagmus*.



Eyes no longer move as one and nystagmus is present in one eye but not the other.



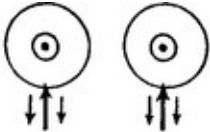
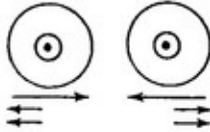
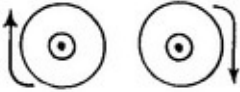
In unilateral medial longitudinal fasciculus lesions the eye fails to

adduct on the affected side.

N.B. Internuclear ophthalmoplegia differs from a bilateral III nerve or nuclear lesion in that the pupil is not affected and when testing eye movements individually, some adduction occurs.

The disorder characteristically occurs in multiple sclerosis but also in brain stem infarction, haemorrhage, trauma, syringobulbia and drug toxicity (phenytoin).

OTHER VARIETIES OF CENTRAL NERVOUS SYSTEM NYSTAGMUS

- | | | |
|---------------------------------|---|---|
| 1. Downbeat nystagmus |  | Occurs with lesions around the aqueduct of Sylvius or cervicomedullary junction. Fast phase is downwards (downbeating nystagmus). |
| 2. Convergence nystagmus |  | Occurs with lesions in upper midbrain region. |
| 3. See-saw nystagmus |  | One eye intorts and moves up while the other extorts and moves down.
Occurs with sellar or parasellar mass lesion. |

A group of confusing terms are used to describe abnormal, involuntary eye movements seen in cerebellar/brain stem disease:

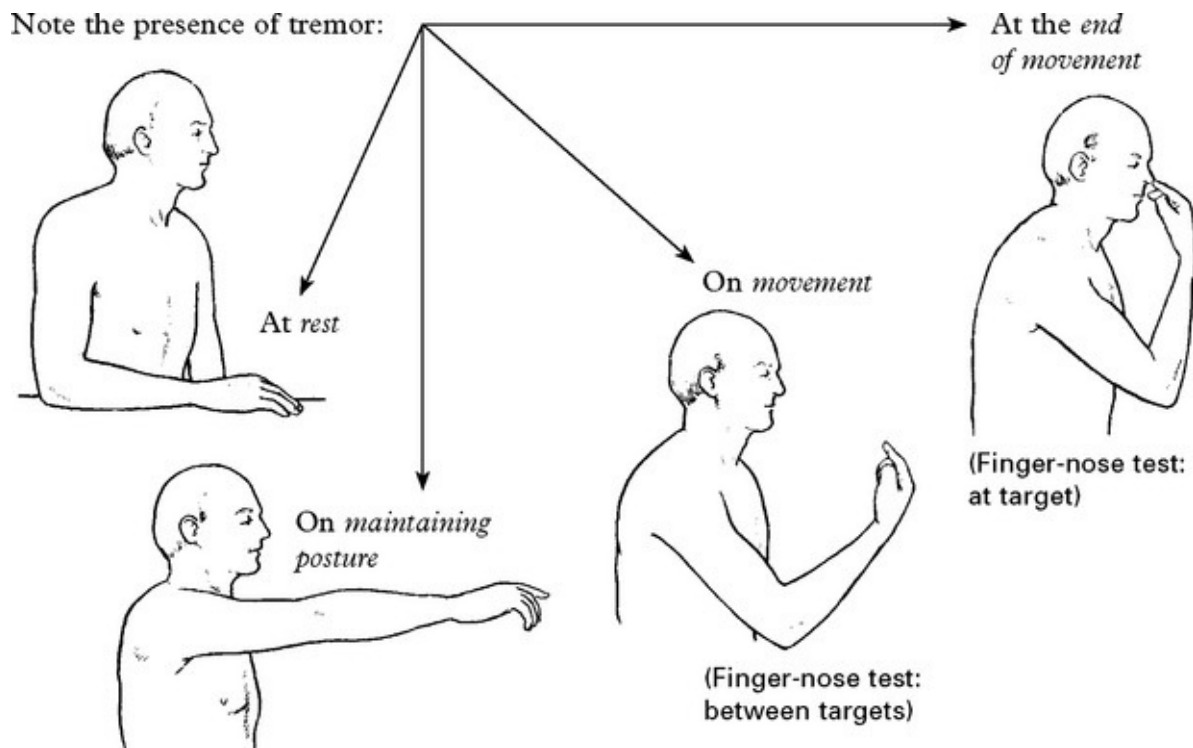
Ocular bobbing – fast drift downwards, slow drift upwards; seen with large pontine lesions. (Horizontal eye movements are absent.)

Opsoclonus – rapid conjugate jerks of eyes; made worse by head movements. The eye movements are random.

Oscillopsia is a term used to describe the patient's awareness of jumping of the environment as a consequence of rapid jerking eye movements.

TREMOR

Tremor is a rhythmic involuntary movement normally affecting the limbs. Diagnosis depends on examination of the character of the tremor as well as the presence of other specific features.



Observe:

- the *rate* (slow, 4–6 Hz), (rapid, 6–12 Hz)
- the *amplitude* (fine or coarse)
- the *distribution*: head, trunk or limbs (distal or proximal)
- *associated features* e.g. disorder of gait or balance

Most tremors disappear during sleep.

Physiological tremor is evident on maintaining a fixed posture, fast

in rate (8–12 Hz), fine in character, distal in distribution and non-disabling. It is enhanced by fatigue, anxiety and drugs *e.g.* caffeine, steroids.

Pathological tremor occurs at rest or with movement, slow in rate, coarse in character, proximal or distal and often asymmetrical in distribution. This tremor is socially and physically disabling.

CHARACTERISTICS OF PATHOLOGICAL TREMOR

Tremor at rest

‘Pill-rolling’ tremor, decreasing with movement.

Rate: 3–7 per second.

Amplitude: coarse.

Distribution: distal limbs.

Usually associated with bradykinesia and rigidity.



PARKINSON’S DISEASE
OR DRUG INDUCED
PARKINSONISM

Tremor on maintaining posture and throughout range of movement

Tremor absent at rest, when the limb is relaxed, but present on maintaining a fixed posture and during movement.

Rate: 6–12 Hz

Amplitude: fine

Slow insidious onset

Distribution: Upper limbs involved, lower limbs rarely.
Titubation (tremor of the head on the trunk) often present.

POSTURAL TREMOR

Specific types of postural tremor are recognised

FAMILIAL TREMOR

– often Mendelian dominant.

ESSENTIAL TREMOR	– no family history
SENILE TREMOR	– develops in old age.

The tremor may progress until handwriting becomes impossible and feeding difficult. Alcohol may temporarily abort the tremor; beta blockers may produce an improvement.

Tremor during and maximal at the end of movement

Tremor absent at rest; present during movement and maximal on approaching target, e.g. finger-nose test.

Rate: 4–6 per second.

Amplitude: coarse.

Distribution: Proximal and distal.

Titubation may occur.

Usually associated with other cerebellar signs.

CEREBELLAR TREMOR
(‘intention tremor’)

Extremely severe tremor – sufficient to interrupt movement and throw patient off balance.

MIDBRAIN TREMOR due to disease involving the cerebellar/red nucleus connections, e.g. *multiple sclerosis*.

MYOCLONUS

Myoclonus is a shock-like contraction of muscles which occur irregularly and asymmetrically. Such jerks occur repetitively in the same muscle groups and range from a flicker in a single muscle to contraction in a group of muscles sufficient to displace the affected limb.

Pathophysiology

The precise nature of myoclonus remains unclear. Several forms exist, some clearly related to epilepsy; others may be associated with damage to inhibitory mechanisms in the brain stem reticular formation. Myoclonus may result from pathological changes affecting a variety of

different sites including the motor cortex, cerebellum and spinal cord.

Clinical features

Myoclonic movements when repetitive vary in frequency between 5–60/minute. The muscles of the face, oral cavity and limbs are preferentially affected. The movements may be accentuated or precipitated by visual, auditory or tactile stimulation. Repetitive stimulation may result in a crescendo of myoclonus which resembles a seizure.

Physiological myoclonus occurs in sleep (hypnic jerks), with anxiety and in infants when feeding.

Causes

Myoclonus occurs in many rare conditions of the nervous system. Five groups of disorder are recognised:

<i>Progressive myoclonus</i>	<i>Metabolic disease associated</i>	<i>Miscellaneous disorders</i>
<i>Familial disorders:</i>	<i>with transient myoclonus</i>	– Cerebral anoxia
– Lafora body disease	– Hyponatraemia	– Vasculitides
– Tay Sach's disease	– Hypocalcaemia	– Sarcoidosis
– Gaucher's disease	– Renal, hypoxic, hepatic	– Paraneoplastic disease
– Ramsay Hunt syndrome	encephalopathy	– Mitochondrial disease
– Benign polymyoclonus	– Non-ketotic hyperglycaemia	– HIV encephalopathy
	– Hypoglycaemia	– Whipple's disease
<i>Degenerative disease:</i>	<i>Epileptic disorders in which myoclonus occurs</i>	
– Subacute sclerosing	Generalised seizures: – associated with petit mal	
panencephalitis	– during prodrome of grand mal	
– Alzheimer's disease	– photosensitive myoclonus	
– Pick's disease		
– Diffuse Lewy body disease	Juvenile myoclonic epilepsy	
– Huntington's disease	Lennox Gastaut syndrome (atypical petit mal,	
– Prion disease	drop attacks and mental retardation)	
– Creutzfeldt–Jakob disease	West's syndrome	

Palatal myoclonus – an unusual myoclonic disorder with rapid regular movements of the soft palate and occasionally of the pharyngeal and

facial musculature. Palatal movements occur at a rate of 120–140/minute. This disorder is associated with degenerative changes in the olivary and dentate nuclei.

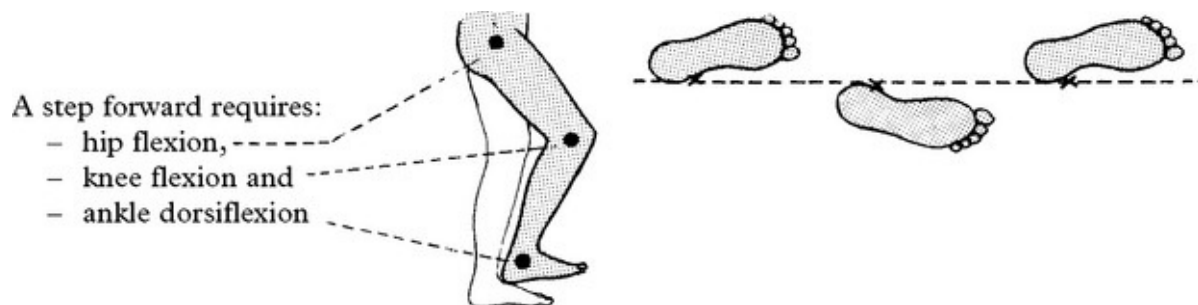
Treatment

Benzodiazepine drugs such as clonazepam may suppress myoclonic movements. Piracetam (G.A.B.A. analogue) and levetiracetam are also used.

An exaggerated startle response can be confused with myoclonus. This is often physiological but can be disabling – hyperekplexia (Startle disease).

DISORDERS OF STANCE AND GAIT

The normal gait is characterised by an erect posture, moderately sized steps and the medial malleoli of the tibia 'tracing' a straight line.



Coordination ensures fluidity of movement.

Antigravity reflexes maintain the erect posture. They depend upon spinal cord and brain stem connections to produce extension.

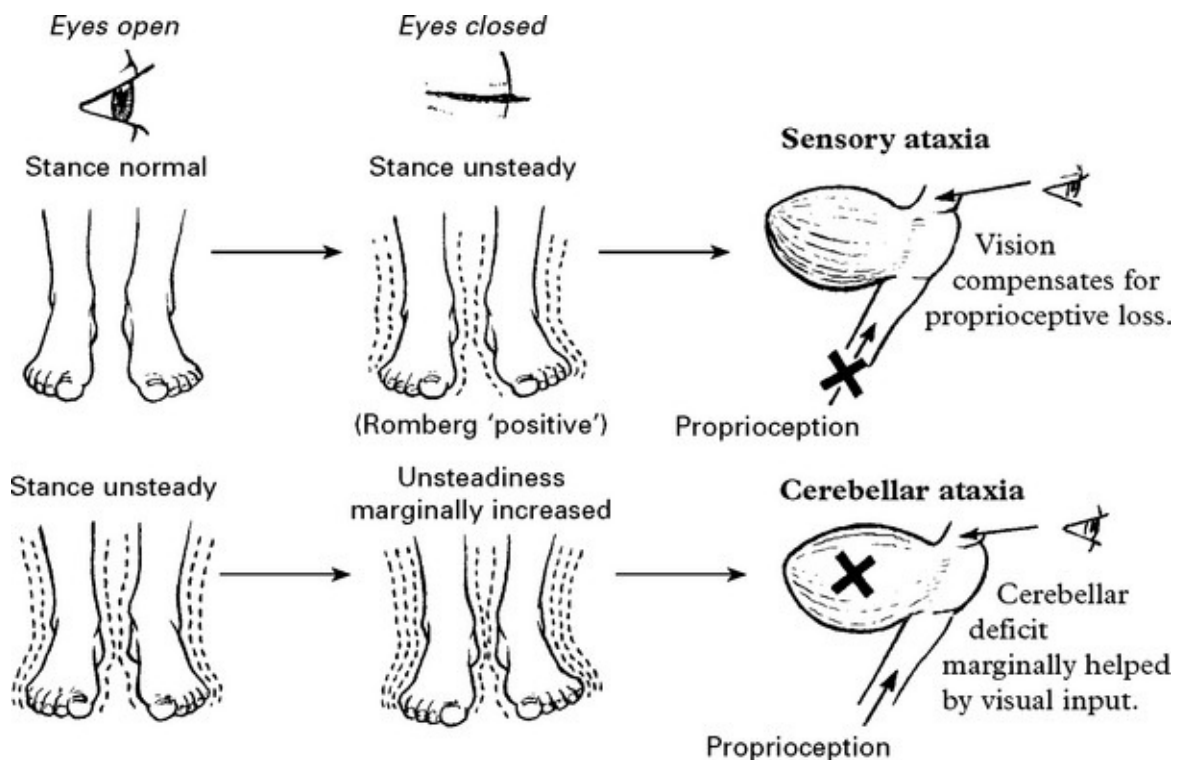
ASSESSMENT OF STANCE AND GAIT

In a patient complaining of disturbance of walking, careful assessment

indicates the likely site of the causative lesion.

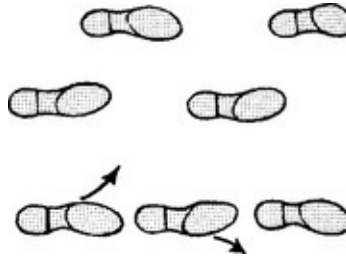
Watch the patient:

- walking
- performing *tandem gait* – heel to toe walking,
- standing with heels together with (a) eyes open, (b) eyes closed – this (Romberg's test) distinguishes cerebellar from sensory ataxia. N.B. You cannot undertake Romberg's test if the patient cannot stand with eyes open.



ATAXIC GAIT

1. **Cerebellar** The feet are separated widely when standing or walking. Steps are jerky and unsure, varying in size. The trunk sways forwards.



In mild cases: Tandem gait (heel-toe walking) is impaired; the patient falling to one or both sides.

2. Sensory

Disturbed conscious or unconscious proprioception due to interruption of afferents in peripheral nerves or spinal cord (posterior columns, spinocerebellar tracts).

The gait appears normal when the eyes are open although the feet usually 'stamp' on the ground. Examination reveals a positive Romberg's test and impaired joint position sensation.

HEMIPLEGIC GAIT

The leg is extended and the toes forced downwards. When walking, abduction and circumduction at the hip prevent the toes from catching on the ground.

In paraplegia, strong adduction at the hips can produce a scissor-like posture of the lower limbs. In mild weakness, the gait may appear normal, but excessive wear occurs at the outer front aspect of the patient's shoe sole.

Hemiplegic gait



PARKINSONIAN (festinating) GAIT



The patient adopts a flexed, stooping posture. To initiate walking, he leans forwards and then hurries (festinates) to 'catch up' on himself. The steps are short and shuffling.

STEPPAGE GAIT

Lower motor neuron weakness of pretibial and peroneal muscles produces this gait disorder. The patient lifts the affected leg high so that the toes clear the ground. When bilateral, it resembles a high-stepping horse.



MYOPATHIC (waddling) GAIT

Characteristic of muscle disease. Trunk and pelvic muscle weakness result in a sway-back, pot-bellied appearance with difficulty in pelvic 'fixation' when walking.



FRONTAL LOBE GAIT

Disturbance of connections between frontal cortex, basal ganglia and cerebellum produces this characteristic disturbance. The gait is wide based (feet wide apart). Initiation is difficult, the feet often seem 'stuck' to the floor. There is a tendency to fall backwards. Power and sensation

are normal.

HYSTERICAL GAIT

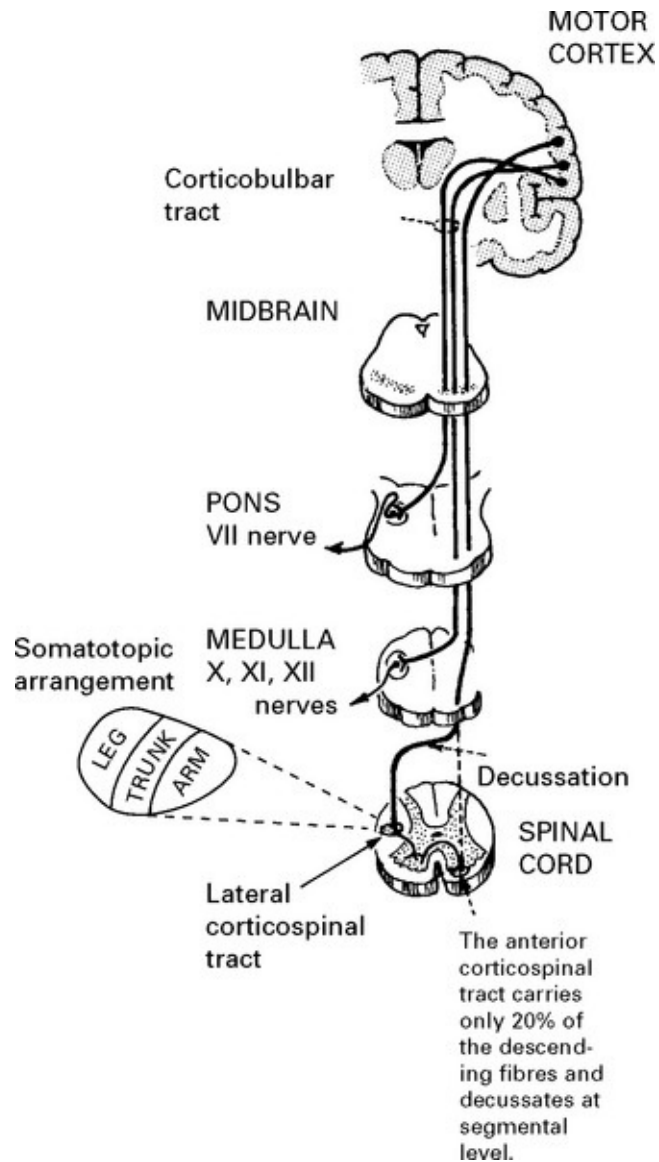
Characterised by its bizarre nature.

Numerous variations are seen. The hallmark is inconsistency supported by the lack of neurological signs. Close observation is essential.

LIMB WEAKNESS

Limb weakness results from damage to the **motor system** at any level from the motor cortex to muscle.

UPPER MOTOR NEURON WEAKNESS



MUSCLE TONE

Hypertonicity develops after a period (a few days or weeks) of 'neural shock'. Passive movements produce a 'clasp knife' quality, *i.e.* sudden 'give' towards the end of movement.

Clonus – present.

MUSCLE FASCICULATION

Absent.

MUSCLE WASTING

Absent – but, in the long term, disuse atrophy results.

REFLEXES

- **Tendon** – exaggerated.
- **Superficial** – depressed or absent (abdominal, cremasteric).
- **Plantar response** – extensor.

DISTRIBUTION

In general, whole limb or limbs are involved, e.g. monoplegia, hemiplegia, paraplegia.

Weakness shows a **PREDILECTION** for certain muscle group in a **PYRAMIDAL DISTRIBUTION**, *i.e.*

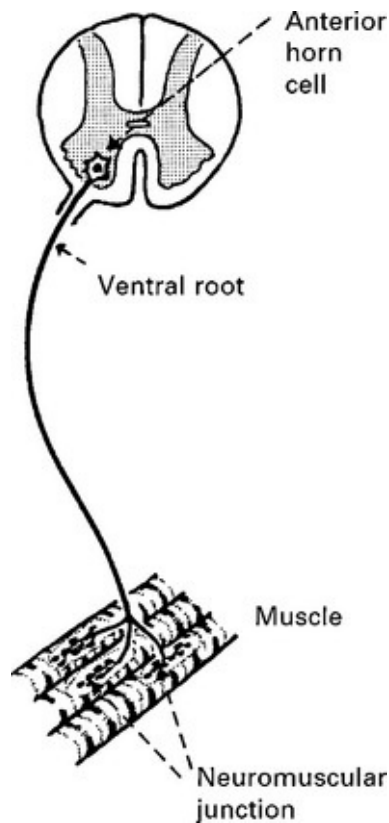
upper limbs	– extensor weakness	>	flexor weakness
lower limbs	– flexor weakness	>	extensor weakness



This results in the ‘*spastic*’ posture with the arm and the wrist flexed

and the leg extended. In upper motor neuron lesions, SKILLED movements, *e.g.* fastening buttons, are always more affected than unskilled movements.

N.B. *Dual innervation* from each hemisphere results in sparing of the upper face, muscles of mastication, the palate and tongue with a unilateral upper motor neuron lesion.



LOWER MOTOR NEURON WEAKNESS

MUSCLE TONE

Hypotonicity with diminished resistance to passive stretch.

Clonus – absent.

MUSCLE FASCICULATION

Present – irregular, non-rhythmical contractions of groups of motor units. More prevalent in anterior horn cell disease than in nerve root damage.

MUSCLE WASTING

Wasting becomes evident in the paretic muscle within 2–3 weeks of the onset.

REFLEXES

- **Tendon** – depressed or absent.
- **Superficial** – rarely affected (abdominal, cremasteric).
- **Plantar response** – flexor.

DISTRIBUTION

Either – muscle groups involved in distribution of a spinal segment/root, plexus or peripheral nerve,

or – generalised limb involvement affecting proximal or distal muscles.

NEUROMUSCULAR JUNCTION WEAKNESS

Muscle tone; muscle bulk; reflexes – all normal

Key feature – weakness fatigues with repetition; most commonly involves ocular muscles, bulbar muscles.

WEAKNESS FROM MUSCLE DISEASE

May be difficult to distinguish from lower motor neuron weakness.

Muscle tone – slightly reduced

Muscle bulk – slightly reduced; no fasciculation

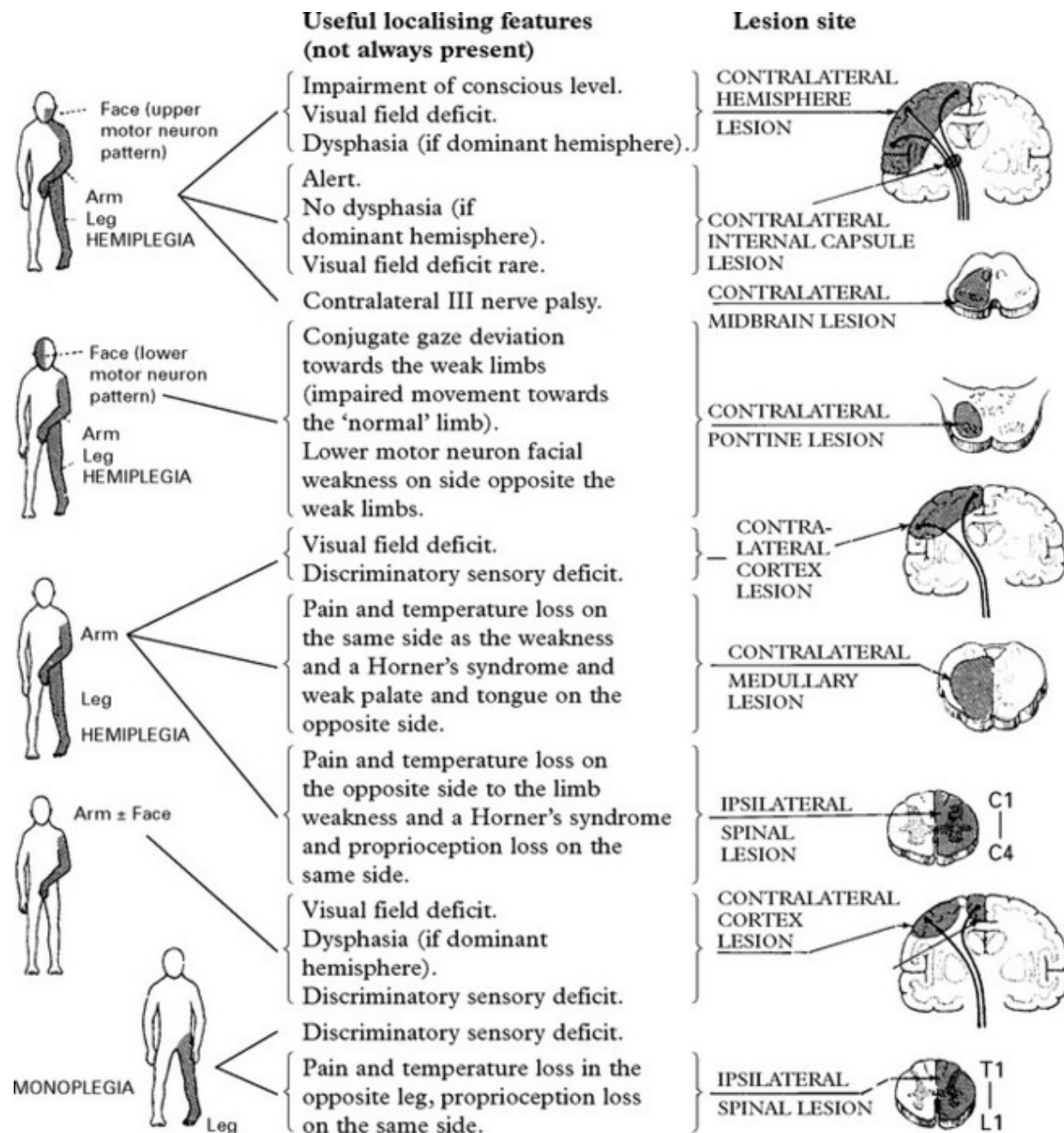
Reflexes – depressed

Distribution – usually proximal weakness, though specific patterns can occur in particular myopathies (e.g. fascioscapulohumeral muscular dystrophy)

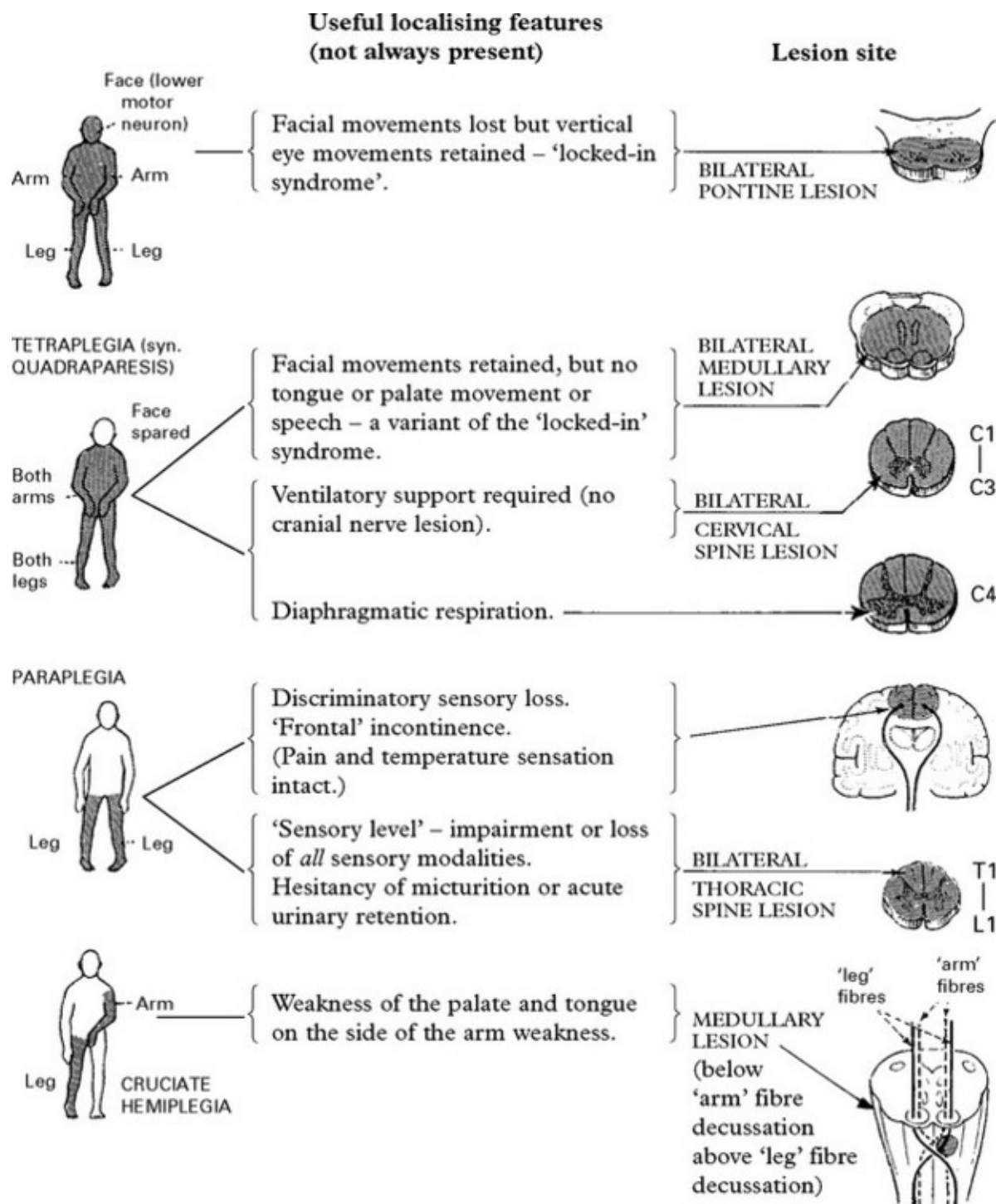
LESION LOCALISATION

The foregoing clinical features readily distinguish weakness of an upper motor neuron, lower motor neuron or mixed pattern. Combining these findings with other neurological signs enables localisation of the lesion site.

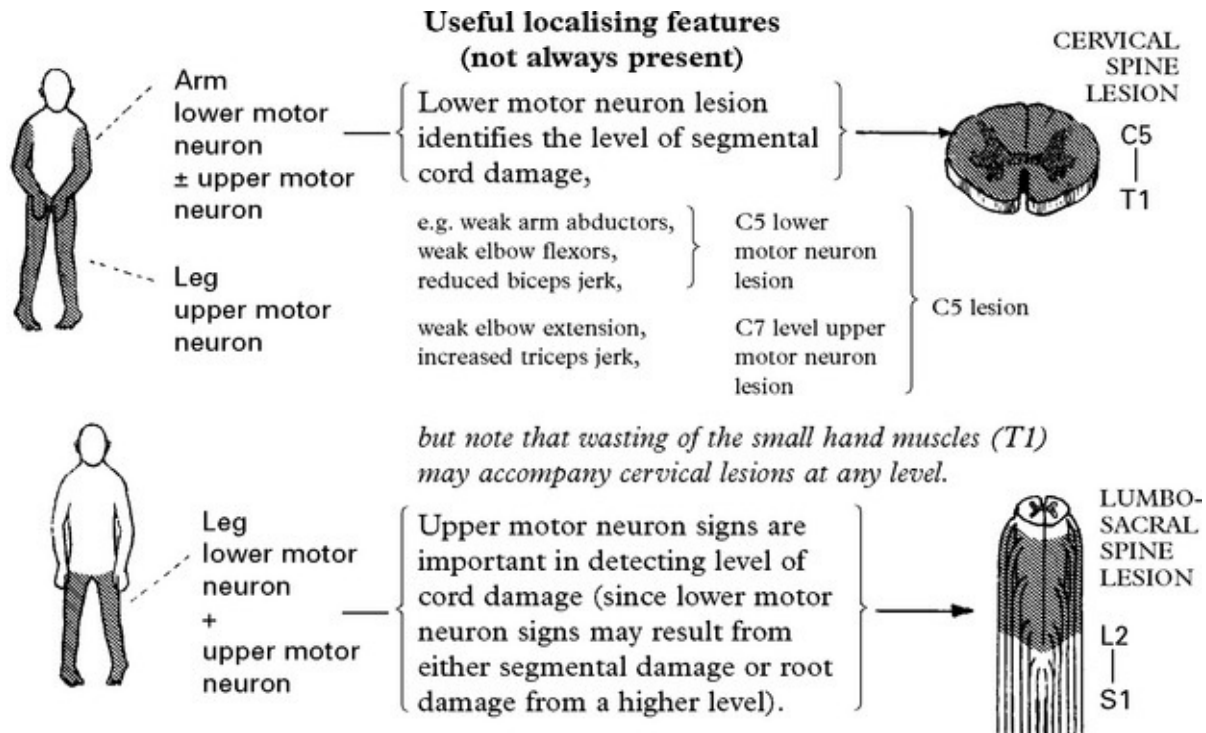
UPPER MOTOR NEURON LIMB WEAKNESS – UNILATERAL



UPPER MOTOR NEURON LIMB WEAKNESS – BILATERAL

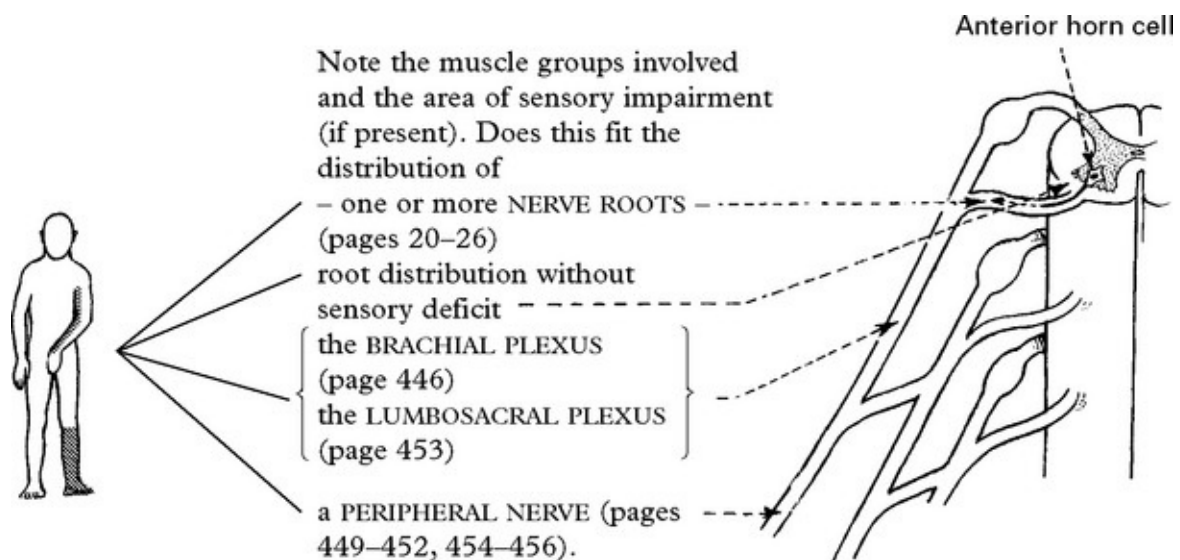


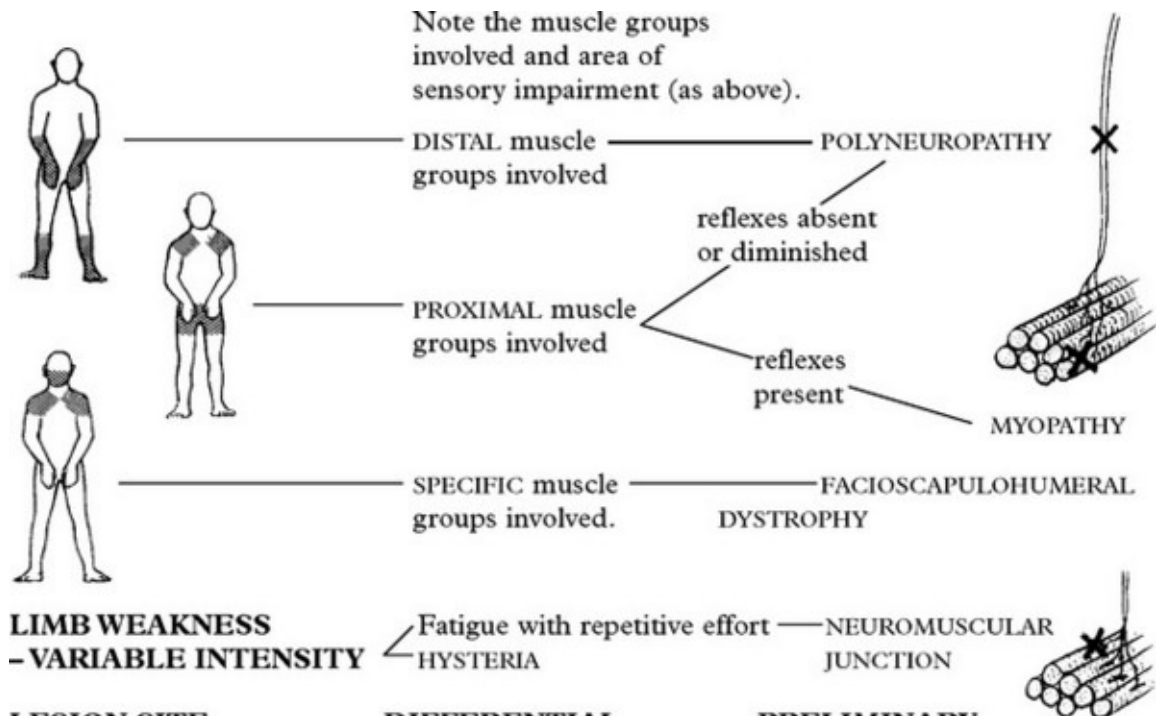
MIXED UPPER AND LOWER MOTOR NEURON WEAKNESS – UNILATERAL OR BILATERAL



N.B. Dual lesions, e.g. cervical + lumbar spondylosis may cause mixed (umnn and lmn) signs in both arm and leg.

LOWER MOTOR NEURON LIMB WEAKNESS – BILATERAL





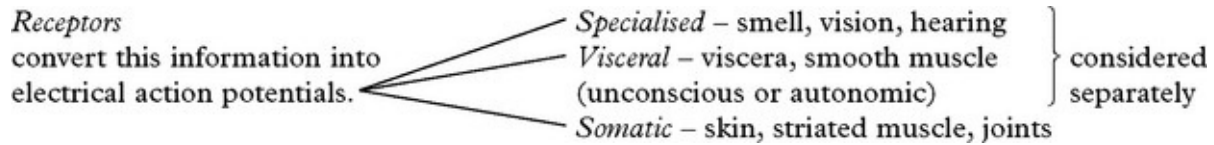
LESION SITE	DIFFERENTIAL DIAGNOSIS	PRELIMINARY INVESTIGATIONS
Cerebral hemispheres, midbrain, pons, medulla	Vascular Tumour Infection	CT scan/MRI
Spinal cord	Demyelination Demyelination Spondylosis/disc disease Tumour Infection Vascular	MRI Visual evoked potentials CSF oligoclonal bands Straight X-ray MRI CT myelography
Anterior horn cell (± spinal cord)	Motor neuron disease (progressive muscular atrophy)	Electromyography (EMG)
Nerve roots	Spondylosis/disc disease Tumour	CT scan/MRI (myelography – cervical roots, radiculography – lumbar roots)
Plexus/peripheral nerves	Peripheral neuropathy Trauma Tumour infiltration	EMG Nerve conduction studies
Neuromuscular junction	Myasthenia gravis Myasthenic syndrome	EMG, Tensilon test
Muscle	Myopathy Dystrophy	EMG, Muscle biopsy

SENSORY IMPAIRMENT

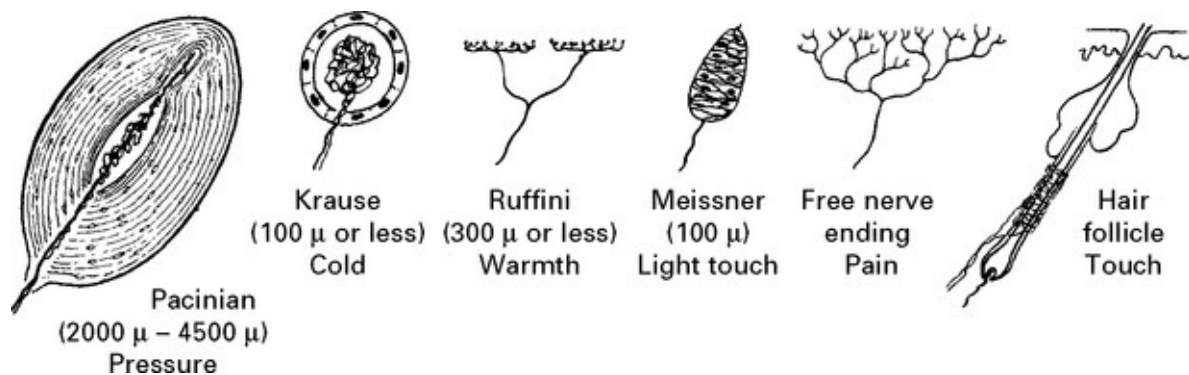
ANATOMY AND PHYSIOLOGY

The sensory system relays information from both the external and the

internal environment.



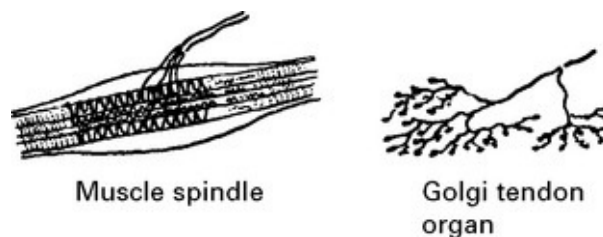
Cutaneous receptors are of several types and, while overlap does occur, each has some specific purpose.



Muscle and tendon receptors

These receptors along with those of pressure and touch provide information on body and limb position – proprioception.

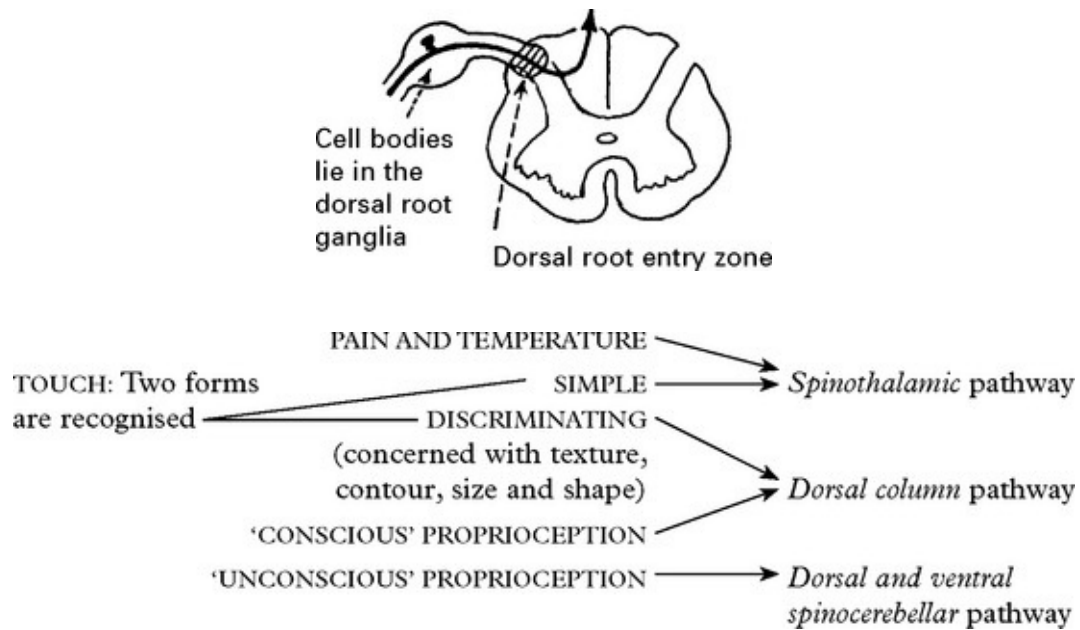
Continual stimulation of most receptors results in a reduction in the action potential frequency – ADAPTATION



CENTRAL CONNECTIONS

Sensory neurons (bipolar cells) relay information to the spinal cord via

the dorsal root to the dorsal root entry zone. The anatomical and physical characteristics of the neurons vary depending on the information they carry, as do the central pathways:

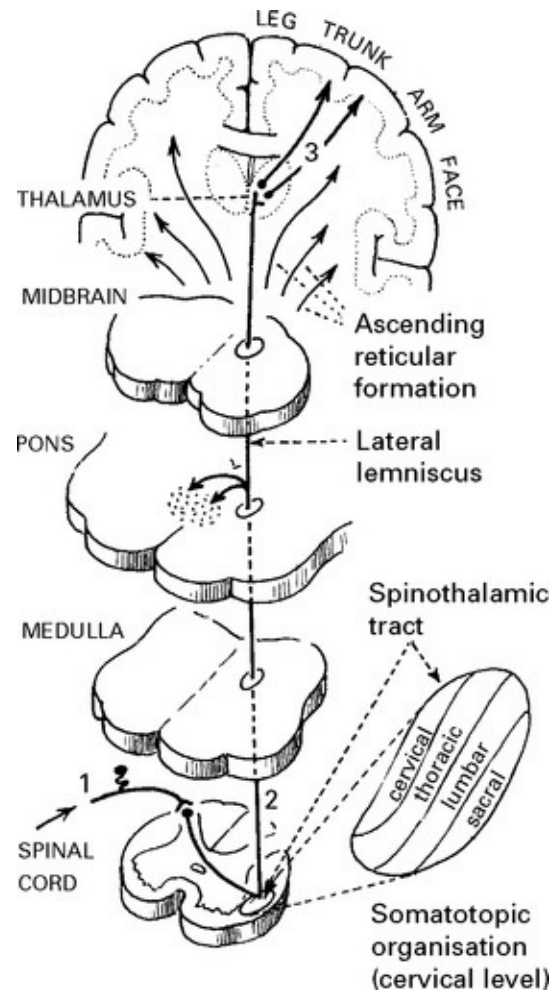


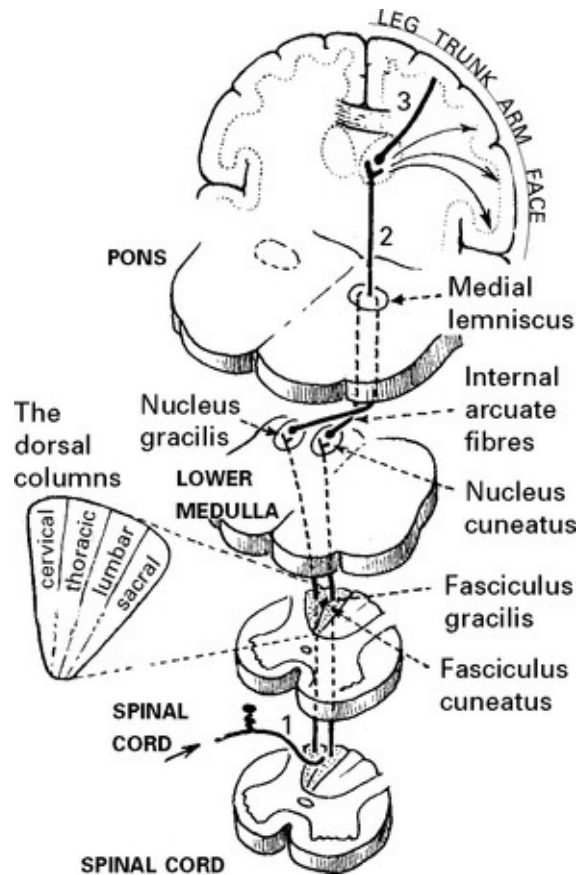
SPINOTHALAMIC PATHWAY

1. Fibres enter the root entry zone and pass up or down for several segments in *Lissauer's* tract before terminating in the dorsal aspect of the dorsal horn.

2. *Second order neurons* synapse locally, cross the midline and run up the *spinothalamic* tract and *lateral lemniscus* to terminate in the posterolateral nucleus of the *thalamus*. Throughout its course, the fibres lie in a *somatotopic arrangement* with sacral fibres outermost. In the brain stem the lateral lemniscus gives off collateral branches to the *reticular formation*, which projects widely to the cerebral cortex and limbic system and is joined by fibres from the contralateral nucleus and tract of the trigeminal nerve.

3. From the thalamus, *third order neurons* project to the *parietal cortex*.





DORSAL COLUMN PATHWAY

1. Fibres enter in the root entry zone and run upwards in the *dorsal columns* to the *lower medulla* where they terminate in the *nucleus gracilis* and *nucleus cuneatus*.
2. *Second order neurons* decussate as the *internal arcuate fibres* and pass upwards in the *medial lemniscus*. Maintaining a *somatotopic arrangement*, they terminate in the ventral posterolateral thalamus.
3. *Third order neurons* arise in the thalamus and project to the *parietal cortex*.

DORSAL AND VENTRAL SPINOCEREBELLAR PATHWAYS: see Cerebellar dysfunction, [page 181](#).

EXAMINATION OF THE SENSORY SYSTEM: see [page 21](#)

CLINICAL FEATURES

Sensory disturbance may result in:

NEGATIVE symptoms:	'a loss of feeling'
	'a deadness'.
POSITIVE symptoms:	'a pins and needles sensation'
	'a burning feeling'.

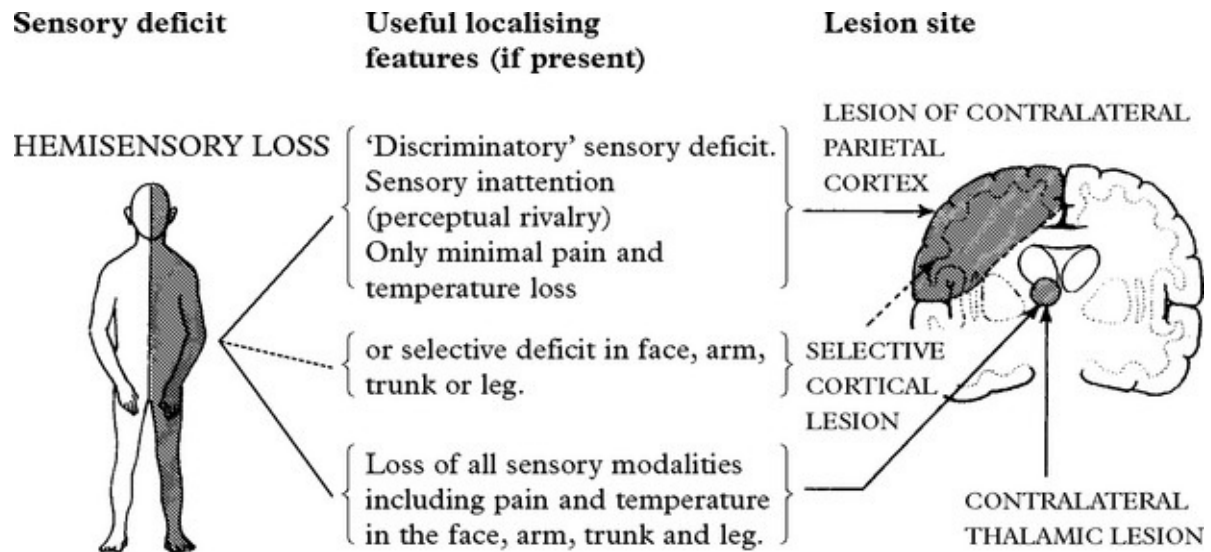
Lesions of the PERIPHERAL NERVES or NERVE ROOTS may produce 'negative' or 'positive' symptoms.

SPINOTHALAMIC TRACT lesions – seldom produce pain but usually a <i>lack of awareness of pain and temperature</i>. This may result in: – trophic changes: cold, blue extremities hair loss brittle nails – painless burns – joint deformation (Charcot's joints).	DORSAL COLUMN lesions – produce a <i>discriminatory type of sensory loss</i>. – impaired two point discrimination – astereognosis (failure to discriminate objects held in the hand). – sensory ataxia (disturbed proprioception).
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Lesions of the PARIETAL CORTEX also produce a *discriminatory type of sensory loss*. Minor lesions produce *sensory inattention* (perceptual rivalry) – with bilateral simultaneous limb stimulation, the stimulus is only perceived on the unaffected side.

LESION LOCALISATION

The pattern of the sensory deficit aids lesion localisation.



Sensory deficit

Useful localising features (if present)

Lesion site

FACIAL
SENSORY
LOSS

HEMISENSORY
LOSS



Loss of all modalities in the limbs (depending on the extent of the lesion)
Loss of pain and temperature on the opposite side of the face with or without 'muzzle' area sparing and a lateral gaze palsy towards that side.

CONTRALATERAL
PONTINE
LESION



(Ipsilateral to the facial sensory loss)

As above – but lateral gaze normal.
Weakness of palate and tongue on side opposite to the limb sensory deficit.

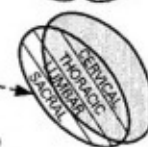
CONTRALATERAL
MEDULLARY
LESION



CONTRALATERAL
SPINOTHALAMIC
TRACT LESION



(Partial
spinothalamic
tract
lesion)



Loss of pain, temperature and light touch below a specific dermatome level (may spare sacral sensation).

Loss of all modalities at one or several dermatome levels.

Loss of pain and temperature below a specific dermatome level.

Loss of proprioception and 'discriminatory' touch up to similar level and limb weakness.

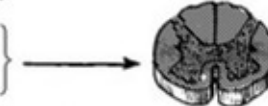
BROWN-SEQUARD
SYNDROME



(Partial cord lesion)

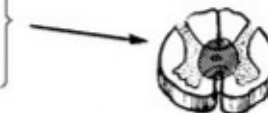


Bilateral loss of all modalities.
Bilateral leg weakness.



COMPLETE CORD LESION

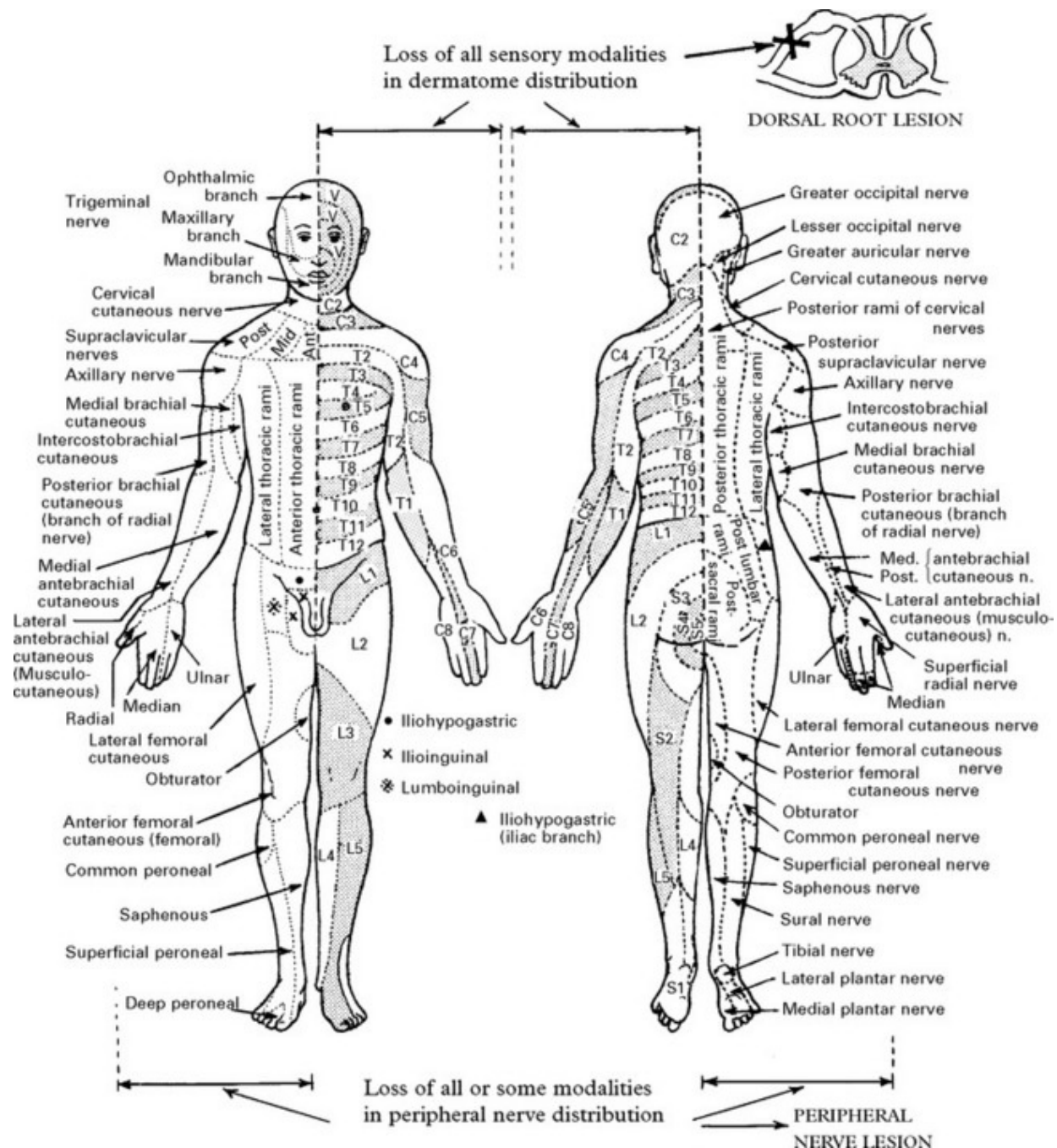
Bilateral loss of pain and temperature.
Preservation of proprioception and 'discriminatory' sensation.



CENTRAL CORD LESION

'SUSPENDED'
SENSORY
LOSS





DIFFERENTIAL DIAGNOSIS –as for limb weakness – [page 198](#)

PAIN

Peripheral receptors of pain – free nerve endings lying in skin or other organs – are the distal axons of sensory neurons. Such unmyelinated or

only thinly myelinated axons are of small diameter. The termination and central connections of these axons are described on [page 200](#).

The type of stimulus required to activate free endings varies, e.g. in muscle – ischaemia, in abdominal viscera – distension.

Certain substances – bradykinins, prostaglandins, histamine – may stimulate free nerve endings.
These substances are released in damaged tissue.



CONTROL OF SENSORY (PAIN) INPUT

The gate control theory

A relay system in the posterior horn of the spinal cord modifies pain input. This involves interneuronal connections within the substantia gelatinosa (a layer of the posterior horn which extends throughout the whole length of the spinal cord on each side).

An afferent impulse arriving at the posterior horn in *thick myelinated fibres* has an inhibitory effect in the region of the substantia gelatinosa.

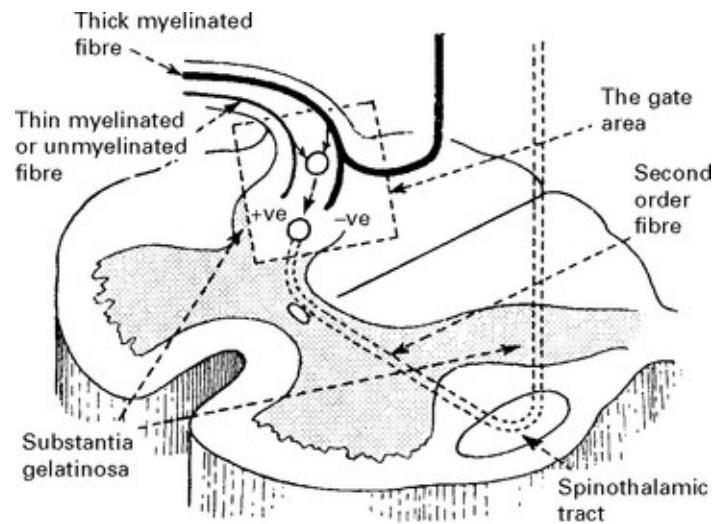
An afferent impulse arriving in *thin myelinated or unmyelinated fibres* (i.e. transmitting pain) has an excitatory effect in the region of the substantia gelatinosa.

The overall interaction of these inhibitory or excitatory effects determines the activity of second order neurons of the spinothalamic pathway.

A reduction in activity of large sensory fibres ‘opens’ the gate. Stimulation of large sensory fibres theoretically ‘closes’ the gate.

In addition to these segmental influences, higher centres also control

the gate region and form part of a feedback loop.



Cross section of the spinal cord: the gate area connections

Pain perception

The awareness of pain is brought about by projection from the thalamus to cerebral cortex. Personality, mood and neuroticism all influence the intensity of pain perception. Diffuse projections through Lissauer's tract and the reticular core of the spinal cord white matter to the reticular formation and limbic system probably contribute to the unpleasant, emotionally disturbing aspects of pain.

NEUROTRANSMITTER SUBSTANCES

Evidence based on both human and animal studies has shown that an endogenous system, lying within the central nervous system can induce a degree of analgesia. Electrical stimulation of certain sites, such as the periaqueductal grey matter, can inhibit pain perception.

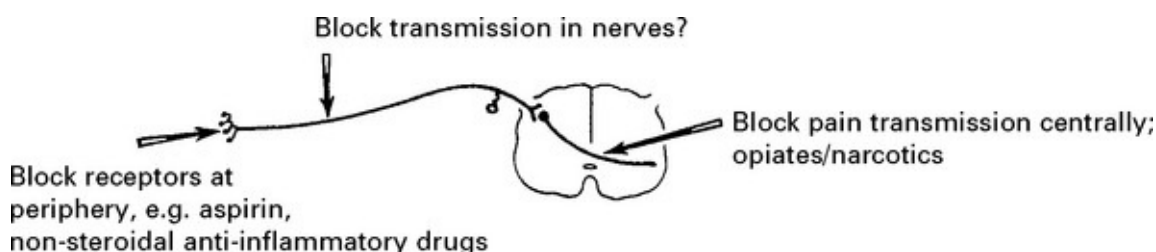
Receptor sites for endogenous opiates have been found in the posterior horns and thalamus as well as at several other sites. The

endogenous substances which bind to these sites are called *encephalins* or *endorphins*.

Substance P, a polypeptide, found predominantly around free nerve ending receptors and in the spinal cord posterior horns, glutamate and calcitonin gene related peptide are the likely primary transmitters of pain.

DRUG TREATMENT

Sites of potential drug action:



Drug selection in pain treatment depends on the severity, cause and the expected duration the pain, i.e. <i>acute</i> pain – less than 2 weeks duration, e.g. postoperative, post-traumatic, renal colic.		
<i>chronic</i> pain	– <i>benign</i> origin, e.g.	postherpetic neuralgia
		phantom limb pain
		chronic back pain.
	– <i>malignant</i> origin.	

1. In acute pain, drug therapy ranges from *mild analgesics* – aspirin, paracetamol – to *narcotic agents* – morphine, heroin. *Tranquillisers* may also help.
2. In chronic pain of benign origin, narcotics and sedatives must be avoided. In these patients, depression usually plays a rôle and the

clinician must not underestimate the value of *tricyclic antidepressants*.

Anticonvulsants – gabapentin and carbamazepine appear to benefit many patients, probably due to their membrane stabilizing effect.

Topical treatment – capsaicin blocks substance P and inhibits pain transmission in the skin. Used for postherpetic neuralgia.

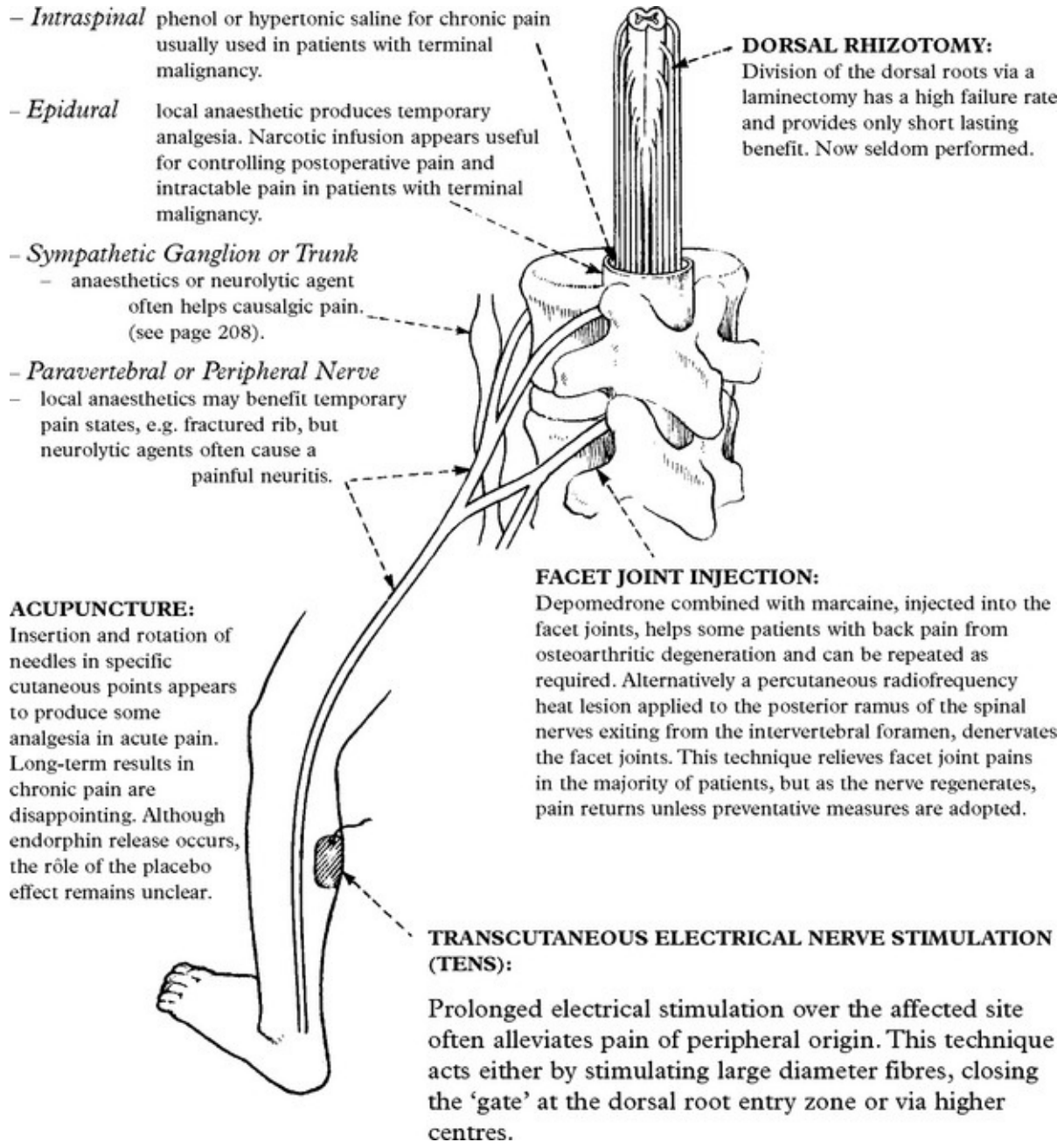
3. In chronic pain from terminal malignancy, patients often require *strong narcotics* – morphine, heroin. Frequent administration of small doses provides the greatest effect.

PAIN – TREATMENT

PERIPHERAL TECHNIQUES

Generally used for more benign conditions and before resorting to central techniques.

NERVE BLOCKS: Injections of agents into peripheral nerves or roots abolishes pain in the appropriate dermatome; motor and sympathetic function are also lost. Local anaesthetics produce a temporary effect; neurolytic agents, *e.g.* phenol, alcohol, give permanent results.



CENTRAL TECHNIQUES

Used primarily in patients with intractable pain from malignancy

PRECENTRAL (MOTOR) CORTEX STIMULATION:

Promising technique in patients suffering hyperpathic pain after stroke or trigeminal territory neuropathic pain.

DEEP BRAIN STIMULATION: Stimulation of implanted electrodes inserted in the periventricular grey matter or sensory relay nucleus of the thalamus may produce relief in patients with neuropathic pain. If successful, a radiocontrolled stimulator is implanted subcutaneously.

HYPOPHYSECTOMY:

By transphenoidal excision or with radioactive yttrium may help pain from metastatic deposits. The mechanism of relief remains uncertain; this is not merely due to tumour regression.

MESENCEPHALOTOMY:

A radiofrequency heat lesion in a stereotactically implanted electrode inserted into the midbrain reticular formation may help patients with head and neck malignancy.

PERCUTANEOUS ANTEROLATERAL CORDOTOMY: A percutaneous radiofrequency heat lesion of the spinothalamic tract now replaces open cordotomy. This produces pain relief in 90% of patients in the contralateral limbs. It is usually applicable in malignant states where simple methods of pain control have failed. Risks (ipsilateral limb weakness and respiratory difficulties) are small.

DORSAL ROOT ENTRY ZONE LESIONS

Following cord exposure, multiple radiofrequency heat lesions of the dorsal root entry zone are produced with a hand held electrode. This may help deafferentation pain, i.e. brachial plexus avulsion, but ipsilateral leg weakness is a potential complication.

MYELOTOMY: Exposure of the cord and division of the decussating pain fibres produces pain relief on a temporary basis, restricting use to patients with terminal malignancy.

SPINAL CORD STIMULATION: Stimulation of electrodes inserted percutaneously or by open surgery into the epidural space may benefit patients with chronic pain, unresponsive to non-invasive techniques, provided the dorsal columns remain at least partially functional, e.g. when lesions are distal to the dorsal root ganglion.

PAIN SYNDROMES

Pain is not primarily a pathological phenomenon, but serves a protective function. Conditions with loss of pain perception exemplify this,

resulting in frequent injuries, burns and subsequent mutilations, *e.g.* syringomyelia, hereditary sensory neuropathy, congenital insensitivity to pain.

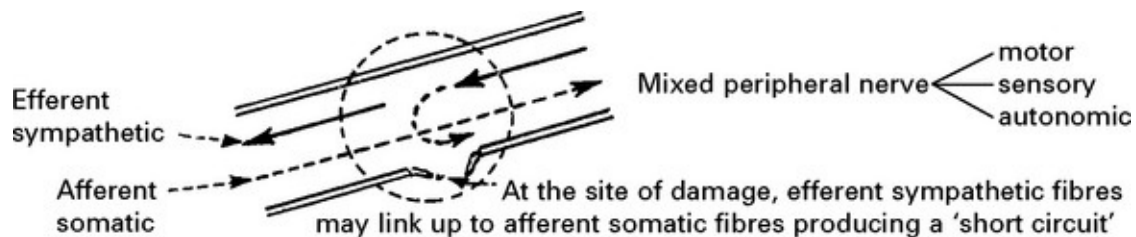
Pathological conditions do, however, cause pain – as a symptom of cancer, injury or other disease.

The following conditions produce characteristic pain syndrome.

CAUSALGIA (Complex Regional Pain Syndrome)

Causalgia is an intense, continuous, burning pain produced by an incomplete peripheral nerve injury. Touching the limb aggravates the pain, and the patient resents any interference or attempt at limb mobilisation. The skin becomes red, warm and swollen.

Theoretical mechanism



Causalgia only occurs with damage to peripheral nerves containing a large number of sympathetic fibres and responds in part to sympathetic blockade (pharmacological or surgical).

POSTHERPETIC NEURALGIA

Following activation of a latent infection with varicella zoster virus lying dormant in the dorsal root or gasserian ganglion, the patient develops a burning, constant pain with severe, sharp paroxysmal twinges over the area supplied by the affected sensory neurons. Touch exacerbates the

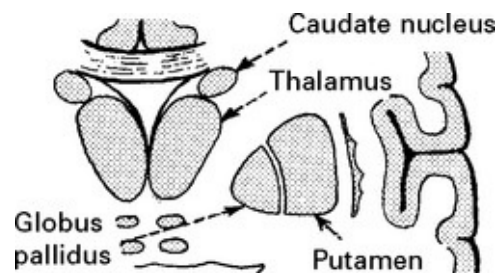
pain. Thick myelinated fibres are preferentially damaged, possibly opening the 'gate'.

Treatment of postherpetic neuralgia is particularly difficult. Carbamazepine and/or antidepressants may help. Ethylchloride spray over the affected area provides temporary relief. Capsaicin, a topical NSAID can be an effective treatment.

THALAMIC PAIN

Thalamic stimulation may produce or abolish pain depending upon the electrode site. A vascular accident which involves the inhibitory portion of the thalamus may result in pain – the thalamic syndrome.

Clinical features: Hemianaesthesia at onset contralateral to the lesion precedes the development of pain. This is burning and diffuse, and exacerbated by the touch of clothing.



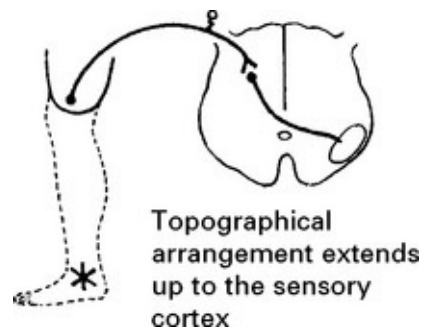
Treatment: Drug treatment gives poor results. A stereotactic procedure although increasing the sensory deficit may help.

Paradoxically the thalamic syndrome may occur following a thalamic stereotactic procedure for movement disorders.

PHANTOM LIMB PAIN

Following amputation of a limb, 10% of patients develop pain with a

continuous persistent burning quality, caused by neuroma formation in the stump. The patient 'feels' the pain arising from some point on the missing limb (the pain input projects through pathways which retain the topographical image of the absent limb).



Treatment: Often responds to simple measures e.g. tricyclic antidepressants.

VISCERAL AND REFERRED PAIN

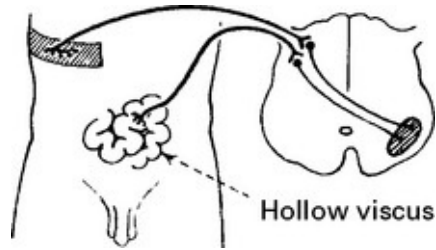
Deep visceral pain is dull and boring; it is the consequence of distension or traction on free nerve endings.

Referred pain of a dull quality relates to a specific area of the body surface – often hypersensitive to touch.

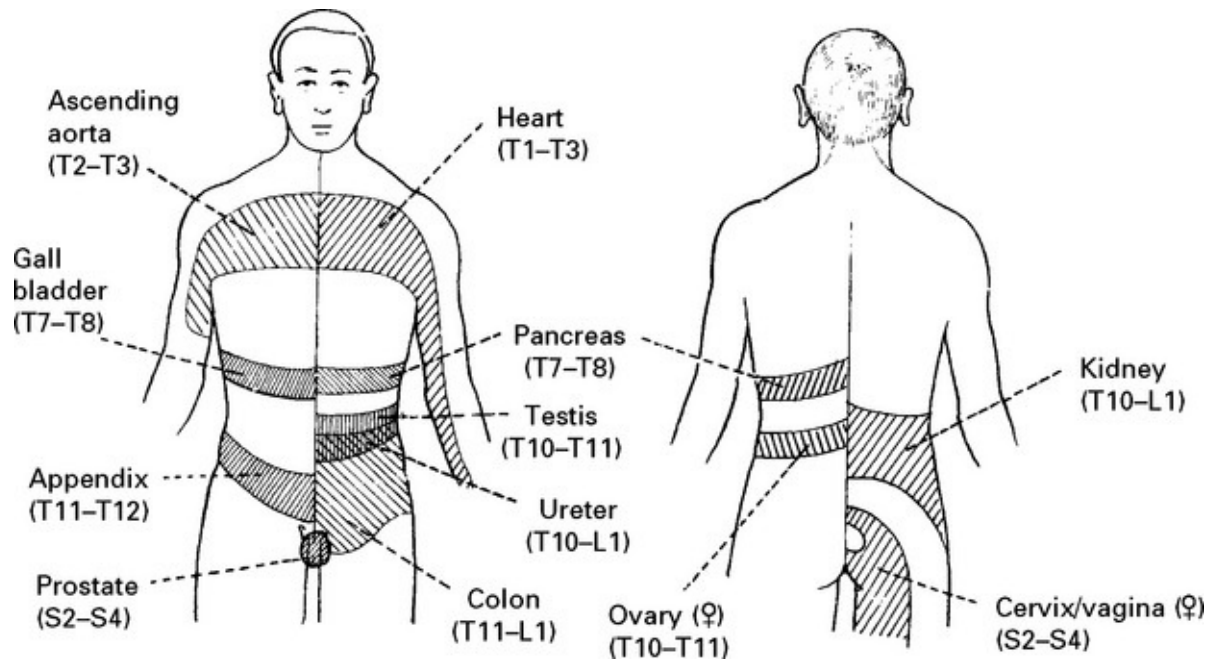
The basis of referred pain

The visceral afferents converge upon the same cells in the posterior horns as the somatic efferents. The patient 'projects' pain from the viscera to the area supplied by corresponding somatic afferent fibres.

A knowledge of the source of referred pain is important in diagnosis and treatment.



SITES OF REFERRED PAIN FROM SPECIFIC ORGANS

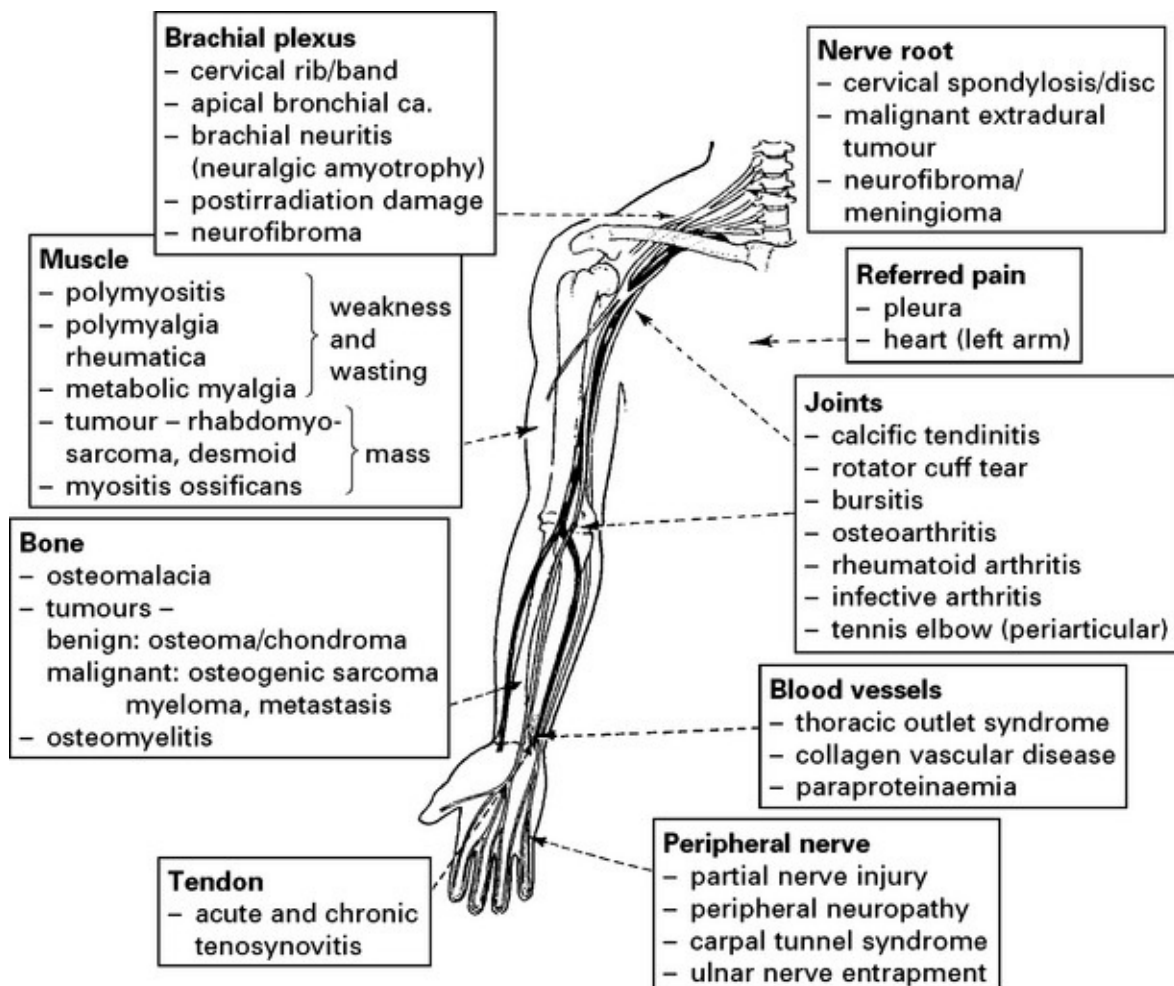


LIMB PAIN

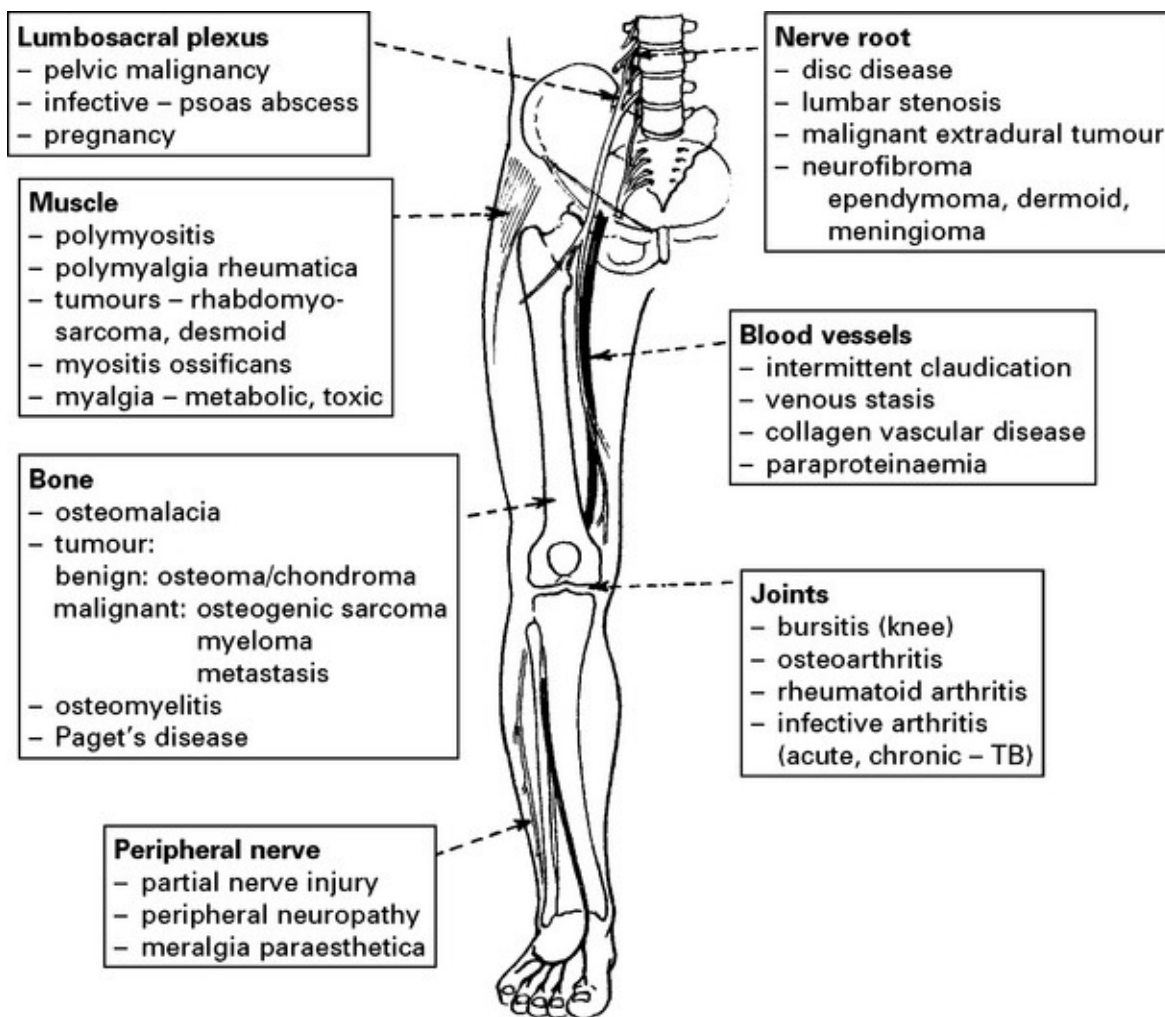
Pain may arise from any anatomical structure within the limb. Each produces characteristic features:

BONE – diffuse, aching pain ± palpable mass.	BLOOD VESSELS – pain brought on by exertion (claudication), relieved by rest. – pain at rest in pale, pulseless limb (occlusion). – pain associated with paraesthesia and digital pallor (Raynaud's).
JOINTS – pain localised to affected joint. – tenderness on palpation. – movements restricted and painful. – wasting of surrounding muscles may follow.	NERVE ROOT – pain increased by coughing or by movement ± associated neurological deficit
MUSCLES – pain localised to specific muscle ± wasting and weakness ± palpable mass.	PLEXUS OR PERIPHERAL NERVE – burning pain ± sweating, cyanosis and oedema of extremity, ± associated neurological deficit.
TENDONS – pain localised to swollen, tender tendon sheath.	

CAUSES OF UPPER LIMB PAIN



CAUSES OF LOWER LIMB PAIN



Meralgia paraesthetica: burning, tingling pain over the outer aspect of the thigh, increased when standing or by walking, due to a localised neuritis of the lateral cutaneous nerve of the thigh.

Restless legs syndrome (syn. Ekbom's syndrome): occurs in about 2% of population. An intolerable tingling, burning sensation or pain in both legs, occurring only when sitting or lying down and relieved by walking; no associated neurological abnormality. Often responds to dopamine agonists (ropinerole and pramipexole), L-dopa and

gabapentin.

Investigation of limb pain depends on the suspected cause and may include straight X-rays, CT scan, MRI, nerve conduction studies and EMG.

MUSCLE PAIN (MYALGIA)

Muscle pain is a common medical complaint. There are many causes and clinical evaluation and appropriate investigation is often difficult. The physiological mechanisms producing such a symptom are limited.

Mechanical pain results from excessive muscle tension or contraction and is 'cramp like'.

Inflammatory pain results from disruption of muscle fibres, inflammatory exudate and fibre swelling.

Ischaemic pain results from metabolic change, usually in response to exercise and is deep and aching.

Muscle pain may be physiological – as a consequence of extreme exercise or pathological – as a consequence of muscle, soft tissue or systemic illness.

DIAGNOSTIC APPROACH TO MUSCLE PAIN

History

Is muscle pain – present at rest?	– Polymyalgia rheumatica – Fibromyalgia – Parkinson's disease – Collagen vascular disease

present with exercise?	– Physiological – Metabolic myopathies – Benign myalgic encephalomyelitis (ME)
localised?	– Muscle haematoma, abscess, tumour or fibromyalgia
generalised?	– Polymyalgia rheumatica – Parkinson's disease – Metabolic myopathies – Inflammatory myopathies – Benign myalgic encephalomyelitis (ME)
family history?	– Metabolic myopathies
exposure to toxins?	– Drug induced myopathies – Alcoholic myopathy

Examination

Is there	– wasting/weakness?	– Inflammatory myopathies – Metabolic myopathies – Drug induced myopathies – Alcoholic myopathy
	skin rash?	– Inflammatory myopathy (dermatomyositis) – Collagen vascular disease
	stiffness or spasms?	– Tetanus

		– Tetany – Spasticity – Neuroleptic malignant syndrome – Malignant hyperthermia
	muscle swelling?	– Muscle abscess, tumour – Metabolic myopathy

Investigations

<i>Serum creatine kinase</i> (muscle enzyme) – elevated in muscle necrosis, high levels result in myoglobinuria	<i>Muscle biopsy</i> (needle or open) Essential in diagnosis of – inflammatory myopathies – metabolic myopathies Helpful in collagen vascular disease
<i>Imaging</i> (occasionally used) Ultrasound, MR or CT in suspected muscle haematoma, abscess or tumour. Radionuclide (Gallium or in suspected muscle abscess, Technitium)	<i>Ischaemic lactate test</i> Measurement of post exercise changes in serum lactate Reduced response in – metabolic myopathies (disorders of glycolytic pathway)
<i>Electromyography</i> (EMG) Will confirm presence of myopathy (rarely more specific)	

Following extensive investigation, in a significant number of cases no cause of myalgia is found.

Most disorders are covered in relevant sections. Those that are not are briefly described.

<i>Fibromyalgia</i>	<i>Polymyalgia rheumatica</i>
A common condition of uncertain pathology in which generalised muscle pain with localised tender areas occurs without objective clinical or laboratory abnormalities. Psychiatric symptoms commonly co-exist.	Proximal muscle pain encountered in the elderly and often associated with giant cell arteritis. The ESR is elevated and the EMG is normal. Muscle biopsy shows type 2 fibre loss. Steroids are effective.
	<i>Muscle tumours</i>
<i>Malignant hyperpyrexia</i>	These are rare. Mixed pathological and of varying degrees of malignancy
Characterised by a sudden rise in body temperature whilst undergoing general anaesthesia, usually with halothane or succinylcholine. Certain hereditary myopathic disorders, e.g. myotonic dystrophy, central core disease – are unduly prone to this condition.	
	<i>Chronic fatigue syndrome/myalgic encephalomyelitis (CFS/ME)</i>
	An idiopathic disorder that may follow viral illness, is often associated with exercise induced muscle pain and associated with fatigue. No clear underlying pathology has been found and diagnosis is based on symptoms and exclusion of other pathology.
<i>Muscle abscess</i>	
Commonly Staphylococcal due to local trauma or blood-borne in debilitated persons.	
	May respond to graded exercise, tricyclic antidepressants or cognitive behavioural therapy.

OUTCOME AFTER BRAIN DAMAGE

Outcome after brain damage has major social and financial implications for both patients and their families. In a welfare state, society may carry most, if not all of the financial burden, particularly with more severe disability. The greater the disability, the greater the support required. Conditions causing brain damage do not respect age; survivors may need long-term care.

A variety of methods have been devised to categorise outcome. Such classifications provide end-points for audit and research, and a means of assessing therapeutic intervention. They permit prediction based on clinical and investigative findings early in the course of the disease. Most outcome scales have been developed with a particular disease in mind (e.g. Bartel/Rankin – stroke, Karnofsky – tumour). In 1975 Jennett and Bond developed the Glasgow Outcome Scale (GOS) for the assessment of head injured patients, and this is now widely applied in the assessment of patients with other causes of brain damage.

The Glasgow Outcome Scale	Five categories exist –
1. Death	
2. Persistent Vegetative State	– see below.
3. Severe Disability	– <i>dependent</i> for some support in every 24 hour period.
4. Moderate Disability	– <i>independent</i> but disabled. May or may not be capable of return to work.
5. Good recovery	– <i>good</i> , but not necessarily complete recovery e.g. cranial nerve deficit.

The Vegetative State

Severe bilateral hemisphere damage may result in a state in which the patient has no awareness of themselves or of their environment. Although periods of eye opening and closure may occur suggesting sleep/wake cycles, along with spontaneous movements of the face, trunk and limbs, the patient does not communicate or interact with others in any way.

The vegetative state becomes 'permanent' when irreversibility can be established with a high degree of certainty, *i.e.* > 6 months after non-traumatic coma and > 12 months after traumatic coma. At one month after *trauma*, about 1/3 of patients in the vegetative state will show some improvement over the subsequent year. After *non-traumatic coma*, outcome is much worse; only about 7% show some improvement and have severe disability.

Outcome Prediction

Outcome from **non-traumatic** coma depends on a variety of factors including the patient's *age*, the *duration* and *depth* of the coma, and the *cause* of the damage provided this is not drug induced.

		Poor outcome (GOS 1-3)	Favourable outcome (GOS 4-5)
Duration	Infective/metabolic	65%	35%
	Hypoxic – ischaemic	90%	10%
	> 6 hours	85%	15%
Depth	Absent pupillary response at 24 hours	100%	0%
	Speaking, eye movements and reactive pupils at 2 hours	0%	100%

Outcome from **traumatic coma** see [page 238](#).

BRAIN DEATH

The advent of improved intensive care facilities and more aggressive resuscitation techniques has led to an increase in numbers of patients with irreversible brain damage in which tissue oxygenation is maintained by a persistent heart beat and artificial ventilation.

A government working party has published guidelines for the diagnosis of brain death which, when fulfilled, indicate that recovery is impossible. In these patients, organs may be removed for transplantation before discontinuing ventilation.

The tests are designed to detect failure of *brain stem* function, but certain *preconditions* must first be met.

Preconditions

Depressant drugs must not contribute towards the patient's clinical state – if in doubt allow an adequate time interval to elapse to eliminate any possible persistent effect.

Hypothermia must not be a primary cause – ensure that temperature is not less than 35°C.

Severe *metabolic* or *endocrine* disturbance must be excluded as a possible cause of the patient's condition.

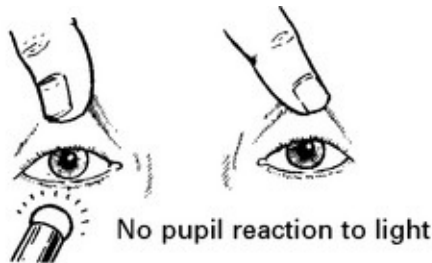
The patient must be on a ventilator as a result of inadequate spontaneous respiration or respiratory arrest – if a neuromuscular blocking drug has been used, exclude a prolonged effect by observing a muscle twitch on nerve stimulation, e.g. electrical stimulation of the median nerve should cause a thumb twitch.

The cause of the patient's condition must be established and this

must be compatible with irreversible brain damage, e.g. severe head injury, spontaneous intracerebral haematoma. *If in doubt, delay brain death testing.*

BRAIN DEATH TESTS

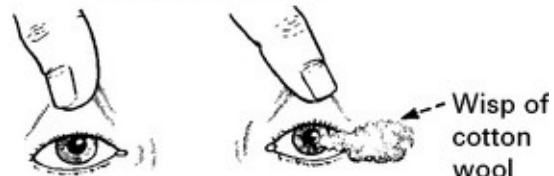
PUPIL RESPONSE



No pupil reaction to light

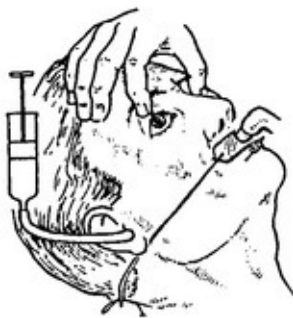
N.B. Ensure light intensity is adequate.

CORNEAL REFLEX



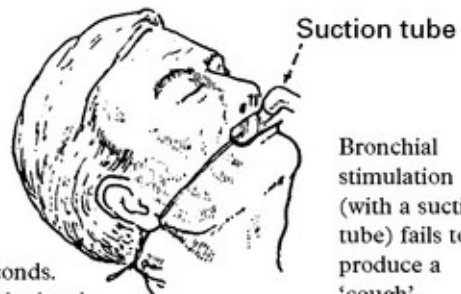
No orbicularis oculi contraction in response to corneal stimulation.

VESTIBULO-OCULAR REFLEX



No eye movements occur when 50 ml of iced water are slowly injected into the external meatus. (Ensure that the external meatus is not occluded with wax or blood.) In coma with preserved brain stem function, the eyes tonically deviate towards the tested ear after a delay of 20 seconds. Maximal response is obtained with the head raised 30° from the horizontal.

GAG REFLEX



Suction tube
Bronchial stimulation (with a suction tube) fails to produce a 'cough' response.

MOTOR RESPONSE

No motor response in the face or in the muscle supplied by cranial nerves in response to a painful stimulus, e.g. supraorbital pain.



N.B. Limb responses are of no value in testing brain stem integrity. Movements can occur in response to limb or trunk stimulation (especially in the legs), and tendon reflexes may persist in a patient with brain stem death but intact cord function. Conversely, limb movements and reflexes may be absent in a patient with an intact brain stem and

spinal cord damage.

RESPIRATORY MOVEMENTS

No respiratory movements are observed when the patient is disconnected from the ventilator. During this test, anoxia is prevented by passing 6 litres O₂ per minute down the endotracheal tube. This should maintain adequate *PO*₂ levels for up to 10 minutes. N.B. Ensure that apnoea is not a result of a low *PCO*₂. This should be greater than 6.65 kPa (50 mmHg).

Clinician's status

The British recommendations state that these tests should be carried out by two doctors, both with expertise in the field; one of consultant status, the other of consultant or senior trainee status. The doctors may carry out the tests individually or together.

Test repetition and timing

The test should be repeated but the interval should be left to the discretion of the clinician. The initial test may be performed within a few hours of the causal event, but in most instances is delayed for 12–24 hours, or longer if there is any doubt about the preconditions.

Timing of death

Certification of death occurs when brain death is established, *i.e.* at the time of the second test. Old concepts of death occurring at the time the heart ceases to beat are no longer applicable.

Supplementary investigations

Electroencephalography (EEG) is of no value in diagnosing brain death.

Some patients with the potential to recover show a 'flat' trace; in others with irreversible brain stem damage, electrical activity can occasionally be recorded from the scalp electrodes.

Similarly, angiography or cerebral blood flow measurement are of no additional value to the clinical tests described above, provided the preconditions are fulfilled.

LOCALISED NEUROLOGICAL DISEASE AND ITS MANAGEMENT A. INTRACRANIAL

HEAD INJURY

INTRODUCTION

Many patients attend accident and emergency departments with head injury. Approximately 300 per 100000 of the population per year require hospital admission; of these 9 per 100 000 die, *i.e.* 5000 patients per year in Britain. Some of these deaths are inevitable, some are potentially preventable.

The principal causes of head injury include road traffic accidents, falls, assaults and injuries occurring at work, in the home and during sports. The relative frequency of each cause varies between different age groups and from place to place throughout the country.

Head injuries from road traffic accidents are most common in young males; alcohol is frequently involved. Road traffic accidents, although only constituting about 25% of all patients with head injury, are the cause of more serious injuries. This cause contributes to 60% of the deaths from head injury; of these, half die before reaching hospital.

In many countries preventative and punitive measures controlling alcohol levels and the use of seat belts, air bags and crash helmets have reduced the incidence. Once a head injury has occurred, nothing can alter the impact damage. The aim of head injury management is to minimise damage arising from secondary complications.

PATHOLOGY

Imaging permits the categorisation of brain damage into *focal* and *diffuse*, although often both types coexist. Alternatively brain damage can be classified as *primary* occurring at impact, or *secondary* from ongoing neuronal damage, haematoma, brain swelling, ischaemia or infection.

FOCAL DAMAGE

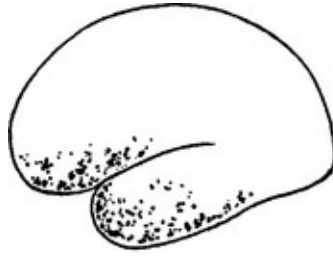
Cortical contusions and lacerations

These may occur under or opposite (contre-coup) the site of impact, but most commonly involve the frontal and temporal lobes. Contusions are usually multiple and may occur bilaterally. Multiple contusions do not in themselves contribute to depression of conscious level, but this may arise when bleeding into the contusions produces a space-occupying haematoma.

Intracranial haematoma

Intracranial bleeding may occur either outside (extradural) or within the dura (intradural).

Intradural lesions usually consist of a mixture of both subdural and intracerebral haematomas although pure subdurals occur in a proportion. Brain damage is caused directly or indirectly as a result of tentorial or tonsillar herniation.



Incidence of haematoma:
Extradural – 27%
Intradural:
Pure subdural – 26%
Intracerebral.
± subdural – 38%
Extra- + Intradural – 8%



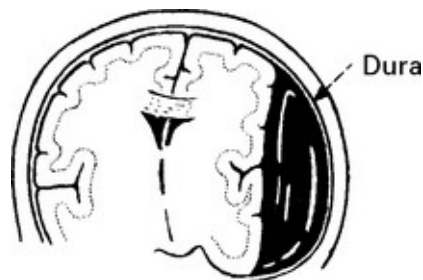
Intracerebral ± subdural (burst lobe)

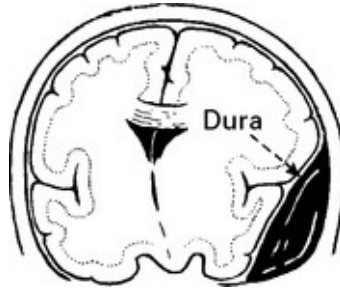
Contusions in the frontal and temporal lobes may bleed into the brain substance, or onto the brain surface producing an overlying subdural haematoma.

'Burst lobe' is a term sometimes used to describe the appearance of intracerebral haematoma mixed with necrotic brain tissue, rupturing out into the subdural space.

Subdural

In some patients impact may rupture bridging veins from the cortical surface to the venous sinuses producing a pure subdural haematoma with no evidence of underlying cortical contusion or laceration.



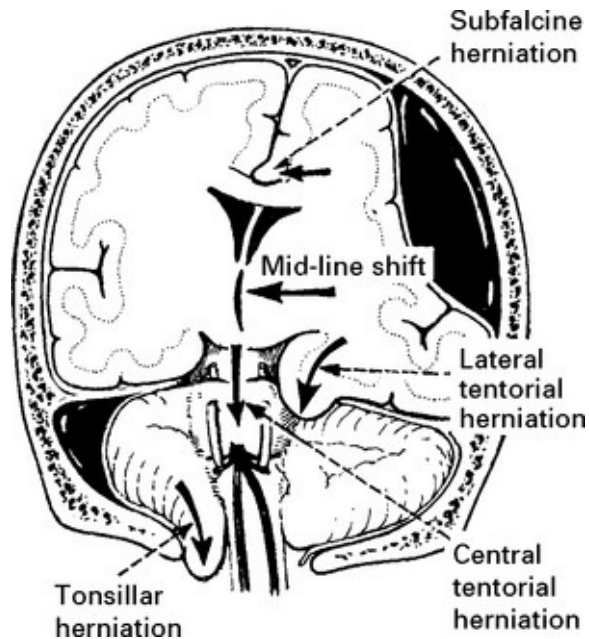


Extradural

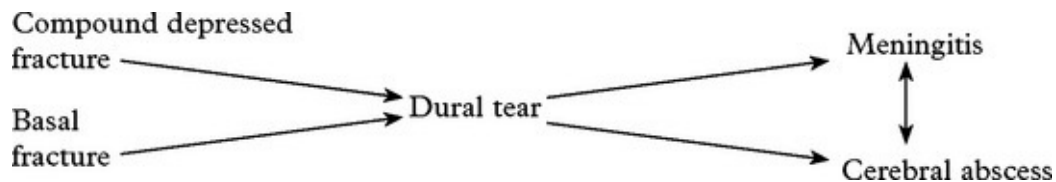
A skull fracture tearing the middle meningeal vessels bleeds into the extradural space. This usually occurs in the temporal or temporoparietal region. Occasionally extradural haematomas result from damage to the sagittal or transverse sinus.

Tentorial/tonsillar herniation (syn. ‘cone’)

It is unlikely that high intracranial pressure alone directly damages neuronal tissue, but brain damage occurs as a result of tonsillar or tentorial herniation (see [page 81](#)). A progressive increase in intracranial pressure due to a supratentorial haematoma initially produces midline shift. Herniation of the medial temporal lobe through the tentorial hiatus follows (*lateral tentorial herniation*), causing midbrain compression and damage. Uncontrolled lateral tentorial herniation or diffuse bilateral hemispheric swelling will result in *central tentorial herniation*. Herniation of the cerebellar tonsils through the foramen magnum (*tonsillar herniation*) and consequent lower brain stem compression may follow central tentorial herniation or may result from the infrequently occurring traumatic posterior fossa haematoma.



Infection



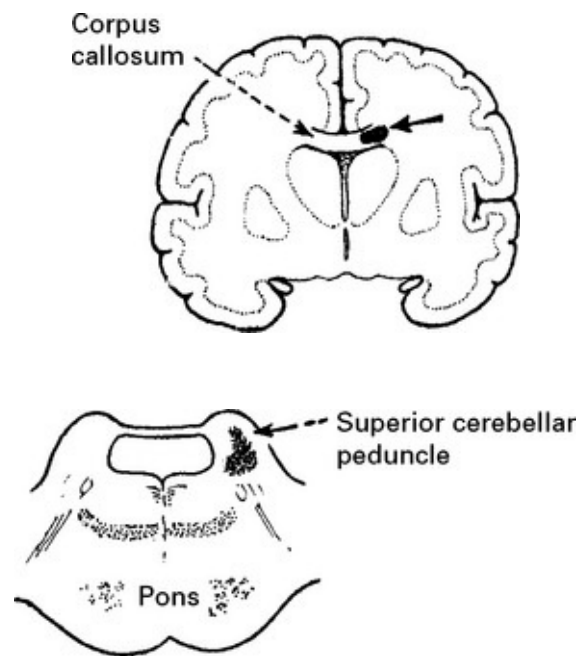
The presence of a dural tear provides a potential route for infection. This seldom occurs within 48 hours of injury. Meningitis may develop after several months or years.

DIFFUSE DAMAGE

Diffuse axonal injury

Shearing forces cause immediate mechanical damage to axons. Over the subsequent 48 hours, further damage results from release of excitotoxic neurotransmitters which cause Ca^{2+} influx into cells and triggers the phospholipid cascade ([page 246](#)). Genetic susceptibility conferred by the presence of the APOE^ε4 gene may also play a part. Depending on the

severity of the injury, effects may range from mild coma to death.



The *macroscopic* appearance may appear entirely normal but in some patients pathological sections reveal small haemorrhagic tears, particularly in the corpus callosum or in the superior cerebellar peduncle.

Microscopic evidence of neuronal damage depends on the duration of survival and on the severity of the injury. After a few days, retraction balls and microglial clusters are seen in the white matter.



Retraction balls reflect axonal damage. Note only axons in one plane are involved, indicating the direction of the 'shear'

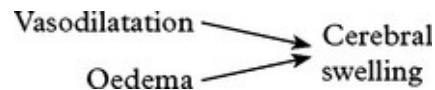


Microglial clusters (hypertrophied microglia) are found diffusely throughout the white matter

If the patient survives 5 weeks or more after injury then appropriate staining demonstrates Wallerian degeneration of the long tracts and white matter of the cerebral hemispheres. Even a minor injury causing a

transient loss of consciousness produces some neuronal damage. Since neuronal regeneration is limited, the effects of repeated minor injury are cumulative.

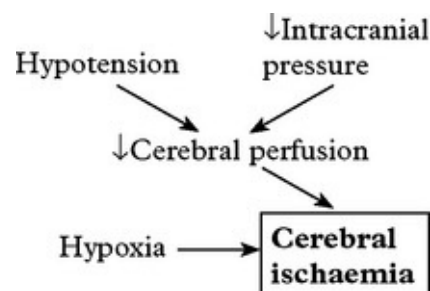
Cerebral swelling



This may occur with or without focal damage. It results from either vascular engorgement or an increase in extra-or intracellular fluid. The exact causative mechanism remains unknown

Cerebral ischaemia

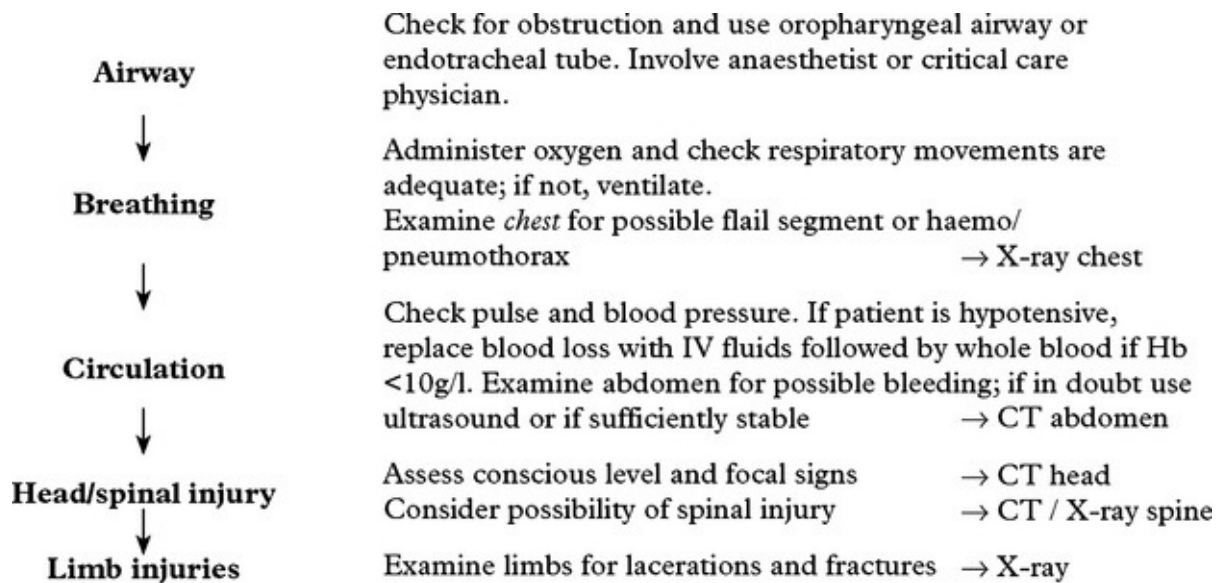
Cerebral ischaemia commonly occurs after severe head injury and is caused by either hypoxia or impaired cerebral perfusion. In the normal subject, a fall in blood pressure does not produce a drop in cerebral perfusion since ‘autoregulation’ results in cerebral vasodilatation. After head injury, however, autoregulation is often defective and hypotension may have more drastic effects. Glutamate excess and free radical accumulation may also contribute to neuronal damage (see [page 246](#)).



MULTIPLE INJURY – PRIORITIES OF ASSESSMENT

Patients admitted in coma with multiple injuries require urgent care and the clinician must be aware of the priorities of assessment and

management.



When intracranial haematoma is suspected, a CT scan is essential, especially before clinical signs are masked by a general anaesthetic required for the management of limb or abdominal injuries. However, if difficulty occurs in maintaining blood pressure, then urgent laparotomy or thoracotomy would take precedence over investigation of a possible intracranial haematoma.

HEAD INJURY – ASSESSMENT

Some patients may describe the events leading to and following head injury, but often the doctor depends on descriptions from witnesses.

Points to determine:

Period of loss of consciousness: relates to severity of diffuse brain damage and may range from a few seconds to several weeks.

Period of post-traumatic amnesia: the period of permanent amnesia occurring after head injury. This reflects the severity of damage and in severe injuries may last several weeks.

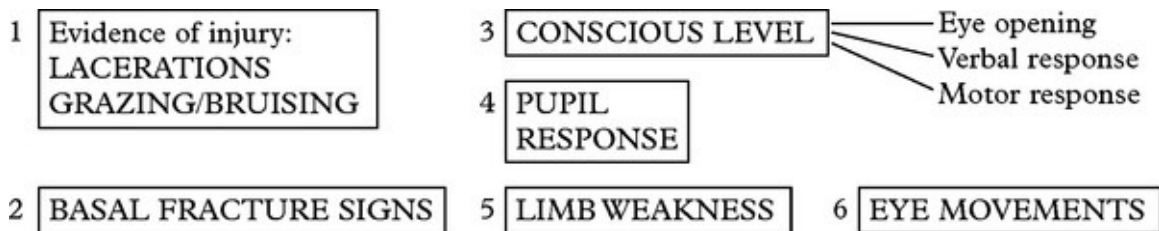
Period of retrograde amnesia: amnesia for events before the injury.

Cause and circumstances of the injury: the patient may collapse, or crash his vehicle as a result of some preceding intracranial event, e.g. subarachnoid haemorrhage or epileptic seizure. The more “violent” the injury, the greater the risk of associated extracranial injuries.

Presence of headache and vomiting: these are common symptoms after head injury. If they persist, the possibility of intracranial haematoma must be considered.

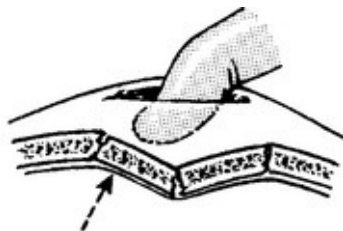
HEAD INJURY – CLINICAL ASSESSMENT

EXAMINATION



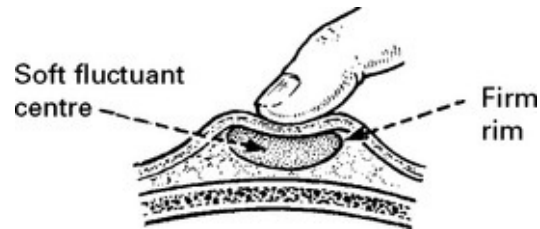
1. Lacerations and bruising

The presence of these features confirms the occurrence of a head injury, but traumatic intracranial haematoma can occur in patients with no external evidence of injury.



Always explore deep lacerations with a gloved finger for evidence of a *depressed fracture*.

Beware of falling into the trap of diagnosing a depressed fracture when only scalp haematoma is present.



Consider the possibility of a hyperextension injury to the cervical spine if frontal laceration or bruising is present.

2. Basal skull fracture

Clinical features indicate the presence of a basal skull fracture which may be hard to detect on CT scan or skull X-ray. If present, a potential route of infection exists with the concomitant risk of meningitis.

ANTERIOR FOSSA FRACTURE
CSF rhinorrhoea



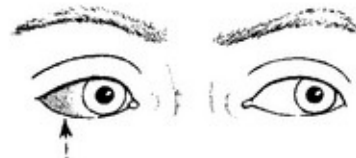
If the nasal discharge contains glucose, then the fluid is CSF rather than mucin.

Bilateral periorbital



Bruising limited to the orbital margins indicates blood tracking from behind.

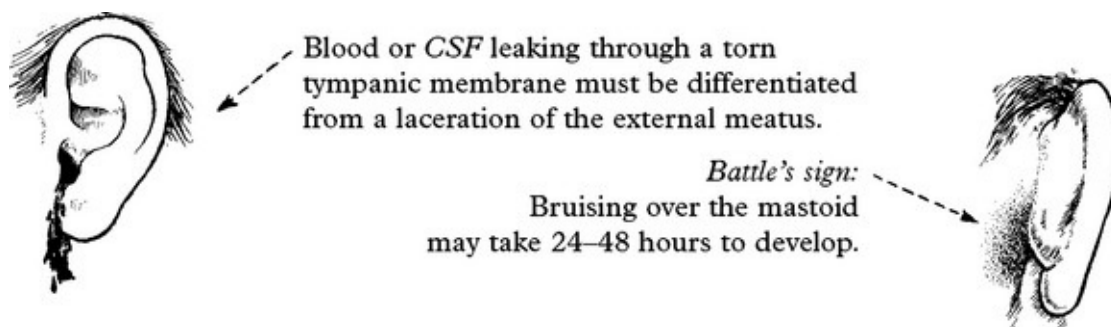
*Subconjunctival haemorrhage
haematoma*



Bruising under conjunctiva extending to posterior limits of the sclera indicates blood tracking from orbital cavity.

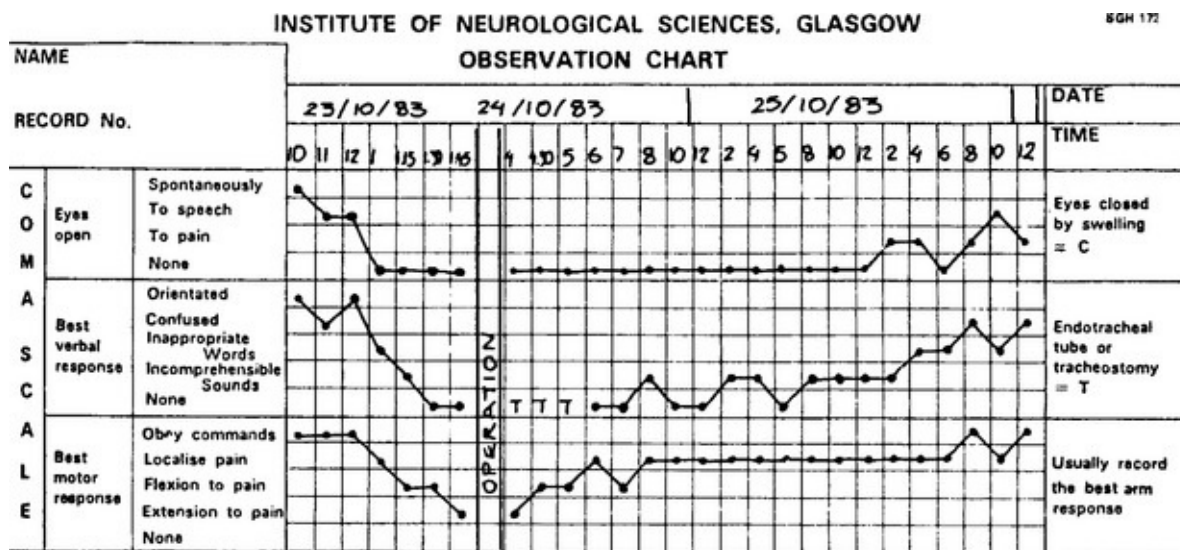
PETROUS FRACTURE

Bleeding from the external auditory meatus or CSF otorrhoea:



3. Conscious level – Glasgow Coma Score (GCS)

Assess patient's conscious level in terms of *eye opening*, *verbal* and *motor response* on admission (see [page 5](#)) and record at regular intervals thereafter. An observation chart incorporating these features is essential and clearly shows the trend in the patient's condition. Deterioration in conscious level indicates the need for immediate investigation and action where appropriate.



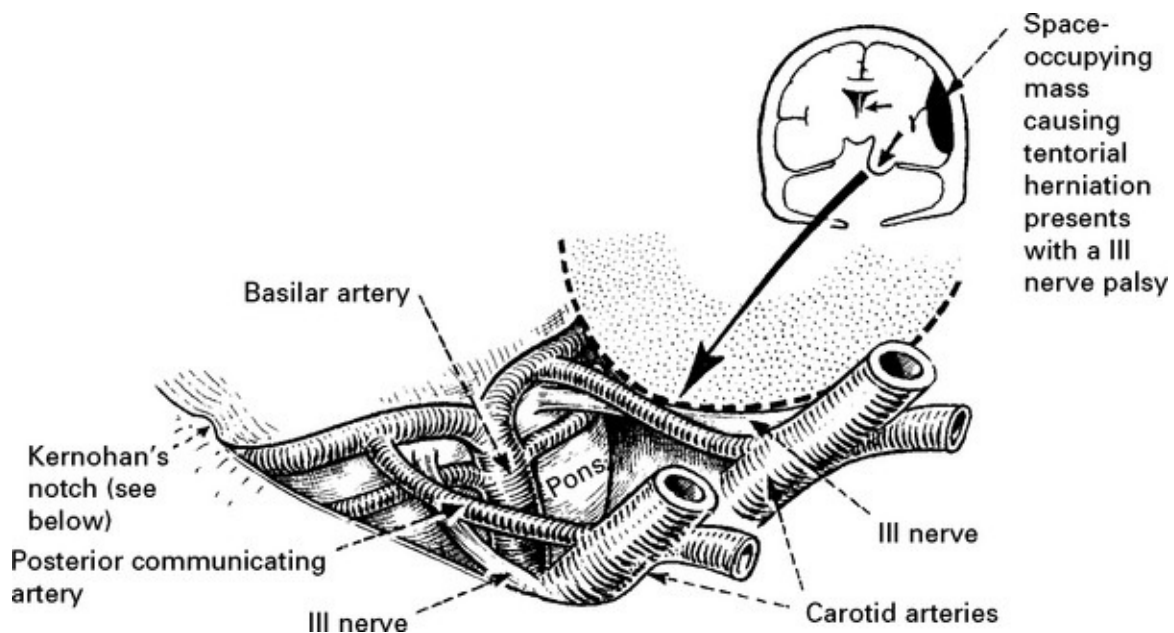
Reproduced by permission of the Nursing Times

Note: This chart shows a '14 point scale' with a maximum score of '14' in a fully conscious patient. Many centres use a 15 point coma scale where 'Flexion to pain' is divided into 'normal' or 'spastic' flexion (see

page 29).

4. Pupil response

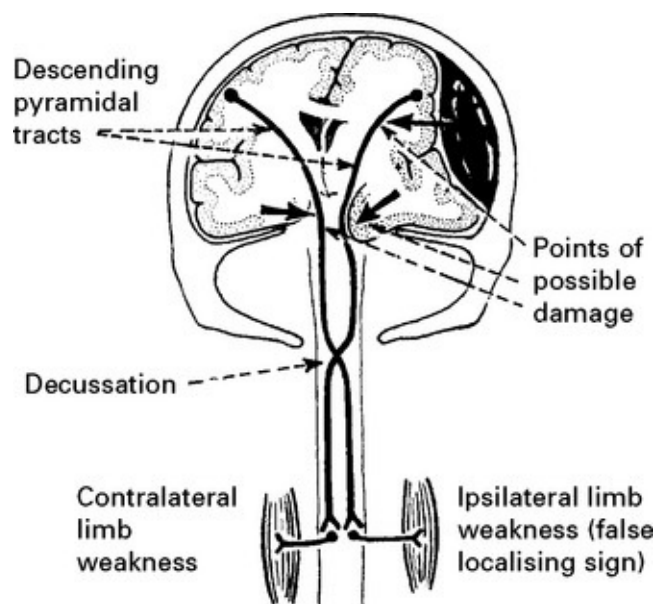
The light reflex ([page 142](#)) tests optic (II) and oculomotor (III) nerve function. Although II nerve damage is important to record and may result in permanent visual impairment, it is the III nerve function which is the most useful indicator of an expanding intracranial lesion. Herniation of the medial temporal lobe through the tentorial hiatus may damage the III nerve directly or cause midbrain ischaemia, resulting in pupil dilatation with impaired or absent reaction to light. The *pupil dilates on the side of the expanding lesion* and is an important localising sign. With a further increase in intracranial pressure, bilateral pupillary dilatation may occur.



5. Limb weakness

Determine limb weakness by comparing the response in each limb to painful stimuli ([page 30](#)). Hemiparesis or hemiplegia usually occurs in the limbs contralateral to the side of the lesion. Indentation of the

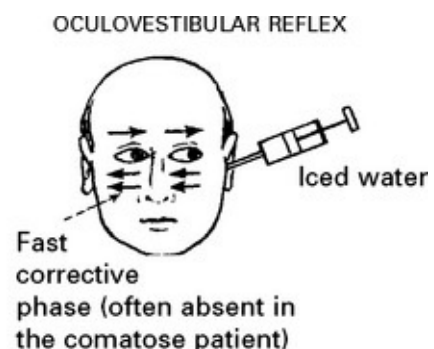
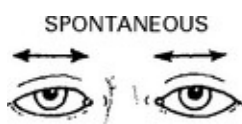
contralateral cerebral peduncle by the edge of the tentorium cerebelli (Kernohan's notch) may produce an ipsilateral deficit, a false localising sign more often seen with chronic subdural haematomas. Limb deficits are therefore of limited value in lesion localisation.



6. Eye movements

Evaluation of eye movements does not help in immediate management, but provides a useful prognostic guide.

Eye movements may occur spontaneously, or can be elicited reflexly ([page 30](#)) by head rotation (oculocephalic reflex) or by caloric stimulation (oculovestibular reflex).



Abnormal eye movements may result from: brain stem dysfunction, damage to the nerves supplying the extraocular muscles or damage to the vestibular apparatus. Absent eye movements relate to low levels of responsiveness and indicate a gloomy prognosis.

Vital signs

At the beginning of the century, the eminent neurosurgeon Harvey Cushing noted that a rise in intracranial pressure led to a rise in blood pressure and a fall in pulse rate and produced abnormal respiratory patterns. In the past, much emphasis has been placed on close observation of these vital signs in patients with head injury, but these changes may not occur and when present are usually preceded by deterioration in conscious level. Close observation of consciousness is therefore more relevant.

Cranial nerve lesions

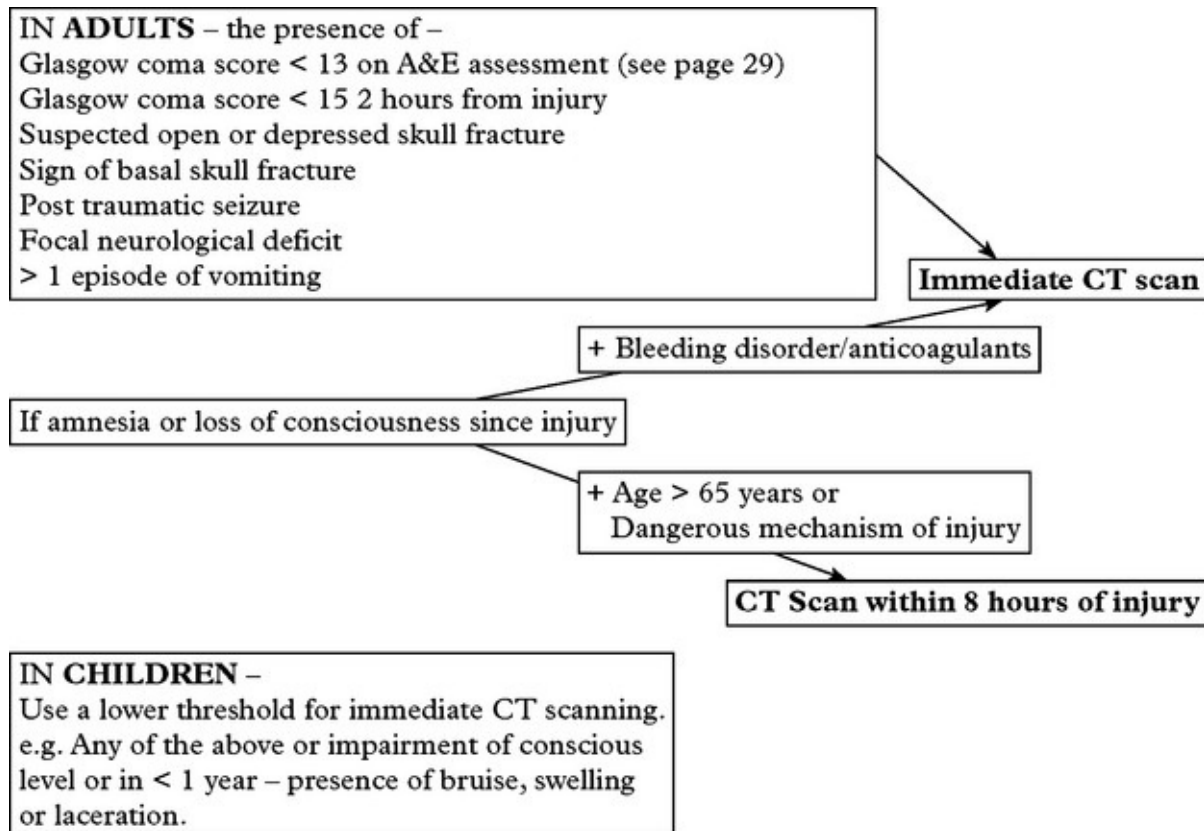
Basal skull fracture or extracranial injury can result in damage to the cranial nerves. Evidence of this damage must be recorded but, with the exception of a III nerve lesion, does not usually help immediate management. Full cranial nerve examination is difficult in the comatose patient and this can await patient co-operation.

Clinical assessment cannot reliably distinguish the type or even the site of intracranial haematoma, but is invaluable in indicating the need for further investigation and in providing a baseline against which any change can be compared.

HEAD INJURY – INVESTIGATION AND REFERRAL CRITERIA

IN THE ACCIDENT AND EMERGENCY DEPARTMENT (A&E)

Various guidelines now exist. The following are based on those from the National Institute for Health and Clinical Excellence (NICE). (Further details available at <http://www.nice.org.uk>).



Admit to Hospital

- New abnormalities on imaging
- Glasgow coma score < 15 (even if imaging normal)
- Persistent vomiting or severe headache
- Fits criteria for a CT scan within 8 hours
- Other concerns, e.g. drugs, alcohol intoxication, other injuries, shock, meningism, CSF leak, suspected non-accidental injury

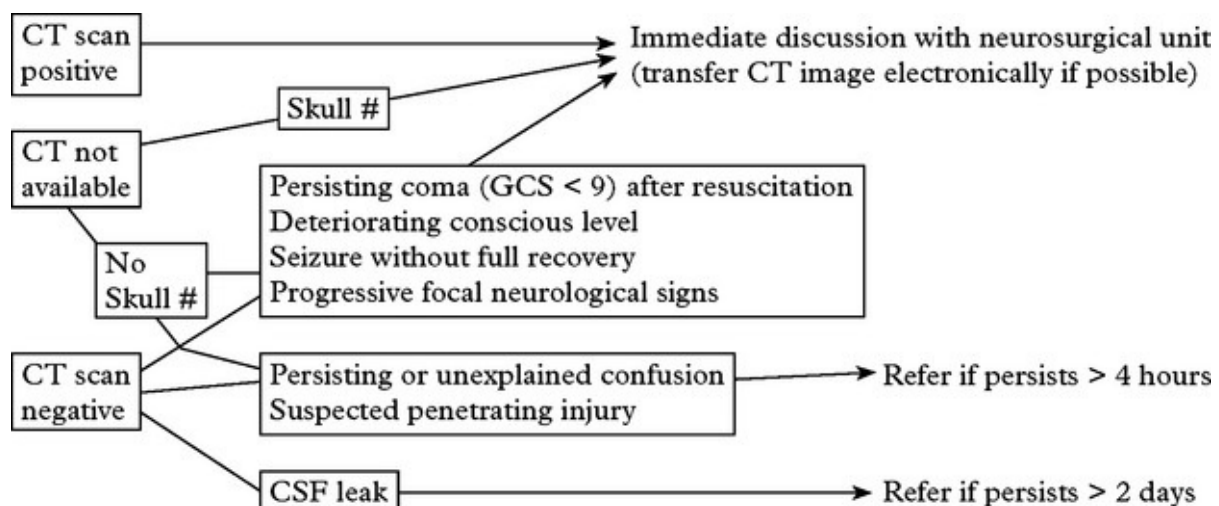
Discharge from A&E

- If Glasgow coma score = 15
AND
- Appropriate supervision at home
- CT not indicated or
- Normal imaging head and spine
- All symptoms and signs resolved

Admit to Hospital	Discharge from A&E
• New abnormalities on imaging	• If Glasgow coma

• Glasgow coma score < 15 (even if imaging normal) • Persistent vomiting or severe headache • Fits criteria for a CT scan within 8 hours • Other concerns, <i>e.g.</i> drugs, alcohol intoxication, other injuries, shock, meningism, CSF leak, suspected non-accidental injury	score = 15 AND • Appropriate supervision at home • CT not indicated or • Normal imaging head and spine • All symptoms and signs resolved
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Referral to Neurosurgical Unit



Transfer to the neurosurgical unit

Prior to the transfer, ensure that resuscitation is complete, and that more immediate problems have been dealt with (see [page 221](#)). Insert an oropharyngeal airway. Intubate and ventilate if the patient is in coma or if the blood gases are inadequate ($PO_2 < 8$ kPa on air, 13 kPa on O_2 or $CO_2 > 6$ kPa). If the patient's conscious level is deteriorating, an

intravenous bolus infusion of 100 ml of 20% mannitol should 'buy time' by temporarily reducing the intracranial pressure.

NOTE: for comatose patients with an unstable systemic state from multiple injuries, a negative CT scan in the local hospital may avoid a dangerous transfer to the neurosurgical unit.

Cervical spine injury may accompany head injury. Guidelines also exist with criteria for investigation. (For full details see <http://www.nice.org.uk/guidance/index.jsp?action=download&o=36259>).

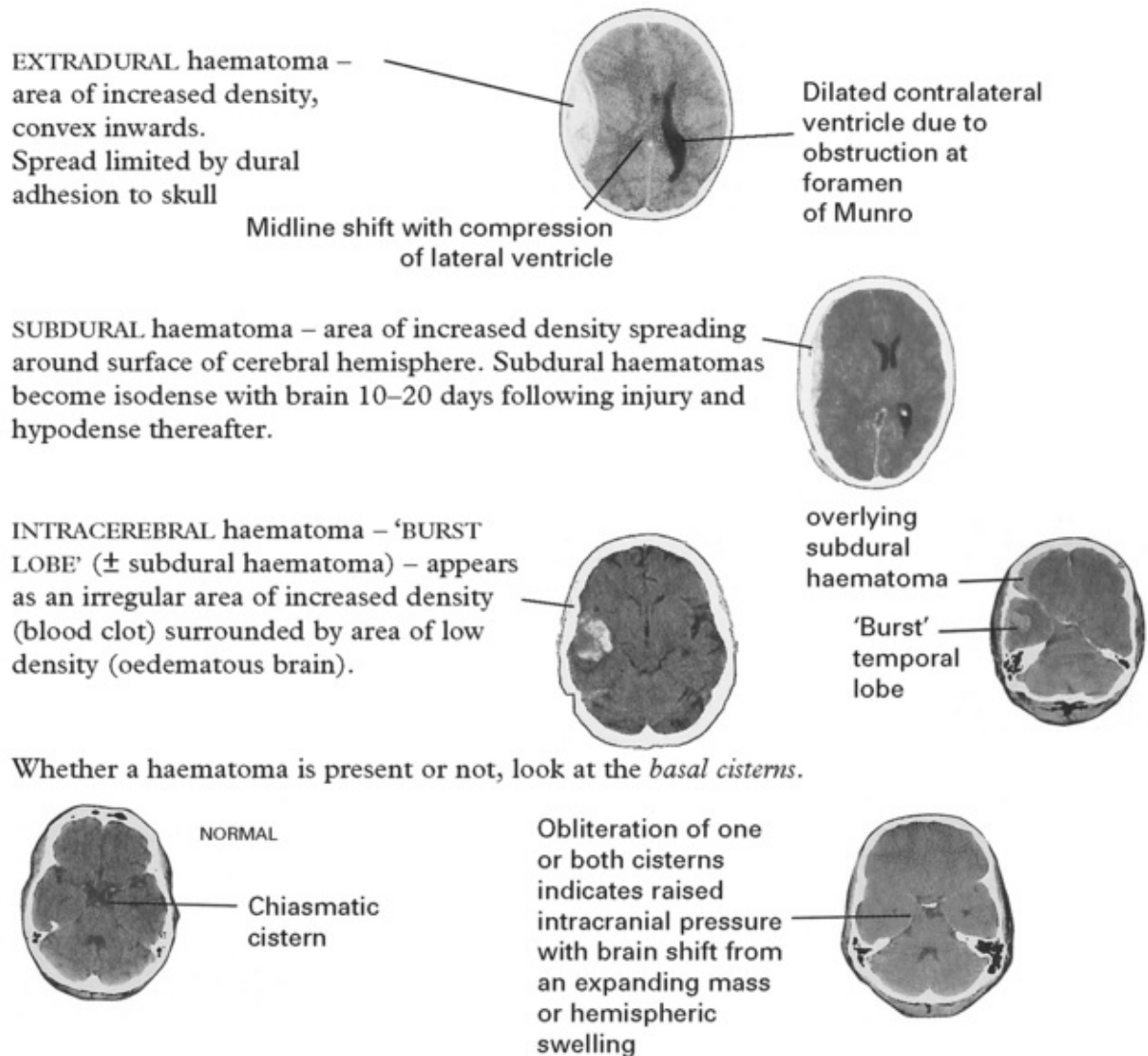
For ADULTS and CHILDREN 10 years and over

<i>AP, Lateral and Odontoid Peg X-rays if</i>	<i>CT cervical spine if</i>
• Impaired neck rotation to right or left • No indication for CT scanning • Not safe to assess clinically • Neck pain/midline tenderness + $\geq$ 65 years or dangerous mechanism of injury • To exclude injury urgently e.g. prior to surgery	• Patient intubated • Continued suspicion despite X-rays • Inadequate X-rays • Undergoing CT scanning for another reason e.g. Glasgow coma score < 13 or multi-region trauma

HEAD INJURY – INVESTIGATION

CT scan: the investigation of choice for head injury (and cervical spine injury in certain circumstances – see [page 227](#)).

Scans must extend from the posterior fossa to the vertex, otherwise haematomas in these sites will be missed.



With *diffuse axonal shearing injuries*, small haematomas may be seen on CT scan scattered throughout the white matter, particularly in the corpus callosum, the subcortical white matter and in the brain stem adjacent to the cerebellar peduncles.

If *hydrocephalus* is present on the upper scan cuts, look carefully for a haematoma (extradural, subdural or intracerebral) in the posterior fossa, compressing and obstructing the 4th ventricle.

Further investigation may be required to exclude other coincidental or contributory causes of the head injury, e.g. drugs, alcohol, postictal state,

encephalitis (Cause of coma, see [page 86](#)).

If a CT scan is not available, a skull fracture on X-ray identifies those at high risk of intracranial haematoma. In those patients, referral to a neurosurgical unit for a CT scan is essential (see [page 227](#)).

Risk of intracranial haematoma (requiring removal) in adults attending A & E departments after head injury.

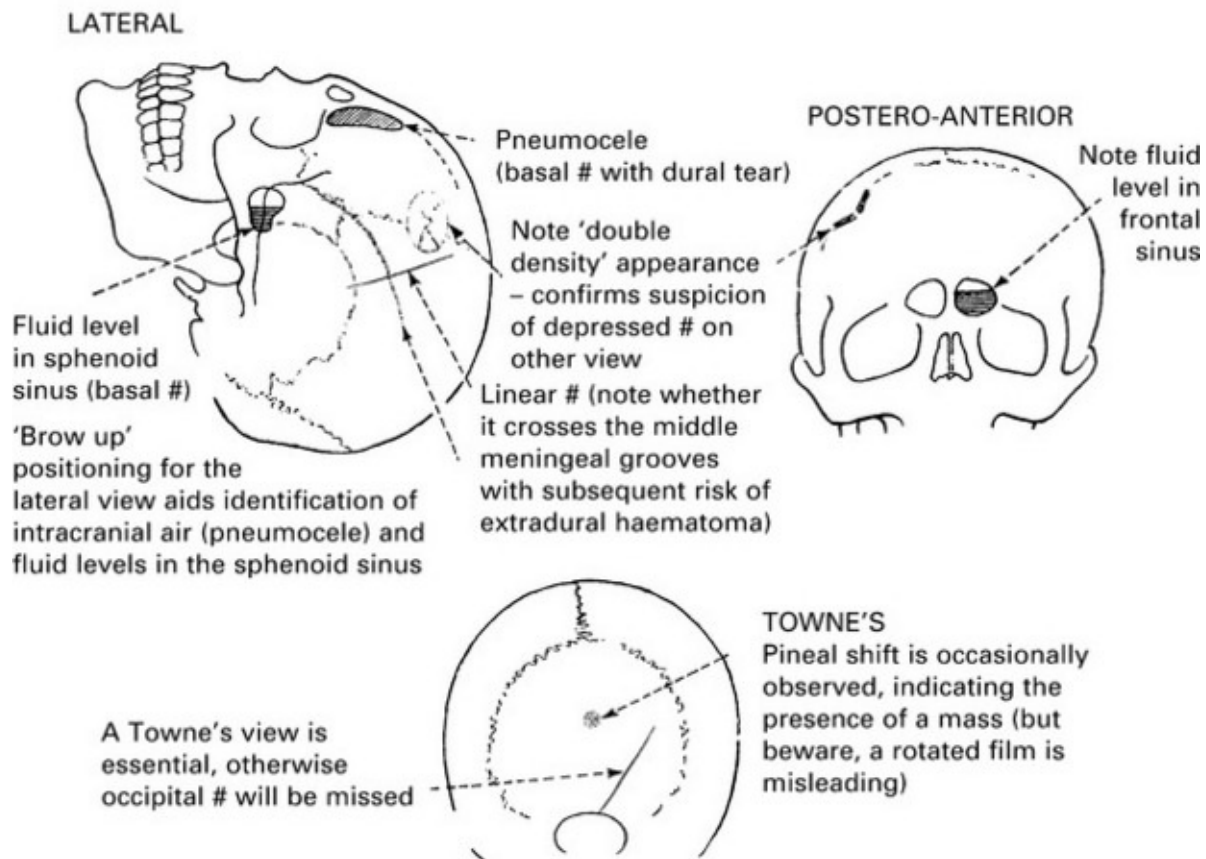
No skull #	– orientated	1 in 6000
No skull #	– not orientated	1 in 120
Skull #	– orientated	1 in 32
Skull #	– not orientated	1 in 4

Adapted with permission Mendelow et al 1983
ii: 1173–1176 British Medical Journal

X-ray the skull if CT not available and:

(plus cervical spine, chest, abdomen, pelvis and limbs if required)

- conscious level is impaired at the time of examination or if the patient has lost consciousness at any time since the injury
- neurological symptoms or signs are present
- CSF leak from the nose (rhinorrhoea) or ear (otorrhoea)
- penetrating injury is suspect
- significant scalp bruising or swelling
- patient assessment is difficult (e.g. alcohol intoxication).



HEAD INJURY – MANAGEMENT

Management aims at *preventing the development of secondary brain damage* from intracranial haematoma, ischaemia, raised intracranial pressure

with tentorial or tonsillar herniation and infection.

- Ensure the *airway is patent* and that *blood oxygenation is adequate*. *Intubation* is advisable in patients ‘flexing to pain’ or worse. *Ventilation* may be required if respiratory movements are depressed or lung function is impaired, e.g. ‘flail’ segment, aspiration pneumonia, pulmonary contusion or fat emboli. Hypoxia can cause direct cerebral damage, but in addition causes vasodilatation resulting in an increase in cerebral blood volume with subsequent rise in ICP.

- A *space-occupying haematoma requires urgent evacuation* (see over). If the patient’s conscious level is deteriorating, give an initial or repeat i.v. bolus of mannitol (100 ml of 20%). Coagulation should be checked and any deficits corrected.

- Scalp *lacerations* require cleaning, inspection to exclude an underlying depressed fracture and suturing.

- *Correct hypovolaemia* following blood loss – but avoid fluid overload as this may aggravate cerebral oedema. In adults, 2 litres/day of fluid is sufficient. Commence nasogastric fluids or oral fluids when feasible.

- *Anticonvulsants* (e.g. phenytoin) must be given intravenously if seizures occur; further seizures and in particular status epilepticus significantly increase the risk of cerebral anoxia.

- *Monitor intracranial pressure (ICP), blood pressure and cerebral perfusion pressure (CPP)* in selected patients with diffuse swelling or after evacuation of an intracranial haematoma. Maintain CPP either by raising blood pressure if low or by treating raised intracranial pressure.

- *Brain protective agents* include corticosteroids, free radical scavengers, calcium channel blockers, and glutamate antagonists. The evolution of

axonal damage after a diffuse shearing injury provides a potential window of opportunity for treatment. Despite experimental animal studies revealing encouraging results, trials of these agents in head-injured patients have failed to show efficacy, perhaps due to insufficient patient numbers or a failure to target treatment at appropriate patients. A recent study of corticosteroids involving 10 000 patients has shown a worse outcome in the treatment group (the CRASH study).

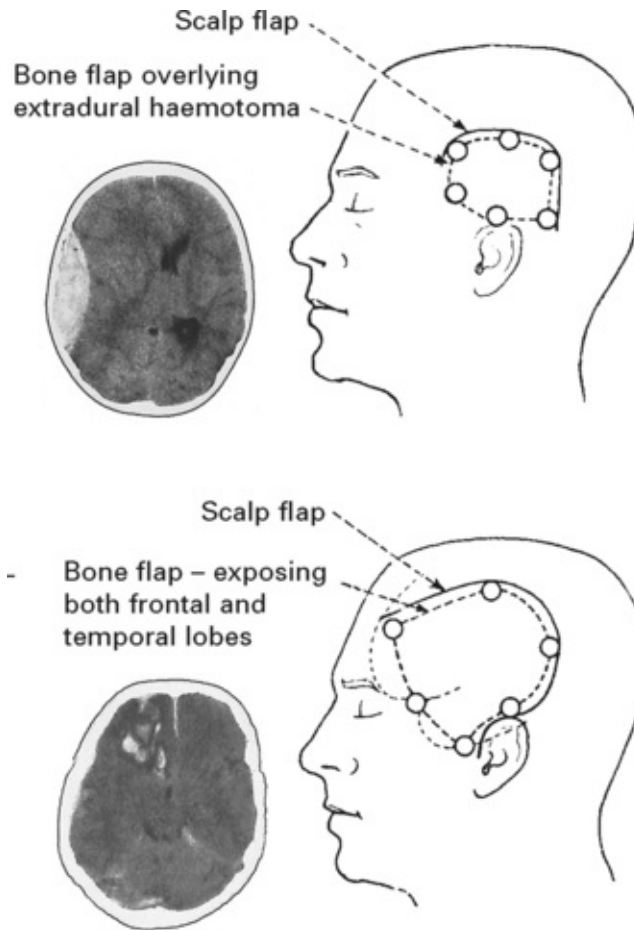
– *Operative repair of a dural defect* is required if CSF leak persists for more than 7 days. (Many still use *prophylactic antibiotics* in patients with a CSF leak, but there is no conclusive evidence of their efficacy and they may do more harm than good by encouraging the growth of resistant organisms.) The development of meningitis requires prompt treatment with an empirical antibiotic.

INTRACRANIAL HAEMATOMA

Most intracranial haematomas require urgent evacuation – evident from the patient's clinical state combined with the CT scan appearance of a space-occupying mass.

Extradural haematoma

Using the CT scan the position of the extradural haematoma is accurately delineated and a 'horse shoe' craniotomy flap is turned over this area, allowing complete evacuation of the haematoma. For low temporal extradural haematomas, a 'question mark' flap may be more suitable. If patient deterioration is rapid, a burr hole and craniectomy positioned centrally over the haematoma may provide temporary relief, but this seldom provides adequate decompression.



Subdural/intracerebral haematoma ('burst lobe')

Subdural and intracerebral haematomas usually arise from lacerations on the undersurface of the frontal and/or temporal lobes. Again the CT scan is useful in demonstrating the exact site. A 'question mark' flap permits good access to both frontal and temporal 'burst' lobes. The subdural collection is evacuated and any underlying intracerebral haematoma is removed along with necrotic brain.

N.B. Burr holes are insufficient to evacuate an acute subdural haematoma or to deal with any underlying cortical damage.

Conservative management of traumatic intracranial haematomas

Not all patients with traumatic intracranial haematomas deteriorate. In some, the haematomas are small and clearly do not require evacuation. In others, however, the decision to operate proves difficult, *e.g.* the CT scan may reveal a moderate-sized haematoma with minimal or no mass effect in a conscious but confused patient.

If conservative management is adopted, careful observation in a neurosurgical unit is essential. Any deterioration indicates the need for immediate operation. In this group of patients, intracranial pressure monitoring may serve as a useful guide. An intracranial pressure of 25 mmHg or more suggests that haematoma evacuation is required as the likelihood of subsequent deterioration with continued conservative management would be high.

TREATMENT OF RAISED INTRACRANIAL PRESSURE (ICP)

Raised ICP in the absence of any easily treatable condition (*e.g.* intracranial haematoma or raised $p\text{CO}_2$) requires careful management. The various techniques used to lower ICP have already been described ([pages 83–84](#)) but these must not be applied indiscriminately.

Recent studies show that even in a modern ITU head injured patient are still at risk of sustaining potentially harmful “insults” to the brain in the first few days after head injury from high ICP, low BP, low cerebral perfusion pressure (CPP), hypoxaemia, hypoglycaemia or raised temperature.

Most believe that both raised ICP and reduced cerebral perfusion pressure (CPP) can exacerbate brain damage. What is less clear is whether treatment should focus on lowering ICP or increasing CPP. When autoregulation is impaired, raising CPP beyond 70 mmHg could cause harm. The blind use of hyperventilation in the past to lower ICP by causing vasoconstriction and reduced intracranial blood volume has

now been recognised to produce worse outcomes by aggravating cerebral ischaemia.

Patient selection for ICP monitoring: Monitoring ICP and CPP is most relevant in patients with a flexion response to painful stimuli or worse (a response of 'localising to pain' signifies a milder degree of injury and spontaneous recovery is likely). Such patients may have already undergone removal of an intracranial haematoma or may have had no mass lesion on CT scan (i.e.: diffuse injury or contusional damage). Each neurosurgical unit is likely to have its own policy for ICP monitoring but the following outline may serve as a guide *for patients with no intracranial mass lesion* –

If ICP high (e.g. > 25 mmHg) & CPP low (e.g. < 60 mmHg) [Note: CPP = BP – ICP]

Is BP low? (e.g. mean < 100 mmHg) → measure central venous pressure (CVP) and/or cardiac output / vascular resistance

If hypovolaemia

→ give plasma volume expanders, e.g. starch solutions (do not use mannitol)

If low cardiac output/normovolaemia

→ give inotropic agents and/or vasopressors
aim for CPP of 50–70 mmHg

If BP normal → give hypnotics e.g. Propofol, morphine, Midazolam (see page 84)
Maintain CPP between 50–70 mmHg
Consider Mannitol
Do *not* hyperventilate

If ICP high (e.g. > 25 mmHg) & CPP normal (e.g. > 60 mmHg)

→ give hypnotics e.g. Propofol, morphine, Midazolam (see page 84)
Consider Mannitol

If ICP high (e.g. > 25 mmHg) despite above measures

→ Consider decompressive craniectomy (currently under evaluation in randomised trial (see page 84))

DIFFUSE BRAIN DAMAGE/NEGATIVE CT SCAN

A proportion of patients have no intracranial haematoma on CT scan or have only a small haematoma or contusion without mass effect.

In these patients, coma or impairment of conscious level may be due to:

- *diffuse axonal injury* – suspect if conscious level impaired from impact.
 - *cerebral ischaemic damage*
 - *cerebral swelling*
 - *fat emboli*
 - *meningitis*
- } suspect if deterioration is delayed – a patient who talks after impact does not have a significant shearing injury.

Several of these factors may coexist and contribute to brain damage in patients with intracranial haematoma.

The management principles outlined above apply; in particular it is essential to ensure that respiratory function is adequate and that cerebral perfusion pressure is maintained.

Fat emboli usually occur a few days after injury and may be related to fracture manipulation; deterioration of respiratory function usually accompanies cerebral damage and most patients require ventilation.

Meningitis may occur several days after injury in the presence of basal fractures.

Cerebral swelling may occur at any time after injury and cause a rise in intracranial pressure.

REPEAT CT SCANNING

Indications:

- Delayed deterioration in clinical state
 - Maintained rise in ICP
 - or
 - Failure to improve after 48 hours
- } in patients with diffuse injury or following evacuation of an intracranial haematoma

Occasionally, small areas of 'insignificant' contusion on an initial CT scan may develop into a space-occupying haematoma requiring evacuation. Following haematoma evacuation, recollection may occur in 5–10% of cases.

Occasionally, small areas of 'insignificant' contusion on an initial CT

scan may develop into a space-occupying haematoma requiring evacuation. Following haematoma evacuation, recollection may occur in 5–10% of cases.

DEPRESSED SKULL FRACTURE

This injury is caused by a blow from a sharp object. Since diffuse ‘deceleration’ damage is minimal, patients seldom lose consciousness.

SIMPLE DEPRESSED FRACTURE (closed injury)

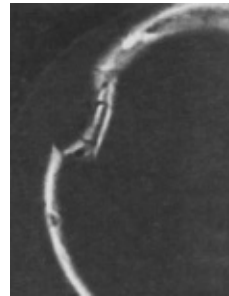
There is no overlying laceration and no risk of infection. Operation is not required except for cosmetic reasons. Removal of any bone spicules imbedded in brain tissue does not reverse neuronal damage.

COMPOUND DEPRESSED FRACTURE (open injury)

A scalp laceration is related to (but does not necessarily overlie) the depressed bone segments. A compound depressed fracture with an associated dural tear may result in meningitis or cerebral abscess.

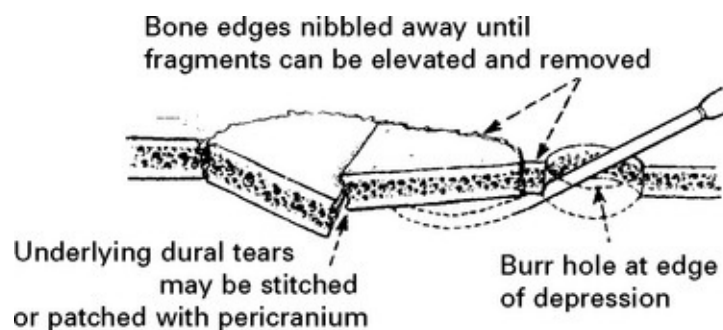
Investigation

Double density appearance on *skull X-ray* suggests depression but tangential views may be required to establish the diagnosis. Impairment of conscious level or the presence of focal signs indicate the need for a *CT scan* to exclude underlying extradural haematoma or severe cortical contusion. Selecting bone window levels on CT scan will clearly demonstrate any depressed fragments.



'Bony' window levels on CT scan clearly demonstrate the depressed fragments

Management



Treatment aims to minimise the risk of infection. The wound is debrided and the fragments elevated within 24 hours from injury. Bone fragments are either removed or replaced after washing with antiseptic. Antibiotics are not essential unless the wound is excessively dirty.

If the venous sinuses are involved in the depressed fracture, then operative risks from excessive bleeding may outweigh the risk of infection and antibiotic treatment alone is given.

Complications

Most patients make a rapid and full recovery, but a few develop complications:

Infection: May lead to meningitis or abscess formation. Some believe that operation does not reduce the infection risk and advocate a conservative approach unless contamination is severe.

Epilepsy: Early epilepsy (in the first week) occurs in 10% of patients with depressed fracture. Late epilepsy develops in 15% overall, but is especially common when the dura is torn, when focal signs are present, when post-traumatic amnesia exceeds 24 hours or when early epilepsy has occurred (the risk ranges from 3 to 60%, depending on the number of the above factors involved). Elevation of the bone fragments does not alter the incidence of epilepsy.

DELAYED EFFECTS OF HEAD INJURY

POST-TRAUMATIC EPILEPSY

Early epilepsy (occurring within the first week from injury)

Early epilepsy occurs in 5% of patients admitted to hospital with non-missile (i.e. deceleration) injuries. It is particularly frequent in the first 24 hours after injury. Focal seizures are as common as generalised seizures. Status epilepticus occurs in 10%.

The risk of early epilepsy is high in

- children under 5 years.
- patients with prolonged post-traumatic amnesia
- patients with an intracranial haematoma
- patients with a compound depressed fracture.

Late epilepsy (occurring after the first week from injury)

Late epilepsy also occurs in about 5% of all patients admitted to hospital after head injury. It usually presents in the first year, but in some the first attack occurs as long as 10 years from the injury. Late epilepsy is prevalent in patients with

- early epilepsy (25%)

- intracranial haematoma (35%)
- compound depressed fracture (17%).

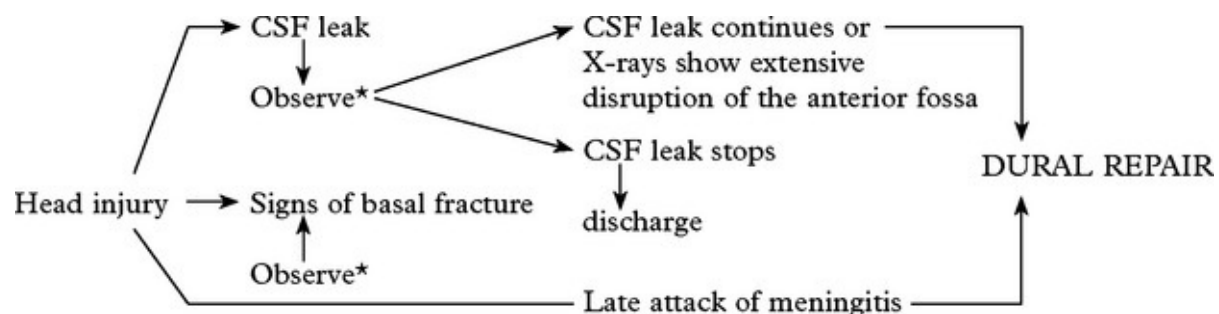
Prophylactic anticonvulsants appear to be of little benefit in preventing the development of an epileptogenic focus. Management is discussed on [page 102](#).

CEREBROSPINAL FLUID (CSF) LEAK

After head injury a basal fracture may cause a fistulous communication between the CSF space and the paranasal sinuses or the middle ear. Profuse CSF leaks (rhinorrhoea or otorrhoea) are readily detectable, but brain may partially plug the defect and the leak may be minimal or absent. Patients risk developing meningitis particularly in the first week, but in some this occurs after several years. When this is associated with anterior fossa fractures, it is usually pneumococcal; when associated with fractures through the petrous bone, a variety of organisms may be involved.

Clinical signs of a basal fracture have previously been described ([page 222](#)). The patient may comment on a ‘salty taste’ in the mouth. Anosmia suggests avulsion of the olfactory bulb from the cribriform plate.

Management



*A Cochrane Review has concluded that evidence does not support the use of prophylactic antibiotics. Prophylactic antibiotics only encourage

resistance and late attacks of meningitis may still occur despite their use.

(Lancet (1994) 344:1547–1551)

Preoperative investigations

Coronal high definition CT scanning should identify the fracture site.

CT cisternography – CT scanning after running contrast injected into the lumbar theca, up to the basal cisterns may identify the exact site of the leak.

CSF isotope infusion studies combined with pledget insertion into the nasal recesses may also be of value, but results can be misleading.

Operation

As fractures of the anterior fossa often extend across the midline, a bifrontal exploration is required. The dural tear is repaired with fascia lata, pericranium or synthetic dural substitute. A CSF leak through the middle ear requires a subtemporal approach.

Failure to repair a CSF fistula may result from impaired CSF absorption with an intermittent or persistent elevation of ICP. In these patients a CSF shunt may be required.

POSTCONCUSSIONAL SYMPTOMS

Even after relatively minor head injury, patients may have persistent symptoms of:

- headache, dizziness and increased irritability
- difficulty in concentration and in coping with work
- fatigue and depression.

This condition was once thought to have a purely psychological basis, but it is now recognised that in an injury of sufficient severity to cause loss of consciousness, or a period of post-traumatic amnesia, some neuronal damage occurs; studies show a distinct delay in information processing in these patients, requiring several weeks to resolve. Vestibular ‘concussion’ (end-organ damage) may contribute to the symptomatology (‘dizziness’ and vertigo).

CUMULATIVE BRAIN DAMAGE

The effects of repeated neuronal damage are cumulative; when this exceeds the capacity for compensation, permanent evidence of brain damage ensues. The ‘punch-drunk’ state is well recognised in boxers; dementia may also occur from repeated head injury in jockeys.

CRANIAL NERVE DAMAGE

Cranial nerve damage occurs in about one-third of patients with severe head injury, but treatment is seldom of benefit. These lesions may contribute towards the patient’s residual disability.

Nerve	Cause of damage	Clinical problem	Management	Prognosis
I	Usually associated with anterior fossa fracture and CSF rhinorrhoea	Anosmia	Nil	Recovery often occurs in a few months
II	Optic nerve usually damaged in the optic foramen Chiasmal damage occasionally occurs. [N.B. Visual loss may also occur from damage to the globe, occipital cortex or optic radiations]	Visual loss or field defect in one eye Bitemporal hemianopia.	Nil Local eye/orbital damage may need treatment	Recovery seldom occurs
III IV VI	III nerve damage usually results from tentorial herniation but can also occur in fractures involving the superior orbital fissure or cavernous sinus IV nerve damage is uncommon VI nerve damage is usually associated with fractures of the petrous or sphenoid bones	Pupil inequality, ptosis and disturbance of ocular movements	Nil [other than removing cause of tentorial herniation]	Recovery usually occurs
V	Occasionally follows petrous or sphenoid fractures	Facial numbness	Nil	Usually permanent
VII	Associated with petrous fracture	Immediate or delayed facial palsy	Otologists occasionally recommend decompression. Early steroid therapy may benefit	Immediate lesions have a poor prognosis; delayed lesions usually recover
VIII	Petrous fracture may damage: – nerve – cochlea – ossicles Haemotympanum may result	Vertigo, 'dizziness', hearing loss, tinnitus	Ossicular damage may benefit from operation	Vestibular symptoms usually improve after several weeks. Nerve deafness is usually permanent. Conductive deafness from haemotympanum should gradually improve
IX, X XI, XII	Associated with very severe basal fractures or extracranial injury	Patient seldom survives primary damage		

OUTCOME AFTER SEVERE HEAD INJURY

Head injury remains a major cause of disability and death, especially in the young. Of those patients who survive the initial impact and remain in coma for at least 6 hours, approximately 40% die within 6 months. The extent of recovery in the remainder depends on the severity of the injury. Residual disabilities include both mental (impaired intellect,

memory and behavioural problems) and physical defects (hemiparesis and dysphasia). Most recovery occurs within the first 6 months after injury, but improvement may continue for years. *Physiotherapy* and *occupational therapy* play an important role not only in minimising contractures and improving limb power and function but also in stimulating patient motivation.

Outcome is best categorised with the *Glasgow Outcome Scale* (GOS – see [page 214](#)) which uses *dependence* to differentiate between intermediate grades. After severe injury, about 40% regain an independent existence and may return to premorbid social and occupational activities. Inevitably some remain severely disabled requiring long term care, but few (< 2%) are left in a vegetative state with no awareness or ability to communicate with their environment (see [page 214](#)). Prognosis in this group is marginally better than for non-traumatic coma – with about one-third of those vegetative at *one month* regaining consciousness within one year; of those who regain consciousness, over two-thirds either subsequently die or remain severely disabled. Of those vegetative at *3 months* after the injury, none regain an independent existence.

Prognostic features following traumatic coma

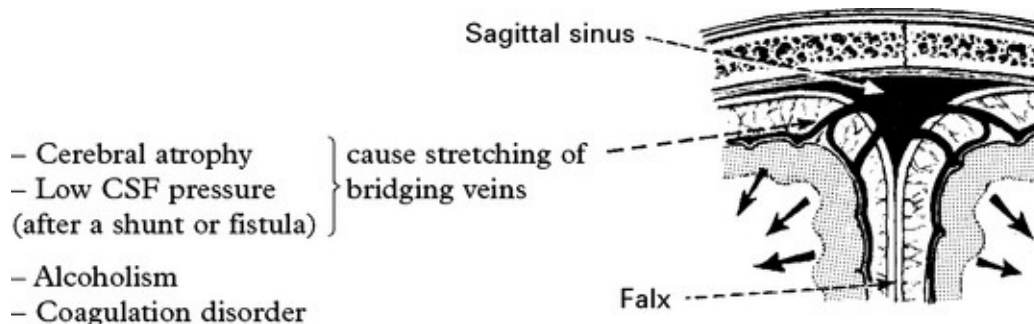
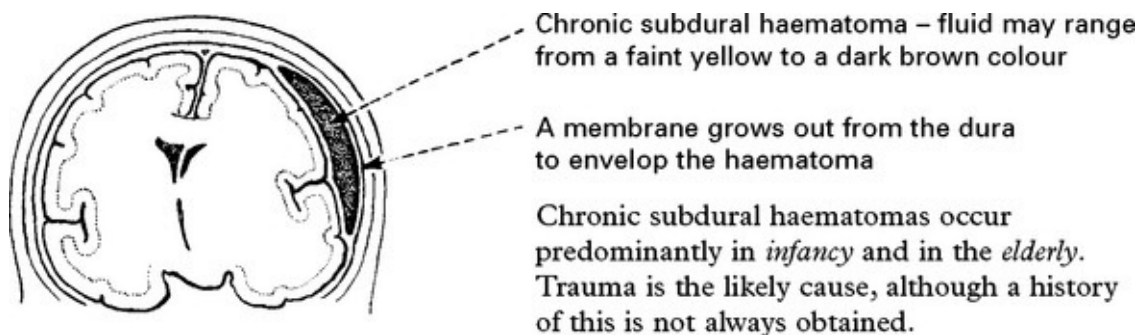
The duration of coma relates closely to the severity of injury and to the final outcome, but in the early stages after injury the clinician must rely on other features – age, eye opening, verbal and motor responses, pupil response and eye movements.

	Poor outcome (GOS 1–3)	Favourable outcome (GOS 4–5)
Patients in coma for > 6 hours	61%	39%
Best <i>Glasgow Coma Score</i> > 11	18%	82%

Best <i>Glasgow Coma Score</i> 8–10	32%	68%
Best <i>Glasgow Coma Score</i> < 8	73%	27%
<i>Pupillary response</i> – reacting	50%	50%
<i>Pupillary response</i> – non-reacting	96%	4%
Age < 20 years	41%	59%
Age > 60 years	94%	6%

CHRONIC SUBDURAL HAEMATOMA

Subdivision of subdural haematomas into acute and subacute forms serves no practical purpose. Chronic subdural haematoma however is best considered as a separate entity, differing both in presentation and management.



Predisposing factors

Breakdown of protein within the haematoma and a subsequent rise in

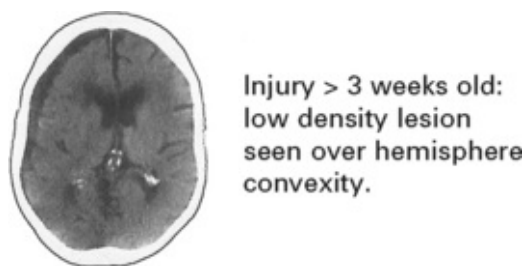
osmotic pressure was originally believed to account for the gradual enlargement of the untreated subdural haematoma. Studies showing equality of osmotic pressures in blood and haematoma fluid cast doubt on this theory and recurrent bleeding into the cavity is now known to play an important role.

Clinical features tend to be non-specific.

- Dementia.
- Deterioration in conscious level, occasionally with fluctuating course.
- Symptoms and signs of raised ICP.
- Focal signs occasionally occur, especially limb weakness. This may be ipsilateral to the side of the lesion, *i.e.* a false localising sign (see [page 224](#)).

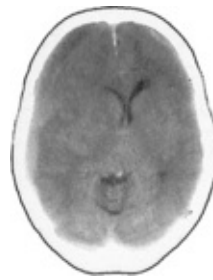
Diagnosis

CT Scan appearances depend on the time between the injury and the scan.



With injuries 1–3 weeks old, the subdural haematoma may be isodense with brain tissue. In this instance, i.v. contrast enhancement may delineate the cortical margin.

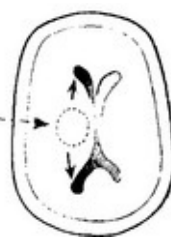
Beyond 3 weeks subdural haematomas appear as a low density lesion.



Isodense lesion causing midline shift. Note the shape of the ventricles

If CT scan shows midline shift without any obvious extra-or intracerebral lesion, look at the shape of the ventricles.

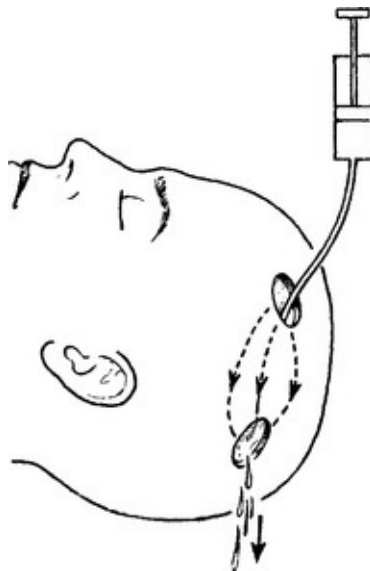
Separation of the frontal and occipital horns suggests an intrinsic lesion, e.g. encephalitis rather than a surface collection



Extracerebral collection, i.e. chronic subdural haematoma, causes approximation of frontal and occipital horns



Management



Adult

The haematoma is evacuated through two or three burr holes and the cavity is irrigated with saline. Drains may be left in the subdural space and nursing in the head-down position may help prevent recollection.

Craniotomy with excision of the membrane is seldom required.

In patients who have no depressed conscious level, conservative treatment with steroids over several weeks may result in resolution.

Infants

The haematoma is evacuated by repeated needle aspiration through the anterior fontanelle.

Persistent subdural collections require a subdural peritoneal shunt. As in adults, craniotomy is seldom necessary.

CEREBROVASCULAR DISEASES

Vascular diseases of the nervous system are amongst the most frequent causes of admission to hospital. The annual incidence in the UK varies

regionally between 150–200/100 000, with a prevalence of 600/100 000 of which one-third are severely disabled.

Better control of hypertension, reduced incidence of heart disease and a greater awareness of all risk factors have combined to reduce mortality from stroke. Despite this, stroke still ranks third behind heart disease and cancer as a cause of death in affluent societies.

RISK FACTORS

Prevention of cerebrovascular disease is more likely to reduce death and disability than any medical or surgical advance in management. Prevention depends upon the identification of risk factors and their correction. Increasing age is the strongest risk factor (but is not amenable to correction).

Hypertension

Hypertension is a major factor in the development of thrombotic cerebral infarction and intracranial haemorrhage.

There is no critical blood pressure level; the risk is related to the height of blood pressure and increases throughout the whole range from normal to hypertensive. A 6 mmHg fall in diastolic blood pressure is associated in relative terms with a 40% fall in the fatal and non-fatal stroke rate.

Systolic hypertension (frequent in the elderly) is also a significant factor and not as harmless as previously thought.

Cardiac disease

Cardiac enlargement, failure and arrhythmias, as well as rheumatic heart disease, patent foramen ovale and, rarely, cardiac myxoma are all associated with an increased risk of stroke.

Diabetes

The risk of cerebral infarction is increased twofold in diabetes. More effective treatment of diabetes has not reduced the frequency of atherosclerotic sequelae.

Heredity

Close relatives are at only slightly greater risk than non-genetically related family members of a stroke patient. Diabetes and hypertension show familial propensity thus clouding the significance of pure hereditary factors.

Blood lipids, cholesterol, smoking, diet/obesity

These factors are much less significant than in the genesis of coronary artery disease.

Race

Alterations in life style, diet and environment probably explain the geographical variations more than racial tendencies.

Haematocrit

A high blood haemoglobin concentration (or haematocrit level) is associated with an increased incidence of cerebral infarction. Other haematological factors, such as decreased fibrinolysis, are important also.

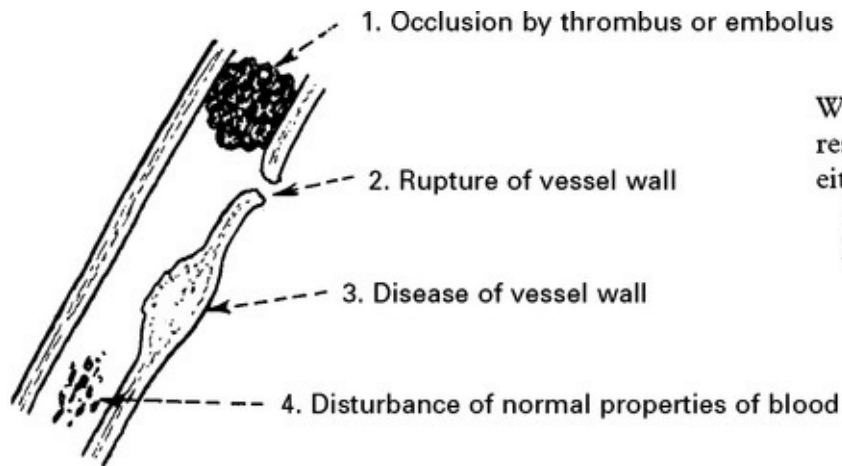
Oral contraceptives

Combined oral contraception (COC) containing high dose oestrogen increased the risk of thrombosis, including stroke. The effect of low dose oestrogen COC is less clear.

CEREBROVASCULAR DISEASE – MECHANISMS

‘Stroke’ is a generic term, lacking pathological meaning. Cerebrovascular

diseases can be defined as those in which brain disease occurs secondary to a pathological disorder of blood vessels (usually arteries) or blood supply.



Whatever the mechanism, the resultant effect on the brain is either:
ischaemia/infarction, or
haemorrhagic disruption.

Of all strokes: – 85% are due to INFARCTION
– 15% are due to HAEMORRHAGE

CEREBROVASCULAR DISEASE – NATURAL HISTORY

Approximately one-third of all ‘strokes’ are fatal. The age of the patient, the anatomical size of the lesion, the degree of deficit and the underlying cause all influence the outcome.

Immediate outcome

In cerebral haemorrhage, mortality approaches 50%.

Cerebral infarction fares better, with an immediate mortality of less than 20%, fatal lesions being large with associated oedema and brain shift.

Embolic infarction carries a better outcome than thrombotic infarction.

Fatal cases of infarction die either at onset, within a few days because of cytotoxic cerebral oedema or later from cardiovascular or respiratory complications.

The level of consciousness on admission to hospital gives a good indication to immediate outcome. The deeper the conscious level the graver the prognosis.

Long-term outcome

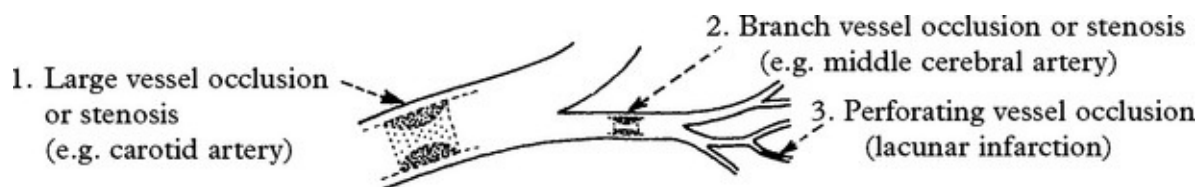
The prognosis following infarction due to thrombosis or embolisation from diseased neck vessels or heart is dependent on the progression of the underlying atherosclerotic disease. Recurrent cerebral infarction rates vary between 5% and 15% per year. Symptoms of coronary artery disease and/or peripheral vascular disease may also ensue. Five year mortality is 44% for males and 36% for females.

The long-term prognosis following survival from haemorrhage depends upon the cause and the treatment.

CEREBROVASCULAR DISEASE – CAUSES

OCCLUSION (50%)

Atheromatous/thrombotic



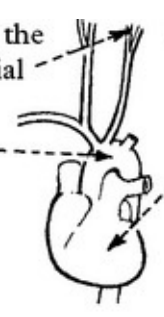
Non-atheromatous diseases of the vessel wall

1. Collagen disease e.g. rheumatoid arthritis systemic lupus erythematosus (SLE)
2. Vasculitis e.g. polyarteritis nodosa temporal arteritis
3. Granulomatous vasculitis e.g. Wegener's granulomatosis

4. Miscellaneous *e.g.* trauma fibromuscular dysplasia syphilitic vasculitis

EMBOLISATION (25%) from:

1. Atheromatous plaque in the intracranial or extracranial arteries or from the aortic arch.



The diagram shows a frontal view of the heart. Arrows point from the aortic arch to the carotid and vertebral arteries, and from the left ventricle into the aorta. Another arrow points from the right atrium to the left atrium, indicating a patent foramen ovale. Dashed lines connect the text labels to these specific anatomical locations.

2. The heart:

 - valvular heart disease
 - arrhythmias
 - ischaemic heart disease
 - bacterial and non-bacterial endocarditis
 - atrial myxoma
 - prosthetic valves
 - paradoxical emboli via patent foramen ovale
 - cardiomyopathy

3. Miscellaneous:

 - fat emboli
 - air emboli
 - tumour emboli.

DISEASES OF BLOOD

e.g. Coagulopathies or Haemoglobinopathies

CEREBRAL VENOUS THROMBOSIS

Thrombosis of cerebral veins may occur with infection, dehydration or in association with oestrogen excess, either post-partum or combined oral contraceptive use.

DECREASED CEREBRAL PERFUSION

Hypotension, from cardiac arrhythmia or GI bleed, can lead to infarction in the watershed between arterial territories.

HAEMORRHAGE (20%)

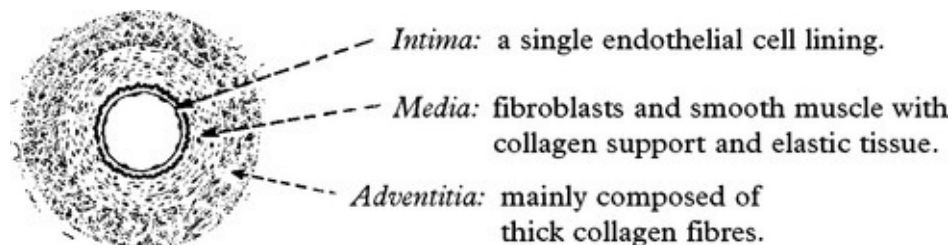
Into the brain substance – parenchymal (15%) and/or subarachnoid space (5%)	Neoplasm
	Coagulation disorder <i>e.g.</i> haemophilia
Hypertension	Anticoagulant therapy
Amyloid vasculopathy	Vasculitis

Aneurysm	Drug abuse e.g. cocaine
Arteriovenous malformation	Trauma

OCCLUSIVE AND STENOTIC CEREBROVASCULAR DISEASE

PATHOLOGY

The normal vessel wall comprises:

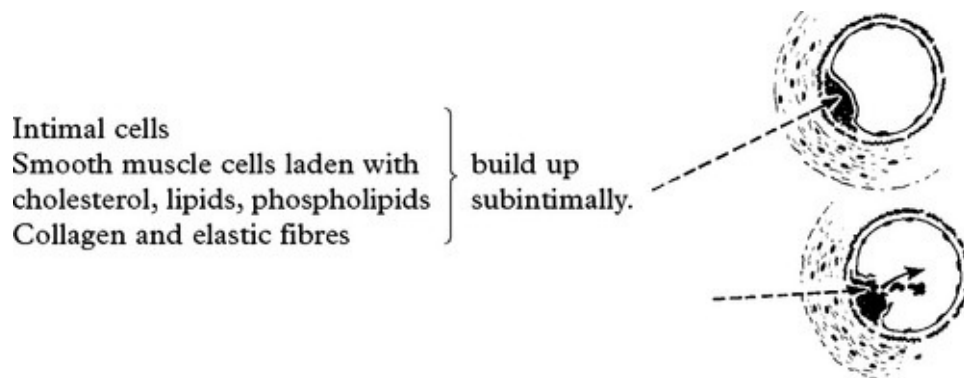


Within brain and spinal cord tissue the adventitia is usually very thin and the elastic lamina between media and adventitia less apparent.

The intima is an important barrier to leakage of blood and constituents into the vessel wall. In the development of the atherosclerotic plaque, damage to the endothelium of the intima is the primary event.

The atherosclerotic plaque

Following intimal damage:

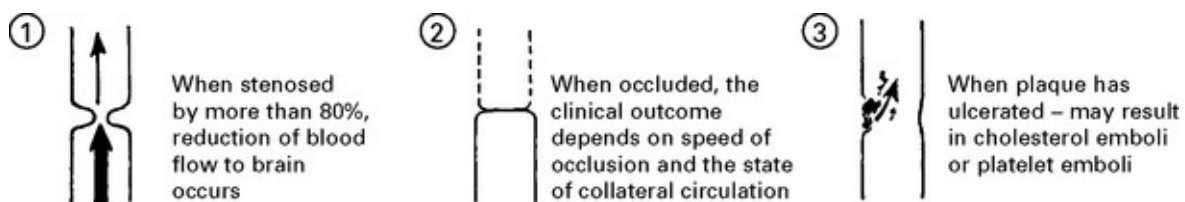


Haemorrhage may occur within the plaque or the plaque may ulcerate

into the lumen of the vessel forming an intraluminal mural thrombus. Either way, the lumen of the involved vessel is narrowed (stenosed) or blocked (occluded).

The plaque itself may give rise to emboli. Cholesterol is present partly in crystal form and fragments following plaque rupture may be sufficiently large to occlude the lumen of distal vessels. The cholesterol esters, lipids and phospholipids each play a role in the aggregation of such emboli.

The carotid bifurcation in the neck is a frequent site at which the atheromatous plaque causes stenosis or occlusion.



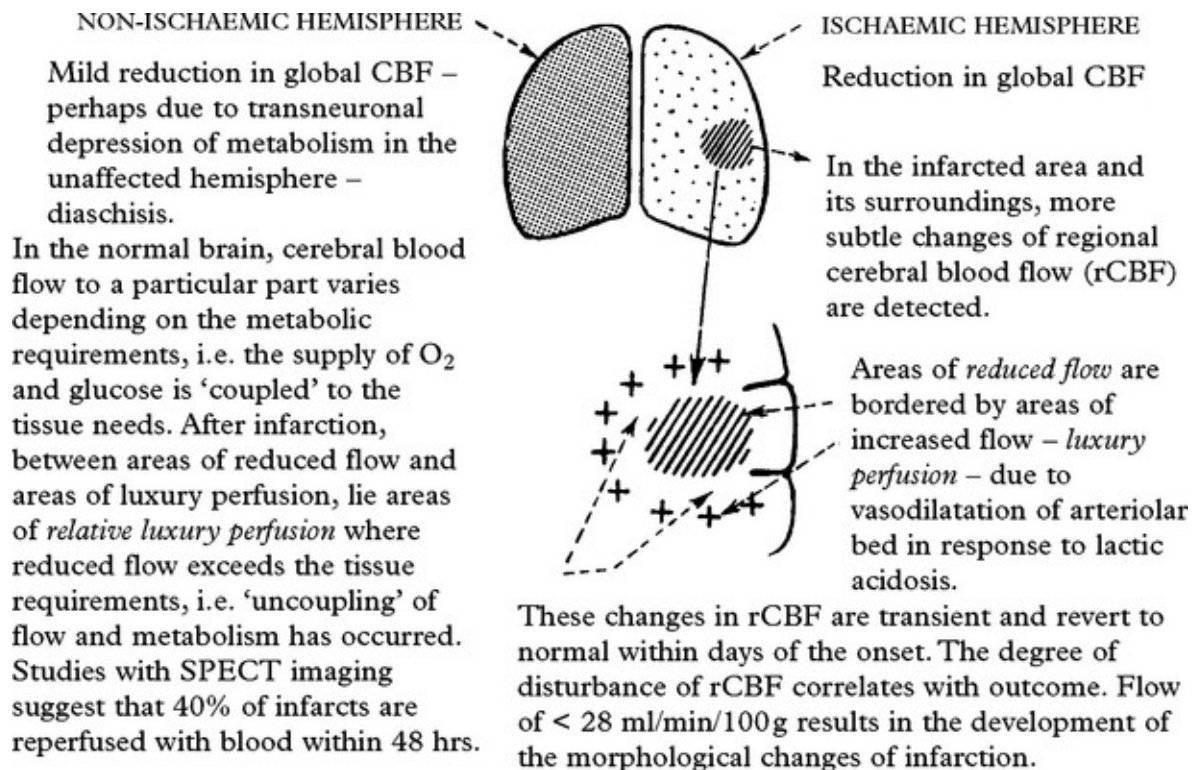
Platelet emboli arise from thrombus developed over the damaged endothelium. This thrombus is produced partly by platelets coming into contact with exposed collagen fibres. Endothelial cells synthesise PROSTACYCLIN which is a potent vasodilator and inhibitor of platelet aggregation. THROMBOXANE A₂, synthesised by platelets, has opposite effects. In thrombus formation these two PROSTAGLANDINS actively compete with each other.

CEREBROVASCULAR DISEASE – PATHOPHYSIOLOGY

Standard techniques of cerebral blood flow (CBF) measurement provide information on both global and regional flow in patients with cerebral ischaemia or infarction. Recent availability of positron emission tomography (PET), recording oxygen and glucose metabolism, as well as

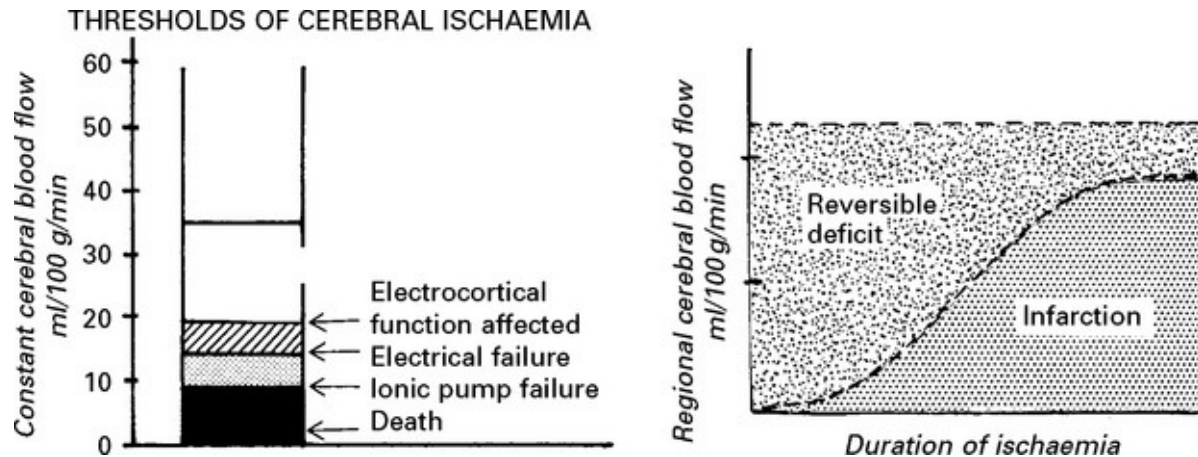
blood flow and blood volume, gives a more detailed and accurate understanding of pathophysiological changes after stroke.

Changes in cerebral infarction



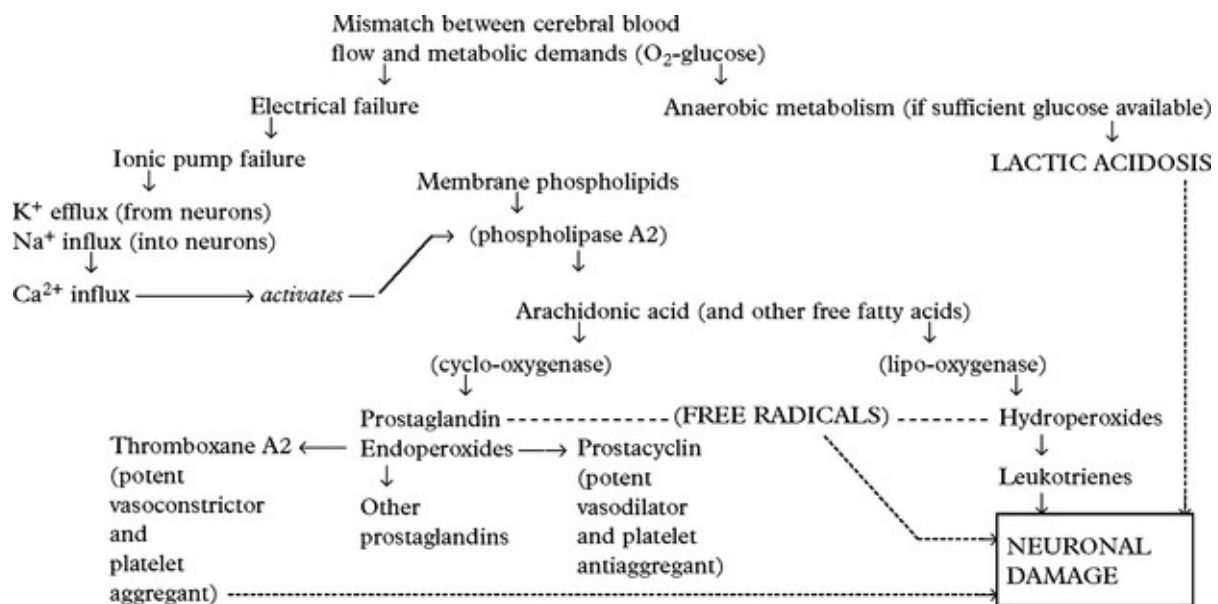
Pathophysiology of ischaemia

Progression from reversible ischaemia to infarction depends upon the degree and duration of the reduced blood flow.



Ischaemic cascade

A significant fall in cerebral blood flow produces a cascade of events which, if unchecked, lead to the production and accumulation of toxic compounds and apoptosis (programmed cell death).



Role of neurotransmitters

In addition to the cascade outlined above one of the amino acid excitatory neurotransmitters, Glutamate, in excess is a powerful neurotoxin, which plays an important role in ischaemic brain damage.

There have been numerous agents that interfere with different steps in this complicated series of interactions with an ultimate aim of providing *neuroprotection* and limiting the size of the stroke. Despite numerous trials of more than 100 different agents no drug has been developed that provides neuroprotection in man.

TRANSIENT ISCHAEMIC ATTACKS (TIAs)

Transient ischaemic attacks are episodes of focal neurological symptoms due to inadequate blood supply to the brain. Attacks are sudden in onset, resolve within 24 hours or less and leave no residual deficit. These attacks are important as warning episodes or precursors of cerebral infarction.

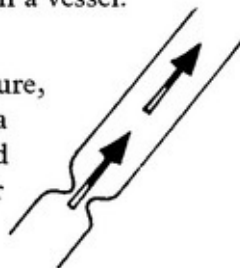
Before diagnosing TIAs, consider other causes of transient neurological dysfunction – migraine, partial seizures, hypoglycaemia, syncope and hyperventilation.

The pathogenesis of transient ischaemic attacks

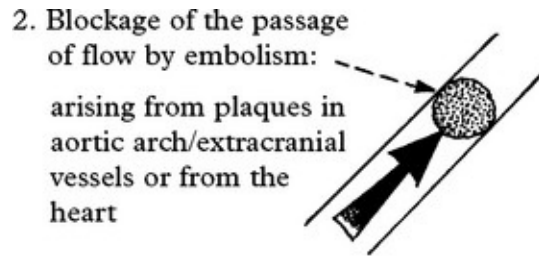
A reduction of cerebral blood flow below 20–30 ml/100 g/min produces neurological symptoms. The development of infarction is a consequence of the *degree* of reduced flow and the *duration* of such a reduction. If flow is restored to an area of brain within the critical period, ischaemic symptoms will reverse themselves. TIAs may be due to:

1. Reduced flow through a vessel:

a fall in perfusion pressure,
e.g. cardiac dysrhythmia
associated with localised
stenotic cerebrovascular
disease



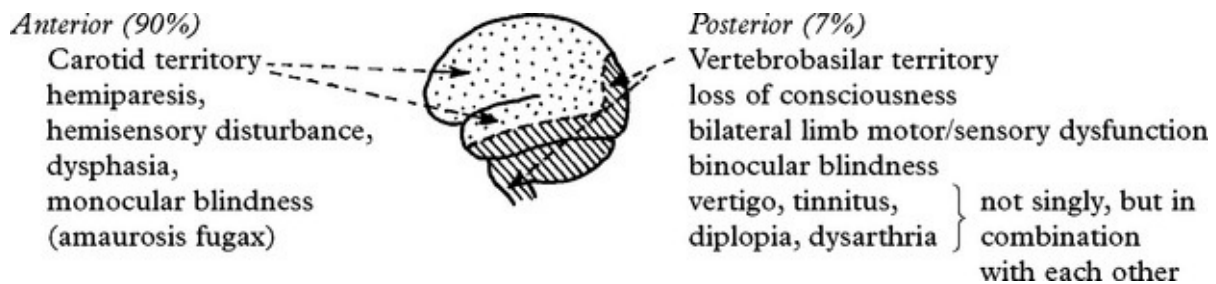
– the *haemodynamic* explanation.



– the *embolic* explanation.

Both mechanisms occur. Emboli are accepted as the cause of the majority of TIAs.

The symptomatology of TIAs



A small number of transient ischaemic attacks are difficult to fit convincingly into either anterior or posterior circulation, *e.g.* dysarthria with hemiparesis.

The natural history of TIAs

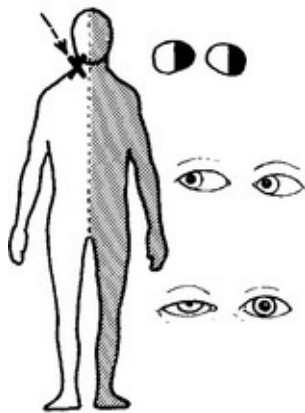
Following a TIA, 5% of patients will develop infarction within 1 week and 12% within 3 months. The risk of infarction is greatest in older patients with more risk factors (hypertension and diabetes) who have had longer hemispheric TIAs. About 10% of patients who have a stroke have had a warning TIA.

CLINICAL SYNDROMES – LARGE VESSEL OCCLUSION

OCCLUSION OF THE INTERNAL CAROTID ARTERY – may present in a

‘stuttering’ manner due to progressive narrowing of the lumen or recurrent emboli.

The degree of deficit varies – occlusion may be asymptomatic and identified only at autopsy, or a catastrophic infarction may result.



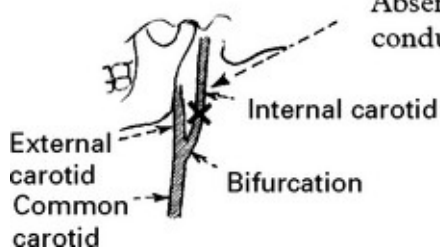
In the most extreme cases there may be:

- Deterioration of conscious level
- Homonymous hemianopia of the contralateral side
- Contralateral hemiplegia
- Contralateral hemisensory disturbance
- Gaze palsy to the opposite side – eyes deviated to the side of the lesion

A partial Horner's syndrome may develop on the side of the occlusion (involvement of sympathetic fibres on the internal carotid wall).

Occlusion of the dominant hemisphere side will result in a global aphasia.

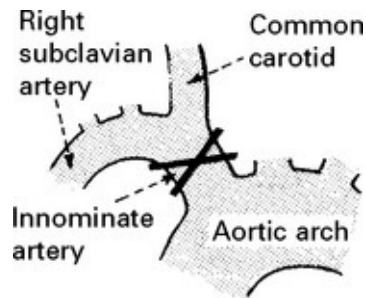
Examination of the neck will reveal:



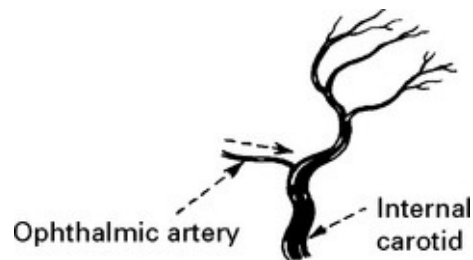
Absent carotid pulsation at the angle of the jaw with poorly conducted heart sounds along the internal carotid artery.

Prodromal symptoms prior to occlusion may take the form of monocular blindness – AMAUROSIS FUGAX and transient hemisensory or hemimotor disturbance (see page 258).

The origins of the vessels from the aortic arch are such that an *innominate artery occlusion* will result not only in the clinical picture of carotid occlusion but will produce diminished blood flow and hence blood pressure in the right arm.

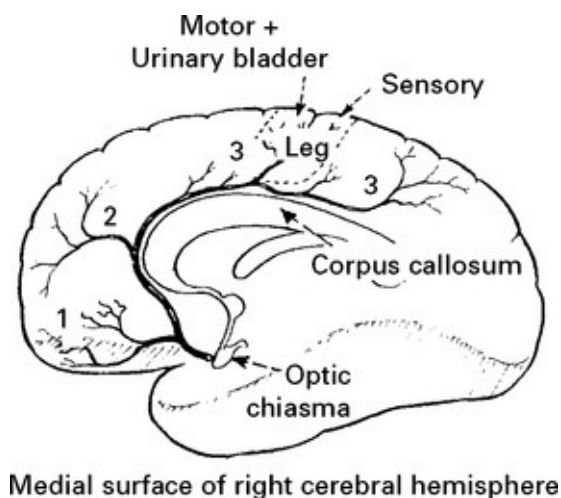


The outcome of carotid occlusion depends on the collateral blood supply primarily from the circle of Willis, but, in addition, the external carotid may provide flow to the *anterior and middle cerebral arteries* through meningeal branches and retrogradely through the ophthalmic artery to the *internal carotid artery*.



ANTERIOR CEREBRAL ARTERY

Anatomy



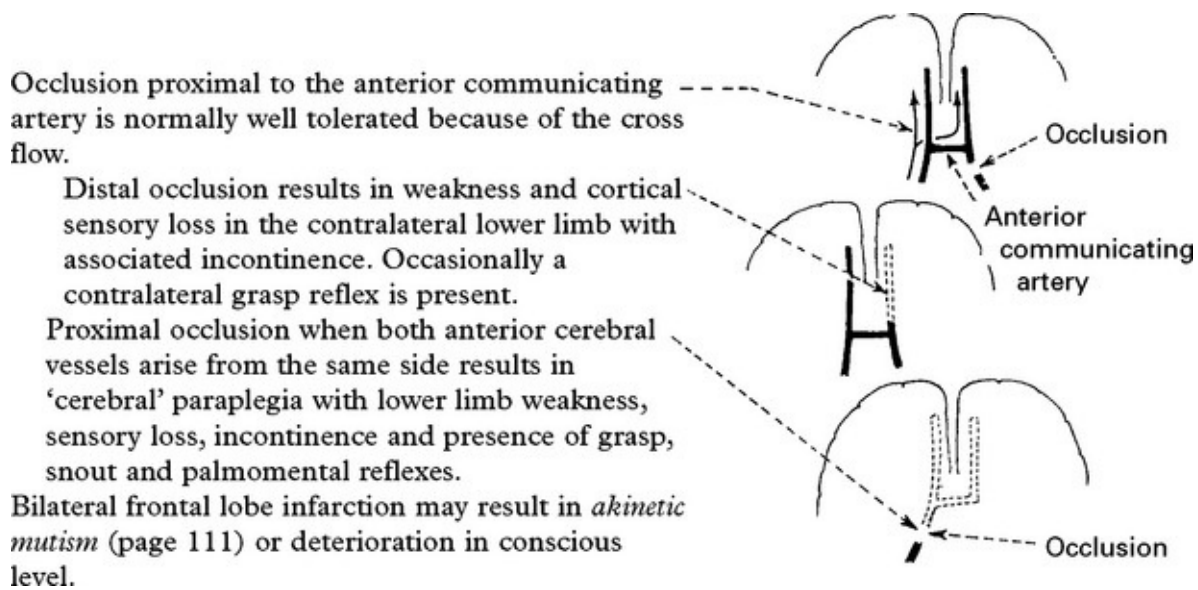
The anterior cerebral artery is a branch of the internal carotid and runs above the optic nerve to follow the curve of the corpus callosum. Soon after its origin the vessel is joined by the anterior communicating artery. Deep branches pass to the anterior part of the internal capsule and basal nuclei.

Cortical branches supply the medial surface of the hemisphere:

1. Orbital
2. Frontal
3. Parietal

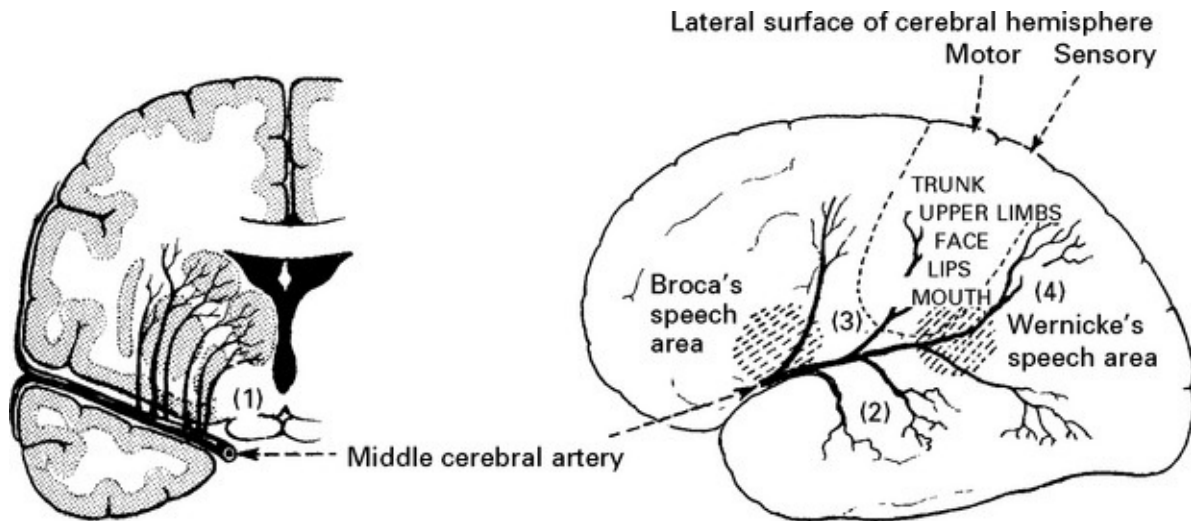
Clinical features

The anterior cerebral artery may be occluded by embolus or thrombus. The clinical picture depends on the site of occlusion (especially in relation to the anterior communicating artery) and anatomical variation, e.g. both anterior cerebral arteries may arise from one side by enlargement of the anterior communicating artery.



MIDDLE CEREBRAL ARTERY

Anatomy



The middle cerebral artery is the largest branch of the internal carotid artery. It gives off (1) deep branches (perforating vessels – lenticulostriate) which supply the anterior limb of the internal capsule and part of the basal nuclei. It then passes out to the lateral surface of the cerebral hemisphere at the insula of the lateral sulcus. Here it gives off cortical branches (2) temporal, (3) frontal, (4) parietal.

Clinical features

The middle cerebral artery may be occluded by embolus or thrombus. The clinical picture depends upon the site of occlusion and whether dominant or non-dominant hemisphere is affected.

Occlusion at the insula

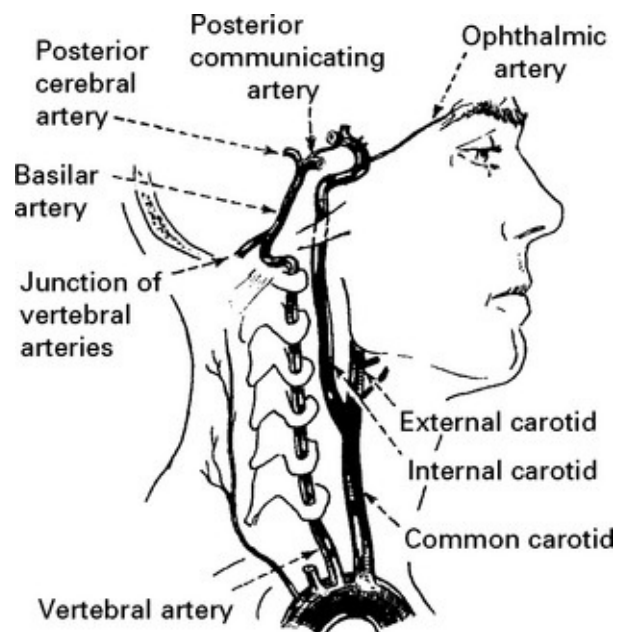
All cortical branches are involved –	Contralateral hemiplegia (leg relatively spared)
	Contralateral hemianesthesia and hemianopia
	Aphasia (dominant)
	Neglect of contralateral limbs
	Dressing difficulty
	} (non-dominant)

When cortical branches are affected individually, the clinical picture is less severe, *e.g.* involvement of parietal branches alone may produce Wernicke's dysphasia with no limb weakness or sensory loss.

The deep branches (perforating vessels) of the middle cerebral artery may be a source of haemorrhage or small infarcts (lacunes – see later).

VERTEBRAL ARTERY OCCLUSION

Anatomy



The vertebral artery arises from the subclavian artery on each side. Underdevelopment of one vessel occurs in 10%.

The vertebral artery runs from its origin through the foramen of the transverse processes of the mid-cervical vertebrae. It then passes laterally through the transverse process of the axis, then upwards to the atlas accompanied by a venous plexus and across the suboccipital triangle to the vertebral canal. After piercing the dura and arachnoid matter, it enters the cranial cavity through the foramen magnum. At the

lower border of the pons, it unites with its fellow to form the basilar artery.

The vertebral artery and its branches supply the medulla and the inferior surface of the cerebellum before forming the basilar artery.

Clinical features

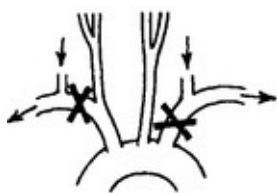
Occlusion of the vertebral artery, when low in the neck, is compensated by anastomotic channels.

When one vertebral artery is hypoplastic, occlusion of the other is equivalent to basilar artery occlusion.

Only the posterior inferior cerebellar artery (PICA) depends solely on flow through the vertebral artery. Vertebral artery occlusion may therefore present as a PICA syndrome ([page 255](#)).

The close relationship of the vertebral artery to the cervical spine is important. Rarely, damage at intervertebral foramina or the atlanto-axial joints following subluxation may result in intimal damage, thrombus formation and embolisation.

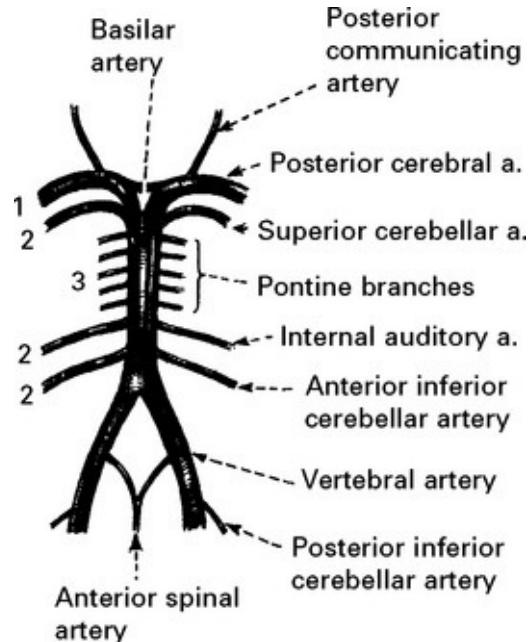
Vertebral artery compression during neck extension may cause symptoms of intermittent vertebrobasilar insufficiency.



X Stenosis of the proximal left or right subclavian artery may result in retrograde flow down the vertebral artery on exercising the arm. This is commonly asymptomatic and demonstrated incidentally by Doppler techniques or angiography. Occasionally symptoms of vertebrobasilar insufficiency arise – *subclavian 'steal' syndrome*. Surgical reconstruction or bypass of the subclavian artery may be indicated.

BASILAR ARTERY OCCLUSION

Anatomy



The basilar artery supplies the brain stem from medulla upwards and divides eventually into posterior cerebral arteries as well as posterior communicating arteries which run forward to join the anterior circulation (circle of Willis).

Branches can be classified into:

1. Posterior cerebral arteries
2. Long circumflex branches
3. Paramedian branches.

Clinical features

Prodromal symptoms are common and may take the form of diplopia, visual field loss, intermittent memory disturbance and a whole constellation of other brain stem symptoms:

- vertigo
- ataxia

- paresis
- paraesthesia

The *complete basilar syndrome* following occlusion consists of:

- impairment of consciousness → coma
- bilateral motor and sensory dysfunction
- cerebellar signs
- cranial nerve signs indicative of the level of occlusion.

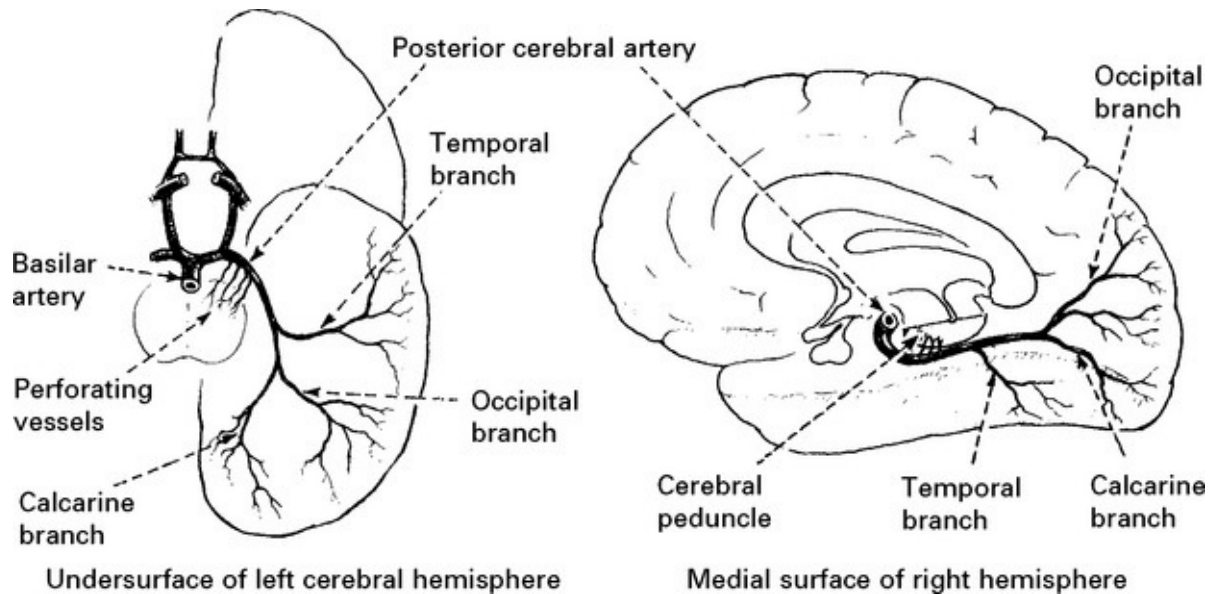
The clinical picture is variable. Occasionally basilar thrombosis is an incidental finding at autopsy.

‘Top of basilar’ occlusion: This results in lateral midbrain, thalamic, occipital and medial temporal lobe infarction. Abnormal movements (hemiballismus) are associated with visual loss, pupillary abnormalities, gaze palsies, impaired conscious level and disturbances of behaviour.

Paramedian perforating vessel occlusion gives rise to the ‘LOCKED-IN’ SYNDROME ([page 256](#)) and LACUNAR infarction ([page 257](#)).

POSTERIOR CEREBRAL ARTERY

Anatomy



The posterior cerebral arteries are the terminal branches of the basilar artery. Small perforating branches supply midbrain structures, choroid plexus and posterior thalamus. Cortical branches supply the undersurface of the temporal lobe – temporal branch; and occipital and visual cortex – occipital and calcarine branches.

Clinical features

Proximal occlusion by thrombus or embolism will involve perforating branches and structures supplied:

Midbrain syndrome – III nerve palsy with contralateral hemiplegia – WEBER's SYNDROME

Thalamic syndromes – chorea or hemiballismus with hemisensory disturbance.

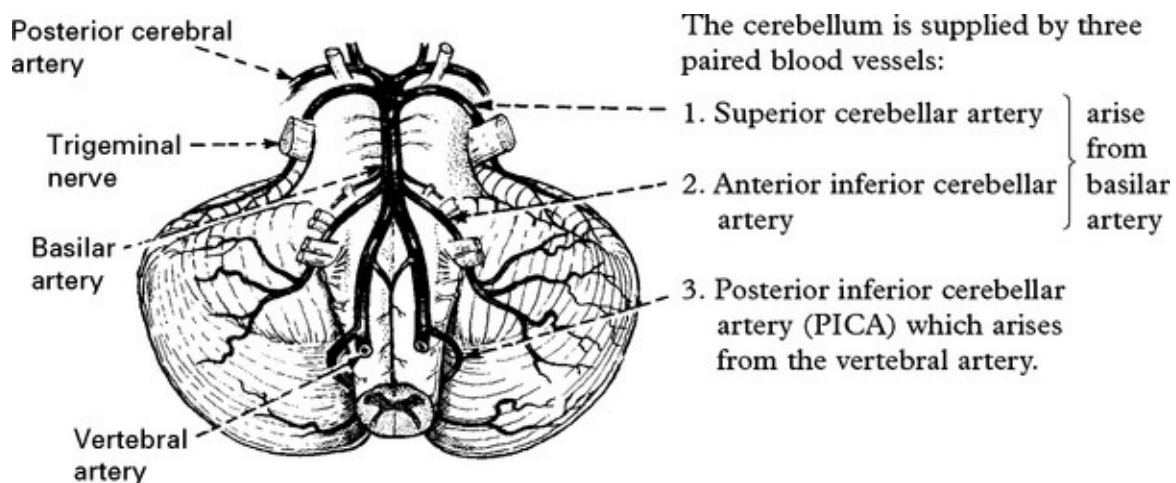
Occlusion of cortical vessels will produce a different picture with visual field loss (homonymous hemianopia) and sparing of macular vision (the posterior tip of the occipital lobe, *i.e.* the macular area, is also supplied by the middle cerebral artery).

Posterior cortical infarction in the dominant hemisphere may produce problems in naming colours and objects.

CLINICAL SYNDROMES – BRANCH OCCLUSION

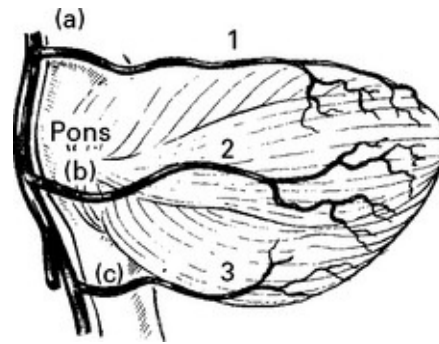
BASILAR ARTERY – LONG CIRCUMFLEX BRANCH OCCLUSION

Anatomy



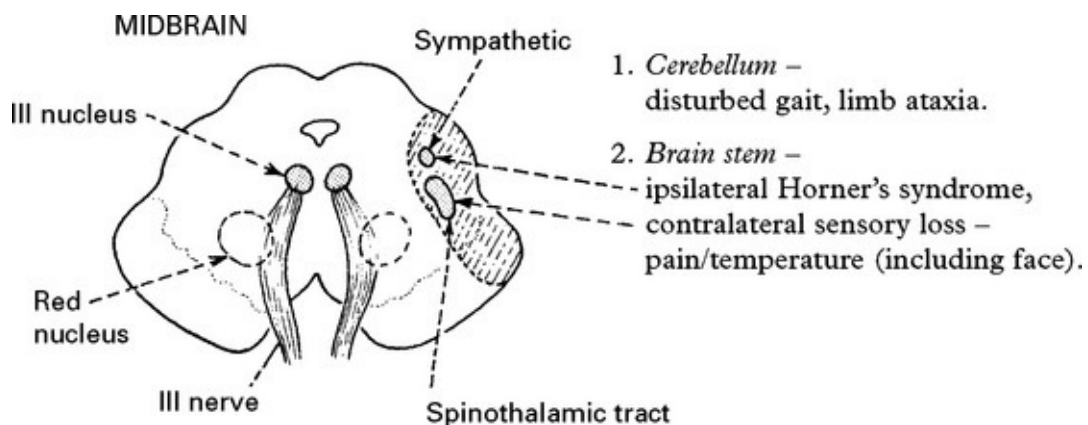
It can be seen that a vascular lesion in the territory of these vessels will produce, not only cerebellar, but also brain stem symptoms and signs localising to:

- (a) superior pontine,
- (b) inferior pontine and
- (c) medullary levels.

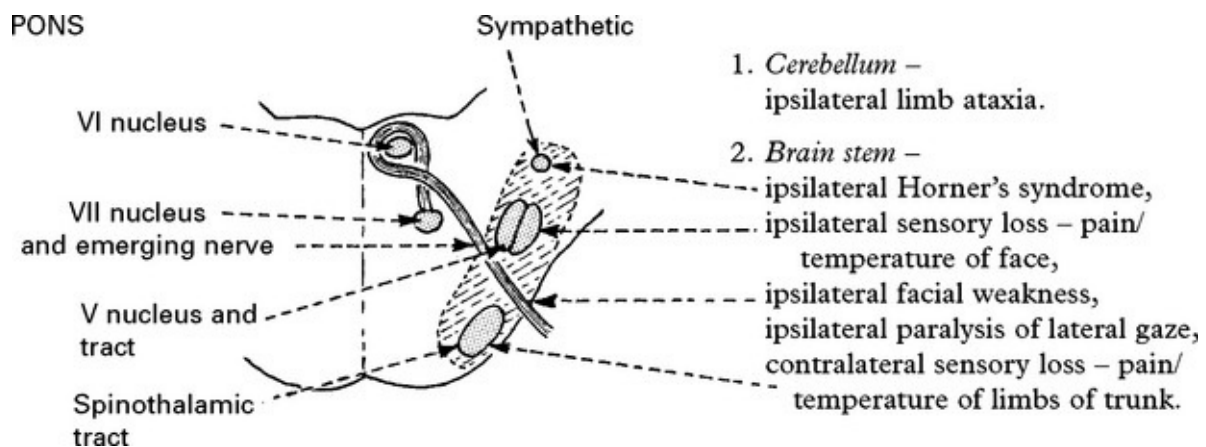


Clinical features

Superior cerebellar artery syndrome results in:



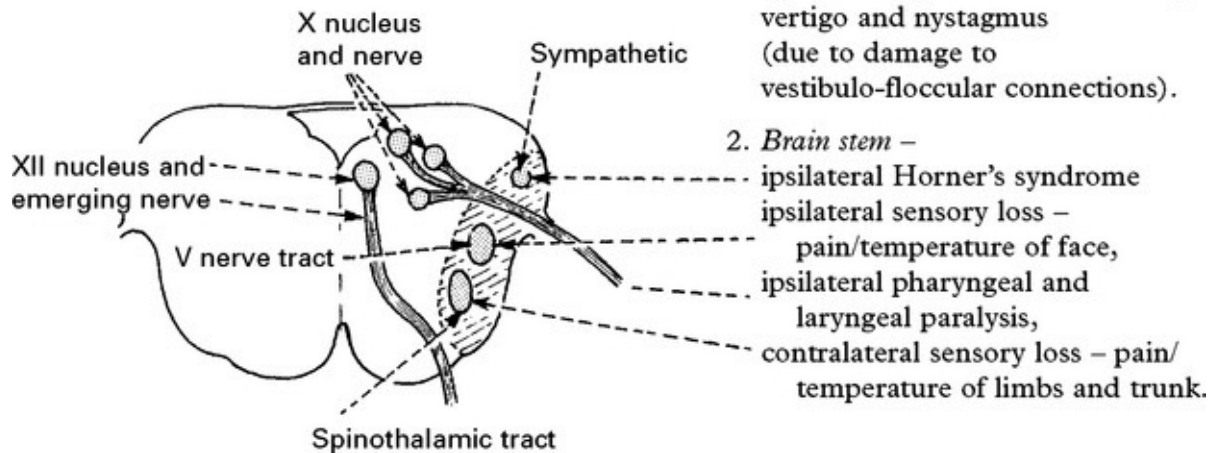
Anterior inferior cerebellar artery syndrome results in:



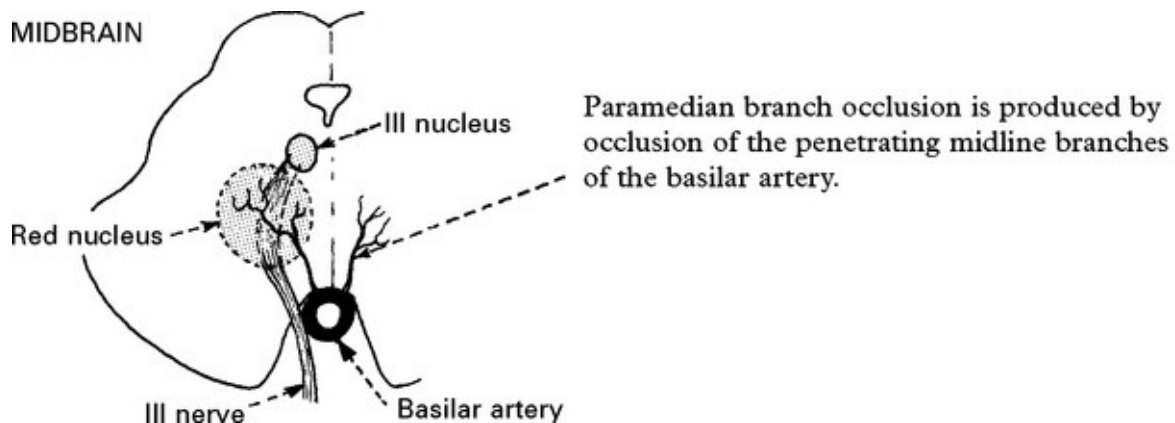
Posterior inferior cerebellar artery syndrome (lateral medullary

syndrome) results in:

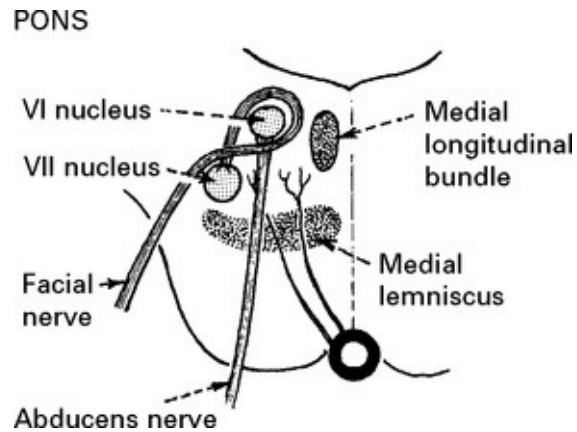
MEDULLA



BASILAR ARTERY – PARAMEDIAN BRANCH OCCLUSION

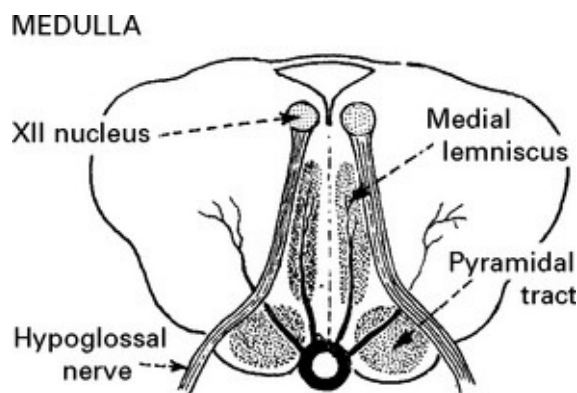


At the midbrain level damage to the nucleus or the fasciculus of the oculomotor nerve (III) will result in a complete or partial III nerve palsy; damage to the red nucleus (outflow from opposite cerebellar hemisphere) will also produce contralateral tremor – referred to as BENEDIKT'S SYNDROME.



At the pontine level an abducens nerve (VI) palsy will occur with ipsilateral facial (VII) weakness and contralateral sensory loss – light touch, proprioception (medial lemniscus damage) when the lesion is more basal.

Abducens and facial palsy may be accompanied by contralateral hemiplegia – MILLARD-GUBLER SYNDROME.



At the medullary level, bilateral damage usually occurs and results in the 'LOCKED-IN' SYNDROME. The patient is paralysed and unable to talk, although some facial and eye movements are preserved.

Spinothalamic sensation is retained, but involvement of the medial lemniscus produces loss of 'discriminatory' sensation in the limbs. The syndrome usually follows basilar artery occlusion and carries a grave

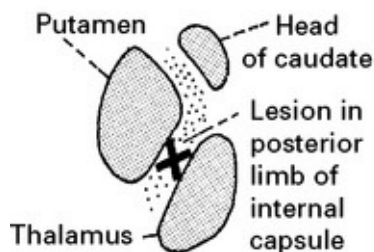
prognosis.

CLINICAL SYNDROMES – LACUNAR STROKE (LACI)

Occlusion of deep penetrating arteries produces subcortical infarction characterised by *preservation* of cortical function – language, other cognitive and visual functions.

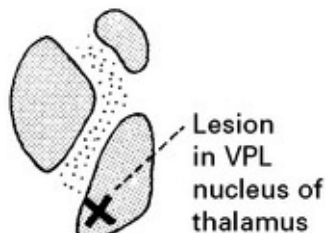
Clinical syndromes are distinctive and normally result from long-standing hypertension. In 80%, infarcts occur in periventricular white matter and basal ganglia, the rest in cerebellum and brain stem. Areas of infarction are 0.5 – 1.5cm in diameter and occluded vessels demonstrate lipohyalinosis, microaneurysm and microatheromatous changes. Lacunar or subcortical infarction accounts for 17% of all thromboembolic strokes and knowledge of commoner syndromes is essential.

1. Pure motor hemiplegia



Clinical: Equal weakness of contralateral face, arm and leg with dysarthria
Vessel(s): Lenticulostriate A.

2. Pure sensory stroke



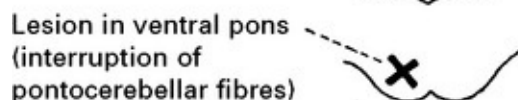
Clinical: Numbness and tingling of contralateral face and limbs. Sensory examination may be normal
Vessel(s): Thalamogeniculate A.

3. Dysarthria/clumsy hand



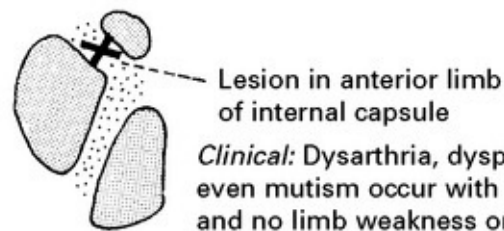
Clinical: Dysarthria due to weakness of ipsilateral face and tongue associated with clumsy but strong contralateral arm.
Vessel(s): Perforating branch of Basilar A.

4. Ataxic hemiparesis



Clinical: Mild hemiparesis with more marked ipsilateral limb ataxia
Vessel(s): Perforating branch of Basilar A. (This syndrome can also be produced by anterior capsular lesions)

5. Severe dysarthria with facial weakness



Clinical: Dysarthria, dysphagia and even mutism occur with mild facial and no limb weakness or clumsiness.
Vessel(s): Lenticulostriate A.

Sensorimotor syndromes are common although anatomical basis is obscure. A recent Stroke Data Bank survey showed the commonest presentations to be:

Pure motor hemiplegia 57%

Sensorimotor 20%

Ataxic hemiparesis 10%

Pure sensory 7%

Dysarthria/Clumsy hand 6%

Investigations MRI is superior to CT demonstrating lacunae, although either may occasionally misdiagnose a small resolving haematoma. Confirmation of lacunar stroke may save patients from unnecessary investigations for carotid and cardiac embolic source.

Prognosis For all syndromes this is encouraging. Careful control of blood pressure and the use of aspirin usually prevents recurrence. Multiple lacunar infarctions – ‘état lacunaire’ – results in shuffling gait, pseudobulbar palsy and subcortical dementia.

CLASSIFICATION OF SUBTYPES OF CEREBRAL INFARCTION

A recently devised classification of infarction has proved simple and of practical value in establishing *diagnosis* and in *predicting outcome* –

	Clinical features	Outcome
Total Anterior Circulation	motor and sensory deficit, hemianopia and disturbance of higher cerebral function	Poor
Syndrome (TACS)		

Partial Anterior Circulation	any two of above	Variable
Syndrome (PACS)	or isolated disturbance of cerebral function	
Posterior Circulation	signs of brain stem dysfunction	Variable
Syndrome (POCS)	or isolated hemianopia	
Lacunar Anterior Circulation	pure motor stroke	Good
Syndrome (LACS)	or pure sensory stroke	
	or pure sensorimotor stroke	
	or ataxic hemiparesis	

EMBOLISATION

Emboli consist of friable atheromatous material, platelet-fibrin clumps or well formed thrombus.

The **diagnosis** of embolic infarction depends on:

- The identification of an embolic source, *e.g.* cardiac disease.
- The clinical picture of sudden onset.
- Infarction in the territory of a major vessel or large branch.

Clinical picture – depends on the vessel involved. Emboli commonly produce *transient ischaemic attacks (TIA)* as well as *infarction*.

Symptoms are referable to the eye (retinal artery) and to the anterior and middle cerebral arteries, and take the form of:

Visual loss – transient, *i.e. amaurosis fugax* or permanent.

Hemisensory and hemimotor disturbance.

Disturbance of higher function, *e.g.* dysphasia.

Focal or generalised seizures – may persist for some time after the ischaemic episode.

Depression of conscious level if major vessel occlusion occurs.

Emboli less frequently affect the posterior circulation.

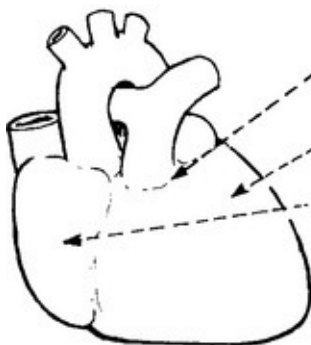
EMBOLI FROM THE INTERNAL CAROTID ARTERY AND AORTA

Emboli from these sources are commonest outwith the heart. The majority of all cerebral emboli arise from ulcerative plaques in the carotid arteries (see [page 244](#)).

Emboli arising from the aorta (atheromatous plaque or aortic aneurysm) often involve both hemispheres and systemic embolisation (e.g. affecting limbs) may coexist.



EMBOLI OF CARDIAC ORIGIN



The heart represents a major source of cerebral emboli.

Valvular heart disease: rheumatic heart disease e.g. mitral stenosis with atrial fibrillation or mitral valve prolapse.

Ischaemic heart disease: myocardial infarction with mural thrombus formation.

Arrhythmias: Non-rheumatic (non-valvular) atrial fibrillation is the most common cause of cardioembolic stroke

Bacterial endocarditis may give rise to septic cerebral embolisation with ischaemia → infection → abscess formation.

Neurological signs will occur in 30% of all cases of bacterial endocarditis, *S. aureus* and *streptococci* being the offending organisms in the majority.

Non-bacterial endocarditis (marantic endocarditis): associated with malignant disease due to fibrin and platelet deposition on heart valves.

Atrial myxoma is a rare cause of recurrent cerebral embolisation. Bihemisphere episodes with a persistently elevated ESR should arouse suspicion which may be confirmed by cardiac ultrasound.

Patent foramen ovale may result in paradoxical embolisation; suspect in patient with deep venous thrombosis who develops cerebral infarction. Emboli can also arise from intracardiac thrombus.

New cardiac imaging techniques especially Transoesophageal Echocardiography (TOE) allow a more accurate detection of potential embolic source. Transcranial Doppler (TCD) may characterise emboli by analysing their signals and help quantify risk of recurrence.

EMBOLI FROM OTHER SOURCES

Fat emboli: following fracture, especially of long bones and pelvis, fat appears in the bloodstream and may pass into the cerebral circulation, usually 3 –6 days after trauma. Emboli are usually multiple and signs are diffuse.

Air emboli follow injury to neck/chest, or follow surgery. Rarely, air emboli complicate therapeutic abortion. Again the picture is diffuse neurologically. Onset is acute; if the patient survives the first 30 minutes, prognosis is excellent.

Nitrogen embolisation or decompression sickness (the ‘bends’) produces a similar picture, but if the patient survives, neurological disability may be profound.

Tumour emboli result in metastatic lesions; the onset is usually slow and progressive. Acute stroke-like presentation may occur, followed weeks or months later by the mass effects.



STENOTIC/OCCLUSIVE DISEASE – INVESTIGATIONS

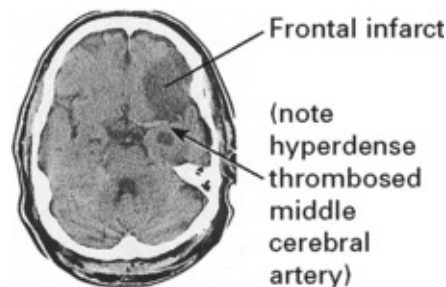
1. CONFIRM THE DIAGNOSIS

Computerised tomography (CT scan)

All patients should have a CT scan, urgently if

- conscious level depressed
- diagnosis uncertain
- on anticoagulants
- before commencing/resuming antithrombotics
- if thrombolysis is considered.
- severe headache at onset.

Infarction is evident as a low-density lesion which conforms to a vascular territory, *i.e.* usually wedge shaped. Subtle changes occur within 3 hours in some patients; most scans become abnormal within 48 hours.



CT scan also identifies:

- the site and size of the infarct, providing a prognostic guide
- the presence of haemorrhagic infarction where bleeding occurs into the infarcted area
- intracerebral haemorrhage or tumour.

Magnetic resonance imaging (MRI)

T2 prolongation (hyperintensity in relation to white and grey matter) occurs within hours of onset of ischaemic symptoms. Advanced techniques, diffusion weighted imaging (DWI) and perfusion imaging (PWI) show respectively early infarction (cytotoxic oedema) and ischaemic tissue at risk (the ischaemic penumbra). These advanced techniques are valuable predictors of outcome and guide treatments directed as 'ischaemic salvage' e.g. thrombolysis.

2. DEMONSTRATE THE SITE OF PRIMARY LESION

(a) Non-invasive investigation

Ultrasound – Doppler/Duplex scanning: assesses extra-and intracranial vessels (page 44). A normal study precludes the need for angiography.

Cardiac ultrasound (transthoracic or transoesophageal): this often reveals a cardiac embolic source in young people with stroke, e.g. prolapsed mitral valve, patent foramen ovale.

Magnetic resonance angiography (MRA)

'Time of flight' or contrast enhanced techniques are used. Whilst of value in patients with heavily calcified carotid plaques, resistant to Doppler, it tends to overestimate the severity of stenosis. When assessing the carotid arteries it is best used in combination with Doppler. Its non-invasive

nature makes it helpful in investigating the intracranial circulation.

Computed tomographic angiography (CTA)

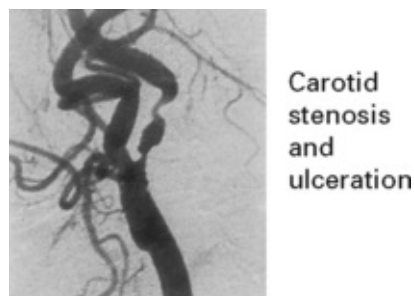
Dynamic helical CT, following bolus injection of non-ionic contrast, can be used to investigate both intracranial and extracranial vasculature. CTA compared with DSA correctly classifies the degree of carotid stenosis in 96% of cases but is insensitive to ulcerative plaque. Again it is best used in conjunction with Doppler.

(b) Digital intravenous subtraction angiography (DSA)

The combination of the above techniques has decreased the need for invasive investigation but *cerebral angiography* may still be required to make a definitive diagnosis.

Indications of angiography

1. In those patients with anterior circulation TIA or minor stroke if non-invasive techniques have not clarified the nature and degree of carotid stenosis.
2. In patients where unusual aetiologies are suspected and less invasive imaging has not been diagnostic – for example young patients or when cerebral vasculitis is suspected.



3. IDENTIFY FACTORS WHICH MAY INFLUENCE

TREATMENT AND OUTCOME

General investigations identify conditions which may predispose towards premature cerebrovascular disease. These are essential in all patients.

Chest x-ray – cardiac enlargement – hypertension/valvular heart disease

ECG – ventricular enlargement and/or arrhythmias – hypertension/embolic disease recent myocardial infarct – embolic disease sinoatrial conduction defect – embolic disease/output failure

Blood glucose – diabetes mellitus

Serum lipids and cholesterol – hyperlipidaemia

<i>ESR</i> –	} vasculitis/collagen vascular disease	} See inflammatory vasculitis and blood diseases (pages 267–271)
<i>Auto-antibodies</i> –		
<i>Urine analysis</i> – polyarteritis, thrombocytopenia		
<i>Full blood count</i> – polycythaemia, thrombocytopenia		

VDRL-TPHA – neurosyphilis

Prothrombin time – circulating auto-anticoagulants

Partial thromboplastin time (PTT) – prolonged by lupus anticoagulant

Note drug history – oral contraceptives, amphetamines, opiates

Following the interpretation of these preliminary investigations, more detailed studies may be required, e.g.

– Echocardiography	— structural or cardiac embolic source
– 24 hour cardiac monitor	— occult AF
– blood cultures	— subacute bacterial endocarditis
– HIV screen	— AIDS
– sickle cell screen	} haematological disorder
– plasma electrophoresis	
– viscosity studies	

- anticardiolipin antibodies – antiphospholipid syndrome
- muscle biopsy – mitochondrial disease

CEREBRAL INFARCTION – MANAGEMENT

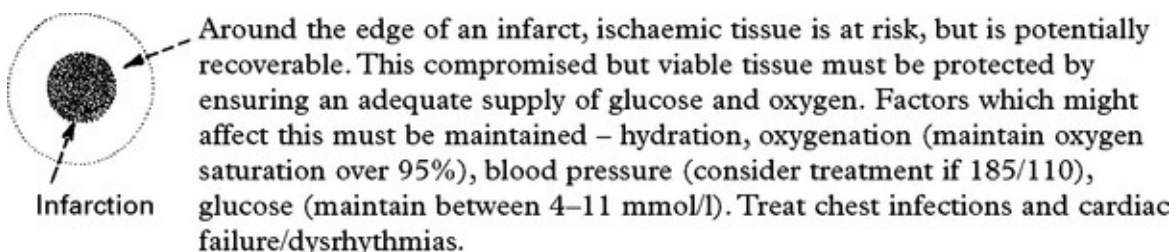
THE ACUTE STROKE

Clinical history, examination and investigation will separate infarction and haemorrhage. Once the nature of the ‘stroke’ has been confidently defined, treatment should be instigated. The treatment of stroke has been the subject of many clinical trials and the following is a digest of the current advice based on those studies.

Treatment aims

- Recanalise blocked vessels
- Prevent progression of present event
- Prevent immediate complication
- Prevent the development of subsequent events
- Rehabilitate the patient.

General measures



Specific measures

Thrombolysis

Intravenous recombinant tissue plasminogen activator (alteplase) given within 3 hours of an anterior circulation ischaemic stroke improves outcome despite the increased risk of iatrogenic intracranial haemorrhage. Thus patients who might be candidates need urgent assessment and CT scanning to exclude cerebral haemorrhage.

Key contraindications to thrombolysis –

Uncertain time of onset

Spontaneously improving

Head injury or previous stroke in last 3 months

GI surgery in last 21 days

BP > 180/110

On anticoagulant

Seizure

Hypodensity on CT

Patients not eligible for thrombolysis

Give aspirin 300 mg daily for 2 weeks or clopidogrel in aspirin intolerant patients. Anticoagulants should be avoided if possible as they increase the risk of deterioration from haemorrhagic transformation.

Transfer to stroke unit

There is good evidence that multidisciplinary care on a stroke unit improves the outcome of patients with stroke.

Assess swallow

Aspiration pneumonia is a significant complication after stroke. Minimise this risk by assessing swallowing and using a nasogastric tube for fluids and food if swallowing unsafe.

Early mobilisation

Help patients sit up when possible and mobilise early.

Special situations

Decompressive hemicraniectomy

A small number of young patients (<60) with large middle cerebral artery strokes deteriorate after 24–72 hours from massive cytotoxic cerebral oedema which is resistant to medical therapy. Surgical decompression can save life, allowing many patients to make reasonable recoveries.

Other neurosurgical interventions

Patients with large cerebellar infarcts can deteriorate 24–48 hours after their stroke when oedema leads to compression of the posterior fossa and associated hydrocephalus. Posterior fossa decompression can be life saving and many patients then make good recoveries.

Prevention of further stroke

The recognition of risk factors and their correction to minimise the risk of further events forms a necessary and important step in long-term treatment.

The strategies here are the same as those used for treatment for patients with TIA (see below).

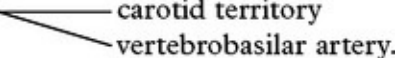
- Control hypertension
- emphasise the need to stop cigarette smoking

- Correct lipid abnormality
- Give platelet antiaggregation drugs (aspirin or in selected cases Dipyridamole or Clopidogrel) to reduce the rate of reinfarction
- Remove or treat embolic source (long term anticoagulation in atrial fibrillation). Defer anticoagulation in disabling stroke for 2 weeks as risk of haemorrhage outweighs benefit.
- Treat inflammatory or vascular inflammatory diseases
- Stop thrombogenic drugs, e.g. oral contraceptives.

TIAs AND MINOR INFARCTION – MANAGEMENT

The *aim of treatment* is to prevent subsequent cerebral infarction:

Establish diagnosis and exclude other pathologies causing transient neurological symptoms, e.g. migraine.

Establish which vessel is involved 

Correct predisposing condition.

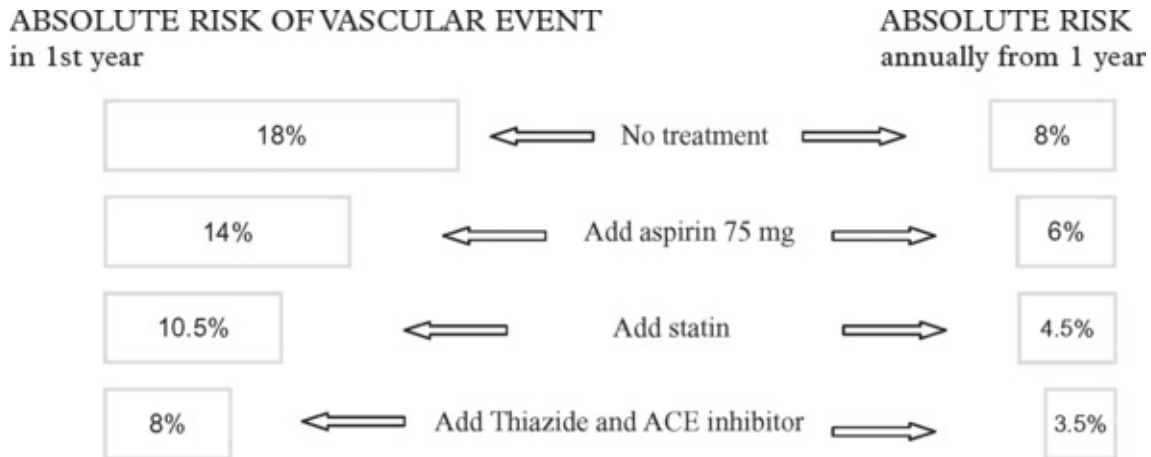
Examine patient for evidence of extracranial vascular disease:

Palpate carotids, upper limb pulses. Auscultate the neck for bruits.

Check blood pressure in both arms. Examine heart.

Medical treatment

Prevention of a further cerebrovascular event (secondary prevention) depends on the cause, which for almost all patients is atherosclerosis – Stop smoking, control diabetes, reduce, cholesterol or blood pressure, even if the levels are in the normal range.



Adding dipyridamole with aspirin reduces the risk further. Clopidogrel has a similar effect to aspirin.

Special situations:

Atrial fibrillation – high risk of recurrence (12%) is reduced to approximately 4% by oral anticoagulation.

Surgical and other interventional treatments

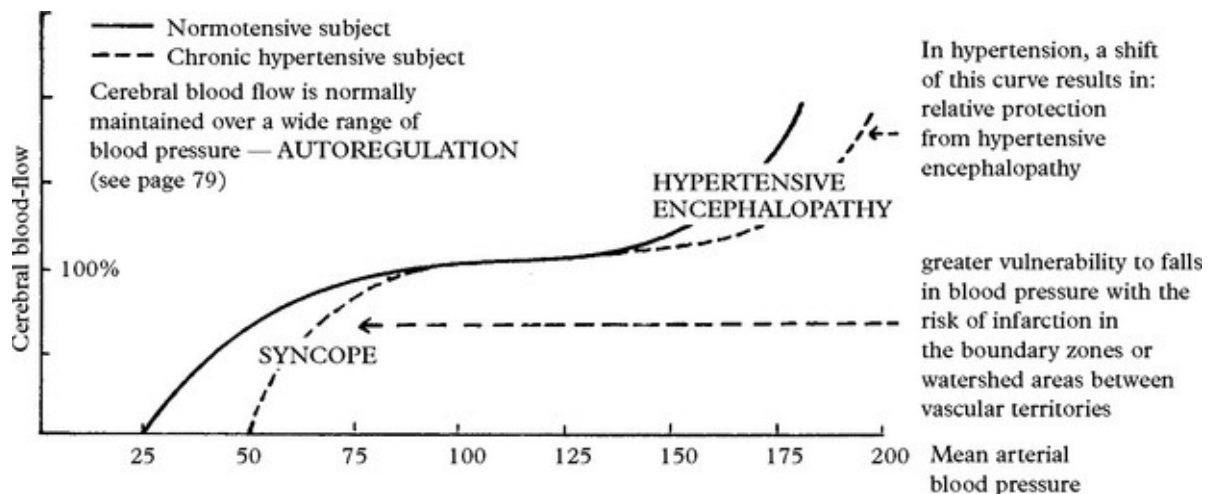
Carotid stenosis – high quality surgical studies demonstrated that patients with carotid stenosis >70% (though not with occlusion) and a TIA or small stroke in carotid territory have a lower risk of further stroke if they undergo a carotid endarterectomy. This benefit depends on the procedure being done by an experienced surgeon with low rate of complications. The risk of stroke, and thus the benefit of surgery is highest with higher grades of stenosis and in patients with hemisphere TIA (as opposed to amaurosis fugax). The risk of complications for patients with lower degrees of stenosis outweighs the benefit.

Carotid angioplasty and stenting is an alternative to carotid endarterectomy in patients with stenosis of >70% but recent studies have found a higher risk of late recurrence than for surgery, The role for these interventions in vertebrobasilar stenosis is not yet established.

Other surgical interventions, for example the superficial temporal to middle cerebral artery bypass provide no benefit.

HYPERTENSION AND CEREBROVASCULAR DISEASE

Next to age, the most important factor predisposing to cerebral infarction or haemorrhage is hypertension. The risk is equal in males and females and is proportional to the height of blood pressure (diastolic and systolic).



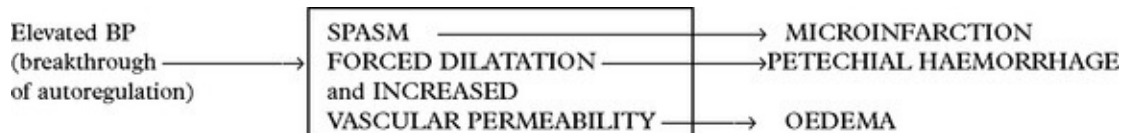
The pathological effects of sustained hypertension are:

- Charcot Bouchard microaneurysms → INTRACEREBRAL HAEMORRHAGE (from perforating vessels)
- Accelerated atheroma and thrombus formation → INFARCTION (large vessels)
- Hyalinosis and fibrin deposition → INFARCTION (lacunes – small vessels)

HYPERTENSIVE ENCEPHALOPATHY

An acute, usually transient, cerebral syndrome precipitated by sudden severe hypertension. The excessive blood pressure may be due to *malignant hypertension* from any cause, or uncontrolled hypertension in *glomerulonephritis*, *pregnancy* (eclampsia) or *phaeochromocytoma*.

The mechanism is complex: Cerebral resistance vessels



Clinical features: Headache and confusion precede convulsions and coma. Papilloedema with haemorrhages and exudates are invariably found. Proteinuria and signs of renal and cardiac failure are common.

Diagnosis: CT scanning shows a widespread white matter low attenuation and excludes other pathology. MRI confirms increased brain water content and SPECT shows hyperperfusion adjacent to these changes.

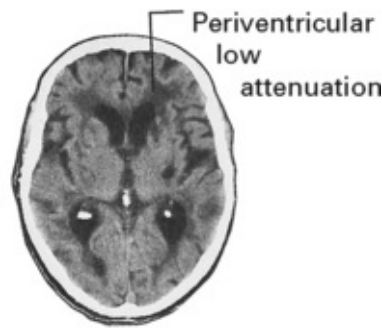
Treatment: a precipitous fall in blood pressure can result in retinal damage and watershed infarction. Gradually reduce blood pressure with i.v. nitroprusside or hydralazine. Reserve peritoneal dialysis for resistant cases.

N.B. *With treatment full recovery is usual. Without treatment death occurs.*

BINSWANGER'S ENCEPHALOPATHY (Subcortical arteriosclerotic encephalopathy – SAE)

A rare disorder in which progressive dementia and pseudobulbar palsy are associated with diffuse hemisphere demyelination. The CT scan shows areas of periventricular low attenuation, often also involving the external capsule. The pathological changes were previously attributed to

chronic diffuse oedema, but the recent finding of a high plasma viscosity in these patients suggests that this, in conjunction with hypertensive small vessel disease, could produce chronic ischaemic change in central white matter.



Subclinical forms of this disease may exist as this CT scan appearance is occasionally found in asymptomatic patients.

MRI appears more sensitive in establishing radiological diagnosis.

NON-ATHEROMATOUS CEREBROVASCULAR DISEASE

ABNORMALITIES OF EXTRACRANIAL VESSELS

SPONTANEOUS AND TRAUMATIC ARTERIAL DISSECTION

Extracranial and intracranial dissections are an underdiagnosed cause of stroke in young persons. Spontaneous dissections occur in Marfan's syndrome or collagen disorders (Ehlers–Danlos), but more frequently in patients with no clear risk factors. Whilst there may be a clear history of neck trauma, often the trauma is minor (eg a sneeze). This may lead to dissection and stenosis or occlusion. The vertebral arteries are particularly susceptible to trauma in view of their close relationship to the cervical spine at intervertebral foramina, the atlanto-axial joint and the occipito-atlantal joint. Carotid dissection may present with a painful

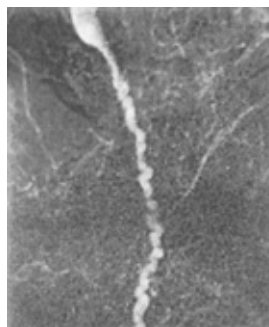
isolated Horner's syndrome or lower cranial nerve palsies. Angiography or CT/MR angiography will confirm. Treatment is with anticoagulation or antiplatelet agents. No studies are available to determine the best treatment strategy.

Internal carotid
artery dissection



FIBROMUSCULAR DYSPLASIA

This disease involves intracranial as well as extracranial vessels which appear like a 'string of beads'. The patient presents with infarction as a result of thrombotic occlusion or from an associated saccular aneurysm, of which there is an increased risk. Transluminal angioplasty can be used to dilate a stenotic segment.



Dilated
and narrowed
segments of
vessel. Produced
by fibrosis of the
lamina media.

INFLAMMATION VESSEL OCCLUSION

Infection in structures close to the carotid artery can result in inflammatory change in the vessel wall and secondary thrombosis. In

children, infection in the retropharyngeal fossa (tonsillar infection) may cause cerebral infarction. Meningitis (especially pneumococcal) may result in secondary arteritis and occlusion of intracerebral vessels as they cross the subarachnoid space.

MOYAMOYA DISEASE

Bilateral occlusion of the carotid artery at the siphon is followed by the development of a fine network of collateral arteries and arterioles at the base of the brain. This may be a congenital or acquired disorder associated with previous meningitis, oral contraception or granulomatous disease (e.g. sarcoidosis). Children present with alternating hemiplegia, adults with subarachnoid haemorrhage. There is no specific treatment though some use surgical revascularisation procedures.

DISEASES OF THE VESSEL WALL

VASCULITIS AND COLLAGEN VASCULAR DISEASES

These disorders have systemic as well as neurological features. Occasionally only the nervous system is diseased. All are rare causes of stroke but need different treatments.

Collagen vascular diseases:

- Systemic lupus erythematosus
- Rheumatoid arthritis
- Other connective tissue disorders.

Vasculitis

- Vasculitis associated with connective tissue disease.

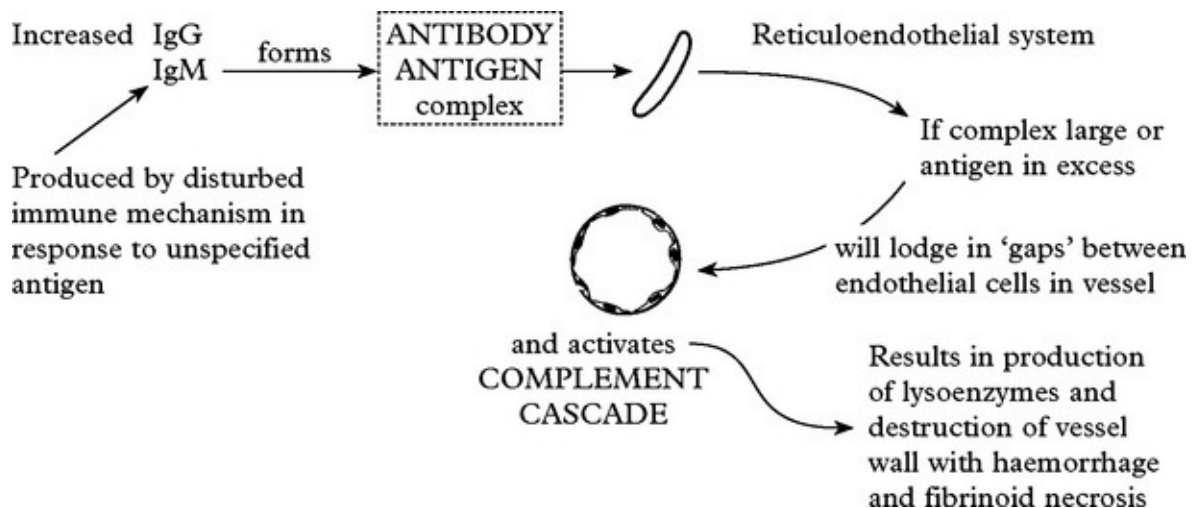
- Micropolyangiitis (previously called polyarteritis nodosa).
- Allergic angiitis (hypersensitivity vasculitis).
- Takayasu's arteritis.
- Isolated angiitis of the central nervous system (IAC).
- Giant cell arteritis/Temporal arteritis
- Churg-Strauss angiitis.

All the above conditions can result in infarction or haemorrhage.

Granulomatous vasculitis *e.g.* Wegener's granulomatosis.

Mechanism

An immune basis for these disorders is likely.



This is termed IMMUNE COMPLEX VASCULITIS.

Indirect immunofluorescent microscopy on biopsy material will demonstrate the presence of immune complexes.

In giant cell arteritis and granulomatous vasculitis, cellular immune mechanisms are probably to blame and vessels are directly attacked. A

reaction of antigen with sensitised lymphocytes results in lymphokine release – attracted mononuclear cells release lysosomal enzymes with resultant granuloma formation.

In all vasculitides affecting predominantly large and medium size vessels, angiography is important in establishing diagnosis. On MRI the presence of bilateral cortical and subcortical infarction is suggestive.

SYSTEMIC LUPUS ERYTHEMATOSUS: in 75% of patients nervous system involvement occurs and may predate systemic manifestation.

- Psychiatric change
- Dementia
- Seizures
- HEMIPLEGIA
- Cranial or peripheral nerve involvement
- SPINAL stroke
- Involuntary movements.

Pathology

The predominant CNS finding is microvascular injury with hyalinisation, perivascular lymphocytosis, endothelial proliferation and thrombosis. Active vasculitis is rare. Cardiogenic embolism and coagulopathy (antiphospholipid antibodies) are alternative mechanisms of stroke.

Investigations

Blood

Elevated ESR and C-reactive protein

Circulating antibodies to nucleoproteins *e.g.* anti-DNA(ANA)

Elevated immunoglobulins

Depressed serum complement levels

Prolonged prothrombin time and antiphospholipid antibodies (60%)

Other

EEG – diffuse disturbance

CT/MRI – multiple small intraparenchymal haemorrhages or infarcts

CSF – protein elevated (Ig), mononuclear cells

Angiography – vessels have beaded appearance

Treatment

Corticosteroids in moderate dosage. In patients with severe or fulminant disease, immunosuppressants and plasma exchange may help.

POLYARTERITIS NODOSA

Neurological involvement is common (80%): Small and medium size arteries are affected.

- HEMIPLEGIA – microinfarction
- INTRACRANIAL HAEMORRHAGE – aneurysm formation
- SPINAL INFARCTION or HAEMORRHAGE
- Peripheral nerve involvement (mononeuritis multiplex)

– ‘Cogan’s’ syndrome $\left\{ \begin{array}{l} \text{Interstitial keratitis} \\ \text{deafness} \\ \text{vertigo} \end{array} \right\}$ progressing to
→ seizures/stroke/coma

Hypertension and renal involvement are common.

Investigations

<i>Blood</i>	<i>Other</i>
Elevated ESR and C-reactive protein	Biopsy – renal or peripheral nerve
Anaemia	– necrotic vessel
Leukocytosis	– lumen diminished
Eosinophilia	– leucocytes and eosinophils in necrotic media and adventitia
Antinuclear cytoplasmic antibodies (ANCA)	CT/MRI as in systemic lupus erythematosus
Circulating immune complexes	Angiography. Multiple irregularities and microaneurysm formation.
IgM rheumatoid factor	These changes can be visible on MRA

Treatment

Steroids and immunosuppressant therapy have dramatically improved outcome (60% 5-year survival).

Plasmapheresis is successful in acute cases.

ALLERGIC ANGIITIS (Hypersensitivity vasculitis)

Intercurrent illnesses (infection or neoplasia) trigger immune complex deposition and basement membranes of capillaries and venules. Systemic symptoms – rash, fever and arthralgia are associated with multiorgan involvement. Neurological features – neuropathy, stroke-like syndromes – occur in 30% of patients. Investigations suggest systemic upset – elevated ESR, anaemia, leukopaenia. Skin biopsy confirms peri-venular inflammation. Treatment of underlying infection and steroids produce rapid improvement.

TAKAYASU'S (PULSELESS) DISEASE

A giant cell arteritis involving the aorta and its major branches. Predominantly affects Asian females in third or fourth decades.

<i>Symptoms:</i> – Non-specific – fever, arthralgias and myalgia – Vascular – myocardial ischaemia, peripheral vascular disease – Neurological vascular TIAs (including subclavian steal), strokes and dementia.	<i>Diagnosis:</i> Steroids are useful initially. The role of surgical reconstruction of occluded vessels is uncertain
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ISOLATED ANGIITIS OF CENTRAL NERVOUS SYSTEM

Systemic symptoms and laboratory evidence of generalised vasculitis are absent.

Presentation with headaches/seizures/encephalopathy and stroke

<i>Diagnosis:</i> Condition should be borne in mind in atypical stroke – CSF shows lymphocytes – MRI, multiple ischaemic changes – Angiography, beading (multiple narrow segments) on intracranial arteries – Meningeal biopsy.	<i>Treatment:</i> Prognosis often dismal. Steroids and cyclophosphamide may produce remission.
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GIANT CELL ARTERITIS (see [page 73](#))

CHURG-STRAUSS ANGIITIS

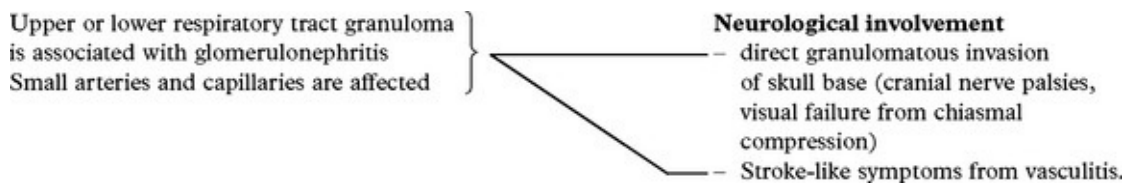
A distinctive syndrome of eosinophilia, pulmonary infiltrates,

neuropathy and encephalopathy or stroke. Related to polyarteritis nodosa, steroid responsive. Other immunosuppressants e.g. cyclophosphamide in resistant cases.

GRANULOMATOUS GRANULOMATOSIS

VASCULITIS/WEGENER'S

A rare disorder, most frequent in males aged 20–50 years.

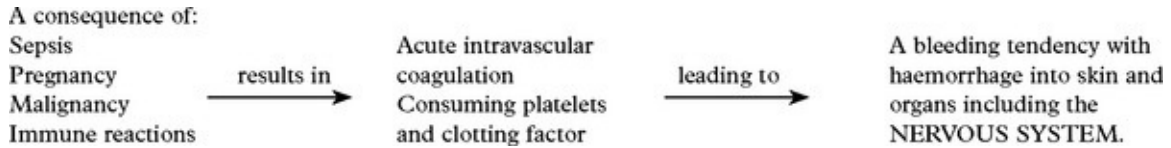


<i>Diagnosis:</i> – Elevated ESR and C-reactive protein (CRP) – Elevated immunoglobulins – Impaired renal function – Radiological findings: Chest and sinuses: granuloma mass MRI (cranium): granuloma mass or vasculitis.	<i>Treatment:</i> – Immunosuppression: steroids and cyclophosphamide – Surgical decompression of granulomas occasionally required.
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DISEASES OF THE BLOOD

Disorders of the blood may manifest themselves as ‘stroke-like’ syndromes. Examination of the peripheral blood film is an important investigation in cerebrovascular disease. Where indicated, more extensive haematological investigation is necessary.

DISSEMINATED INTRAVASCULAR COAGULATION (DIC)



Neurological involvement – a diffuse fluctuating encephalopathy, subarachnoid or subdural haemorrhage.

Diagnosis confirmed by – low platelet count – prolonged prothrombin time, elevated fibrin degradation products and reduced fibrinogen levels.

Treatment

Heparin. Fresh frozen plasma/vitamin K. Treatment of underlying cause.

HAEMOGLOBINOPATHIES

These are genetically determined disorders in which abnormal haemoglobin is present in red blood cells.

Sickle cell disease

This disorder is common in people of African origin but also occurs sporadically throughout the Mediterranean and Middle East region.

The patient is of small stature, usually with chronic leg ulcers, cardiomegaly and hepatosplenomegaly. When arterial oxygen saturation is reduced, ‘sickling’ will occur, manifested clinically by abdominal pain/bone pain.

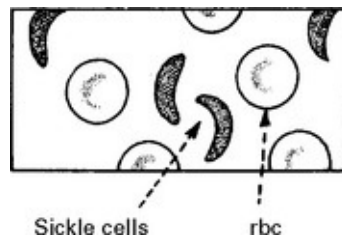
Neurological involvement – hemiparesis, optic atrophy, subarachnoid haemorrhage.

Diagnosis is confirmed in vitro by the ‘sickling’ of cells when O₂ tension is reduced and by haemoglobin electrophoresis.

Treatment

Analgesics for pain O₂ therapy, or hyperbaric O₂. Exchange transfusion

should be carried out for those with a severe or progressive deficit.



ANTIPHOSPHOLIPID ANTIBODIES

These IgG or IgM antibodies prolong APTT and appear to be associated with thrombotic stroke. There remains uncertainty as to whether they are caused by or represent a transient non-specific 'acute phase' reaction to illness. Such antibodies can be found in patients with systemic lupus erythematosus.

ANTITHROMBIN III, PROTEIN C AND PROTEIN S DEFICIENCY

Deficiency of any of these circulating antithrombotic fibrinolytic agents can result in deep venous thrombosis, pulmonary embolism or cerebral venous sinus thrombosis.

POLYCYTHAEMIA

Both polycythaemia rubra vera (primary) and secondary polycythaemia may result in *neurological* involvement – increased viscosity results in reduced cerebral blood flow and an increased tendency towards thrombosis.

Headaches, visual blurring and vertigo are common neurological symptoms.

Transient ischaemic attacks and thrombotic cerebral infarction occur.

Diagnosis

Hb and PCV are elevated.

Primary polycythaemia is confirmed by increased red cell count, white blood count and platelets.

Secondary polycythaemia – respiratory, renal or congenital heart disease are causal.

Treatment

Venesection with replacement of volume with low molecular weight dextran.

Antimitotic drugs may also be used when polycythaemia is due to myeloproliferative disease.

HYPERGAMMAGLOBULINAEMIA

An increase in serum gamma globulin may arise as a primary event or secondary to leukaemia, myeloma, amyloid.

Neurological involvement develops in 20% of cases – due to increased viscosity.

Clinical features are similar to those of polycythaemia – peripheral nervous system involvement may also occur.

Diagnosis is confirmed by protein electrophoresis.

Treatment – underlying cause – plasmapheresis.

THROMBOTIC THROMBOCYTOPENIC PURPURA (syn: Moschkowitz's syndrome)

This is a fibrinoid degeneration of the subintimal structures of small blood vessels. Lesions occur in all organs including the brain.

Clinical features – fever with purpura and multiorgan involvement and

neurological features of diffuse encephalopathy or massive intracranial haemorrhage.

Haemolytic anaemia, haematuria and thrombocytopenia are the main laboratory features.

Treatment

Heparin, steroids and platelet inhibitors may be of value.

THROMBOCYTOPENIA

Whether idiopathic, drug-induced or due to myeloproliferative disorders, this condition may be associated with intracranial haemorrhage.

THROMBOCYTOSIS

This is an elevation in platelet count above 600 000 per mm³. It may be part of a myeloproliferative disorder, or 'reactive' to chronic infection. Patients present with recurrent thrombotic episodes.

Treatment

Aspirin in mild cases; plasmapheresis and antimitotic drugs if more severe.

HYPERFIBRINOGENAEMIA

Serum fibrinogen is occasionally elevated in people with cerebrovascular disease. This enhances coagulation and raises blood viscosity. Infection, pregnancy, malignancy and smoking all raise fibrinogen and may explain in part the increased risk of cerebral infarction. Arvin (Malayan viper venom) acutely lowers serum levels.

METABOLIC DISORDERS

HOMOCYSTEINURIA

A recessively inherited disorder. Accumulation of homocystine in blood damages endothelium and induces premature occlusive arterial disease. The significance of the heterozygote state is uncertain.

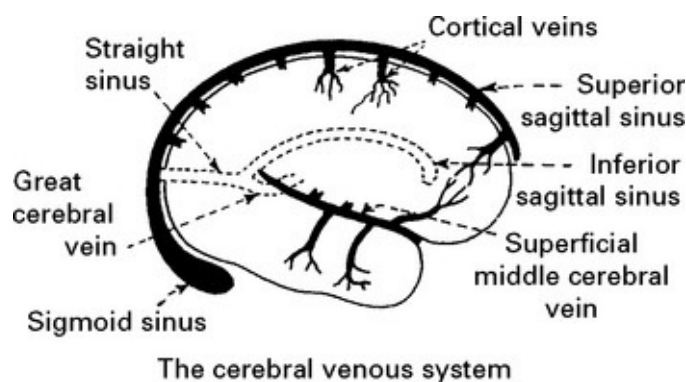
MELAS

See Mitochondrial disorders ([page 481](#)).

CEREBROVASCULAR DISEASE – VENOUS THROMBOSIS

The venous sinuses are important in CSF absorption, with arachnoid villi invaginating the sagittal sinus in particular. Thrombotic occlusion of the venous system occurs with

- infection (especially ear or sinus infection)
- dehydration
- pregnancy, puerperium and pill
- coagulation disorders
- malignant meningitis
- miscellaneous disorders *e.g.* sarcoid, Behçets



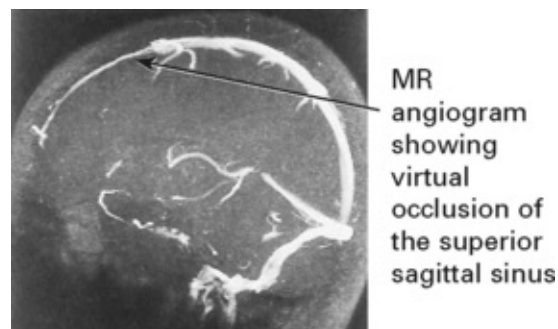
Improved imaging (MRI) has resulted in increased recognition. Venous

infarction accounts for 1% of all 'strokes'.

Superior sagittal and lateral sinus thrombosis (85% of cases)

Impaired CSF drainage results in headache, papilloedema and impaired consciousness. Venous infarction produces seizures and focal deficits (e.g. hemiplegia).

Diagnosis is suggested by venous (nonarterial territory) infarction and 'empty delta' sign (following contrast the wall of the sinus enhances but not the central thrombus on CT) and confirmed by occlusion of filling deficit on MR or CT venography. Outcome is variable; intracranial hypertension may develop (p. 378). A thorough search for causation – coagulation screen, drug history and underlying systemic illness – essential.



Treatment:

Correct causative factors (dehydration/infection etc)

Anticoagulation with heparin or alternative.

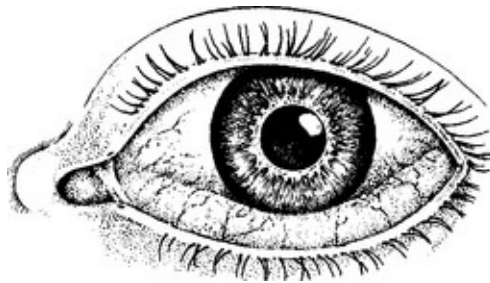
Deep cerebral venous thrombosis (10% of cases)

This produces venous infarction of the basal ganglion and other subcortical structures. Presentation with similar features; diagnosis can

only be established by imaging (CT/MRI and MRV). *Treatment* as above.

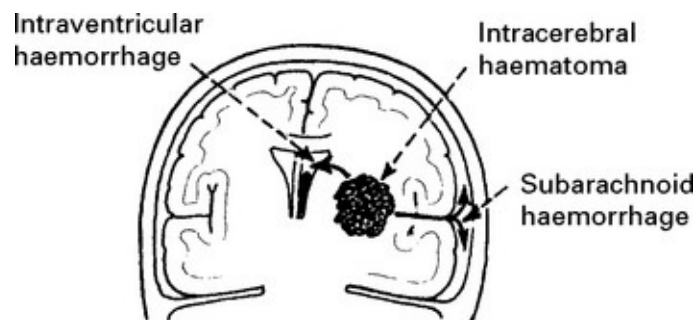
Cavernous sinus thrombosis (5% of cases)

Commonly results from infection spreading from the jaw through draining veins or paranasal sinuses. Painful ophthalmoplegia, proptosis and chemosis with oedema of periorbital structures are associated with facial numbness and fever. The disorder may be bilateral. Base diagnosis on clinical suspicion supported by venography. Treatment with antibiotics and if indicated, sinus drainage.



CEREBROVASCULAR DISEASE – INTRACEREBRAL HAEMORRHAGE

By definition, ‘intracerebral haemorrhage’ occurs within the brain substance, but rupture through to the cortical surface may produce associated ‘subarachnoid’ bleeding. When the haemorrhage occurs deep in the hemisphere, rupture into the ventricular system is common.



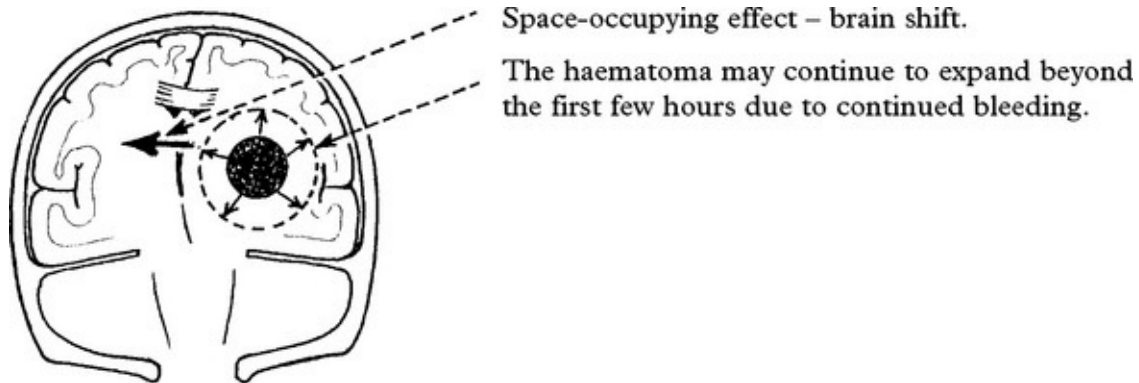
CAUSES

CAUSES

Hypertension	} 80%	Anticoagulants
Cerebral amyloid angiopathy		Vasculitis
Aneurysm		Drug abuse e.g. cocaine
Vascular malformation/fistula		Intracranial venous thrombosis
Neoplasm		Haemorrhagic infarction
Coagulation disorders e.g. haemophilia		Idiopathic

In autopsy series, hypertension accounts for 40–50% of patients dying from non-traumatic haematomas. In hypertensive patients, degenerative changes weaken the walls of small intraparenchymal perforating vessels. Rupture usually occurs near a vessel bifurcation. The ‘microaneurysms’ originally described by Charcot and Bouchard are more likely to be small subadventitial haemorrhages or extravascular clots. In normotensive patients without any evident underlying pathology the cause remains unknown, but cryptic arteriovenous malformations are suspect especially in younger patients (i.e. less than 40 years) and when the haematoma is ‘lobar’ (i.e. frontal, temporal, parieto-occipital). In these patients, the haematoma may temporarily or permanently obliterate the lesion. Reinvestigation following haematoma resolution occasionally reveals previously undetected malformations. In the normotensive elderly patient, subcortical haematomas are commonly associated with amyloid vasculopathy, a degenerative disorder affecting the walls of arteries.

PATHOLOGICAL EFFECTS



Within 48 hours the blood and plasma act on surrounding brain causing disruption of the blood–brain barrier, vasogenic and cytotoxic oedema, neuronal damage and necrosis.

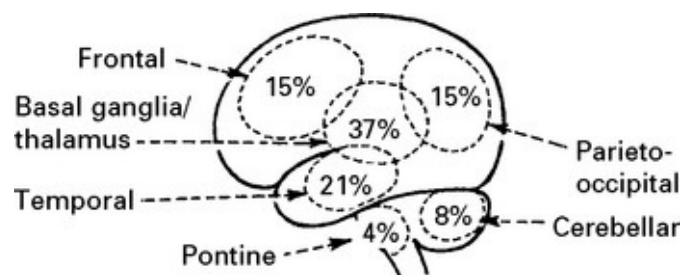
Haematoma resolution occurs in 4–8 weeks, leaving a cystic cavity.

INTRACEREBRAL HAEMORRHAGE

SITES

In *hypertensive* patients, up to 70% occur in the basal ganglia/thalamic region.

In *normotensive* patients:

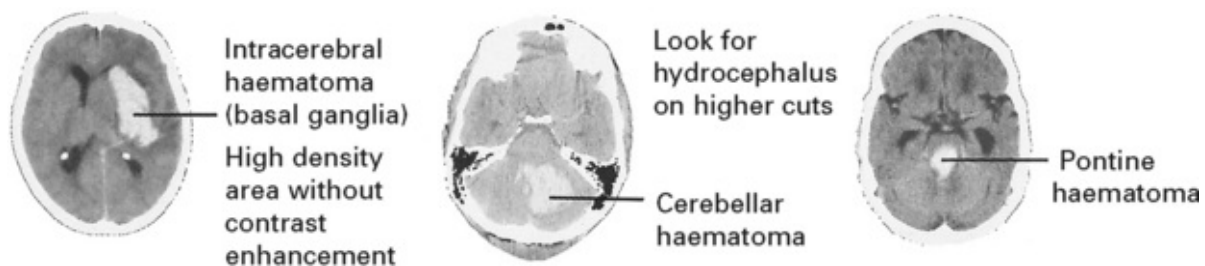


CLINICAL EFFECTS

SUPRATENTORIAL HAEMATOMA	- Mass effect: Sudden onset of headache followed by either a rapid loss of consciousness or a gradual deterioration in conscious level over 24–48 hours. - Focal signs: Hemiparesis, hemisensory loss and homonymous hemianopia are common. The patient may be aware of limb weakness developing prior to losing consciousness. A III nerve palsy indicates transtentorial herniation.
CEREBELLAR HAEMATOMA	- Sudden onset of headache with subsequent effects developing either acutely or subacutely – - Cerebellar and brain stem symptoms and signs, e.g. severe ataxia, dysarthria, nystagmus, vertigo and vomiting - CSF obstruction → hydrocephalus with symptoms and signs of ↑ ICP.
PONTINE HAEMATOMA	- Sudden loss of consciousness - Quadraplegia - Respiratory irregularities → slowed respiration - Pinpoint pupils, pyrexia - Skewed/dysconjugate eye movements - Death often follows.

INVESTIGATIONS

A **CT scan** determines the exact site and size of the haematoma and excludes other pathologies.



Angiography/CT angiography

- Performed immediately if clinical state requires urgent operation, to identify a secondary cause *i.e.* arteriovenous malformation, aneurysm or vasculitis.
- Otherwise delayed until condition improves and the haematoma resolves, unless age and medical condition preclude further management.

In patients with negative angiography, a late MRI may demonstrate a CAVERNOUS ANGIOMA (see [page 299](#)).

MANAGEMENT	PROGNOSIS
For patients with intracerebral haemorrhage on anticoagulants, reverse using intravenous prothrombin concentrate and vitamin K. <i>Supratentorial haematoma</i> There is still no evidence to show that early evacuation of an intracerebral haematoma improves outcome.	Poor prognostic features – Increasing age – Large, deep lesions (basal ganglia/thalamic) – Intraventricular blood – Depth of conscious level (flexion or extension to painful stimuli).
In general, haematoma evacuation is indicated in patients who deteriorate gradually from the ‘mass’ effect, especially when the lesion lies superficially; operation will not benefit moribund patients, <i>i.e.</i> patients extending to painful stimuli with no pupil reaction.	Good prognostic factors – Small superficial lesions (<i>i.e.</i> frontal, temporal or parieto-occipital) – Conscious patients or patients localising to painful stimuli. The overall mortality ranges from 25–60% (90% if the patient is in coma) and is improved by an integrated ‘Stroke Unit’

Cerebellar haematoma:

Small haematomas causing minimal effects may be managed conservatively. Otherwise, urgent evacuation through a suboccipital craniectomy is required. Relief of brain stem compression may be life

saving and operative morbidity is low.

The overall mortality is approximately 30%.

Pontine haemorrhage

The mortality from pontine haemorrhage is high. A conservative approach is usually adopted although some advocate operative exploration.

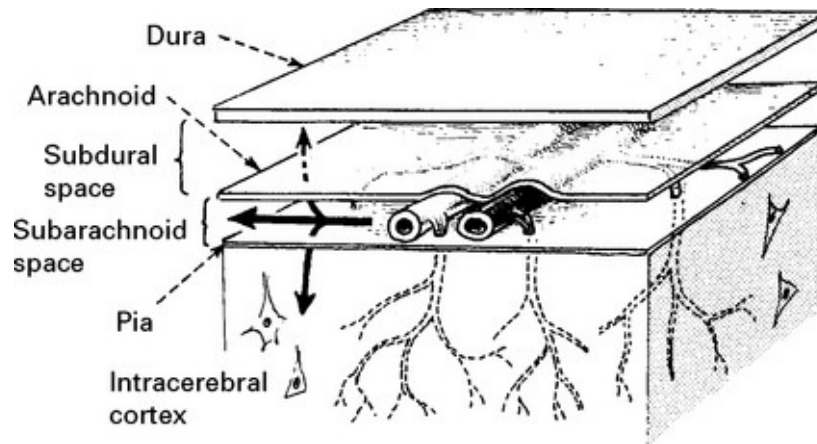
INTRAVENTRICULAR HAEMORRHAGE

Haemorrhage into the ventricles causes a sudden loss of consciousness. With a large bleed, death may follow from the pressure transmission from within the ventricular system. Blood in the ventricles does not in itself cause damage and, following clot resolution, complete recovery may occur.

No treatment is required; attempts at flushing out the ventricles usually fail. If the blood 'cast' causes obstructive hydrocephalus, then ventricular drainage (although hampered by the presence of blood) is indicated. Infusion of thrombolytic agents awaits evaluation.

SUBARACHNOID HAEMORRHAGE (SAH)

Intracranial vessels lie in the *subarachnoid* space and give off small perforating branches to the brain tissue. Bleeding from these vessels or from an associated aneurysm occurs primarily into this space. Some intracranial aneurysms are embedded within the brain tissue and their rupture causes intracerebral bleeding with or without subarachnoid haemorrhage.



Occasionally the arachnoid layer gives way and a subdural haematoma results.

INCIDENCE

Subarachnoid haemorrhage occurs in approximately 8–10 per 100 000 per year.

CAUSE

Cerebral aneurysms are the most frequent cause of subarachnoid haemorrhage, with arteriovenous malformations accounting for up to 5%.

In about 20% of patients detailed investigation fails to reveal a source of the haemorrhage. Small thrombosed or undetected aneurysms or cryptic arteriovenous malformations may account for some.

CAUSES OF SAH	
	Approx. Incidence
Aneurysm	70–75%
Perimesencephalic haemorrhage	10%
A-V malformations	< 5%
Bleeding diathesis	
Anticoagulants	
Tumours	
Vasculitis	< 5%
Undefined	
	10%

SYMPTOMS AND SIGNS

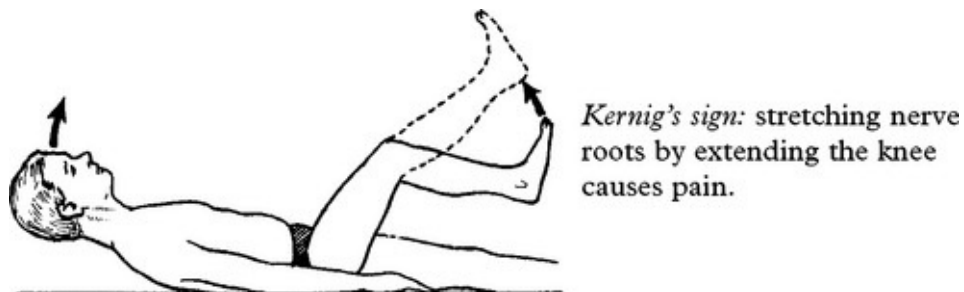
The severity of the symptoms is related to the severity of the bleed.

Usually the *headache* is severe and the onset usually instantaneous (often described as a ‘blow to the head’). A transient or prolonged *loss of consciousness* or *epileptic seizure* may immediately follow. Nausea and *vomiting* commonly occur. Symptoms continue for many days.

Occasionally, the headache is mild (although still sudden onset) and may represent a ‘warning leak’ of blood before a major bleed.

Signs of meningism develop after 3–12 hours

Neck stiffness is present in most patients on passive neck flexion.



Kernig's sign: stretching nerve roots by extending the knee causes pain.

Coma or depression of conscious level may result from the direct effect of

the subarachnoid haemorrhage or from the mass effect of an associated intracerebral haematoma.

Focal damage from a haematoma will produce *focal signs*, e.g. limb weakness, dysphasia. The presence of a III nerve palsy indicates either transtentorial herniation or direct nerve damage from a posterior communicating artery aneurysm (or rarely from a basilar artery aneurysm).

Seizures frequently occur and may mask other features.

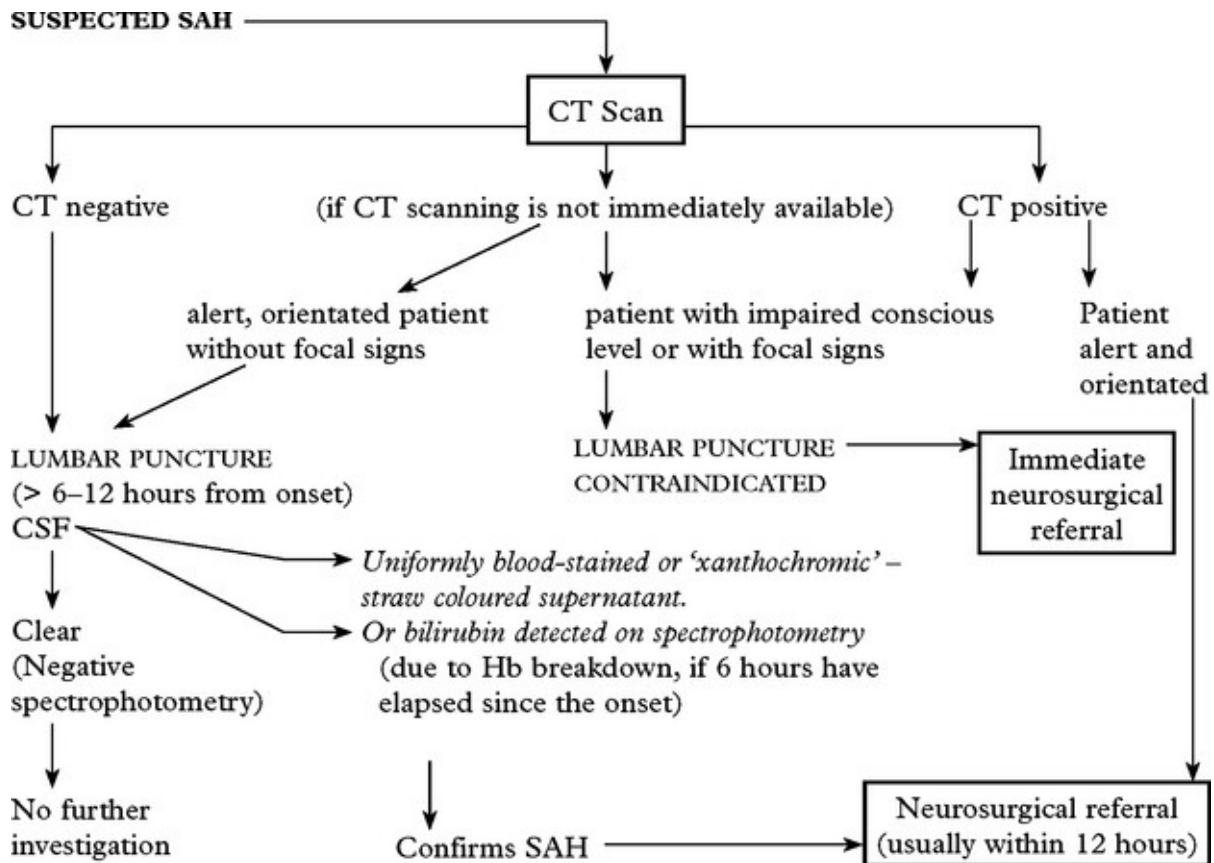
Fundus examination may reveal *papilloedema* or a *subhyaloid or vitreous haemorrhage* caused by the sudden rise in intracranial pressure.

A '*reactive hypertension*' commonly develops, i.e. a rise in BP in patients with no evidence of pre-existing hypertension, and takes several days to return to normal levels.

Pyrexia is also a common finding; if severe and fluctuating, it may reflect ischaemic hypothalamic damage.

INVESTIGATIVE APPROACH

CT scan is the investigation of choice, performed as soon as possible after the headache onset. Lumbar puncture establishes the diagnosis of subarachnoid haemorrhage, but in patients with a mass lesion, lumbar puncture could precipitate transtentorial herniation.



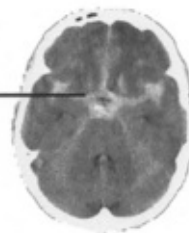
Age limit for neurosurgical referral: Although mortality and morbidity increase with age, with the option of endovascular aneurysm treatment, age limitations no longer apply provided the patient's clinical state is satisfactory.

CT scan

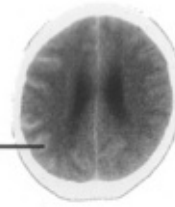
Confirms the diagnosis of SAH in 95% (if within 48 hours of the bleed).

Blood may be *widely distributed*

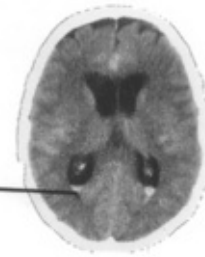
– throughout the
basal cisterns,
Sylvian and
interhemispheric
fissures



- over the
cortical sulci

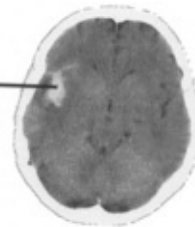


- within the
ventricular system

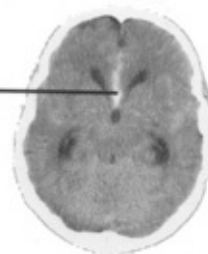


or *more localised* aiding identification of the site of the ruptured aneurysm

- e.g. – within
a Sylvian fissure –
middle cerebral
aneurysm



- within the
interhemispheric
fissure –
anterior
communicating
aneurysm

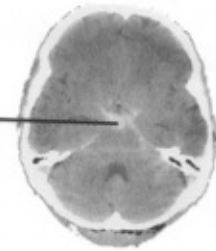


CT also identifies other associated lesions

- hydrocephalus
- intracerebral haematoma
- tumour

- arteriovenous malformation

Blood restricted to the interpeduncular region and not extending into the lateral Sylvian or interhemispheric fissures (i.e. a '*perimesencephalic*' pattern) is usually associated with a negative angiogram, but angiography is still required to exclude a basilar aneurysm.



MRI scan

Not routinely used, but in patients with multiple aneurysms, MRI performed several days after the bleed may provide greater sensitivity than CT in detecting small areas of subarachnoid clot and help determine the particular lesion responsible.

N.B. Spinal arteriovenous malformations can also cause SAH – if the patient's pain begins in the back before spreading to the head, or if any features of cord compression exist, then MRI of the cervical or thoracic spine should be the preliminary investigation (see [page 424](#)).

CT/MR angiography



These non invasive techniques, particularly when combined with 3-D or 4-D imaging (colour as the 4th dimension), will detect up to 95% of intracranial aneurysms, but those < 3mm in diameter may be missed.

Both MRA and in particular CTA can provide more information than conventional angiography about the aneurysm shape and size of the neck.

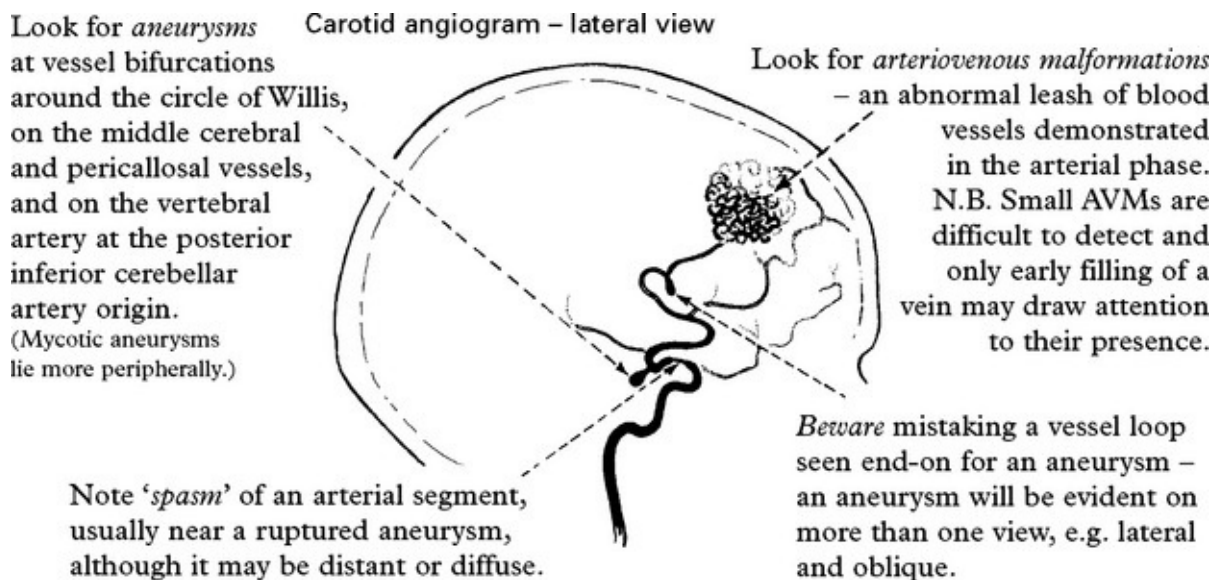
Demonstration of an aneurysm which matches the distribution of blood on standard CT, permits planning of treatment on the assumption that this is the source of the haemorrhage.

CT angiogram showing carotid aneurysm

Digital angiography

Four-vessel angiography is performed in patients with a negative CTA or MRA, in patients where further clarification of the vessels or the aneurysm is required to aid decision making or immediately prior to endovascular treatment.

With the latest equipment, 3-D rotational techniques are employed, combined with cropping of unwanted data, to permit the radiologist to focus on specific regions. Without this aid, *antero-posterior, lateral and oblique* views are required for each vessel.



Negative angiography

Angiography fails to reveal a source of the subarachnoid haemorrhage in approximately 20% of patients. In the presence of arterial spasm, reduction in flow may prevent the demonstration of an aneurysm and repeat angiography may be required at a later date.

Prognosis: In patients with a 'perimesencephalic' pattern of haemorrhage on CT scan and with negative angiography, the outlook is excellent; those patients with an 'aneurysmal' pattern with blood lying in the interhemispheric or Sylvian fissure still run a risk of rebleeding.

CEREBRAL ANEURYSMS

INCIDENCE

At autopsy intracranial aneurysms are found in approximately 2% of the population.

Aneurysm rupture occurs in 6–8 per 100 000 per year

Female:male = 3.2; but this ratio varies with age: < 40 years, male > females

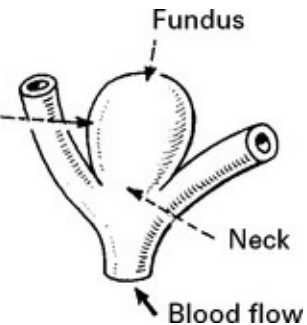
> 40 years, females > males

Risk factors: atherosclerotic diseases (2.3x), family history (6x), polycystic kidney disease (4.4x).

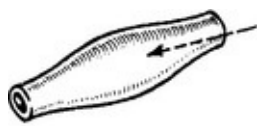
Inheritance: investigations reveal aneurysms in 10% of relatives with two or more affected 1st degree family members. The genetic basis remains unknown. Procollagen III deficiency may play a role in some patients.

MORPHOLOGY

Intracranial aneurysms are usually *saccular*, occurring at vessel bifurcations. Size varies from a few millimetres to several centimetres. Those over 2.5 cm are termed 'giant' aneurysms.



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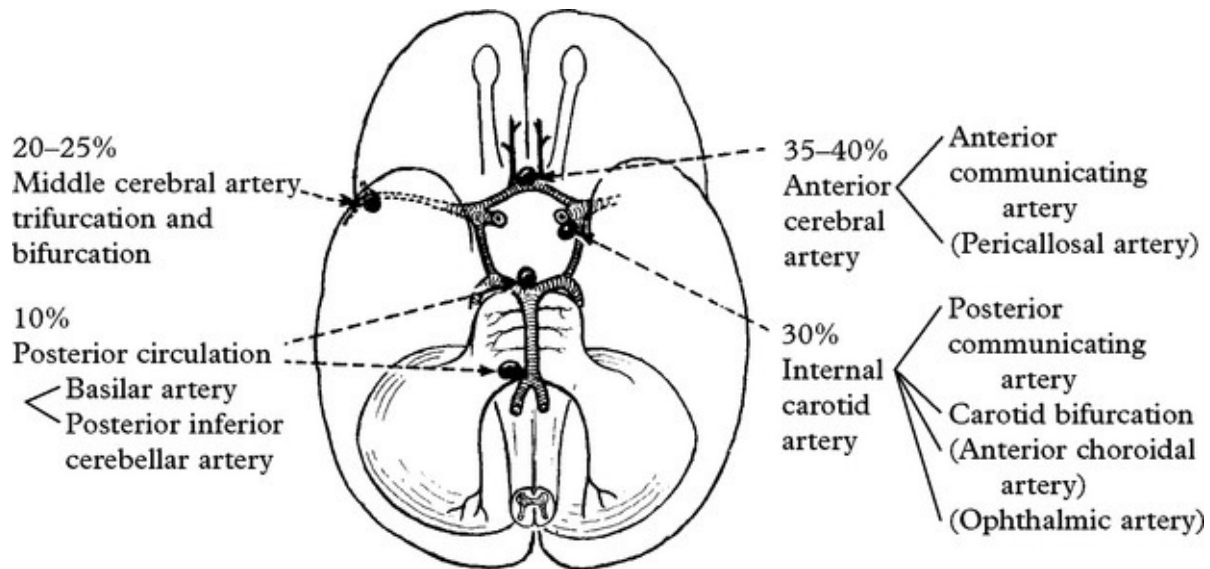


Fusiform dilatation and ectasia of the carotid and the basilar artery may follow atherosclerotic damage. These aneurysms seldom rupture.

Mycotic aneurysms, secondary to vessel wall infection, arise from haematogenous spread, e.g. subacute bacterial endocarditis.

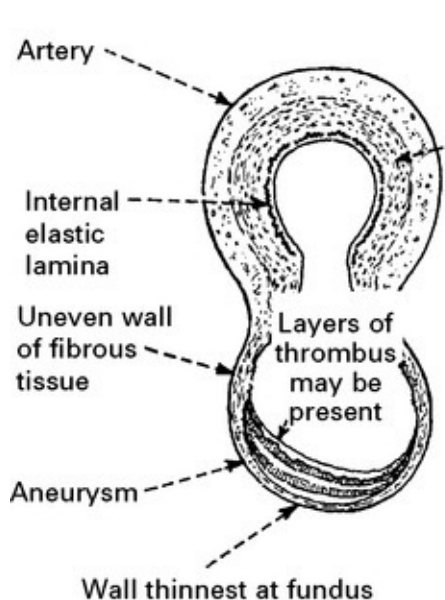
Aneurysm rupture: usually occurs at the fundus of the aneurysm and the risk is related to size. Smoking, hypertension and alcohol excess also play a part. In some patients, rupture occurs during exertion, straining or coitus, but in most there is no associated relationship.

Sites of saccular aneurysm



Multiple aneurysms: in approximately 30% of patients with aneurysmal SAH, more than one aneurysm is demonstrated on angiography.

PATHOGENESIS



The cause of aneurysm formation may be multifactorial with acquired factors combining with an underlying genetic susceptibility.

Aneurysms were once thought to be 'congenital' due to the finding of *developmental defects in the tunica media*. These defects occur at the apex of vessel bifurcation as do aneurysms, but they are also found in many extracranial vessels as well as intracranial vessels; saccular aneurysms in contrast are seldom found outwith the skull. Tunica media defects are often evident in children, yet aneurysms are rare in this age group. It now appears that *defects of the internal elastic lamina* are more important in aneurysm formation and these are probably related to arteriosclerotic damage.

Aneurysms often form at sites of *haemodynamic stress* where for example, a congenitally hypoplastic vessel leads to excessive flow in an adjacent artery. It is not known whether they form rapidly over the space of a few minutes, or more slowly over days, weeks, or months.

Hypertension may play a role; more than half the patients with ruptured

aneurysm have pre-existing evidence of raised blood pressure.

CLINICAL PRESENTATION

Of those patients with intracranial aneurysms presenting acutely, most have had a subarachnoid haemorrhage. A few present with symptoms or signs due to compression of adjacent structures. Others present with an aneurysm found incidentally.

1. Rupture

The features of SAH have already been described in detail ([page 276](#)); they include sudden onset of headache, vomiting, neck stiffness, loss of consciousness, focal signs and epilepsy.

Since the severity of the haemorrhage relates to the patient's clinical state and this in turn relates to outcome, much emphasis has been placed on categorising patients into 5 level grading systems, *e.g.* Hunt and Hess. A scale incorporating the Glasgow Coma scale ([page 29](#)) has been adopted by the World Federation of Neurosurgical Societies:

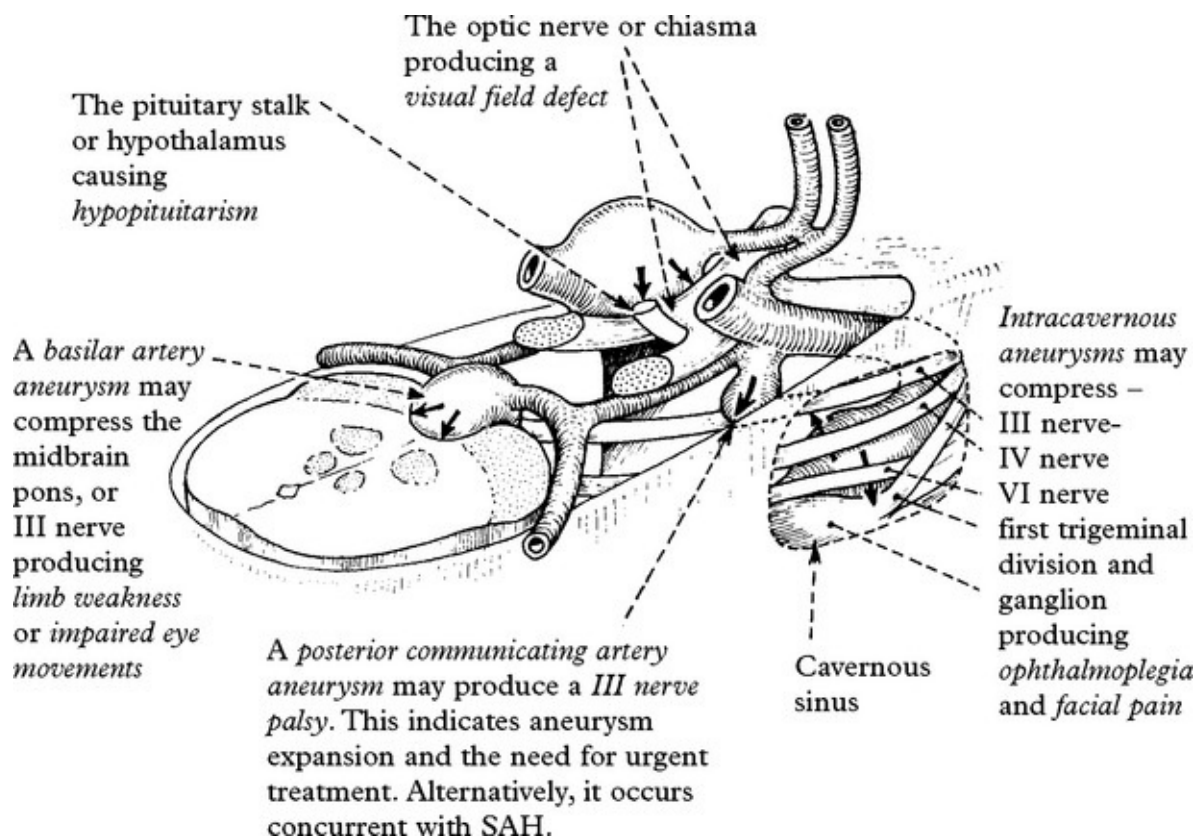
WFNs Grade	Glasgow Coma Score	Motor deficit	Glasgow Coma Score		
I	15	absent	eye opening	1-4	e.g. no eye opening (1)
II	14-13	absent	verbal response	1-5	no verbal response (1)
III	14-13	present	motor response	1-6	spastic flexion to pain (3)
IV	12-7	present or absent		<u>3-15</u>	= 5
V	6-3	present or absent			

This grading scale correlates well with final outcome and provides a prognostic index for the clinician. In addition, it enables matching of patient groups before comparing the effects of different management techniques.

2. Compression from aneurysm sac

A large *internal carotid artery aneurysm* (or *anterior communicating artery*

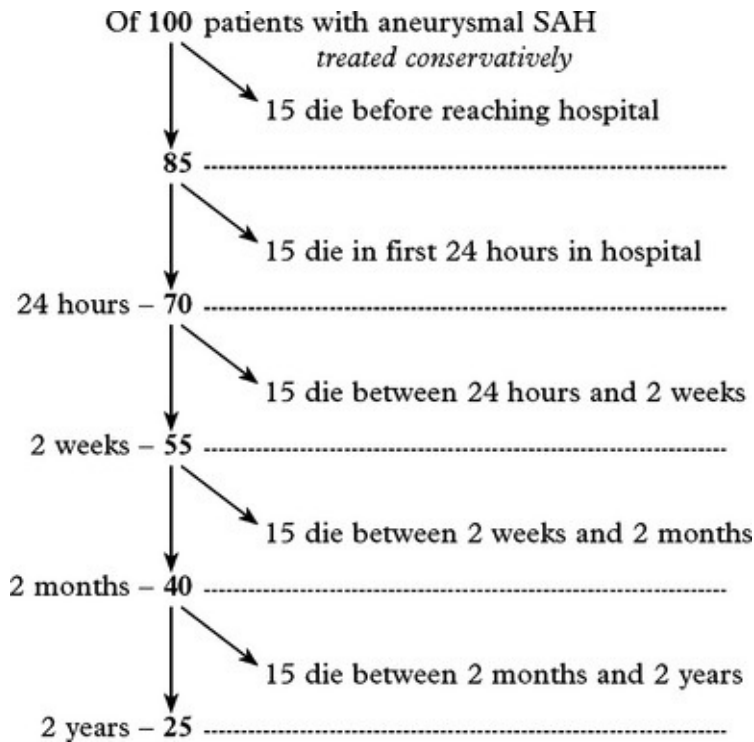
aneurysm) may compress –



3. Incidental finding

The improved availability of sensitive high quality, non-invasive MR or CT imaging techniques has greatly increased the number of patients in whom an intracranial aneurysm is detected incidentally, during investigation for other disease.

NATURAL HISTORY OF RUPTURED ANEURYSM



SAH from ruptured aneurysm carries a high initial mortality risk which gradually declines with time. Of those who survive the initial bleed, rebleeding and cerebral infarction (see below) are the major causes of death.

These figures are based on studies of conservative treatment carried out in the 1960s, at a time when the risks of operation were greater and benefits uncertain.

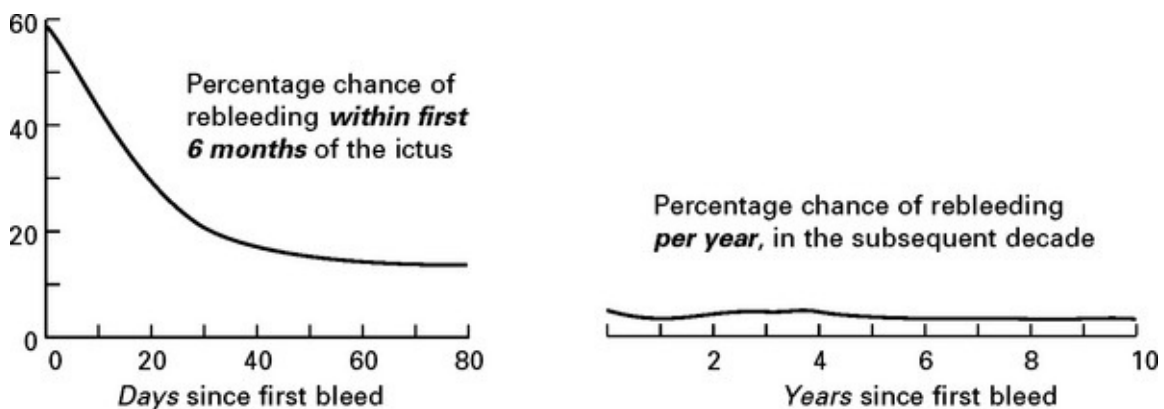
COMPLICATIONS OF ANEURYSMAL SAH

INTRACRANIAL	EXTRACRANIAL
– Rebleeding – Cerebral ischaemia/infarction – Hydrocephalus – ‘Expanding’ haematoma – Epilepsy.	– Myocardial infarction – Cardiac arrhythmias – Pulmonary oedema – Gastric haemorrhage (stress ulcer).

CEREBRAL ANEURYSMS – COMPLICATIONS

REBLEEDING

Rebleeding is a major problem following aneurysmal SAH. In the first 28 days (in *untreated* patients), approximately 30% of patients would rebleed; of these 70% die. In the following few months the risk gradually falls off but it never drops below 3.5% per year.



Adapted from Winn, Richardson, Jane 1977 Annals of Neurology

If, for example, a patient survives the first 30 days after a bleed, there is still a 20% chance of a rebleed occurring in the next 5 months. Even if patients survive the 'high risk' period in the first 6 months, there is still a considerable chance of rebleeding and death in the subsequent years.

The clinical picture of rebleeding is that of SAH, but usually the effects are more severe than the initial bleed. Most patients lose consciousness; the risk of death from a rebleed is more than twice that from the initial bleed.

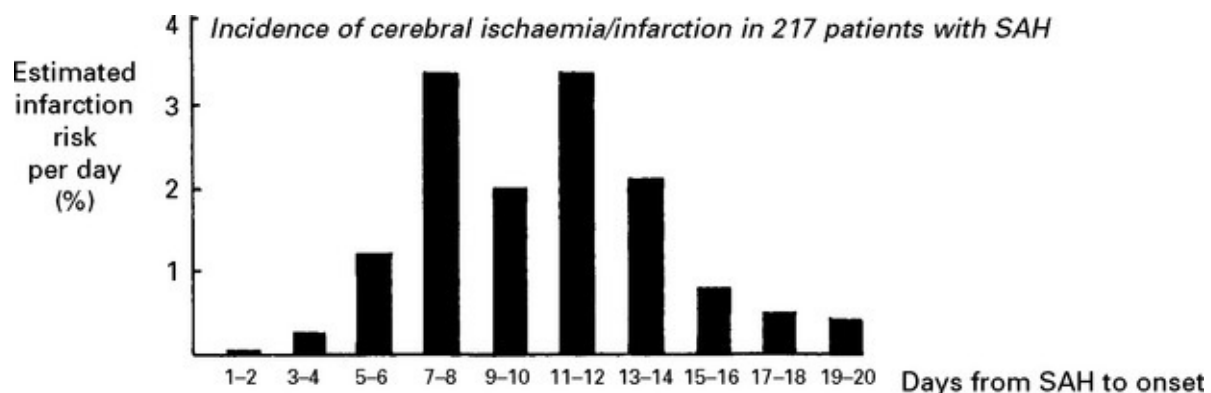
Investigation

All patients deteriorating suddenly require a CT scan. This helps in establishing the diagnosis of rebleeding and excludes a remediable cause

of the deterioration, e.g. acute hydrocephalus.

CEREBRAL ISCHAEMIA/INFARCTION

Following subarachnoid haemorrhage, patients are at risk of developing cerebral ischaemia or infarction and this is an important contributory factor to mortality and morbidity. Cerebral ischaemia/infarction may occur as an immediate and direct result of the haemorrhage, but more often develops 4–12 days after the onset, either before or after operation – hence the term ‘delayed cerebral ischaemia’. Approximately 25% of patients develop clinical evidence of delayed ischaemia/infarction; of these 25% die as a result. About 10% of the survivors remain permanently disabled.



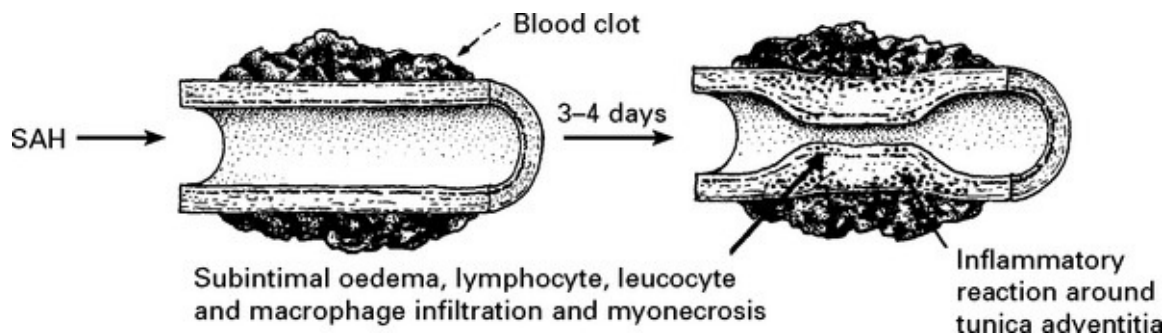
Adapted from Vermeulen, Lindsay, Murray et al 1984 New England Journal of Medicine 311: 432–437

Aetiology of cerebral ischaemia/infarction

Several factors probably contribute to the development of cerebral ischaemia or infarction: ‘*Vasospasm*’: arterial narrowing on angiography occurs in up to 60% of patients after SAH and is either focal or diffuse. The development of ‘vasospasm’ shows a similar pattern of delay to that of cerebral ischaemia.

The angiogram appearance was initially thought to result from arterial

constriction; this may be so, but the pathogenesis of 'vasospasm' now seems more complex. Many vasoconstrictive substances either released from the vessel wall or from the blood clot appear in the CSF after SAH, e.g. serotonin, prostaglandin, oxyhaemoglobin, endothelin-1 and endothelial synthesis of the vasodilator nitric oxide is reduced, but numerous studies with vasoconstrictor antagonists have failed to reverse the angiographic narrowing. This failure may be a result of the *arteriopathic changes* which have been observed in the vessel wall. Only calcium antagonists appear to have a beneficial effect (see [page 291](#)).



The greater the amount of blood in the basal cisterns (as shown on CT scan), the higher the incidence of arterial narrowing and associated ischaemic deficits.

Hypovolaemia

Hyponatraemia develops after SAH in many patients due to excessive renal secretion of sodium rather than a dilutional effect from inappropriate antidiuretic hormone secretion. Fluid loss and a fall in plasma volume lead to a raised blood viscosity with an increased risk of developing cerebral ischaemia.

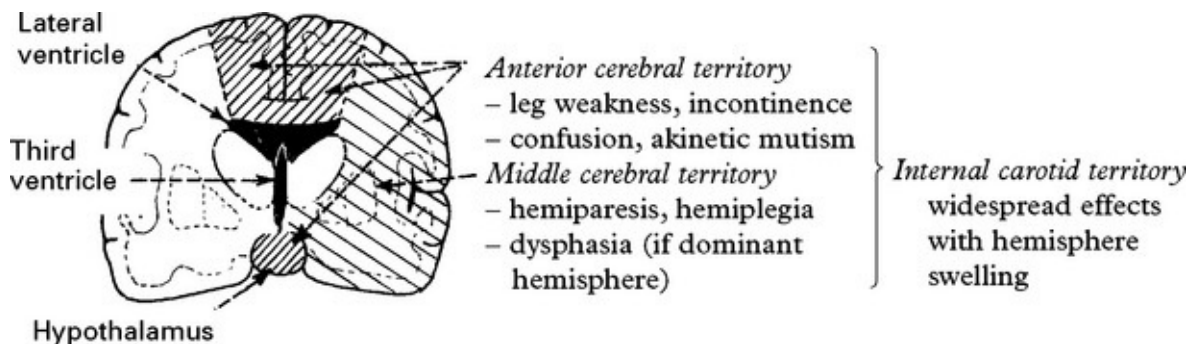
Reduced cerebral perfusion pressure

Following SAH, intracranial haematoma or hydrocephalus may cause a

rise in intracranial pressure (ICP). Since cerebral perfusion pressure = mean BP – ICP, a subsequent reduction in cerebral perfusion may occur.

Clinical effects of cerebral ischaemia/infarction

This may affect one particular arterial territory producing characteristic signs:



Commonly the ischaemia occurs in multiple areas, often in both hemispheres. This correlates with the pattern of arterial ‘spasm’.

Transcranial Doppler: a significant increase in flow velocity within an intracranial vessel may indicate developing ‘vasospasm’, even before clinical problems develop, and allow the early introduction of prophylactic measures (see [page 291](#)).

HYDROCEPHALUS

Following SAH, cerebrospinal fluid drainage may be impaired by:

- blood clot within the basal cisterns
 - obstruction of the arachnoid villi
 - blood clot within the ventricular system
- These first two are grouped under ‘communicating’ hydrocephalus (see page 374). The third is ‘obstructive’ hydrocephalus.

Acute hydrocephalus occurs in about 20% of patients, usually in the first few days after the ictus; occasionally this is a late complication. In only one-third are symptoms of headache, impaired conscious level,

dementia, incontinence, or gait ataxia severe enough to warrant treatment.

In a further 10% of patients, hydrocephalus develops late – months or even years after the haemorrhage.

‘EXPANDING’ INTRACEREBRAL HAEMATOMA

Brain swelling around an intracerebral haematoma may aggravate the mass effect of the haematoma; this may cause a progressive deterioration in conscious level or progression of focal signs.

EPILEPSY

Epilepsy may occur at any stage after SAH, especially if a haematoma has caused cortical damage.

Seizures may be generalised or partial (focal).

EXTRACRANIAL COMPLICATIONS

Myocardial infarction/cardiac arrhythmias: electrocardiographic and pathological changes in the myocardium are occasionally evident after SAH, and ventricular fibrillation has been recorded. These problems are likely to occur secondarily to catecholamine release following ischaemic damage to the hypothalamus.

Pulmonary oedema: this occasionally occurs after SAH, probably as a result of massive sympathetic discharge; note the ‘pink, frothy’ sputum and typical auscultatory and chest X-ray findings.

Gastric haemorrhage: bleeding from gastric erosions occasionally occurs after SAH but rarely threatens life.

CEREBRAL ANEURYSMS – MANAGEMENT FOLLOWING SAH

Headache requires *analgesia* – codeine or dihydrocodeine. Stronger analgesics may depress conscious level and mask neurological deterioration. Management is otherwise aimed at preventing complications –

PREVENTION OF REBLEEDING

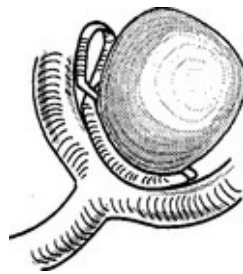
Bed rest: Often enforced after SAH, although there is no evidence that this reduces the rebleed risk. Allowing gentle mobilisation and using the toilet may induce less ‘stress’ than using a bedpan.

Aneurysm repair: Both surgical (clipping of the aneurysm neck) and endovascular (coil embolisation of the aneurysm sac) techniques are used. Aneurysm repair, whether coiling or clipping, is performed within 48 hours of the bleed, but this is not always feasible. Operative risks are greater the earlier the procedure, but the greater the delay, the greater the risk of rebleeding. Despite this, in past years operation was often deferred in patients in poor clinical condition. Endovascular techniques appear to be less dependent on timing and avoid potentially harmful effects of brain retraction and vessel dissection. This supports their use in the elderly or in patients in poor clinical condition, but other factors must also be considered – see [page 290](#). Once the aneurysm is clipped, aggressive methods of treating ischaemia with induced hypertension can be applied – see [page 291](#).

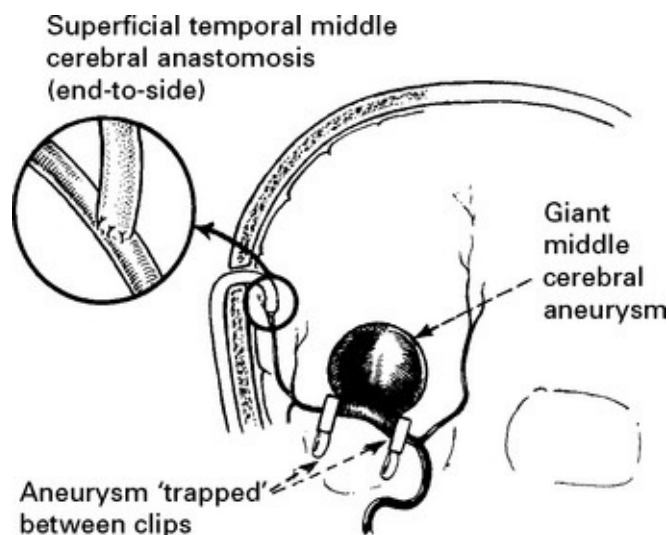
OPERATIVE TECHNIQUES

Direct clipping of the aneurysm neck: Through a craniotomy and using the operating microscope, the surgeon dissects out the aneurysm and applies a clip across the neck without compromising the proximal or distal vessels. Clipping prevents rebleeding; clip slippage rarely occurs. If any part of the neck lies outwith the clip, this may occasionally lead to

recurrent growth.



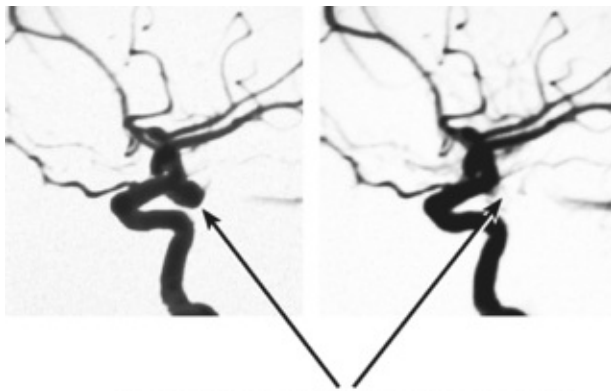
Wrapping: If the width of the aneurysm neck or its involvement with adjacent vessels prevents clipping, then muslin gauze may be wrapped around the fundus. This provides some protection, but rebleeding may still occur.



Trapping: Used for giant aneurysms (>25mm diameter) where other methods have failed due to the width of the aneurysm neck. Should be combined with cerebral revascularisation to minimize the risk of ischaemia. Techniques include either superficial temporal–middle cerebral artery anastomosis or insertion of a saphenous vein or radial artery bypass graft from the carotid to the middle cerebral artery.

ENDOVASCULAR TECHNIQUES

Coil embolisation: this endovascular technique, where multiple helical platinum coils are packed into the aneurysm fundus, has been developed and refined over the last two decades. A tracker catheter is inserted via a femoral puncture and guided up through the arterial system into the aneurysm sac.



Posterior communicating aneurysm
before and after coil embolisation

Posterior communicating

aneurysm before and after coil embolisation

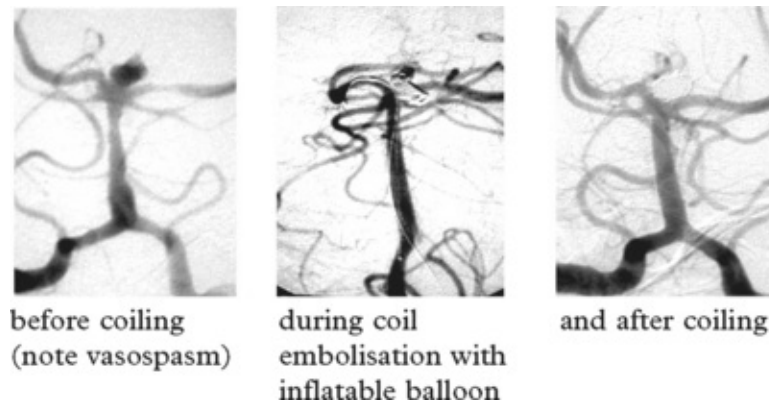
The coil attached to the end of a delivery wire is then guided into the fundus and after accurate placement, the passage of an electric current causes electrochemical release. On average, 4–5 coils are required to pack each aneurysm.

The radiologist aims to completely obliterate the fundus, but this is not always feasible and to avoid occluding the adjacent vessel, a portion of the neck may remain. In either case, a small risk of rebleeding persists, even when completely obliterated. Thrombotic complications may also occur during the procedure.

Balloon remodelling: the wider the aneurysm neck, the greater the risk that coils will project into and occlude the vessel lumen. A balloon is

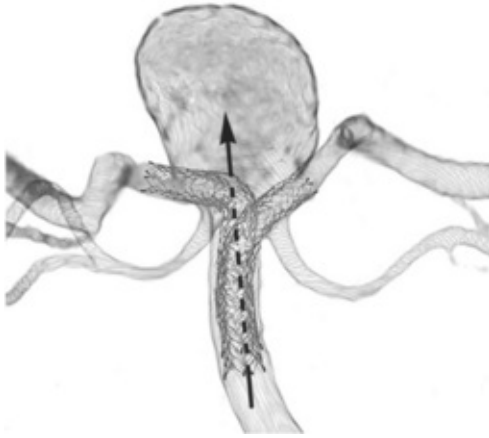
attached to a second catheter and periodically inflated across the aneurysm neck during coil insertion to preserve the vessel lumen.

Basilar bifurcation aneurysm with wide neck –



Stent assisted coil embolisation: for very wide-necked aneurysms or for those where balloon remodelling has failed, one or more stents can be manoeuvred through the parent vessel alongside the aneurysm neck. Coils are then packed into the fundus via a tracker catheter passed through the interstices of the stent. Such patients require long-term antiplatelet therapy to prevent thrombotic complications and this may create difficulties in the acute phase of SAH management.

Balloon occlusion: On rare occasions the above operative or endovascular techniques fail to treat ‘giant’ or fusiform aneurysms arising from the carotid artery. Temporary balloon occlusion of the carotid artery for a 30 minute period tests whether the patient’s collateral circulation from the Circle of Willis is sufficient to sustain flow through the hemisphere. Similarly temporary occlusion of the vertebrobasilar system is possible. If tolerated, intra-arterial inflation of a detachable balloon can provide permanent occlusion. Intra-arterial balloon inflation can also provide temporary intra-operative protection when proximal control is difficult to achieve.



Wide necked basilar aneurysm with one stent inserted into the left posterior cerebral artery, and another stent passing through the interstices of the first and inserted into the right posterior cerebral artery. Coils will be passed beyond the stent into the aneurysm fundus.

SELECTION OF TREATMENT

Until recent years direct surgical clipping was the standard method of aneurysm treatment. Coil embolisation was reserved for aneurysms technically difficult to repair, particularly those in the posterior circulation. A large multicentre randomized trial of clipping vs. endovascular treatment (the International Subarachnoid Aneurysm Trial – ISAT) demonstrated a 23% reduction in the proportion of patients with a poor outcome (dependent or dead) at one year in those patients undergoing coil embolisation, despite more rebleeds occurring in this group. Following publication in 2002, the proportion of patients undergoing coil embolisation as the first line of treatment dramatically increased, reaching 85–90% in some centres. This swing occurred despite the trial being weighted towards small anterior circulation aneurysms in patients in good clinical condition. Concern persisted that coil treatment would not eliminate rebleeding. Long-term follow up (mean 9 years after treatment) of the trial patients has shown that

although rebleeding was higher in the coil treatment group, the risk of death was still significantly lower in coiled patients. In young patients *e.g.* 30–40 years of age, with 40 years of expectant life, the rebleeding concern after coil treatment persists and clipping may be the preferred option.

Aneurysm treatment requires a team approach involving interventional radiologists and neurosurgeons. Treatment selection must take a variety of factors into account including the nature and location of the aneurysm, the relative difficulties of the endovascular or operative approach and the patient's age and clinical condition. Some patients *e.g.* those with basilar, carotid and anterior cerebral aneurysms, the elderly or those in poor clinical condition are more likely to require coil embolisation, whereas others, such as those with large middle cerebral aneurysms are more likely to require direct operative treatment. Unfortunately aneurysms that are difficult to treat with one technique are often difficult to treat with both methods. Patients undergoing coil treatment require check angiography/CT angiography follow-up, *e.g.* 6 months and 2 years, to ensure recanalisation has not occurred. Up to 10% will require retreatment.

PREVENTION OF CEREBRAL ISCHAEMIA/INFARCTION

Despite considerable clinical and experimental research, cerebral ischaemia is still a major cause of morbidity and mortality after subarachnoid haemorrhage. In recent years some advances have proved beneficial.

Calcium antagonists: several large studies and a meta-analysis have confirmed that Nimodipine reduces the incidence of cerebral infarction by about one third and improves outcome. Whether this acts by improving collateral circulation, by reducing the harmful effect of

calcium flooding into brain cells or by reducing cerebral 'vasospasm' remains uncertain.

Avoidance of antihypertensive therapy: after SAH, autoregulation ([page 79](#)) is often impaired; a drop in BP causes a reduction in cerebral blood flow with a subsequent risk of cerebral ischaemia. Patients on long-term antihypertensive treatment can continue with this therapy, but 'reactive' hypertension should not be treated.

High fluid intake (haemodilution): maintenance of a high fluid input (3 litres per day) may help prevent a fall in plasma volume from sodium and fluid loss. If hyponatraemia develops do *not* restrict fluids (this significantly increases the risk of cerebral infarction). If sodium levels fall below 130 mmol/l, give hypertonic saline or fludrocortisone.

Plasma volume expansion (hypervolaemia): expanding the plasma volume with colloid, e.g. plasma proteins, dextran 70, Haemacel, increases blood pressure and improves cerebral blood flow. This should be given either prophylactically in high risk patients (heavy cisternal blood load on CT scan or with high Doppler velocities) or at the first clinical sign of ischaemia. If clinical evidence of ischaemia develops despite this treatment, then (if the aneurysm has been repaired) combine with:

Hypertensive therapy: treatment with inotropic agents, e.g. dobutamine, increases cardiac output and blood pressure. Since cerebral autoregulation commonly fails after subarachnoid haemorrhage, increasing blood pressure increases cerebral blood flow. Up to 70% of ischaemic neurological deficits developing after aneurysm operations can be reversed by inducing hypertension; often a critical level of blood pressure is evident.

Early recognition and treatment of a developing neurological deficit

may prevent progression from ischaemia to infarction. Delayed treatment may merely aggravate vasogenic oedema in an ischaemic area. This technique of induced hypertension is now widely applied, with good results, but requires careful, intensive monitoring. In view of the risk of precipitating aneurysm rupture, it is reserved until after aneurysm repair.

Transluminal angioplasty/papaverine infusion: this involves balloon dilatation of the vasospastic segment of the vessel. It is usually combined with an intra-arterial infusion of the antispasmodic agent papaverine. Although no controlled studies exist, many small studies report a beneficial effect on cerebral blood flow and on clinical state. Timing is difficult. If used too early, the patient may be unnecessarily exposed to an invasive procedure; if too late, the ischaemia may be irreversible. Consider angiography and angioplasty if other measures (haemodilution/hypervolaemia/hypertension) have failed to reverse a significant clinical deterioration within a few hours.

Brain protective agents: to date, studies of neuroprotective drugs (antioxidants and anti-inflammatory agents) other than calcium antagonists, have failed to demonstrate a beneficial effect. Some recent studies assessing magnesium sulphate infusion, pravastatin and the endothelin-1 antagonist clazosentan have had encouraging results, but await further evaluation.

Antifibrinolytic agents: i.e. tranexamic acid, epsilon aminocaproic acid should *not* be used.

These agents prevent rebleeding by delaying clot dissolution around the aneurysm fundus, but any beneficial effect is offset by an *increased incidence of cerebral ischaemia*.

HYDROCEPHALUS

Hydrocephalus causing acute deterioration in conscious level requires urgent CSF drainage with a ventricular catheter (in 'communicating' hydrocephalus a lumbar drain provides an alternative. Lumbar puncture may provide temporary benefit).

Gradual deterioration or failure to improve in the presence of enlarged ventricles indicates the need for permanent CSF drainage with either a ventriculoperitoneal or lumboperitoneal shunt.

EXPANDING INTRACEREBRAL HAEMATOMA

Intracerebral haematomas from ruptured aneurysms do not require specific treatment unless the 'mass' effect causes a deterioration of conscious level. This necessitates urgent CT or digital angiography followed by evacuation of the haematoma with or without simultaneous clipping of the aneurysm; under these circumstances, operative mortality is high.

OUTCOME AFTER SUBARACHNOID HAEMORRHAGE

The National Study of Subarachnoid Haemorrhage collected information on patients admitted to all neurosurgical units in the UK and Ireland between September 2001 and September 2002 and provides useful information on outcome. The study included 3174 patients, of which 2397 had a confirmed aneurysm. (Published by the Royal College of Surgeons, 2006 – available online)

Of those patients surviving the initial bleed and admitted to the neurosurgical unit with a confirmed aneurysm, 11% died in hospital. Of those undergoing aneurysm repair, 40% made a good recovery; a further 21% had moderate disability and were independent.

Factors associated with unfavourable outcome were: age, clinical condition on admission, quantity of subarachnoid blood on CT scan and

the presence of pre-existing medical illness.

Table showing relationship of admission grade to outcome

Neurological grade on admission (WFNS)	No. of patients undergoing aneurysm repair	Unfavourable outcome – death/severe disability (percentage)
I	1214	24.7
II	378	37.6
III	88	48.9
IV	164	64.0
V	118	71.2

Of all patients with a confirmed aneurysm, 92% underwent repair, 53% by surgical clipping and 38% by coil embolisation. No difference was noted in outcome between the two groups even after case mix adjustment (unfavourable outcome 35% for clipped group: 34% for coiled group).

Comparing different operative or management policies: Comparison of different treatments for ruptured aneurysms is difficult, unless conducted under the confines of a randomised controlled trial. ‘Operative mortality’ provides limited information unless patient groups are carefully matched for age, clinical condition and timing of operation. ‘Management mortality’ (e.g. outcome of all admitted patients up to 3 months from the ictus) is of more practical value, but even then, admission policies require careful scrutiny.

CEREBRAL ANEURYSMS – UNRUPTURED

UNRUPTURED ANEURYSMS

Identification of unruptured aneurysms may result from –

- Investigation of unrelated neurological symptoms with CT, MRI or angiography
- Investigation of symptoms arising as a result of aneurysmal compression of adjacent structures
- Investigation of patients with a family history of aneurysmal SAH (see [page 295](#))
- Investigation of SAH when multiple aneurysms exist (about 25% of patients)

Management: depends on the above circumstances and on balancing the risk of rupture (and death) in future years against the risk of aneurysm repair; the decision is often difficult. The International Study of Unruptured Intracranial Aneurysms (ISUIA), by far the largest study of its kind, examined both the *natural history* and the *results of treatment* of unruptured aneurysms. The data suggested that the risk of rupture related to the size, site and the occurrence of a SAH from a previously treated source. For small aneurysms < 7 mm in diameter and no previous SAH, the annual risk of rupture was 0.1% (far lower than the 1–2% suggested from previous smaller studies). For aneurysms > 12 mm in diameter the annual risk of rupture ranged from 3–10% depending on the site and size. For those treated, the study also reported a combined mortality and morbidity of from 7–10% for the coiled patients and from 10–13% for the operated patients, a figure higher than surgeons had previously liked to admit. The operative risk increased with age, aneurysm size and a site on the posterior circulation.

When determining appropriate management, the following factors must be taken into account

- patient's age, life expectancy and the occurrence of a previous

haemorrhage

- aneurysm size and site
- the possibility of future growth (about 1/3 expand > 3 mm over 20 years)
- the patient's view (a conservative approach can create considerable stress)
- the operative/endovascular results of unruptured aneurysm treatment for that centre

In general treatment would not be recommended for anterior circulation aneurysms < 7 mm in size and without a previous SAH, although this view may change with continued improvement in endovascular techniques.

For those undergoing a conservative approach, it is essential to ensure that they do not smoke, since this doubles the risk of aneurysm rupture.

When aneurysms present with compressive symptoms such as a III nerve palsy, it is assumed that recent expansion has occurred and that rupture could be imminent. Such patients normally receive urgent treatment.

Cumulative 5 year rupture rates for unruptured aneurysms at different sites (ISUIA, Lancet 2003)

	< 7 mm	< 7 mm	7–12 mm	13–24 mm	> 24 mm
	No previous SAH	Previous SAH from another source			
Anterior & middle cerebral, carotid artery aneurysms	0	1.5%	2.6%	14.5%	40%
Posterior circulation & posterior communicating artery	2.5%	3.4%	14.5%	18.4%	50%

CEREBRAL ANEURYSMS – SCREENING

SCREENING FOR INTRACRANIAL ANEURYSMS

When two or more first-degree relatives have a history of cerebral aneurysms or SAH, then other members of that family (over the age of 25 years) have an increased risk of harbouring an intracranial aneurysm (about 10% or 6x greater than the rest of the population). A similar increased risk occurs for patients with a genetic predisposition, e.g. polycystic kidney disease, Type IV Ehlers-Danlos. Before undergoing screening to detect whether such an aneurysm exists, several important facts should be considered –

- We do not know how rapidly aneurysms form. They may develop over a few hours, days or weeks. A negative screening investigation will fail to provide the reassurance that a subarachnoid haemorrhage from a ruptured aneurysm will never occur.
- The ISUIA study described above, suggests that for small aneurysms (< 7 mm), the rupture risk is extremely low. Even if a small aneurysm is found, treatment risks may preclude any action.
- Aneurysm repair, either by direct operation or by coil embolisation, carries a risk of death or disability, which depends on the patient's age and the size and site of the aneurysm.
- The presence of an aneurysm may carry implications for life insurance, mortgage applications and even for future pregnancies (since the risk of aneurysm rupture is increased during pregnancy).
- It is impractical to consider screening in relatives < 20 –25 years (due to the rarity of aneurysms in this age group) and in those > 60 –70 years since the risks outweigh any potential benefit.

– In those with only one affected first-degree relative the low, albeit slightly increased risk of finding an unruptured aneurysm, is still insufficient to justify screening.

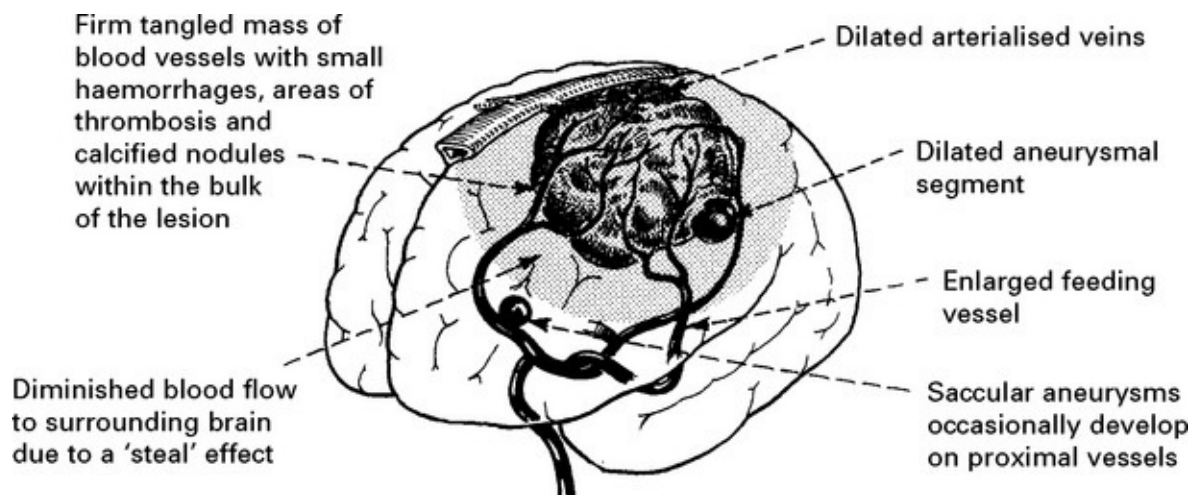
If after consultation and consideration of these issues the patient wishes to proceed with screening, then CT angiography would be the most appropriate technique, accepting that this may fail to detect aneurysms 3 mm or less in diameter. For those who decide not to undergo screening, other measures may minimise the risk of aneurysm formation in the future – avoid smoking and treat elevated blood pressure and cholesterol.

After a negative investigation, the patient may wish to consider the possibility of a further screen in 3–5 years time.

VASCULAR MALFORMATIONS

Vascular malformations vary in size and different forms exist:

Arteriovenous malformations (AVMs) are developmental anomalies of the intracranial vasculature; they are not neoplastic despite their tendency to expand with time and the descriptive term ‘angioma’ occasionally applied.



Dilated arteries feed directly into a tangled mass of blood vessels of varying calibre; they bypass capillaries and shunt oxygenated blood directly into the venous system. Due to high intraluminal pressure, veins may adopt an ‘aneurysmal’ appearance. Arteriovenous malformations occur at any site but are commonest in the middle cerebral artery territory.

Capillary telangiectasis: an area of dilated capillaries, like a small petechial patch on the brain surface – especially in the pons. These lesions are often only revealed at autopsy.

Cavernous malformation/angioma: plum coloured sponge-like mass composed of a collection of blood filled spaces with no intervening brain tissue. No enlargement of feeding or draining vessels.

ARTERIOVENOUS MALFORMATIONS

CLINICAL PRESENTATION

Haemorrhage

About 40–60% of patients with an AVM present with haemorrhage – often with an intracerebral or intraventricular component. In comparison with saccular aneurysms, AVMs tend to bleed in younger patients, *i.e.* 20–40 years, and are less likely to have a fatal outcome. Vasospasm and delayed ischaemic complications rarely develop. Small AVMs, those with high intranidal pressure and those draining exclusively to deep veins have an increased risk of haemorrhage.

Annual risk of haemorrhage: patients with no history of haemorrhage have an annual risk of bleeding of 2–4%. For those presenting with haemorrhage, the risk of rebleeding may be higher, particularly in the first year. One study reported an annual risk of 17%.

Mortality from haemorrhage: in contrast to the high mortality following aneurysm rupture, haemorrhage from an AVM carries the relatively low mortality rate of approximately 10%.

Epilepsy

Generalised or partial seizures commonly occur in patients with arteriovenous malformation, especially if the lesion involves the cortical surface. Of patients presenting with haemorrhage, 30% have a history of epilepsy.

Neurological deficit

Large AVMs, especially those involving the basal ganglia, may present with a slowly progressive dementia, hemiparesis or visual field defect, probably as a result of a 'steal' effect. The infrequent brain stem AVM may also produce a motor or sensory deficit, with or without cranial nerve involvement.

Headache

Attacks of well localised headache – unilateral and throbbing – occur in a proportion of patients subsequently shown to have a large AVM.

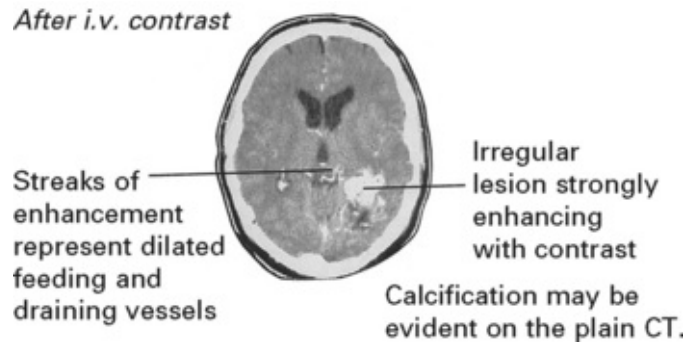
Cranial bruit

Auscultation, especially over the eyeball, occasionally reveals a bruit.

INVESTIGATIONS

CT scan

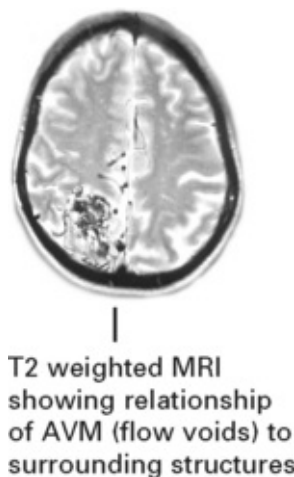
Most AVMs are evident on CT scan unless masked by the presence of an intracranial haematoma. A double dose of intravenous contrast may aid visualisation, especially with small 'cryptic' lesions.



MRI

Conventional MRI will clearly demonstrate the AVM as a region of flow voids, with associated signal change within or around the lesion from areas of old haemorrhage or gliosis.

The MRI provides exact anatomical detail and helps surgical planning. Functional MRI ([page 42](#)) aids identification of any adjacent eloquent areas.

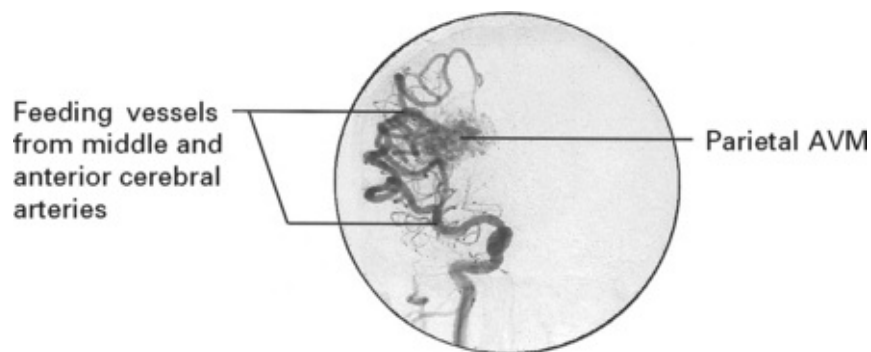


Angiography

Both CT and MR angiography should confirm the presence of an AVM but digital subtraction four-vessel angiography is required to delineate the feeding and draining vessels. Occasionally small AVMs are difficult

to detect and only *early venous filling* may draw attention to their presence.

N.B. In the presence of a haematoma, digital subtraction angiography should be delayed until the haematoma resolves, otherwise local pressure may mask demonstration of an AVM. If the angiogram is subsequently negative, then MRI is required to exclude the presence of a cavernous malformation.



MANAGEMENT

Various methods of treating arteriovenous malformations are available. All risk further damage and a team comprised of the neurosurgeon and neuroradiologist should decide on the optimal method or combination of methods for each patient. The urgency of the patient's clinical condition and the risks of treatment must be weighed against the risk of a conservative approach. The Spetzler-Martin grading system provides a useful guide to operative risk.

Spetzler-Martin grading system

Size of AVM	< 3 cm	1	e.g. 2 cm AVM in non-eloquent area with no deeply draining veins = grade 1
	3–6 cm	2	
	> 6 cm	3	
Eloquence of adjacent brain	non-eloquent	0	4 cm AVM in eloquent area (motor, speech or visual cortex, thalamus, internal capsule, basal ganglia, brain stem) with deep venous drainage = grade 4
	eloquent	1	
Pattern of venous drainage	superficial only	0	
	deep	1	

Indications for intervention

Indications for intervention – ‘Expanding’ haematoma associated with AVM – Progressive neurological deficit – Risk of haemorrhage especially – young patients with many years at risk – AVMs < 3 cm	Take into account operative risk low in – Grade 1–2 – AVMs in ‘non-eloquent’ sites – AVMs < 3 cm diameter – with superficial venous drainage high in – Grade 4–5 – AVMs > 3 cm diameter – AVMs in ‘eloquent’ sites – with drainage to deep veins
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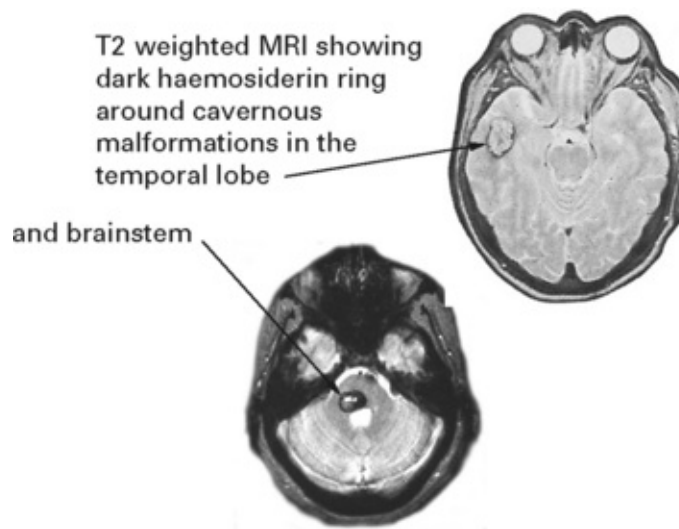
Methods of treatment

<i>Operation:</i>	<i>Excision</i> – complete excision of the AVM (confirmed by per-or postoperative angiography) is the most effective method of treatment particularly for small AVMs in non-eloquent areas. Image guidance (page 386) may aid localisation.
	Larger lesions (> 6 cm) have a greater risk of postoperative hyperperfusion syndrome and brain swelling and carry a 40% risk of permanent neurological deficit.
<i>Stereotactic</i>	
<i>radiosurgery:</i>	Focused beams from multiple cobalt sources or from a linear accelerator (25Gy) obliterates about 75% of AVMs < 3 cm in diameter, but this may take <i>up to 3 years during which time the risk of haemorrhage persists</i> . In smaller lesions < 1 cm the obliteration rate with 25 Gy approaches 100%. For lesions greater than 3 cm, the lower dose required to minimise the damaging effect of local tissue destruction, makes obliteration unlikely. Pre-treatment with embolisation helps only if this produces a segmental reduction in size. Suboptimal embolisation may merely hinder radiosurgical treatment. Despite the delay in action, radiosurgery may prove ideal for small deeply seated lesions.

Embolisation:	Skilled catheterisation permits selective embolisation of feeding vessels with isobutyl-cyanoacrylate, although this technique is not without risk.
	Embolisation may cure up to 40% of AVMs when small particularly if supplied by a single feeding vessel, but filling may persist from collaterals. When used preoperatively, it may significantly aid operative removal.

CAVERNOUS MALFORMATIONS (syn. cavernous angioma, cavernoma)

Cavernous malformations occur in 0.5% of the population. They are occasionally multiple and in a few patients, have a familial basis. A cavernous malformation may present with epilepsy, haemorrhage or with focal neurological signs. *MRI* is the investigation of choice as *cavernous malformations*, are often missed on CT scanning and rarely seen on angiography. Most lesions show marked signal change around this lesion due to a rim of haemosiderin deposition.

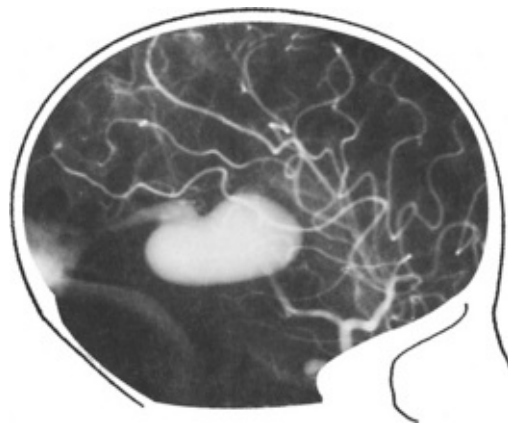


The annual risk of haemorrhage is about 1% per year, but this varies depending on whether the lesion lies deeply (i.e. brain stem or basal ganglia) or superficially. With deep lesions in critical sites, a small bleed causes damage more readily. For deep lesions the risk of a bleed

sufficiently severe to cause neurological signs is about 5% per year, whereas for superficial lesions, this is almost zero. Unfortunately the high risk, deep lesions are more hazardous to surgically remove, although this may be the appropriate management in selected patients.

VEIN OF GALEN MALFORMATION

The term 'vein of Galen' malformation is actually a misnomer since this is a type of arteriovenous malformation in which arteries feed directly into the embryonic precursor of the great vein of Galen (the prosencephalic vein of Markowski) causing massive aneurysmal dilatation. Patients present either in the neonatal period with severe high output cardiac failure due to the associated arteriovenous shunt, in infancy with cranial enlargement due to an obstructive hydrocephalus, or in childhood with subarachnoid haemorrhage. A cranial bruit is always evident. Cardiac failure usually develops in the neonatal period and is usually fatal. In the other groups the treatment of choice is now endovascular obliteration of the feeding vessels followed by ventricular drainage if required. As a result, the high mortality and morbidity experienced with direct operative repair has been considerably reduced.



STURGE-WEBER SYNDROME

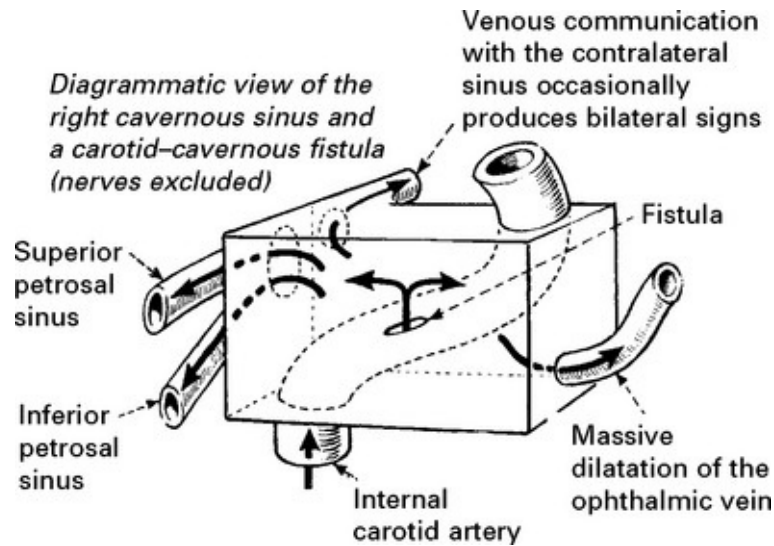
Angiomatosis affecting the facial skin, eyes and leptomeninges produces the characteristic features of the Sturge-Weber syndrome – a capillary naevus over the forehead and eye, epilepsy and intracranial calcification. (See [page 563](#).)

DURAL ARTERIOVENOUS FISTULA

In contrast to AVMs these fistulous communications are usually acquired rather than developmental in origin. Arterial blood drains directly into either a venous sinus, cortical veins or a combination of both (see carotid-cavernous fistula [page 301](#)). The aetiology remains unknown, but sinus thrombosis or trauma may play a part. In a benign form, no reversal of flow occurs and no treatment is required. When retrograde venous flow occurs, venous hypertension results and haemorrhage may follow. For this type, treatment requires ligation and division of the draining vein, often combined with endovascular occlusion.

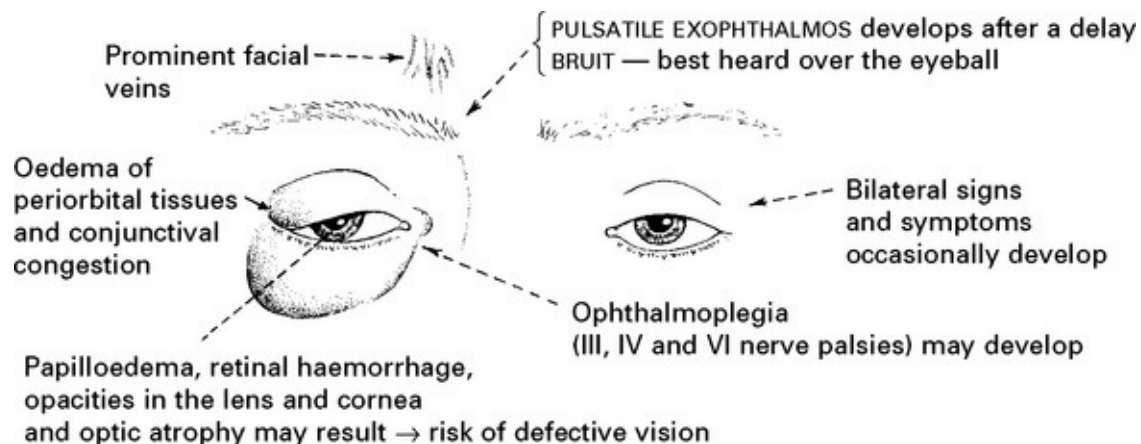
CAROTID–CAVERNOUS FISTULA

A fistulous communication between the internal carotid artery and the cavernous sinus may follow skull base trauma either immediately or after a delay of several days or weeks. Less often carotid-cavernous fistulae occur spontaneously, perhaps from rupture of a small intracavernous meningeal artery or a saccular carotid aneurysm.



Clinical features

Symptoms develop *suddenly* (cf. cavernous sinus thrombosis) – the patient becomes aware of pulsating tinnitus as a ‘noise’ inside the head. Pain may follow. Examination reveals characteristic signs:



Methods of fistula repair

Spontaneous closure occurs in up to 60%. Provided symptoms do not progress, for the first few months, treatment should be conservative.

Endovascular: with *detachable balloon* catheterization, either through the transvenous or intra-arterial route, or by stent assisted coil

embolisation. Recent reports include the use of covered stents alone.

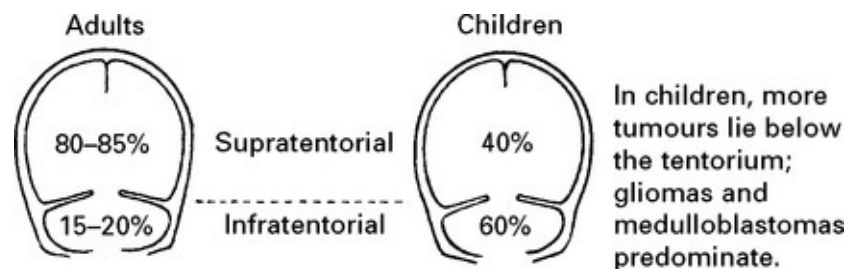
INTRACRANIAL TUMOURS

INCIDENCE

Primary brain tumours occur in approximately 6 persons per 100 000 per year. Fewer patients with metastatic tumours reach a neurosurgical centre, although the actual incidence must equal, if not exceed that of primary tumours. It is estimated that 25% of patients with a malignancy have a CNS metastasis. About 1 in 12 primary brain tumours occur in children under 15 years.

SITE

In adults, the commonest tumours are gliomas, metastases and meningiomas; most lie in the supratentorial compartment.



PATHOLOGY

Intracranial tumours are often described as 'benign' or 'malignant', but these terms cannot be directly compared with their extracranial counterparts:

A *benign* intracranial tumour may have devastating effects if allowed to expand within the rigid confines of the skull cavity. A benign astrocytoma may infiltrate widely throughout brain tissue preventing complete removal, or may occupy a functionally critical site preventing

even partial removal.

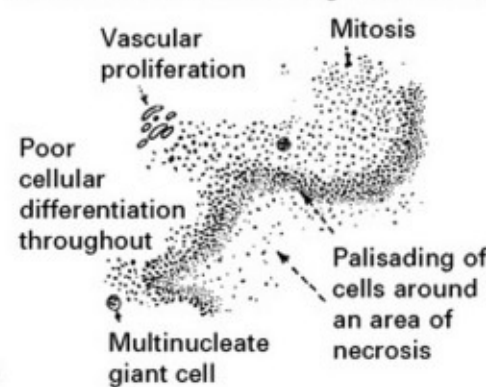
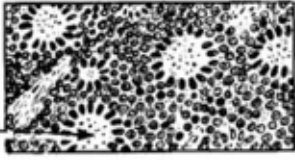
A *malignant* intracranial tumour implies rapid growth, poor differentiation, increased cellularity, mitosis, necrosis and vascular proliferation, but metastases to extracranial sites rarely occur.

Pathological classification

In 2000, the World Health Organization drew up an internationally agreed classification of intracranial tumours based on the tissue of origin. This system avoids the term 'glioma' – previously encompassing astrocytoma, oligodendroglioma, ependymoma and glioblastoma multiforme. The cell origin of the highly malignant glioblastoma is now recognizable as astrocytic rather than embryonal as previously classified.

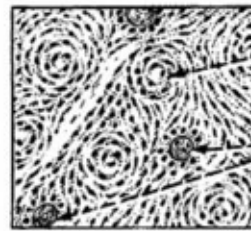
INTRACRANIAL TUMOURS – PATHOLOGICAL CLASSIFICATION

NEUROEPITHELIAL

Astrocytes	→ Astrocytoma: The most common primary brain tumour. Histological features permit separation into four grades depending on the degree of malignancy. Grading only reflects the features of the biopsy specimen and not necessarily those of the whole tumour. The most malignant type – glioblastoma (grade IV) – occurs most frequently and widely infiltrates surrounding tissue. Other types range from the less common low-grade astrocytomas including the pilocytic type (grade I) and diffuse types (fibrillary, protoplasmic and gemistocytic) (grade II) to the anaplastic astrocytomas (grade III).	Composite diagram showing the characteristic features of a glioblastoma. 
Oligodendrocytes	→ Oligodendroglioma: Usually a slowly growing, sharply defined tumour (grade II). Variants include an anaplastic form (grade III) and a 'mixed' oligoastrocytoma (grade II).	
Ependymal cells and choroid plexus	→ Ependymoma: Occurs anywhere throughout the ventricular system or spinal canal, but is particularly common in the 4th ventricle and cauda equina. It infiltrates surrounding tissue and may spread throughout the CSF pathways (grade II). Variants include an anaplastic type (grade III) and a subependymoma arising from subependymal astrocytes (grade I). → Choroid plexus papilloma: Rare tumours and an uncommon cause of hydrocephalus due to excessive CSF production. They are usually benign but occasionally occur in a malignant form.	
Neurons	→ Ganglioglioma/gangliocytoma/neurocytoma: Rare tumours containing ganglion cells and abnormal neurons occurring in varying degrees of malignancy. This classification includes the very low grade dysembryoplastic neuroepithelial tumour (DNET).	
Pineal cells	→ Pineocytoma/pineoblastoma: Extremely rare tumours. The latter are less well differentiated and show more malignant features.	
Embryonal cell origin	→ Primitive Neuroectodermal Tumours (PNET): Small cell malignant tumours of childhood occurring rarely supratentorially, but far more commonly infratentorially where they are called medulloblastomas. These arise in the cerebellar vermis. Small closely packed cells are often arranged in rosettes surrounding abortive axons. May seed through the CSF pathways.	

MENINGES

→ **Meningioma:** Arises from the arachnoid granulations, usually closely related to the venous sinuses but also found over the hemispheric convexity. The tumours compress rather than invade adjacent brain. They also occur in the skull base, spinal canal and orbit. Most are benign (despite their tendency to invade adjacent bone) but some undergo sarcomatous change.



Whorls of meningotheelial and spindle cells
Calcified psammoma bodies

Histological types – meningotheelial, transitional, fibroblastic and angioblastic.

The haemangio-pericytoma is poorly differentiated, aggressive in nature and of uncertain histogenesis.

→ **Meningeal sarcoma** and primary **Meningeal melanoma:** Exceedingly rare tumours.

NERVE SHEATH CELLS

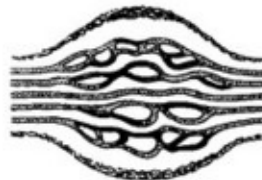
→ **Schwannoma** (Syn. neurilemmoma/neurinoma): a non-invasive, slowly growing tumour of the Schwann cells, surrounding cranial nerve roots (usually the vestibular part of the VIII nerve) or the peripheral nerves. Occurs in neurofibromatosis type 2 (NF2)



Different histological types exist:

Antoni type A } see page 333
Antoni type B }

→ **Neurofibroma:** tumour of Schwann cells, fibroblasts and perineural-like cells producing a fusiform expansion through which nerve fibres run. It involves the spinal nerve roots or peripheral nerves but rarely affects cranial nerves and has a greater tendency to undergo malignant change than schwannoma. Predominant in neurofibromatosis type 1 (NF1), although schwannomas and mixed tumours may also occur (see page 561).

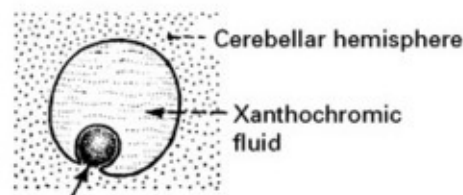


N.B. Many tumours have mixed characteristics in varying proportions.

BLOOD VESSELS

Haemangioblastoma:

Occurs within the cerebellar parenchyma or spinal cord. In 1926, Lindau described a syndrome relating cerebellar and/or spinal haemangioblastomas with similar tumours in the retina and cystic lesions in the pancreas and kidney (Von Hippel-Lindau disease).



Reddish-brown tumour nodule lying in the wall of the cyst. Histology shows a mass of blood vessels separated by clear, foamy vacuolated (stromal) cells

GERM CELLS

- **Germinoma:** Primitive spheroidal cell tumour comparable to seminoma of the testis.
 - **Teratoma:** A tumour containing a mixture of well differentiated tissues – dermis, muscle, bone.
- } uncommon tumours of the *pineal region* (not arising from pineal cells)

TUMOURS OF THE SELLAR REGION

- **Craniopharyngioma:** Arises from embryonic remnants of Rathke's cleft and lies in close relation to the pituitary stalk. Usually a nodular tumour with cystic areas containing greenish fluid and cholesteatomatous material.
- **Pituitary adenoma:** Benign tumour, usually secreting excessive quantities of prolactin, growth hormone, adrenocorticotrophic hormone, thyrotropin or gonadotropin.

CYSTS AND TUMOUR LIKE CONDITIONS

- **Epidermoid/dermoid cysts:** Rare cystic tumours arising from cell rests predetermined to form epidermis or dermis.
- **Colloid cyst:** A cystic tumour arising from an embryological remnant in the anterior roof of the 3rd ventricle.

LOCAL EXTENSION FROM ADJACENT TUMOURS

- **Chordoma:** Rare tumour arising from cell rests of the notochord. May occur anywhere from the sphenoid to the coccyx – but commonest in the basi-occipital and the sacrococcygeal region, invading and destroying bone at these sites.
- **Glomus jugulare tumour** (syn. chemodectoma): Vascular tumour arising from 'glomus jugulare' tissue lying either in the bulb of the internal jugular vein or in the mucosa of the middle ear. The tumour invades the petrous bone and may extend into the posterior fossa or neck.
- Other local tumours include **chondroma**, **chondrosarcoma** and **cylindroma**.

Primary central nervous system lymphoma (PCNSL): Forms around periventricular parenchymal blood vessels. May be solitary or multifocal. It generally occurs in immunocompromised patients, *e.g.* AIDS. Metastatic spread from systemic lymphoma (*e.g.* non-Hodgkin's lymphoma) is less common, involves the meninges and is rarely intraparenchymal.

Metastatic tumours: May arise from any primary site but most commonly spread from the bronchus or breast. Nervous system metastases occur in 25% of patients with disseminated cancer.

Tumour markers

Immunohistochemical techniques permit identification of antigens

specific for certain cell or tissue characteristics and aid the histological diagnosis of tumours.

e.g. Glial fibrillary acidic protein (GFAP) – for astrocytic tumours

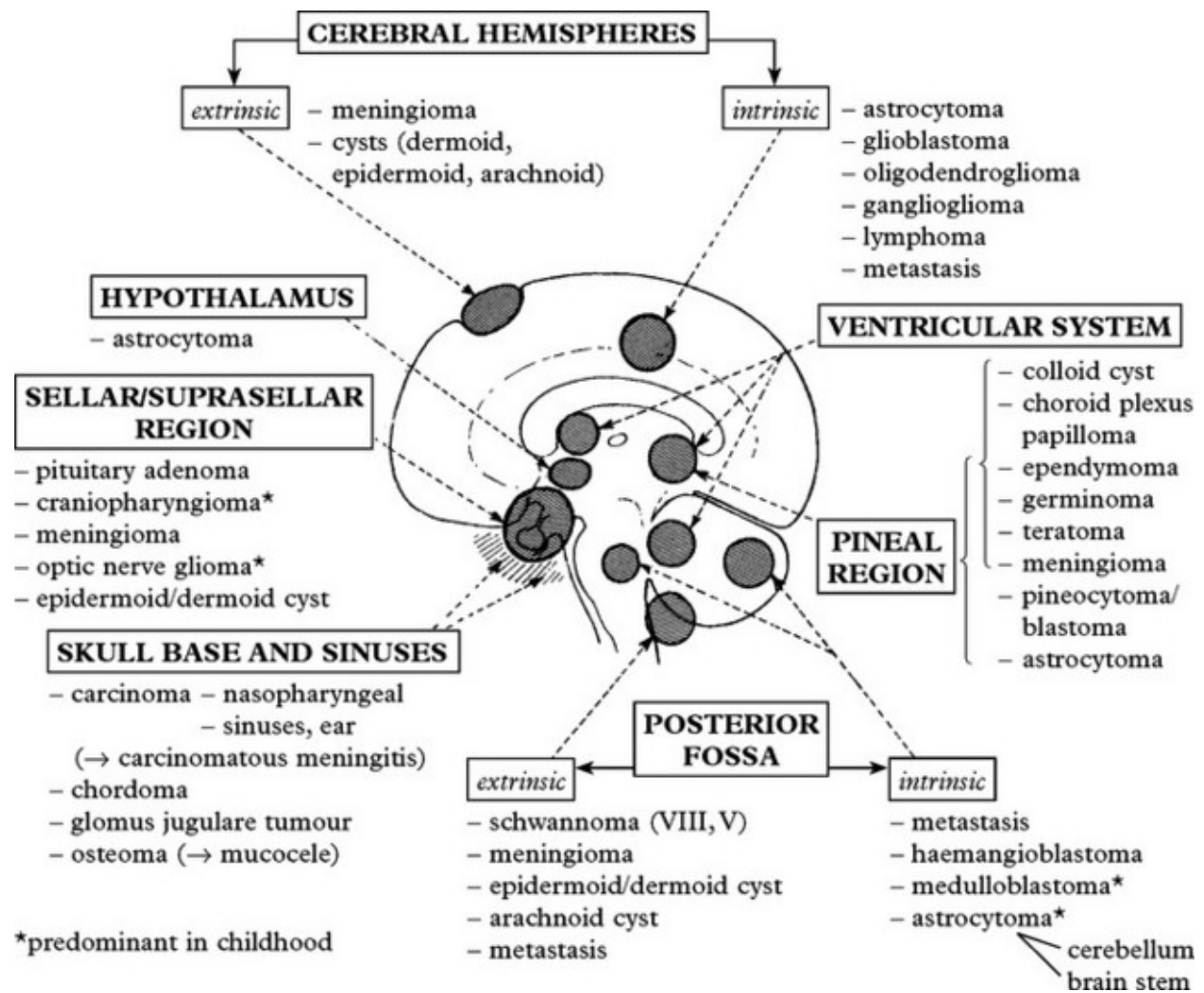
Cytokeratin – for metastatic carcinoma

Synaptophysin – for neuronal tumours

HMB 45 – for malignant melanoma

Some markers also indicate the degree of proliferation in various tumours (e.g. *Ki-67*). The identification of growth factors (e.g. *Epidermal growth factor (EGFR)*) may help distinguish between a primary glioblastoma (arising de novo) from a secondary glioblastoma (dedifferentiating from a previous lower grade tumour). Molecular techniques are increasingly used to identify *loss of heterozygosity e.g. 1p,19q* in oligodendroglioma.

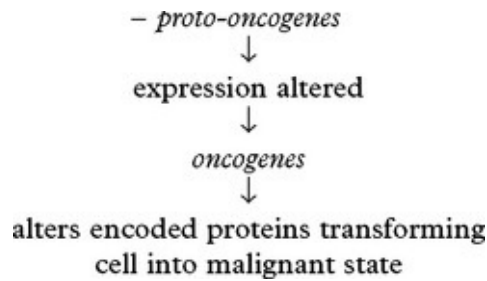
INTRACRANIAL TUMOURS – CLASSIFICATION ACCORDING TO SITE



AETIOLOGY

Genetic factors: Over recent years, the role of genetic factors in tumour development has gained increasing prominence. Transformation of normal cells to malignant growth probably results from a variety of different processes –

(a) Normal cell growth and differentiation controlled by –



INTRACRANIAL TUMOURS – AETIOLOGY/INCIDENCE

(b) Inactivation of expression of *tumour suppressor genes* (e.g. mutation of the p53 gene with loss of heterozygosity on the 17p chromosome in many patients with low grade astrocytoma).

(c) Over expression of genes controlling growth factor (e.g. amplification of EGFR in primary glioblastoma).

Clearly defined inherited factors play a minor role. Only 5% of patients have a family history of brain tumour and with the exception of **tuberous sclerosis** (related to the formation of subependymal astrocytomas) and **neurofibromatosis** (linked to an increased incidence of schwannoma, optic nerve glioma and meningioma) do not fall into an obvious autosomal recessive or dominant pattern. Others include von Hippel-Lindau disease, Cowden's disease and Li-Fraumeni syndrome.

Cranial irradiation: long term follow-up of patients undergoing whole head irradiation for treatment of tinea capitis and childhood leukemia shows an increased incidence of both benign and malignant tumours – e.g. astrocytoma, meningioma.

Immunosuppression: increased incidence of lymphoma.

INCIDENCE

The table below shows the approximate incidence of intracranial

tumours extracted from large series.

Adults	
Glioblastoma	15%
Low grade glioma	5%
Meningioma	25%
Pituitary adenoma	25%
Primary CNS Lymphoma	4%
Peripheral Nerve Sheath Tumour (schwannoma)	8%
Others	18%

Adapted from Louis et al. WHO Classification of Tumours of the CNS, IARC Press 2007

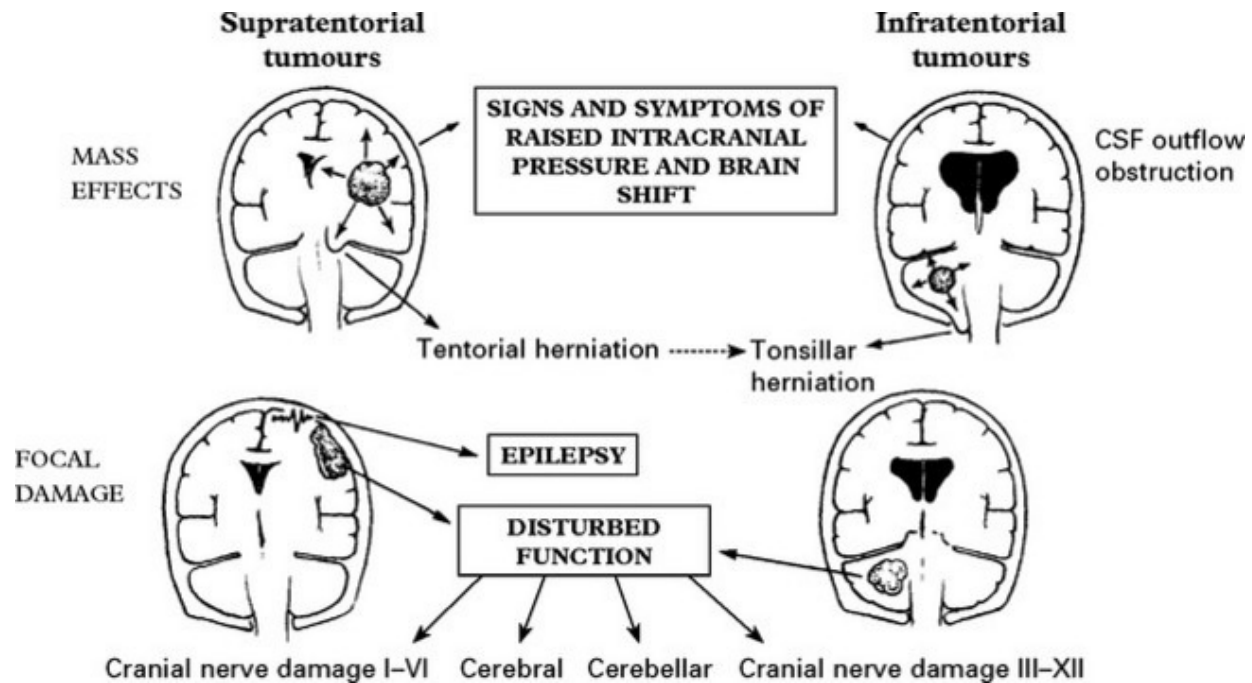
Children	
Medulloblastoma/PNET	16%
Low grade glioma	33%
Malignant glioma	14%
Ependymoma	10%
Craniopharyngioma	6%
Germ cell tumours	2.5%
Meningioma	2.5%
Others	16%

Adapted from Rickert & Paulus Child's Nervous System 2001 17.503-511

INTRACRANIAL TUMOURS – CLINICAL FEATURES

Symptoms tend to develop insidiously, gradually progressing over a few weeks or years, depending on the degree of malignancy (cf. acute onset of a cerebrovascular accident followed by a gradual improvement if the patient survives). Occasionally tumours present acutely due to

haemorrhage or the development of hydrocephalus.



CLINICAL EFFECTS

RAISED INTRACRANIAL PRESSURE – headache, papilloedema
BRAIN SHIFT – vomiting, deterioration of conscious level, pupillary dilatation } (see pages 81–83)

EPILEPSY (see page 92)

- generalised
- partial (simple or complex)
- partial progressing to generalised

(occur in 30% of patients with brain tumours)

Partial motor seizures arise in the motor cortex – tonic or clonic movements in the contralateral face or limbs.

Partial sensory seizures arise in the sensory cortex and cause numbness and tingling in the contralateral face, limbs.

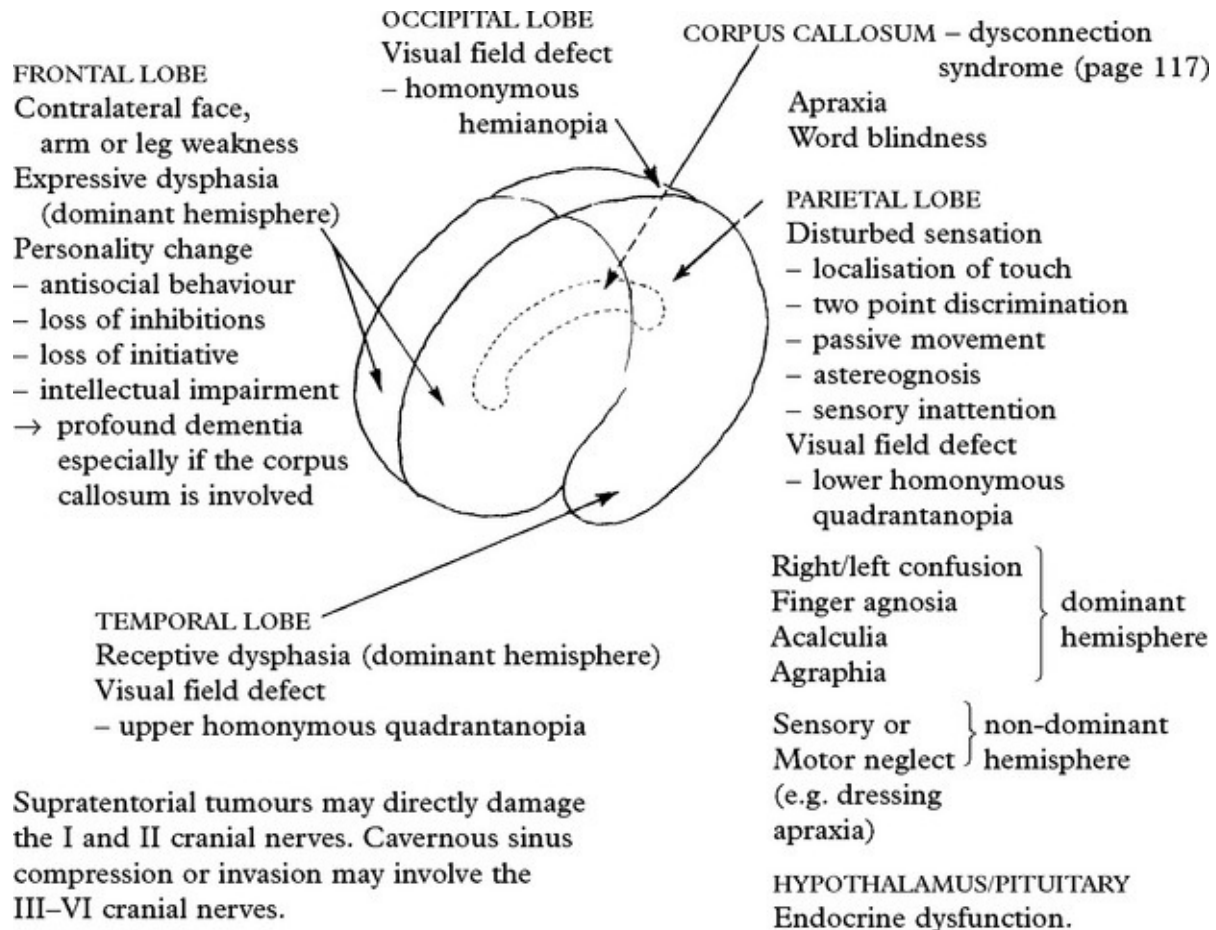
Pure visual (or auditory) seizures are rare.

Complex partial (temporal lobe) seizures arise from the medial temporal lobe – formed visual or auditory hallucinations, awareness of abnormal taste, feelings of fear, déjà vu, unfamiliarity or depersonalisation and automatisms.

Partial seizures help localise the tumour site.

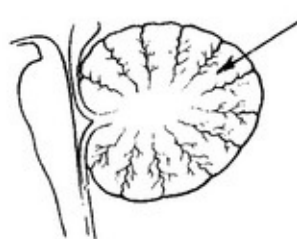
DISTURBED FUNCTION

Supratentorial – see higher cortical dysfunction, [pages 109–117](#)



Infratentorial

MIDBRAIN/BRAIN STEM
 Cranial nerve lesions III–XII
 Long tract signs
 – motor and sensory
 Deterioration of conscious level
 Tremor (red nucleus)
 Impaired eye movements
 Pupillary abnormalities
 Vomiting, hiccough (medulla)



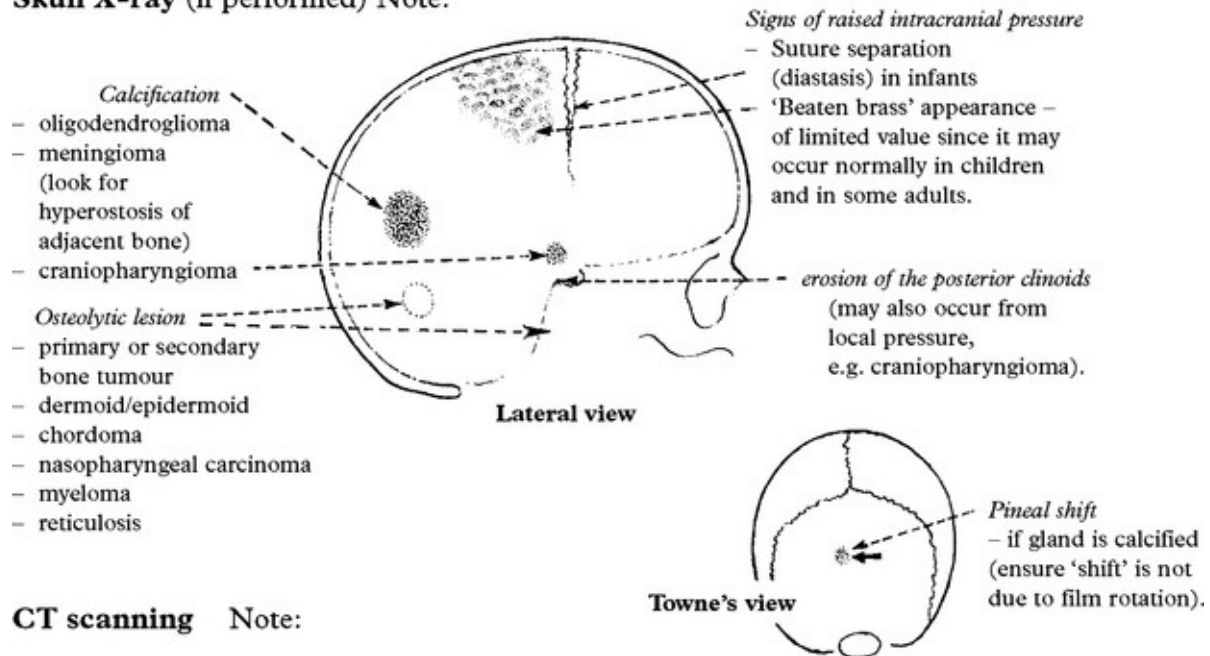
CEREBELLUM – see cerebellar dysfunction, [pages 180–183](#)
 Ataxic gait
 Intention tremor
 Dysmetria
 Dysarthria
 Nystagmus

N.B. Intrinsic brain stem tumours in contrast to extrinsic tumours are more likely to produce long tract (motor and sensory) signs early in the course of the disease.

INTRACRANIAL TUMOURS – INVESTIGATION

Chest X-ray, ESR, CRP } The high incidence of metastatic tumour makes these tests mandatory in patients with suspected intracranial tumour.

Skull X-ray (if performed) Note:



CT scanning Note:

SITE

e.g. frontal, occipital

- extrinsic:** outwith brain substance, e.g. meningioma
- intrinsic:** within brain parenchyma, e.g. astrocytoma.

MASS EFFECT

- midline shift.
- ventricular compression.
- hydrocephalus (secondary to 3rd ventricular or posterior fossa lesion).
- obliteration of basal cisterns



Effect on adjacent bone i.e.
if meningioma → hyperostosis

Single or multiple lesions
if multiple → metastasis

Effect of contrast enhancement
e.g. none – low grade astrocytoma
irregular – malignant astrocytoma
homogeneous – meningioma

HIGH DEFINITION SCANS (1 mm slice width) – useful in the detection of pituitary, orbital and posterior fossa tumours.

**CORONAL AND SAGITTAL
RECONSTRUCTION**

} – useful in demonstrating the vertical extent of a tumour and its relationship with other structures, especially when intraventricular or arising from the pituitary fossa or skull base.

MRI Note: SITE, MASS EFFECT and LESION MULTIPLICITY as for CT scanning.

Of particular value in tumours of the skull base, cranio-cervical junction and brain stem.

Flow voids show the relationship of adjacent blood vessels to the tumour.

Coronal and sagittal scanning provide additional information, showing the exact anatomical relationship of the tumour to the sulci and gyri, the ventricles, the falx and the tentorium cerebelli.



T1 weighted MRI with gadolinium showing a haemangioblastoma involving the medulla.

Paramagnetic enhancement: intravenous gadolinium increases sensitivity of detection and clarifies the site of origin, *i.e.* intrinsic or extrinsic, and may delineate the border between tumour and surrounding oedema.

Single or multiple lesions: MRI appears more sensitive than CT scanning in identifying small tumours and improves the detection of multiple lesions, *e.g.* metastasis.

Angiography/CTA/MRA: although angiography may reveal a

tumour 'blush' or vessel displacement, it is only occasionally required to supplement other investigations. In some patients, it provides useful preoperative information, *e.g.* identifies feeding vessels to a vascular tumour or tumour involvement and constriction of major vessels.

Thallium SPECT: helps identify sites of high grade activity within a tumour. Useful to exclude if proposing conservative management or in planning stereotactic biopsy.

Functional MRI: Shows the relationship of eloquent areas to the tumour and may aid resection.

MR Diffusion Tensor Imaging: used to identify fibre tracts running adjacent to or through the tumour. Of potential value in planning operative resection.

CSF examination: lumbar puncture is contraindicated if the clinician suspects intracranial tumour. If CSF is obtained by another source, *e.g.* ventricular drainage or during shunt insertion, then cytological examination may reveal tumour cells.

Tumour markers: although useful as an aid to histological diagnosis (see [page 305](#)), attempts to find a substance in blood or CSF which reflects growth of a specific tumour have been limited – only the link between elevated alpha fetoprotein and human chorionic gonadotrophins with yolk sac tumours and choriocarcinoma of the third ventricle helps diagnosis.

DIFFERENTIAL DIAGNOSIS OF MASS LESIONS (other than tumour)

Vascular – haematoma
– giant aneurysm
– arteriovenous malformation
– infarct with oedema
– venous thrombosis.

Trauma – haematoma
– contusion.

Infection – abscess
– tuberculoma
– sarcoidosis
– encephalitis.

Cysts – arachnoid
– parasitic (hydatid).

INTRACRANIAL TUMOURS – MANAGEMENT

STEROID THERAPY

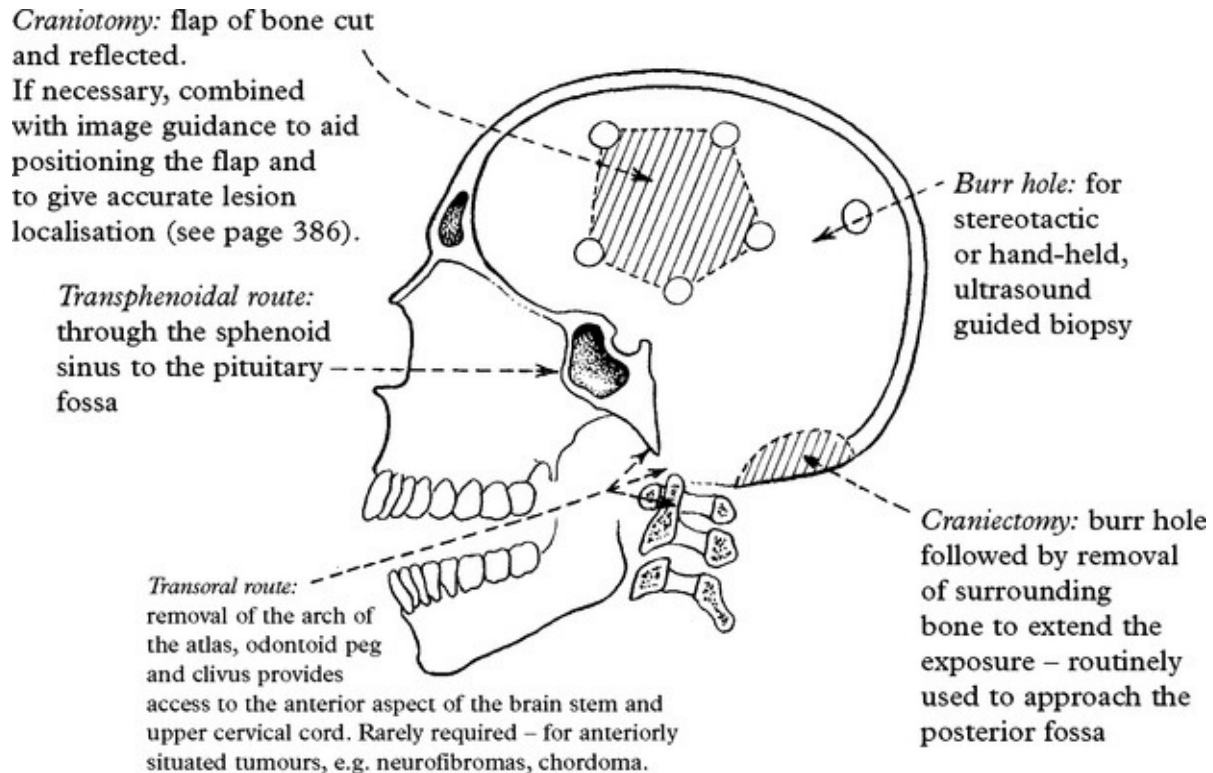
Steroids dramatically reduce oedema surrounding intracranial tumours, but do not affect tumour growth.

A loading dose of 12 mg i.v. dexamethasone followed by 4 mg q.i.d. orally or by injection often reverses progressive clinical deterioration within a few hours. After several days treatment, gradual dose reduction minimises the risk of unwanted side effects.

Sellar/parasellar tumours occasionally present with steroid insufficiency. In these patients, steroid cover is an essential prerequisite of any anaesthetic or operative procedure.

OPERATIVE MANAGEMENT

Most patients with intracranial tumours require one or more of the following approaches:



The subsequent procedure – *biopsy, partial tumour removal/internal decompression or complete removal* – depends on the nature of the tumour and its site. The infiltrative nature of primary malignant tumours prevents complete removal and often operation is restricted to biopsy or tumour decompression. Prospects of complete removal improve with benign tumours such as meningioma or craniopharyngioma; if any tumour tissue is overlooked, or if fragments remain attached to deep structures, then recurrence will result.

Image Guided Surgery

It is essential to accurately identify the tumour site on preoperative imaging and to be able to use this information to guide the surgeon to the tumour whether for biopsy or for resection. Various techniques are available –

Stereotactic surgery: by rigidly attaching the frame to the patient's head and using a CT or MR to identify the position of the locating rods, coordinates are determined for a selected target allowing accurate placement of a biopsy needle to within 1 mm (see [page 384](#)). This technique is routinely used to biopsy selected points within the tumour. It is possible to perform a craniotomy and tumour resection with the frame in place, but the frame tends to impede access and after opening the bone flap, the brain may shift introducing errors of localisation. When a craniotomy is planned, most now use neuronavigation ('frameless' stereotaxy) if available.

Neuronavigation: this technique requires rigid fixation of the head in a standard three pin head holder, but avoids the use of a cumbersome frame (see [page 386](#)). The system accurately detects the position of the handheld probe in relation to the skull and allows the surgeon to see where the probe tip lies in relation to preoperative imaging. Although often routinely used, this technique also fails to take into account brain shift which can occur on opening the bone flap or if cerebrospinal fluid is drained off thus limiting accuracy.

Real-time intraoperative imaging: some centres have now acquired CT or MR imaging available within the operative theatre. Although costly, this real-time imaging overcomes problems encountered with brain shift and not only helps to locate the tumour, but also shows the extent of tumour resection as the operation progresses. Ultrasound has also been combined with neuronavigation to provide real-time imaging at a more realistic expense.

Surgery in Eloquent Areas

When intrinsic tumours lie adjacent to or within eloquent areas within the brain, *i.e.* speech area, motor strip, basal ganglia and internal capsule, resection is potentially hazardous. Various techniques have been developed to try to minimise this risk

fMRI/Diffusion Tensor Imaging (Tractography): Superimposing speech and/or motor strip areas seen on fMRI and white matter tracts seen on tractography on to the standard MR image, demonstrates the relationship of the tumour to these crucial structures. When these images are incorporated into the neuronavigation system it enables the surgeon to avoid extending the tumour resection into these areas and causing irreversible neurological deficits. The reliability of each technique, however, is still in question and benefits remain uncertain.

'Awake' craniotomy: by either performing the surgery wholly under sedation with local anaesthetic, or by giving an anaesthetic for opening and closing the craniotomy and waking the patient up in between, gives the surgeon the opportunity to identify eloquent areas by applying electrical stimulation direct to the cortical surface and observing the functional effect. Studies show that patients tolerate the technique well and maximal tumour resection is possible with a low risk of deficit.

RADIOTHERAPY

Treatment of intracranial tumours with radiotherapy utilises one of the following:

- *megavoltage* X-rays (by far the most common method)
- electron beam from a *linear accelerator* (which can also produce megavoltage X-rays)

– *accelerated particles from a cyclotron, e.g. nuclei of helium, protons* (awaits full evaluation)

– γ rays from *cobalt⁶⁰*.

In contrast to older methods, these modern techniques produce greater tissue penetration and avoid radiation damage to the skin surface. The effect of radiotherapy depends on the total dose – usually up to 60 Gy, and the treatment duration. This must be balanced against the risk to normal structures. Treatment aims to provide the highest possible dose to a specified region whilst minimising irradiation to adjacent normal brain. Various methods have been developed to achieve this –

- *Conformal therapy* where standard radiotherapy is administered, but the beams are shaped by the use of variable collimators or blocks which conform with the shape of the tumour, thereby eliminating normal brain.

- *Stereotactic radiosurgery (SRS)* where multiple converging beams from a linear accelerator or from multiple cobalt⁶⁰ sources are focused on a selected target in a *single* treatment. *Stereotactic radiotherapy (SRT)* uses the same localisation method but with fractionated treatment as used in conventional radiotherapy (see [page 385](#)).

- *Interstitial techniques* where the tumour is treated from within (brachytherapy) by the implantation of multiple radioactive seeds, e.g. iodine¹²⁵.

- *Beam intensity modulated radiotherapy (IMRT)* uses non-uniform beams of varying intensity (in contrast to the conventional uniform dose

intensity) to complex tumour volumes. This helps protect surrounding structures, yet allows a higher dose.

- *Proton therapy* is available in only a few centres. It allows the delivery of high doses of radiation to very localised regions adjacent to vital structures such as the skull base.

Radiotherapy is of particular value in the management of malignant tumours – malignant astrocytoma, metastasis, medulloblastoma and germinoma, but also plays an important part in the management of some benign tumours – pituitary adenoma, craniopharyngioma. With some tumours that seed throughout the CSF pathways, *e.g.* medulloblastoma, *whole neural axis irradiation* minimises the risk of a distant recurrence.

Complications of radiotherapy: following treatment, deterioration in a patient's condition may occur for a variety of reasons:

- Increased oedema – during treatment – reversible.
- Demyelination – after weeks, months – usually reversible.
- *Radionecrosis* – in usually 1–2 years (range 6 months–10 years) – irreversible.
- Cognitive impairment – whole brain irradiation causes dementia, ataxia and incontinence in over 10% at one year. Radiotherapy should be avoided in children under 3 years of age.
- Radiation induced tumours *e.g.* meningioma, may result many years after the treatment.

Oedema, demyelination and radionecrosis may involve the spinal cord after irradiation of spinal tumours. Other harmful effects include

hair loss, skin reactions and endocrine disturbance.

CHEMOTHERAPY

Chemotherapeutic agents have been used for many years in the management of malignant brain tumours, but their benefits remain limited. Historically drugs most commonly used include nitrosoureas (e.g. BCNU, CCNU), procarbazine, vincristine and methotrexate (for lymphoma).

Temozolomide, an oral alkylating agent with excellent blood brain barrier penetration and modest toxicity is established as an alternative treatment for patients with recurrent high grade gliomas. It has also been shown to improve survival for patients with newly diagnosed glioblastoma when given concomitantly with radiotherapy. A combination of maximal safe surgery followed by combined chemoradiotherapy is now the standard of care for good performance patients with glioblastoma. Patients with methylation of the MGMT gene in the tumour appear particularly to benefit. Carmustine impregnated wafers (Gliadel) may also be considered both as a primary treatment or for tumour recurrence (see below).

Patients with *anaplastic oligodendrogliomas* and *oligoastrocytoma* with loss of heterozygosity on chromosomes 1p and 19q have a good prognosis and respond well to both radiation and to alkylating agent based chemotherapy (nitrosoureas, Temozolomide). Chemotherapy may be used either at initial diagnosis or at relapse in these patients. Other tumours where chemotherapy plays an important role include *medulloblastomas*, *primary CNS lymphomas* and *germ cell tumours*.

Traditionally, chemotherapy has had a lesser role in *low grade glial*

tumours but current studies are examining its use in both astrocytomas and oligodendrogliomas as an alternative to radiation in newly diagnosed patients.

Problems of drug administration

Toxicity: The ideal cytotoxic drug selectively kills tumour cells; but tumour cell response relates directly to the dose. High drug dosage frequently causes bone marrow suppression which may limit cytotoxic activity before an adequate therapeutic dose is reached.

Drug access: ‘Toxic’ doses are usually required before sufficient amounts penetrate the blood–brain barrier and gain access to the tumour cells.

Intrinsic resistance: Some tumour cells appear to have an inbuilt resistance to certain drugs. The vast array of available cytotoxic drugs and the infinite permutations of combined therapy creates difficulties in drug selection.

IMPROVING ACCESS: Many attempts to improve the access of cytotoxic drugs have been made with little success. Approaches such as blood brain barrier opening with mannitol, liposome delivery and direct intra-carotid injection have either failed to deliver or proved too toxic. Slow release preparations of BCNU (Gliadel) inserted directly into the surgical cavity has shown a modest increase in survival in some patients with GBM.

TARGETED THERAPY

Greater understanding of the changes in gene expression and their effects on patients with brain tumours has allowed the development of

drugs targeted specifically against these aberrant areas of the cell cycle. Agents directed against epidermal growth factor (EGF), platelet derived growth factor (PDGF) and angiogenetic factors (VEGF) have all shown some activity. Others are now in trial either as single agents or in combination with other targeted agents or conventional therapy. To date only Bevacizumab (Avastin) has been granted a licence in the US.

TUMOURS OF THE CEREBRAL HEMISPHERES – INTRINSIC

Intrinsic tumours arise within the brain substance.

ASTROCYTOMA (and glioblastoma multiforme) Astrocytomas may occur in any age group, but are commonest between 40 and 60 years.

Male:female = 2:1

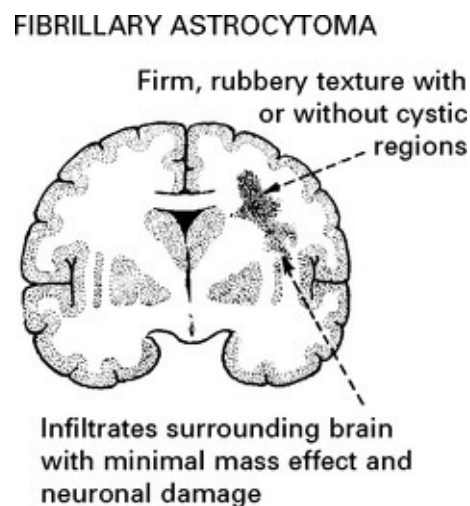
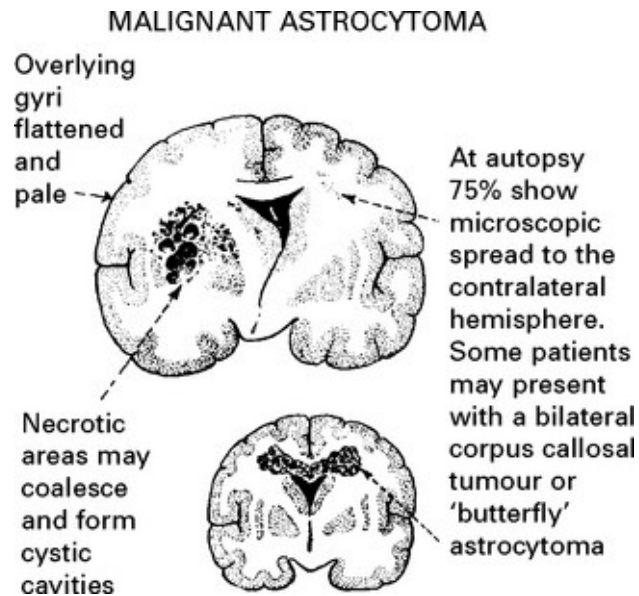
Primary sites: Found in equal incidence throughout the frontal, temporal, parietal and thalamic regions, but less often in the occipital lobe. Microscopic classification defines 4 grades (WHO I–IV), but this is of limited accuracy and gene array technology may play a future role. A practical description divides tumours into either ‘malignant’ or ‘low grade’.

Anaplastic astrocytoma/glioblastoma multiforme

Anaplastic astrocytomas (grade III) and glioblastoma multiforme (grade IV) constitute up to 20% of all primary intracranial tumours. Glioblastoma occurs 4x more commonly than anaplastic astrocytoma. Median age at diagnosis is 64 years and 45 years respectively. These tumours widely infiltrate adjacent brain; growth is rapid. At autopsy, histology often reveals spread to multiple distant sites.

Genetic analysis differentiates ‘primary’ glioblastoma arising de novo

(e.g. amplification of EGFR gene, loss of p16, mutation of PTEN and loss of heterozygosity of 10q), from a 'secondary' glioblastoma where dedifferentiation has occurred from a lower grade tumour (loss of p53, overexpression of PDGFR, loss of heterozygosity of 10q and abnormalities in the p16 and Rb pathways).



'Low grade' astrocytoma

Grade I/II astrocytomas make up 5% of all primary intracranial tumours

in adults. The more frequent grade II tumours occur on average around 35 years. They are diffuse and slowly growing, and composed of well differentiated astrocytic cells subdivided into fibrillary, protoplasmic and gemistocytic types. Up to 90% show loss of the p53 gene. Although benign, these tumours widely infiltrate surrounding brain and lack a definitive edge or capsule.

The pilocytic (grade I) astrocytoma occurs in children and young adults in the hypothalamic region, the optic nerve in association with NF1 ([page 561](#)) and in the cerebellum and brain stem ([pages 331, 332](#)). They grow very slowly, can often stabilise and even regress. Even partial resection can result in a cure.

CLINICAL FEATURES

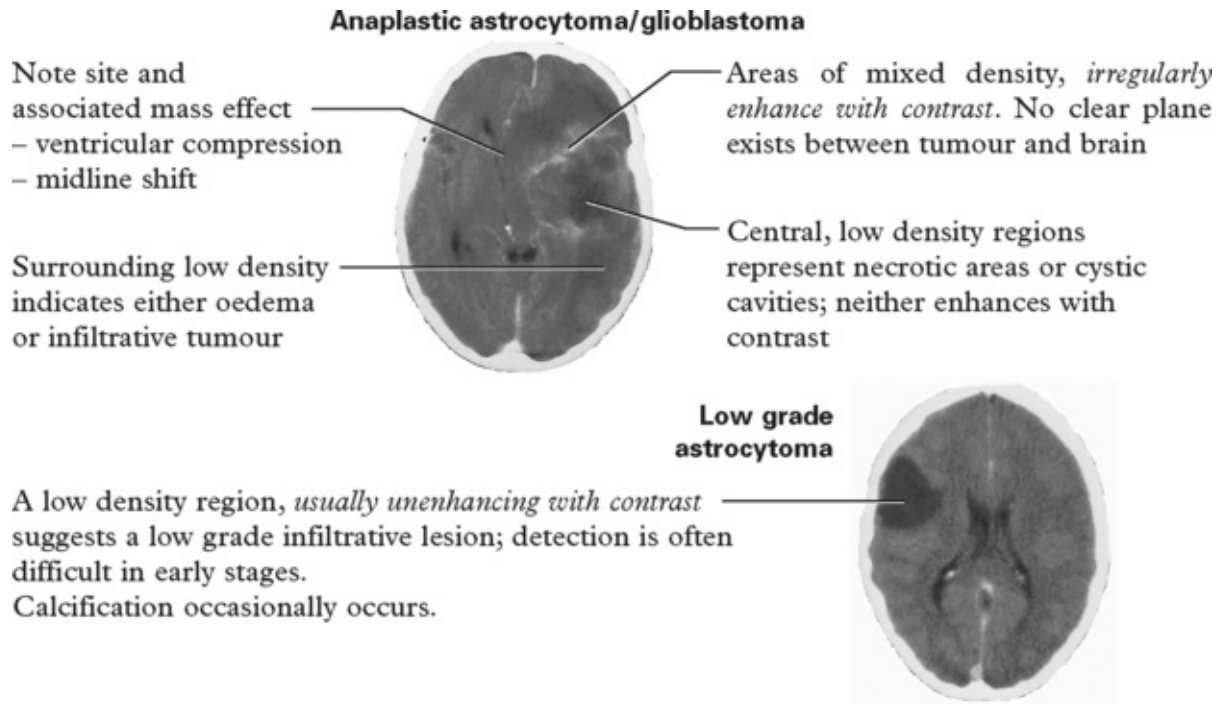
Astrocytomas may present with:

- epilepsy – more common with low grade tumours
- signs and symptoms of focal brain damage – dysphasia, hemiparesis, personality change
- signs and symptoms of raised intracranial pressure – headache, vomiting, depression of conscious level.

Symptoms usually develop gradually, progressing over several weeks, months or years, the rate depending on the degree of malignancy. Sudden deterioration suggests haemorrhage into a necrotic area. In a patient with long standing epilepsy, the rapid development of further symptoms may result from malignant change within a previously 'low grade' lesion.

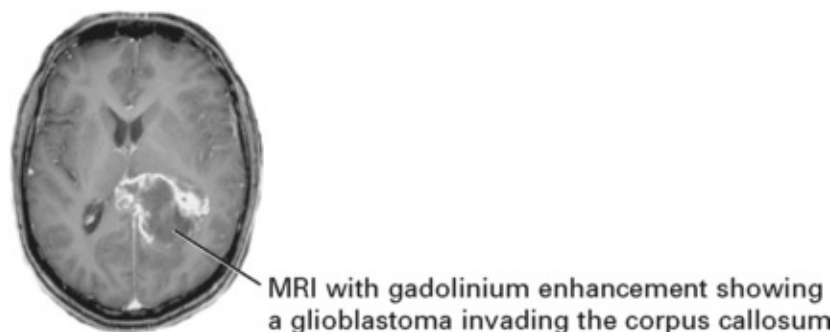
INVESTIGATIONS

CT scan: appearances vary considerably; in general, malignant and low grade lesions show different characteristics:



MRI: with and without gadolinium identifies the tumour location, size and degree of surrounding oedema more clearly. MRI is a more sensitive investigation for detecting low grade astrocytomas.

The *MR spectroscopy* profile ([page 43](#)) may suggest a pathological diagnosis.



MANAGEMENT

The management of glial tumours varies depending on a number of factors –

- the lesion site
- the degree of malignancy
- the presence or absence of a raised ICP
- the degree of disability and the effect of steroid therapy
- the suspected nature of the tumour on imaging
- the patient's age
- the patient's wishes

TREATMENT OPTIONS

Steroid therapy: For patients presenting with symptoms of raised intracranial pressure and/or focal neurological signs, a loading dose of dexamethasone 12 mg i.v. followed by 4mg q.i.d., by injection or orally, reduces surrounding oedema and leads to rapid improvement. Steroid treatment is an essential prerequisite to operation. Its introduction has significantly reduced the perioperative mortality. After several days, a gradual reduction in dosage avoids side effects.

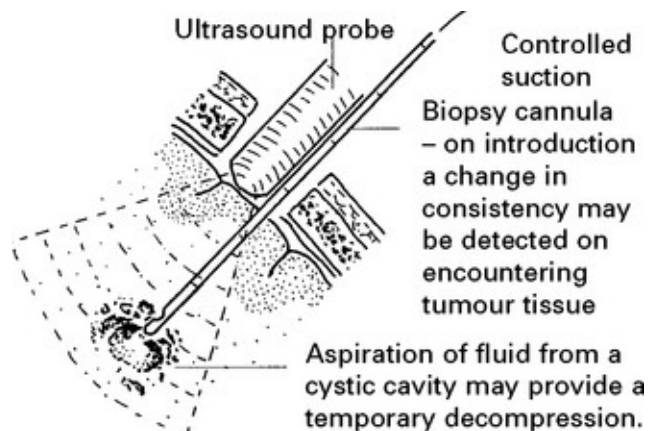
Biopsy: Imaging is insufficient to conclusively establish the diagnosis. If not proceeding to an open operation, failure to confirm the nature of the lesion risks omitting treatment in benign conditions such as abscess, tuberculoma or sarcoidosis. Identification of tumour type and grade

gives a prognostic guide and aids further management.

METHODS:

Framed or frameless stereotactic methods (see [page 386](#)) – permit accurate placement of a fine cannula at a predetermined site selected on CT scan or MRI. Stereotactic guidance is essential for small and/or deep inaccessible lesions (e.g. hypothalamus) and enables biopsy of specific regions in larger tumours e.g. enhancing areas on the MRI or CT scan. Prior selection of the needle path avoids vessels and important structures, thus minimising the risks. Since the degree of malignancy varies from region to region within a single lesion, several samples are taken from different sites to increase accuracy. If findings vary, then the region of greatest malignancy dictates the tumour grade. These techniques are now frequently used, even for more accessible lesions, due to the low mortality and morbidity. Provided patients receive preoperative steroid cover the risks are small, but occasionally biopsy produces or increases a focal deficit or causes a fatal haemorrhage.

Ultrasound guided – a brain cannula inserted into the abnormal region permits aspiration of a small quantity of tissue for immediate (smear and frozen section) and later (paraffin section) examination.



Tumour resection: Often combined with *neuronavigation* (frameless stereotaxy) or ideally *real time CT, MRI* or *ultrasound* if available, to aid tumour localisation and determine the extent of tumour resection as the operation progresses (see [page 313](#)).

Through a craniotomy, the surgeon performs an ‘open’ biopsy under direct vision, or resects as much tumour tissue as is safely feasible. The difficulty lies in the absence of a plane of cleavage between tumour tissue and brain. Neuronavigation can identify the boundaries seen on CT scan or MRI but this is limited by the resolution of the imaging. When adjacent to eloquent regions, the surgeon can merge fMRI or tractography to the neuronavigation image to help guide the resection, but reliability is limited and is still the subject of research. Alternatively performing the procedure in an awake patient and observing the direct effect of electrical stimulation may minimise the risk of causing an irreversible neurological deficit (see [page 313](#)). Large resections are most safely performed in the frontal, occipital or non-dominant temporal lobes. Most believe that the greater the reduction of the tumour mass, *i.e.* the greater the cytoreduction, the greater the effect of adjuvant therapy. A multicentre randomised study has shown that administration of an agent (5-ALA) orally prior to surgery causes tumour tissue to fluoresce during the operative procedure and improves the extent of resection without increasing the risk of deficit.

Radiotherapy: Most effective in rapidly growing tumours – grade III and IV. Radiotherapy extends survival, but does not cure. Studies show a dose–effect relationship – the greater the dose to the tumour area, the longer the survival. Usually up to 60 Gy is delivered in fractionated doses (see [page 314](#)). Methods of ‘conformal’ therapy and ‘interstitial

radiotherapy' (see [page 314](#)) aim to achieve this.

Chemotherapy: Concomitant Temozolomide based chemoradiotherapy is the standard of care following maximal safe resection in patients with glioblastoma of good performance status (WHO 0,1 = can do anything except heavy physical work). This improves survival from 10% to 26% at 2 years compared to radiation alone. The value of combined therapy in anaplastic astrocytoma and oligodendroglioma is under investigation. Chemotherapy, including Gliadel, is also used to palliate patients at recurrence. Bevacizumab is increasingly used as second or third line treatment in the US and some parts of Europe but has not been approved in the UK.

Treatment selection and prognosis

Since treatment cannot cure, the clinician must aim to produce maximal benefit to the patient with minimal burden, taking quality of survival into account as well as the duration.

Malignant astrocytoma/glioblastoma multiforme: Modern techniques have extended survival, but these tumours still carry a grave prognosis. Complete removal is impossible; even the formidable 'hemispherectomy' fails due to interhemispheric spread.

Median survival (months)	
Burr hole biopsy	3–4
Tumour resection	6
Burr hole biopsy + radiotherapy	6–8
Tumour resection + radiotherapy	12–14
Tumour resection + radiotherapy + concomitant chemotherapy	14–18

In patients undergoing surgery, extensive tumour resection extends average survival by only 2 months, but when combined with radiotherapy and concomitant chemotherapy gains a further 12 months. With this treatment, 25% of patients survive 2 years; for patients with silencing of the MGMT gene by promoter methylation, over 40% survive beyond 2 years.

Management policies vary widely, but in general, *maximal tumour resection with radiotherapy and chemotherapy* is considered in most patients except:

- patients over 70 years of age (older patients tolerate radiotherapy less well)
- patients with extensive, deep lesions, *e.g.* involving basal ganglia or corpus callosum
- patients with severe disability, unresponsive to steroid therapy

In such patients, a *diagnostic biopsy* may be the only appropriate treatment.

Tumour recurrence may warrant a further resective procedure, perhaps combined with carmustine wafer implantation or other *chemotherapy*, particularly in younger patients who responded well to the initial treatment.

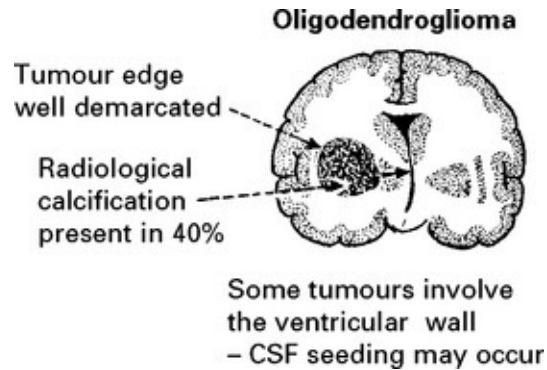
Low grade astrocytoma: A poorly defined region of low density on CT/MR scan without contrast enhancement suggests a low grade tumour (grade I or II) with a better prognosis. As with malignant astrocytomas, since these tumours infiltrate surrounding brain, they tumours cannot be ‘eradicated’ by surgical resection. The dilemma in such patients who

often present with epilepsy and no other symptoms, is whether to proceed with *radical surgery* or to *wait and watch*. The argument for intervention is that at some point in the future the tumour will become malignant; many therefore opt for resecting as much tumour as is safely possible at an early stage in the hope that this defers malignant change. There is however no evidence that active intervention with operation and/or radiotherapy or chemotherapy changes outcome. The issues involved should be discussed with the patient, but if a conservative approach is adopted, the surgeon should advise intervention if subsequent CT or MR scanning shows definitive tumour progression (expansion or contrast enhancement), or if clinical symptoms supervene. A positive Thallium SPECT scan ([page 49](#)) would also indicate the need for action. If proceeding to surgery, the operative techniques used to avoid damaging eloquent regions apply ([page 313](#)).

About 50–60% of patients with grade II astrocytomas survive 5 years; about 40% survive 10 years. Of those patients with pilocytic (grade I) astrocytomas, 80% survive 20 years.

OLIGODENDROGLIOMA

Oligodendrogliomas are far less common than astrocytomas. They occur in a slightly younger age group – 30–50 years, and usually involve the frontal lobes. Occasionally involvement of the ventricular wall results in CSF seeding. Calcification occurs in 40%.



In contrast to astrocytomas, the tumour margin often appears well defined. Both low grade and anaplastic forms exist. Genetic analysis of anaplastic oligodendrogliomas has revealed that almost 80% have 1p and 19q allelic losses (i.e. loss of heterozygosity) and these patients respond well to chemotherapy.

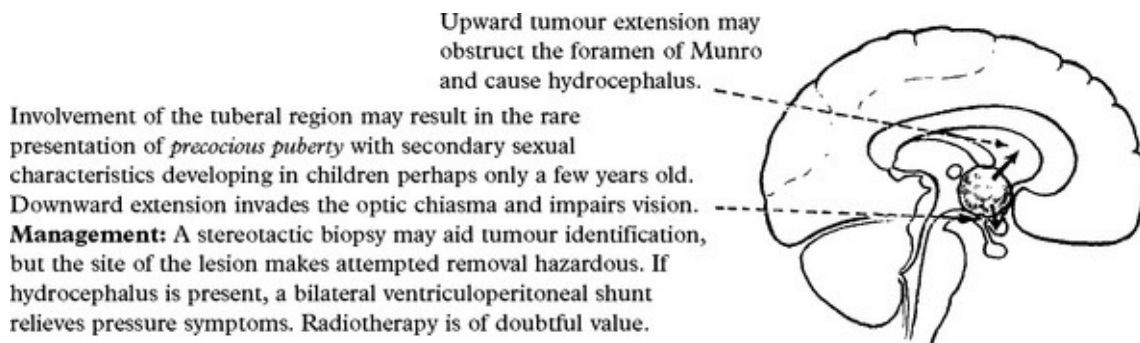
Management and prognosis: when imaging suggests a *low grade tumour*, the approach is similar to astrocytomas with the option of delaying treatment until symptoms appear. Patients can expect to survive for 12–16 years. For patients with *anaplastic oligodendrogliomas*, resection followed by chemotherapy is combined with either immediate or delayed radiotherapy. Ideally treatment should depend on the patient's genetic profile. Those patients with loss of 1p and 19q alleles respond well to chemotherapy and survive over 10 years. The 27% with a genetic profile similar to primary glioblastoma ([page 316](#)) seldom respond to chemotherapy and survive on average about 16 months.

Mixed oligoastrocytoma: those with a mixed form of astrocytoma/oligodendroglioma have a prognosis lying between that for each type.

HYPOTHALAMIC ASTROCYTOMA

Hypothalamic tumours usually occur in children; they are usually

astrocytomas of the pilocytic (juvenile) type. The clinical effect of hypothalamic damage takes different forms. Initially the child *fails to thrive* and becomes *emaciated*. Signs of *panhypopituitarism* may develop. Eventually an anabolic phase results in *obesity* accompanied by *diabetes insipidus* and *delayed puberty*. Disturbance of affect and of sleep–wake rhythms may occur.



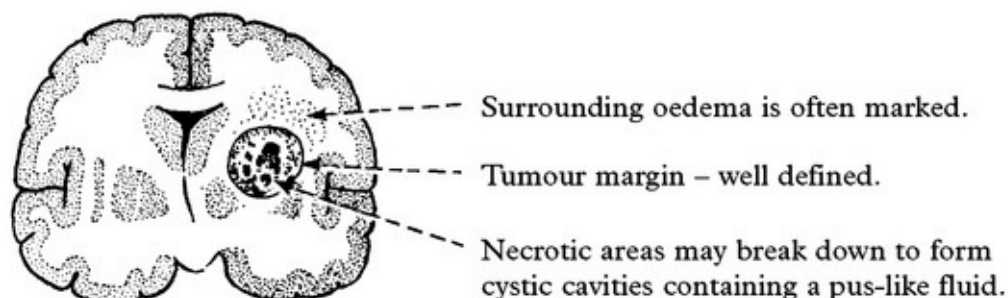
METASTATIC TUMOURS

	<i>Common primary sites</i>
Any malignant tumour may metastasise to the brain. Malignant melanomas show the highest frequency (of those with metastasis, 66% are in the brain); this contrasts with tumours of the cervix and uterus where < 3% develop intracranial metastasis. The most commonly encountered metastatic intracranial tumours arise from the bronchus and the breast; of patients with carcinomas at these sites, 25% develop intracranial metastasis. In over 50% of patients, metastases are multiple.	– bronchus – breast – kidney – thyroid – stomach – prostate – testis

Spread is usually haematogenous to the grey/white matter junction. Occasionally a metastasis to the skull vault may result in a nodule or plaque forming over the dural surface from direct spread.

Intracranial sites $\left\{ \begin{array}{l} \frac{3}{4} \text{ cerebral hemispheres} \\ \frac{1}{4} \text{ cerebellum} \\ \text{(see page 329)} \end{array} \right\}$

In the cerebral hemispheres, metastases often occur at the grey/white matter interface in middle cerebral artery territory. Involvement of the ventricular wall or encroachment into the basal cisterns may result in tumour cells seeding through the CSF pathways – malignant meningitis.

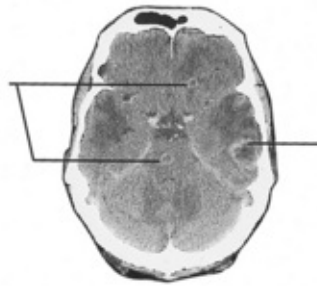


Clinical features

Patients with supratentorial metastatic tumours may present with epilepsy, or with signs and symptoms occurring from focal damage or raised intracranial pressure. Cerebellar metastases are discussed on [page 329](#). Malignant meningitis causes single or multiple cranial nerve palsies and may obstruct CSF drainage (see [page 517](#)). About 10% of diagnosed intracranial metastases are asymptomatic, detected on screening patients with known malignancy.

Investigations

A **CT scan** shows single or multiple well demarcated lesions of variable size. Often an extensive low density area, representing oedema, surrounds the lesion.



Metastatic lesions usually enhance with contrast. A ring-like appearance may resemble an abscess – but the wall is irregular and thickened.

MRI scanning, with and without paramagnetic enhancement, is even more sensitive than CT in detecting small metastatic lesions.

The search for a primary lesion if not already established must include a thorough clinical examination and a chest X-ray. Other investigations including barium studies, intravenous pyelogram (IVP), abdominal CT scans, ultrasound and sputum and urine cytology have questionable value, unless clinically indicated. Whole body PET scanning, if available, is the most sensitive method of detecting the primary lesion.

Management and prognosis:

Corticosteroids (dexamethasone) have a dramatic, rapid effect, producing clinical improvement in most patients.

– *Solitary lesions*: If the tumour lies in an accessible site, complete excision followed by radiotherapy provides good results – survival usually depends on the extent of extracranial disease and its ability to respond to treatment rather than on intracranial recurrences. Stereotactic radiosurgery provides a valuable alternative, particularly for lesions less than 3 cm in diameter and for deep-seated lesions.

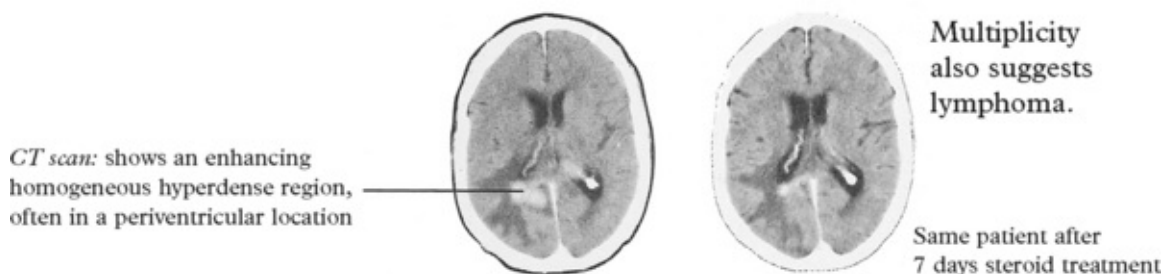
– *Multiple lesions*: Operative removal is seldom practical. Provided no doubt exists about the diagnosis (abscesses or tuberculomata may

resemble metastasis) radiosurgery may be administered to two or even three lesions. For other patients whole brain irradiation may be considered.

Prognosis: Patients < 65 years, with a good performance status and no evidence of systemic metastasis have the best prognosis. In the absence of evidence of systemic cancer, the median survival period approaches 2 years. In those with systemic disease, results are less good with a median survival of 8 months.

PRIMARY CNS LYMPHOMA (PCNSL) (syn. HIGH GRADE NON HODGKIN'S B-CELL LYMPHOMA) Single or multiple lymphomas usually lie deep within the basal ganglia or in the periventricular region. Some are discrete lesions, others extensively invade surrounding brain. Histology shows sleeves of primitive reticulum cells extending outwards from the blood vessels. The incidence is significantly increased in AIDS and in immunocompromised patients.

CSF examination is important; 30% of patients with PCNSL show positive cytology. A positive Epstein-Barr test within tumour cells in CSF is diagnostic of AIDS lymphoma.



Management: In AIDS patients, CT and MRI finding of PCNSL appear similar to toxoplasmosis; antiprotozoal therapy should be tried first. Failure to respond indicates the need for *biopsy*. *Steroids* can cause

dramatic shrinkage, and the imaging should be repeated if any delay occurs prior to biopsy. *Radiotherapy* also has dramatic effects, but with this treatment alone, the median survival period is only 10–12 months. Methotrexate based *chemotherapy* (in patients with a normal immune system) can increase median survival to up to 44 months. Some advocate delaying radiotherapy treatment until a recurrence occurs. AIDS patients, who receive radiotherapy, have a median survival of about 4 months, but chemotherapy can improve this in selected patients.

GANGLIOGLIOMA

This a rare tumour occurring in the younger age group (< 30 years), composed of abnormal neuronal growth mixed with a glial component. The proportion of each component varies from patient to patient. Growth is slow and malignant change uncommon; when this occurs it probably develops in the glial component.

Management follows that of low grade astrocytomas.

NEUROBLASTOMA

Rarely occurs intracranially in children < 10 years. Highly cellular, malignant lesion composed of small round cells, some showing neuronal differentiation.

TUMOURS OF THE CEREBRAL HEMISPHERES – EXTRINSIC

Extrinsic tumours arise outwith the brain substance.

MENINGIOMA

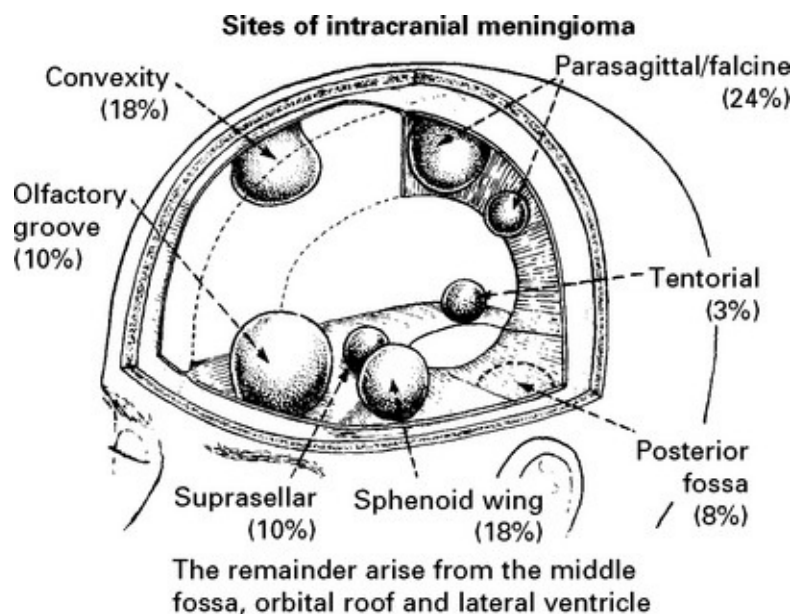
Meningiomas constitute about one-quarter of all primary intracranial tumours. They are slow growing and arise from the arachnoid

granulations. These lie in greatest concentration around the venous sinuses, but they also occur in relation to surface tributary veins. Meningiomas may therefore develop at any meningeal site. Occasionally they are multiple.

Meningiomas present primarily in the 40–60 age group and have a slight female preponderance. They are principally benign tumours, although 1–3% show malignant change.

Pathology

Various histological types are described – syncytial, transitional, fibroblastic and angioblastic; different types may coexist within the same tumour. These distinctions serve little clinical value, although it is important to identify the anaplastic (malignant) form, as this indicates the likelihood of rapid growth and a high rate of recurrence following removal.



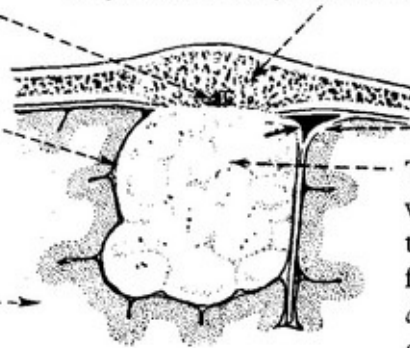
Macroscopic appearance

The dural origin usually incorporates the main arterial supply.

The tumour surface, although often lobulated, is well demarcated from the surrounding brain and attached only by small bridging vessels.

Marked oedema often develops in the surrounding brain.

A reactive *hyperostosis* develops in adjacent bone, forming a swelling on the inner table. Hyperostosis affecting the outer table may produce a palpable lump. Tumour tissue may infiltrate adjacent bone.



Parasagittal tumours may invade and obstruct the sagittal sinus.

Tumour texture and vascularity varies considerably from patient to patient – some are firm and fibrous, others soft. *Calcified deposits* (psammoma bodies) are often found.

En-plaque meningioma: In some patients, rather than developing a spherical form, the meningioma spreads ‘en-plaque’ over the dural surface. This type often arises from the outer aspect of the sphenoid wing.

MENINGIOMA Clinical features:

Approximately a quarter of patients with meningioma present with epilepsy – often with a focal component. In the remainder, the onset is insidious with pressure effects (headache, vomiting, papilloedema) often developing before focal neurological signs become evident.

Notable characteristic features occur, dependent on the tumour site – PARASAGITTAL/PARAFALCINE tumours lying near the vertex affect the ‘foot’ and ‘leg’ area of the motor or sensory strip. *Partial seizures* or a ‘pyramidal’ weakness may develop in the leg (i.e. primarily affecting foot dorsiflexion, then knee and hip flexion). Extension of the lesion through the falx can produce *bilateral leg weakness*. Posteriorly situated parasagittal tumours may present with a *homonymous hemianopia*. Tumours arising anteriorly may grow to extensive proportions before

causing focal signs; eventually minor *impairment of memory, intellect and personality* may progress to a *profound dementia*.

INNER SPHENOIDAL WING tumours may compress the optic nerve and produce *visual impairment*. Examination may reveal a central *scotoma* or other *field defect* with *optic atrophy*.

N.B. The FOSTER KENNEDY syndrome denotes a tumour causing optic atrophy in one fundus from direct pressure and papilloedema in the other due to increased intracranial pressure.

Involvement of the cavernous sinus or the superior orbital fissure may produce *ptosis* and *impaired eye movements* (III, IV and VI nerve palsies) or *facial pain and anaesthesia* (V₁ nerve damage) – see diagram on [page 153](#). *Proptosis* occasionally results from venous obstruction or tumour extension into the orbit.

OLFACTORY GROOVE tumours destroy the olfactory bulb or tract causing unilateral followed by bilateral *anosmia*. Often unilateral loss passes unnoticed by the patient; with tumour expansion, dementia may gradually ensue.

SUPRASELLAR tumours – see [page 348](#).

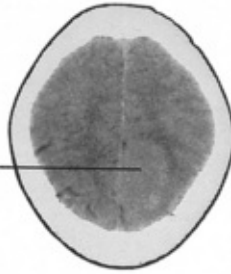
ASYMPTOMATIC TUMOURS: With the increased availability of imaging, small meningiomas are frequently detected incidentally. Conservative management of such patients has shown that over a 5 year period about 40% show expansion and one in six develop symptoms.

Investigations:

CT SCAN

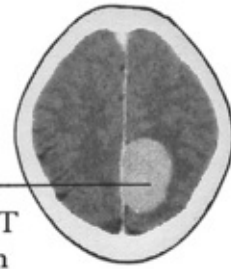
Before i.v. contrast

Meningioma – well circumscribed lesions of a density usually greater than, or equal to brain with a surrounding area of low attenuation (oedema). Calcification may be evident.



After i.v. contrast

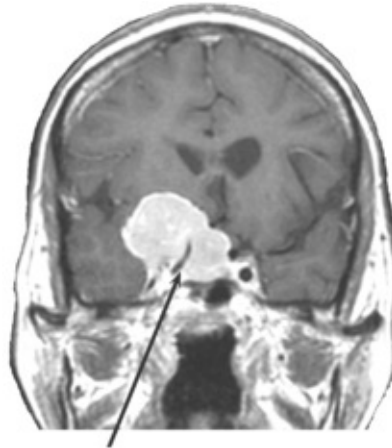
A dense, usually homogeneous enhancement occurs after contrast injection. N.B. Unenhanced CT is more sensitive than unenhanced MRI in detecting meningiomas.



MRI: On T1 weighted images most meningiomas are isointense with brain, but after gadolinium injection, they diffusely and strikingly enhance. T2 weighted images give useful preoperative information by identifying major vessels and showing their relationship with the tumour.

ANGIOGRAPHY: Characteristically shows a highly vascular lesion with a typical tumour 'blush', but with the availability of CT angiography, its main value is in selective catheterisation and embolisation of external carotid feeding vessels to reduce tumour vascularity and diminish operative risks from excessive haemorrhage.

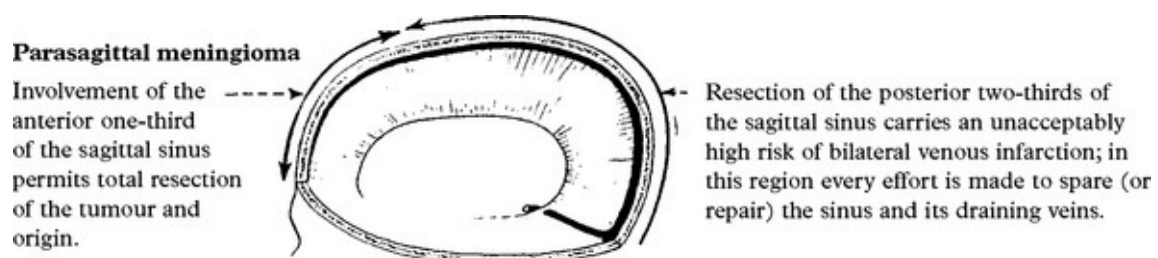
3-D CT ANGIOGRAPHY: can show the relationship of surrounding blood vessels to the tumour.



Coronal MRI with gadolinium enhancement showing meningioma invading cavernous sinus and sella turcica and encasing the carotid artery

Management

Management aims at complete removal of both the tumour and its origin without damaging adjacent brain; but this depends on the tumour site and its nature. Even with 'convexity' tumours, where complete excision of the dural origin is possible, overlooking a small fragment of tumour may result in recurrence. This is more likely with malignant meningiomas where the plane of cleavage is often obscured.



When the tumour is asymptomatic or when the patient's age or the tumour site prevents operation or allows only a limited removal, a *conservative approach* may be more appropriate, only intervening if the tumour progresses or causes disabling symptoms. Alternatively

stereotactic radiosurgery could be considered for small tumours or for residual fragments. Benefits of standard radiotherapy are uncertain unless histology reveals evidence of malignant change.

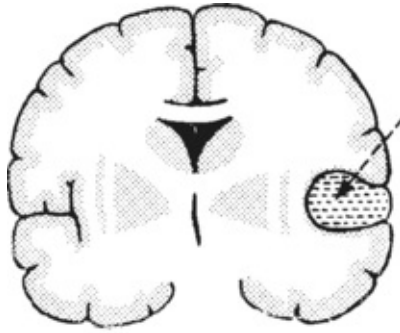
Operative results: with modern techniques, operative mortality has fallen to less than 3%, but this varies depending on the size and position of the tumour. Although *in vitro* studies have demonstrated numerous hormonal receptors (e.g. progesterone and oestrogen) in meningioma tissue, clinical studies of hormonal therapy have failed to show any benefit.

Tumour recurrence: depends predominantly on the completeness of removal and on the duration of follow-up. With 'total' resection, about 20% recur after 10 years. With subtotal resection over 50% require a further operation within 10 years.

HAEMANGIOPERICYTOMA

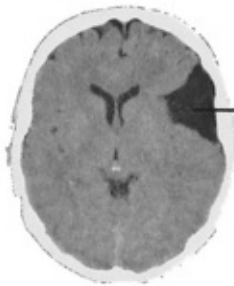
A tumour arising from the meninges, but of uncertain cell of origin. It presents with similar clinical features and CT/MRI appearance to meningiomas, but is fifty times less common. Angiography may show a more prominent vascular supply. Calcification does not occur. Haemangiopericytomas tend to invade adjacent bone and to recur even after apparent complete surgical removal. Postoperative radiotherapy should delay recurrence.

ARACHNOID CYSTS



These cystic collections of CSF-like fluid of developmental origin occur in about 0.3% of the population and are usually asymptomatic. About $\frac{3}{4}$ lie above the tentorium; of these $\frac{2}{3}$ occur in the Sylvian fissure, then often associated with temporal lobe hypoplasia.

Arachnoid cysts may gradually increase in size, either due to CSF being driven in through a valve-like opening or by active secretion of fluid from the cyst wall. Occasionally patients present with mass effects, or in children with asymmetric cranial enlargement, macrocephaly and/or psychomotor retardation. More often they are discovered incidentally on CT or MRI.



CT scan: shows a low density (CSF density) well demarcated lesion, occasionally producing expansion of the overlying bone.

Treatment: These are common findings and in the vast majority, no treatment is indicated. Rarely patients present with mass effect and require *marsupialisation* (via a craniotomy) or *cystoperitoneal shunting*. Some believe that prophylactic treatment in young children aids normal brain development.

EPIDERMOID/DERMOID CYSTS

These cysts, more commonly found in the posterior fossa ([page 337](#)), occasionally develop in the Sylvian or interhemispheric fissure. They are either of congenital or acquired origin due to implantation and sequestration of ectoderm. They may present with epilepsy, features of raised intracranial pressure or with focal neurological signs. Rupture into the subarachnoid space causes a chemical meningitis.

On CT scan, the extreme low attenuation of the cyst contents is characteristic. Symptoms may necessitate operative evacuation of the

cyst contents. Complete removal of the cyst wall is difficult and reaccumulation may occur.

Lipomas

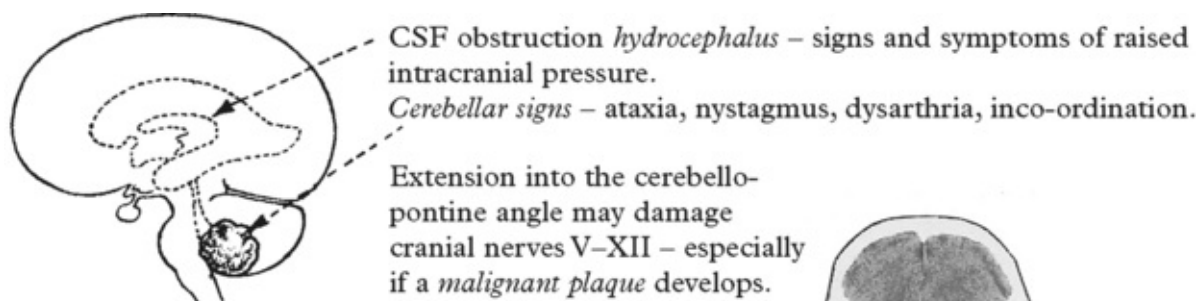
Rarely occur intracranially. They are usually found incidentally on imaging or at autopsy and are often associated with other developmental anomalies such as agenesis of the corpus callosum. They are located in midline structures *e.g.* corpus callosum, dorsal midbrain and cerebellar vermis. They require no treatment.

TUMOURS OF THE POSTERIOR FOSSA – INTRINSIC

CEREBELLAR METASTASIS

In adults, metastasis is the commonest tumour of the cerebellar hemisphere. Primary tumour sites match those of supratentorial lesions ([page 322](#)).

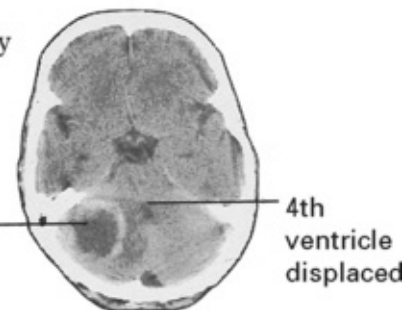
Clinical features: may present acutely or progress over several months.



Investigations

CT scan shows a well-defined solid or cystic lesion lying within the cerebellar hemisphere and enhancing irregularly with contrast.

Obstructive hydrocephalus is often evident on higher scan cuts. As with cerebral metastases *MRI* is more sensitive in detecting small lesions.



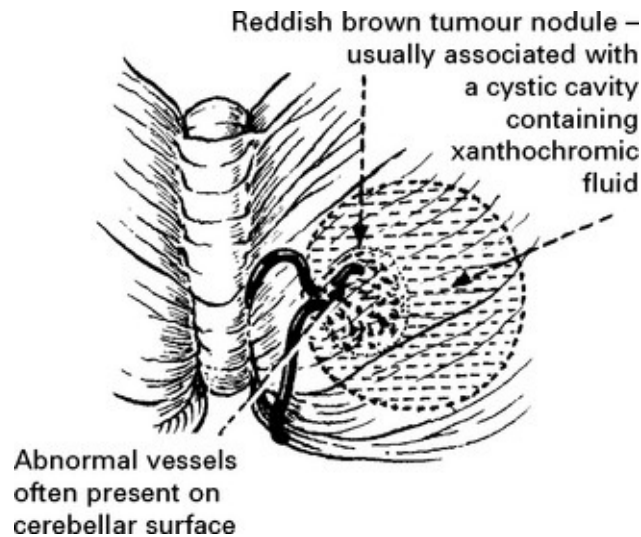
Management

Operative removal of a single metastasis through a suboccipital craniectomy is worthwhile, provided the patient has a reasonable prognosis from the primary tumour. Risks are small – extensive cerebellar hemisphere resection (on one side) seldom produces any significant permanent deficit. A course of *radiotherapy* can follow operation if resection is incomplete. Radiosurgery provides a possible alternative to surgical resection. Persistence of obstructive hydrocephalus requires a ventriculoperitoneal shunt.

HAEMANGIOBLASTOMA

This benign tumour of vascular origin occurs primarily in the middle-aged; it is slightly more prevalent in males and is the commonest primary cerebellar tumour of adults. In some patients, haemangioblastomas occur at other sites, *e.g.* the spinal cord and retina and may be associated with other pathologies *e.g.* polycythaemia and cysts in the pancreas and kidneys – *Von Hippel-Lindau disease* ([page 563](#)).

The tumour is usually highly vascular. In 70% there is an associated cyst, the lining of which does not contain tumour.

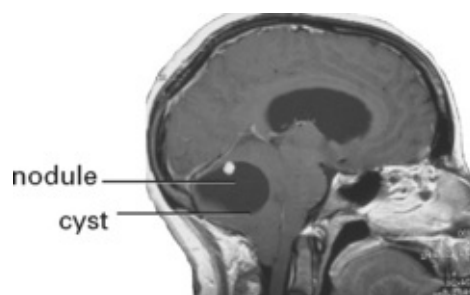


Clinical features

Cerebellar signs and symptoms or the effects of CSF obstruction usually develop insidiously. Occasionally subarachnoid haemorrhage occurs. In female patients, symptoms often appear during pregnancy. Polycythaemia due to increased erythropoietin production is common.

Investigations

CT scan/MRI shows either a strongly enhancing solid tumour in the cerebellum or a tumour nodule lying in the wall of a well defined cystic region. Occasionally, multiple lesions are evident. Enhancing vessels on CT or tortuous flow voids on MRI reflect the high vascularity.

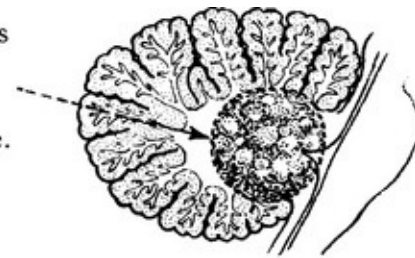


Management

In some patients *operative removal* of the tumour nodule is straightforward, but recurrences (or further tumours at other sites, *e.g.* spine) develop in 20%. Patients with highly vascular solid tumours can present a formidable surgical challenge, particularly if they involve the medulla. Preoperative embolisation may greatly reduce the surgical risks.

MEDULLOBLASTOMA

Medulloblastomas are primitive neuroectodermal tumours (PNETs), which occur predominantly in childhood, peaking at 3–4 yrs and again at 8–9 yrs. They arise in the cerebellar vermis and usually extend into the 4th ventricle. They are highly malignant. In 30%, CSF seeding occurs to the lateral ventricles or the spinal theca. The origin is uncertain but they appear to develop from primitive embryonic cells. Genetic analysis has shown loss of 17p and often duplication of 17q.



Clinical features

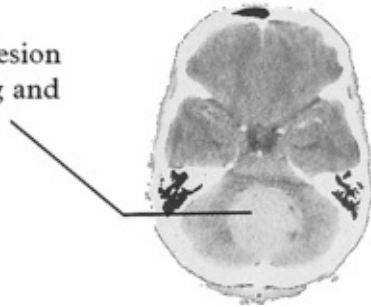
Destruction of the cerebellar vermis causes truncal and gait ataxia often developing over a few weeks.

Alternatively, the patient presents with signs and symptoms of raised intracranial pressure due to blockage of CSF drainage. In the very young, failure to recognise these features has resulted in permanent visual loss from severe papilloedema.

Investigations

CT scan shows an isodense midline lesion in the cerebellar vermis, compressing and displacing the 4th ventricle and enhancing strongly with contrast.

MRI may provide more anatomical detail and more readily detects supratentorial CSF seedlings.



Higher cuts show dilated ventricles, sometimes containing CSF tumour seedlings

Management

Staging is essential due to the high incidence of leptomeningeal spread and bone marrow involvement. Assess this with spinal MRI with gadolinium, CSF analysis and bone marrow examination. CSF obstruction may require urgent relief, preferably by 3rd ventriculostomy.

Operation: The aim is to remove as much tumour as possible (particularly if staging has excluded disseminated disease), without damaging crucial structures in the floor of the 4th ventricle.

Radiotherapy: the most effective postoperative treatment. Whole neural axis irradiation attempts to cover any CSF seeding, but this is unacceptable in children < 3 years due to severe side effects. In this group and in recurrent tumours radiosurgery may help.

Chemotherapy: routinely used, but the extent to which chemotherapy alters the quality or duration of survival is less certain.

Prognosis

Five-year survival ranges from 50–90% depending on the extent of tumour removal, dissemination and age (<3 years poor risk).

CEREBELLAR ASTROCYTOMA

In contrast to astrocytomas of the cerebral hemispheres, cerebellar astrocytomas are usually low grade tumours of the fibrillary or pilocytic

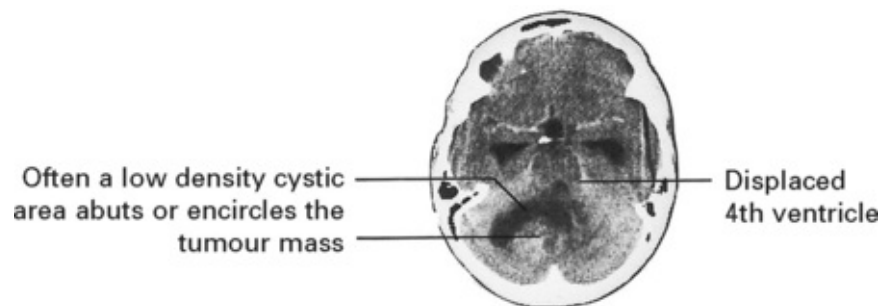
types. They are particularly common in children and carry an excellent prognosis. Occasionally a more diffuse or anaplastic type occurs with a less favourable outcome. They usually lie in the cerebellar hemisphere or vermis but occasionally extend through a peduncle into the brain stem. Many have cystic components.

Clinical features

Cerebellar signs and symptoms tend to develop gradually over many months; if CSF obstruction occurs, the patient may present acutely with headache, papilloedema and deteriorating conscious level.

Investigations

CT scan – density changes and the degree of contrast enhancement are variable.



MRI – may provide more anatomical definition.

Management

Ideally, complete *operative removal* is attempted provided the brain stem is not involved. With pilocytic tumours, 80% survive 20 years. Even after partial removal ‘cures’ have been reported. Persistent hydrocephalus may require 3rd ventriculostomy or a ventriculoperitoneal shunt.

BRAIN STEM ASTROCYTOMA

Rarely, astrocytomas arise within the brain stem. Most are of the fibrillary or pilocytic types and diffusely expand the pontine region although they can be malignant. They develop mainly in children or young adults.

Clinical features

Cranial nerve palsies and long tract signs gradually develop as the tumour progresses. Eventually conscious level is impaired. More malignant gliomas are associated with a rapidly progressing course, often with signs of raised intracranial pressure.

Investigations

CT scan may show low density within the brain stem, with absence of surrounding cisterns and posterior displacement of the 4th ventricle.



MRI scanning is superior to CT scanning in the detection and evaluation of brain stem astrocytoma.



Management

Operative exploration is seldom indicated. Radiotherapy is often administered, usually after a stereotactic biopsy, with occasional palliation of symptoms and uncertain effect on survival. Chemotherapy is of no value.

Prognosis

At best, the 5-year survival following radiotherapy is 35%, although

some patients may survive for up to 20 years with minimum disability.

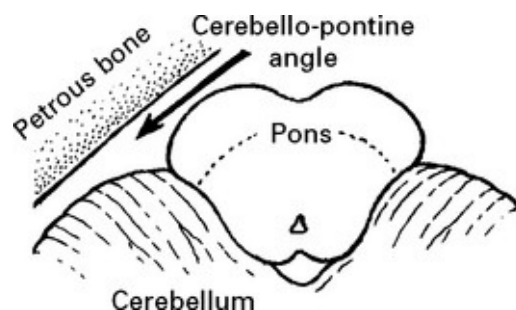
TUMOURS OF THE POSTERIOR FOSSA – EXTRINSIC

VESTIBULAR SCHWANNOMA

Nerve sheath tumours are the commonest infratentorial tumours, constituting 8% of all primary intracranial tumours and 80% of cerebellopontine angle lesions. They usually present in middle age (40–50 years) and occur more frequently in women. Bilateral schwannomas occur in 5% of patients and are characteristic of type 2 neurofibromatosis (NF2) ([page 561](#)).

They are benign, slowly growing tumours which arise primarily from the *vestibular* portion of the VIII cranial nerve and lie in the cerebellopontine angle – a wedge shaped area bounded by the petrous bone, the pons and the cerebellum. Rarely these tumours arise from the V cranial nerve.

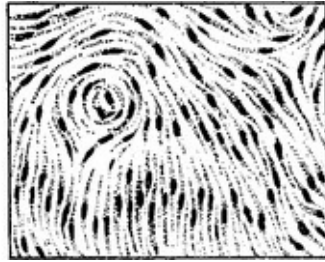
Schwannomas expand at an average rate of 2 mm/year, but about 50% show no growth on serial investigation.



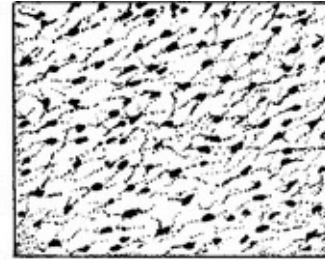
Pathology: The other type of nerve sheath tumour – neurofibroma ([page 304](#)) – does not occur intracranially.

Different histological types exist, often within the same tumour:

Antoni type A – shoals and whorls of tightly packed cells in groups or palisades



Antoni type B – a meshwork of interlinked loosely packed stellate cells.



Clinical features

Patients with acoustic tumours often complain of *occipital pain* on the side of the tumour. In addition:

VIII nerve damage causes a *gradually progressive sensorineural deafness* noted over many months or years. *Vertigo* is rarely troublesome since slow tumour growth readily permits compensation. Similarly *tinnitus* is usually minimal.

V nerve damage can occur with tumours > 2 cm and causes *facial pain, numbness and paraesthesia*. *Depression of the corneal reflex* is an important early sign.

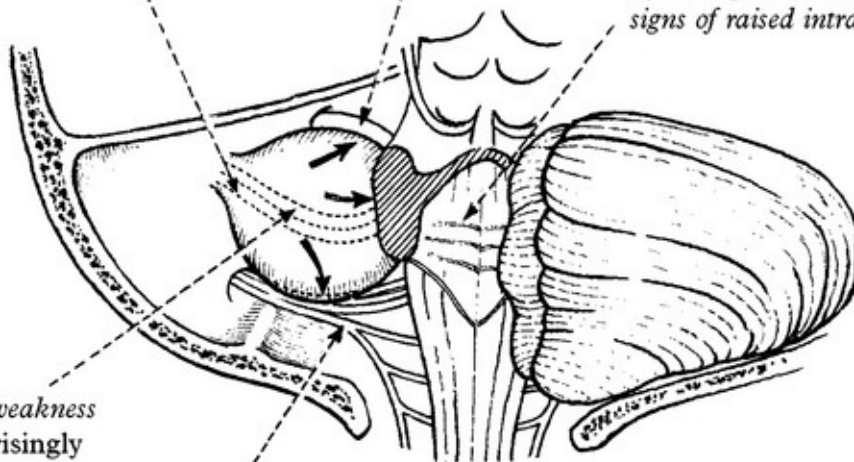
Compression of the aqueduct and the 4th ventricle may result in *hydrocephalus* with *symptoms and signs of raised intracranial pressure*.

Facial weakness is surprisingly *uncommon* despite marked VII nerve compression.

IX, X and XI nerve damage seldom occurs but occasionally large tumours cause *swallowing difficulty, voice change and palatal weakness*.

Cerebellar and pontine damage – large tumours (> 4 cm) may compress the cerebellum causing *ataxia, ipsilateral inco-ordination and nystagmus*. Pontine damage may produce a *contralateral hemiparesis*.

N.B. Left cerebellar hemisphere removed to expose the divided cerebellar peduncles.



Investigations

Neuro-otological test (see pages 62–63)

- audiometry
- speech audiometry
- stapedial reflex decay

help differentiate deafness due to:

conductive deficit

sensorineural

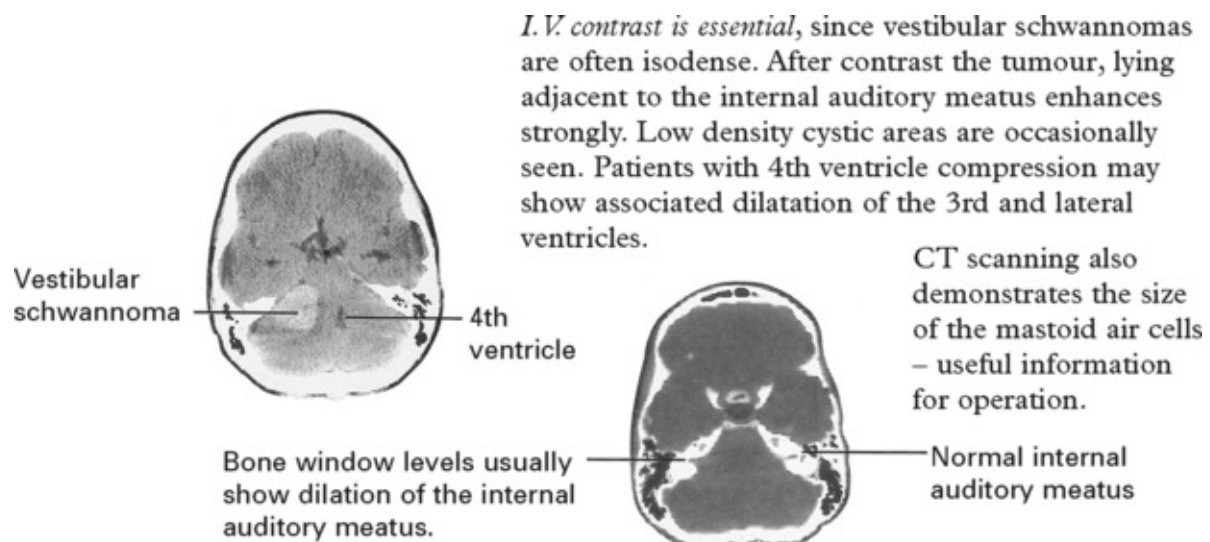
cochlear deficit

retrocochlear deficit

(e.g. vestibular schwannoma)

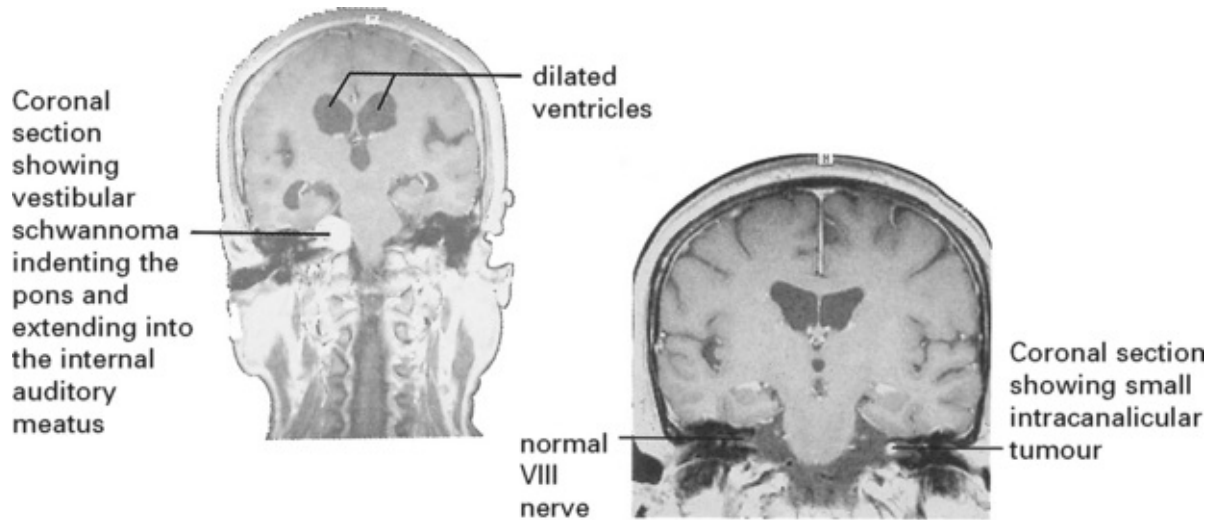
– *brain stem auditory evoked potential (BAEP)* – perhaps the most sensitive of these tests shows a delay of the wave V latency on the affected side.

CT scan



MRI

The investigation of choice, particularly for small intracanalicular tumours. On a T1 weighted image, the lesion enhances strongly after i.v. gadolinium.



MANAGEMENT OPTIONS

Conservative approach

Since about 50% of tumours show no growth on yearly follow-up and since treatment carries risk, a 'wait and watch' policy is a sensible option for small to medium-sized tumours (< 20 mm), particularly in the elderly.

Stereotactic radiosurgery

This single dose technique (see [page 314](#)), initially reserved for elderly patients, is now used more widely for schwannomas up to 3 cm in size. Centres report 'control' of tumour growth in up to 90%, with preservation of hearing in about 75% and facial nerve function in 98%. A 10-year follow-up study suggests that growth control is maintained.

SURGICAL RESECTION

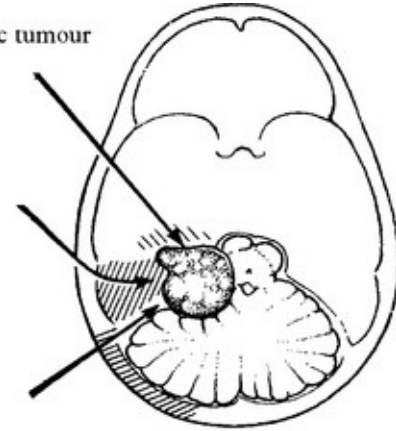
Techniques

MIDDLE FOSSA APPROACH: temporal lobe retraction exposes acoustic tumour and facial nerve from above. The tentorium cerebelli and the superior petrosal sinus are divided if necessary.

TRANSLABYRINTHINE APPROACH: approaching the tumour through the mastoid air cells and the labyrinth, permits early identification of the facial nerve; tumour decompression and removal follows.

SUBOCCIPITAL APPROACH: the cerebellopontine angle is approached from below by removing occipital bone and retracting the cerebellum.

Tumour debulking aids dissection of the tumour capsule from the surrounding structures, including the facial nerve. Drilling away the posterior wall of the internal meatus exposes the tumour and facial nerve lying within the canal.



Results

Outcome relates to tumour size. With a tumour diameter of 5–20 mm, some hearing can be preserved in > 50% and facial nerve function in > 95%. With tumours of > 3 cm, all lose hearing and 25–50% sustain facial nerve damage. When present, incomplete eye closure may require *tarsorrhaphy* to prevent corneal ulceration. When facial nerve palsy persists, *hypoglossal-facial anastomosis* may improve the cosmetic result. Mortality ranges from 1–3% and usually results from damage to important vascular structures (e.g. anterior inferior cerebellar artery), haemorrhage, aspiration pneumonia or pulmonary embolus.

Treatment selection: For tumours < 2 cm in diameter a conservative approach is the most appropriate option. If serial scans show tumour growth or if the tumour is > 2 cm on diagnosis, treatment is required aimed at removal or control of growth, preservation of facial nerve function and retention of useful hearing unless this is already lost. The options of radiosurgery and surgical removal should be discussed with the patient along with the pros and cons (i.e. removal with increased risk or ‘control’ with less risk). For tumours > 3 cm in

diameter only surgical resection is feasible.

TRIGEMINAL SCHWANNOMA

Rarely schwannomas arise from the trigeminal ganglion or nerve root. These lie in the middle fossa or extend into the cerebellopontine angle, compress surrounding structures – cavernous sinus, midbrain and the pons – and erode the apex of the petrous bone.

Clinical features are usually long-standing – facial pain, paraesthesia and numbness. Compression of posterior fossa structures results in nystagmus, ataxia and hemiparesis.

CT scan or MRI with contrast demonstrates an enhancing lesion eroding the petrous apex and extending into the middle and/or posterior fossa.

Management: Operative removal, even if subtotal, should provide long-lasting benefit. The tumour is approached either from above via a subtemporal route across the middle fossa floor, from below via a suboccipital craniectomy, or via a combination of these approaches.

MENINGIOMA

Approximately 8% of all intracranial meningiomas arise in the posterior fossa.

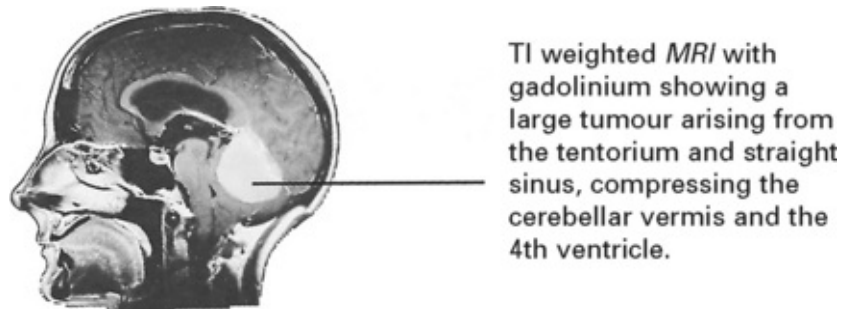
Clinical features

These depend on the exact tumour site. Those arising over the *cerebellar convexity* may not present until the mass obstructs CSF drainage. Meningiomas arising in the *cerebellopontine angle* may involve any cranial nerve from V to XII. A *clivus* meningioma may cause bilateral VI nerve palsies before pontine pressure causes long tract signs.

Tumours growing at the *foramen magnum*, compressing the cervico-medullary junction, produce characteristic effects – pyramidal weakness initially affecting the ipsilateral arm, followed by the ipsilateral leg, spreading to the contralateral limbs with further tumour growth.

Investigations

CT scan with intravenous contrast will identify the tumour site, but *MRI* with gadolinium enhancement shows more anatomical detail.



Management

As with supratentorial meningiomas, treatment aims at complete tumour removal. In the posterior fossa, cranial nerve involvement makes this difficult and exacting; excision of the tumour origin is seldom possible. For some, stereotactic radiosurgery is an alternative method of controlling tumour growth. Radiotherapy or radiosurgery may be considered when residual tumour persists.

EPIDERMOID/DERMOID CYSTS

These rare cysts of embryological origin develop from cells predestined to become either epidermis or dermis. They most commonly arise in the cerebellopontine angle but may also occur around the suprasellar cisterns, in the lateral ventricles and in the Sylvian fissures, often

extending deeply into brain tissue.

Pathology: Depends on cell of origin:

Epidermoid (epidermis) – a thin transparent cyst wall often adheres firmly to surrounding tissues; the contents – keratinised debris and cholesterol crystals – produce a ‘pearly’ white appearance.

Dermoid (dermis) – as above, but thicker walled and, in addition, containing hair follicles and glandular tissue. Midline dermoid cysts lying in the posterior fossa often connect to the skin surface through a bony defect. This presents a potential route for infection.

Clinical features

When lying in the cerebellopontine angle, epidermoid/dermoid cysts often cause *trigeminal neuralgia* (see [page 163](#)). Neurological findings may range from a depressed corneal reflex to multiple cranial nerve palsies. Rupture and release of cholesterol into the subarachnoid space produces a severe and occasionally fatal *chemical meningitis*. The presence of a *suboccipital dimple* combined with an attack of *infective meningitis* should raise the possibility of a posterior fossa dermoid cyst with a cutaneous fistula.

Investigations

CT scan shows a characteristic low density (often 'fat' density) lesion, unchanged after contrast enhancement or showing only slight peripheral enhancement. Calcification may be evident.

T2 weighted *MRI* appears more sensitive than CT in detecting an abnormality, but the hyperintense signal does not differentiate an arachnoid cyst from an epidermoid.



T2 weighted MRI showing lower cranial nerves traversing the lesion.

Treatment

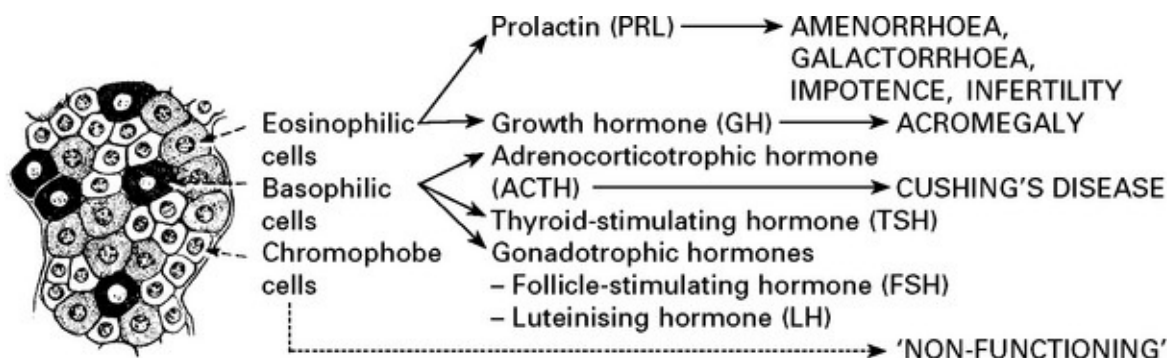
Adherence of the cyst wall to important structures often prevents complete removal, but evacuation of the contents provides symptomatic relief. Aseptic meningitis in the postoperative period requires prompt treatment with steroids. Even when removal is incomplete, recollection of the keratinised debris is uncommon and may take many years.

SELLAR/SUPRASellar TUMOURS – PITUITARY ADENOMA

Tumours of the pituitary gland constitute about 5–10% of intracranial tumours. They arise from the anterior portion of the gland and are usually benign.

'CLASSIC' classification

Previously based on the light microscopic appearance of the tumour cell type.

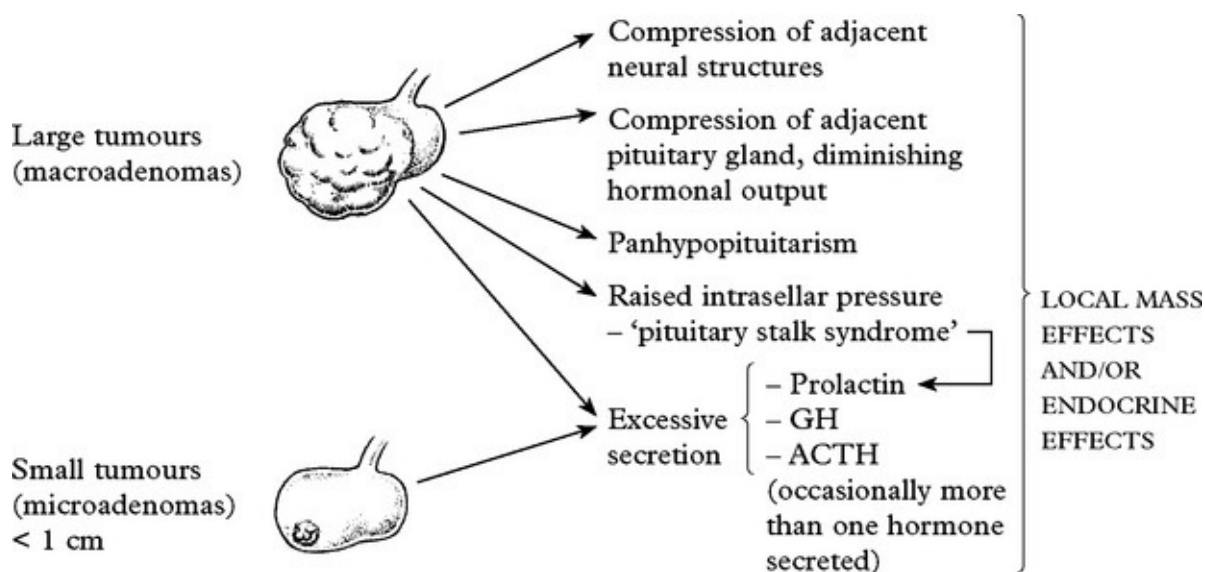


PRESENT classification

Immunohistochemical techniques permit a classification based on the hormone type secreted, but this does not necessarily reflect the active form of the hormone. About half of the 'non-functioning' chromophobe adenomas are shown to secrete prolactin.

	Incidence
– GH secreting tumour	20–25%
– Prolactinoma	25–50%
– ACTH secreting tumour	5–10%
– TSH secreting tumour	} rare
– FSH/LH secreting tumour	
– Inactive	25–40%

CLINICAL PRESENTATION



LOCAL MASS EFFECT

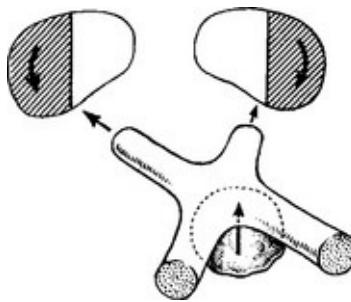
Headache

Occurs in most patients with enlargement of the pituitary fossa. It is not

specific in site or nature.

Visual field defects

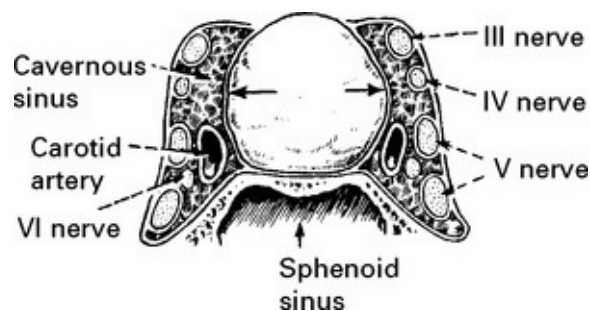
Pressure on the inferior aspect of the optic chiasma usually causes *superior temporal quadrantanopia* initially, with progression to bitemporal hemianopia, but any pattern can occur.



Cavernous sinus compression

In some pituitary tumours, lateral expansion may compress nerves lying within the walls of the cavernous sinus. The III nerve is especially vulnerable.

Rarely vertical expansion obstructs the foramen of Munro causing hydrocephalus and/or hypothalamic compression ([page 346](#)).



ENDOCRINE EFFECT

1. HYPERSECRETION

The clinical syndrome produced is dependent on the hormone secreted.

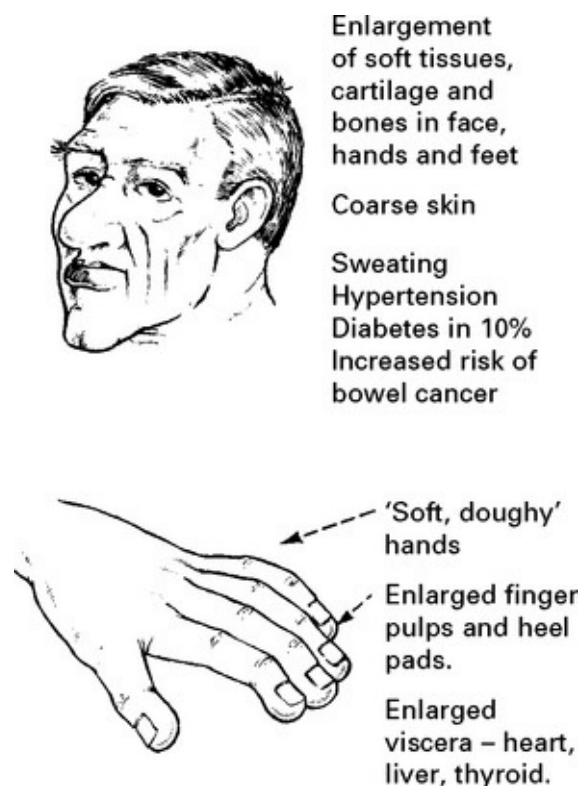
Growth hormone (GH)

Stimulates growth and plays a part in control of protein, fat and carbohydrate metabolism. Excess GH in the adult causes ACROMEGALY.

In childhood, prior to fusion of bone epiphyses, GH excess causes GIGANTISM.

GH levels are usually increased to $> 10 \text{ mU/l}$. Increased serum levels of insulin growth factor-1 enhances the effect of growth hormone on target organs.

Hyperglycaemia normally suppresses GH secretion. GH samples are taken in conjunction with blood glucose during a glucose tolerance test. The lack of GH suppression after glucose administration confirms the presence of a tumour.



Prolactin

This hormone helps promote lactation. Prolactinoma is the commonest type of pituitary tumour. Immunoassay techniques aid early detection.

Female:male ratio – 4:1

This tumour may present with

- INFERTILITY
- AMENORRHOEA
- GALACTORRHOEA

In males, the tumour may present with IMPOTENCE or remain undetected until local pressure effects occur.

In most centres, a serum prolactin of 500 mU/l is considered abnormal, but before assuming the presence of a prolactin secreting tumour, other causes must be excluded.

Causes of hyperprolactinaemia

- Stress
- Pregnancy
- Drugs (phenothiazines, oestrogens)
- Hypothyroidism
- Renal disease
- Pituitary adenoma
- Hypothalamic lesion (e.g. sarcoid, craniopharyngioma) or the pituitary stalk syndrome

- Seizures

Prolactin differs from other anterior pituitary hormones in that it is under tonic *inhibitory* control from the hypothalamus. Hypothalamic lesions or raised intrasellar pressure, compromising hypothalamic–pituitary perfusion (i.e. the '*pituitary stalk syndrome*') produce a rise in serum prolactin, but levels seldom exceed 2000 mU/l. Prolactin levels above 4000 mU/l invariably indicate prolactinoma.

Adrenocorticotrophic hormone (ACTH)

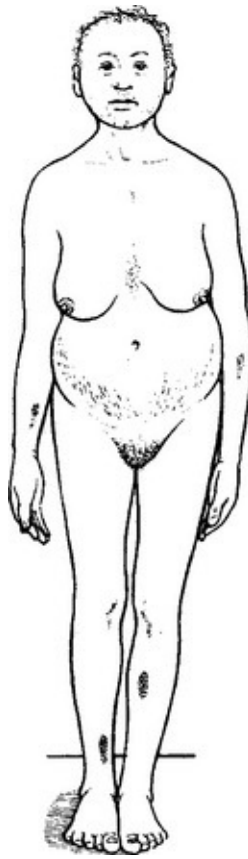
ACTH stimulates secretion of cortisol and androgens. Hypersecretion from a pituitary adenoma or hyperplasia causes CUSHING'S DISEASE (bilateral adrenal hyperplasia) which presents with the characteristic features of CUSHING'S SYNDROME.

This syndrome may also be caused by excessive oral corticosteroids, but also by an adrenal tumour or by ectopic secretion of ACTH from a bronchial carcinoma.

Features of Cushing's syndrome

- Moon face
- Acne
- Hirsutism and baldness
- Buffalo-type obesity
- Purple striae over flanks and abdomen
- Bruising
- Muscle weakness and wasting

- Osteoporosis
- Hypertension
- Increased susceptibility to infection
- Diabetes mellitus



A loss of normal diurnal variation of plasma free cortisol and an increase in 24 hour urinary free cortisol indicates excess secretion. The diagnosis of a pituitary cause is suggested by finding normal or moderately raised ACTH levels which suppress with high doses of dexamethasone.

Ectopic ACTH production does not suppress with dexamethasone and

with adrenal tumours, ACTH levels are virtually undetectable.

Other tests include

- the effect of corticotrophin releasing factor ($\uparrow$ ACTH and cortisol if pituitary origin)
- petrosal versus peripheral venous sampling to identify the source of the ACTH.

Bilateral adrenalectomy for Cushing's syndrome is sometimes followed by the development of *Nelson's syndrome* – high ACTH levels, pituitary enlargement and marked skin pigmentation.

TSH – stimulates thyroid hormone secretion	} Hypersecreting tumours very rare.
FSH – controls growth of ovarian follicles/spermatogenesis	
LH – induces ovulation/testosterone secretion	

2. HYPOSECRETION

Many pituitary tumours are diagnosed before panhypopituitarism develops, but large tumours may cause gradual impairment of pituitary hormone secretion. Growth hormone and the gonadotrophins are first affected, followed by TSH and ACTH. Panhypopituitarism only occurs when more than 80% of the anterior pituitary is destroyed.

Impaired secretion	Adults	Children
GH –	‘Adult GH deficiency syndrome’ – weight gain, loss of libido, fatigue	Pituitary dwarfism – (diminished somatic growth, retarded sexual development, hypoglycaemic episodes, normal intelligence)
Gonadotrophins –	Amenorrhoea, sterility, loss of libido	

ACTH –	Glucocorticoid and androgen deficiency, muscle weakness and fatigue	
TSH –	Secondary hypothyroidism – sensitivity to cold, dry skin, physical and mental sluggishness, coarseness of hair	
Prolactin* –	Failure of lactation	

* Prolactin secretion is most resistant to pituitary damage. Deficiency is seldom evident, usually only presenting after postpartum haemorrhage (Sheehan's syndrome) as a failure of lactation associated with the other features of panhypopituitarism.

Pituitary hormone assay cannot distinguish low 'normal' levels from impaired secretion, *but low levels of pituitary hormone in the presence of low target gland hormones* confirm hyposecretion, e.g. low TSH levels despite a low serum thyroxine. Basal levels guide replacement therapy.

The lack of response to tests designed to increase specific pituitary hormones provides additional confirmation of hypofunction:

1. GH } – *Insulin tolerance test*: Hypoglycaemia acting via the hypothalamic pituitary axis should elevate GH and ACTH levels, the latter causing a significant rise in plasma cortisol.
2. ACTH }
3. Gonadotrophin – *Gonadotrophin releasing hormone (GnRH) injection* should produce a rapid rise in LH and a slower rise in FSH.
4. TSH } – *Thyrotrophin releasing hormone (TRH) injection* should increase plasma levels of both TSH and prolactin.
5. Prolactin }

The above tests can be carried out simultaneously as the *Combined pituitary stimulation test*. Insulin, GnRH and TRH are injected intravenously and all anterior pituitary hormones measured from repeated blood samples taken over a 2-hour period. Glucose levels are also checked to ensure adequacy of the hypoglycaemia.

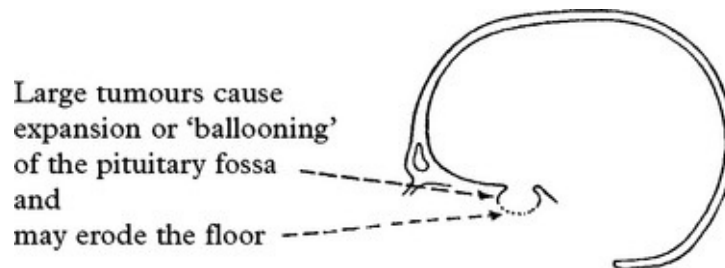
PITUITARY APOPLEXY

This is an uncommon complication of pituitary tumours due to the occurrence of infarction followed by haemorrhage into the tumour. Severe headache of sudden onset simulating subarachnoid haemorrhage, rapidly progressive visual failure and extraocular nerve palsies accompany acute pituitary insufficiency. Death may follow unless urgent steroid treatment is instituted.

NEURORADIOLOGICAL INVESTIGATION

LARGE TUMOURS

Skull X-ray

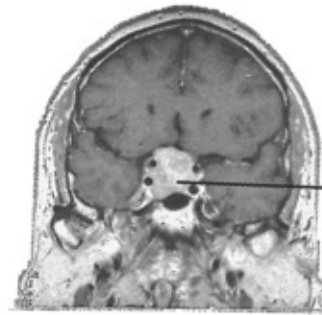


CT scan with contrast enhancement demonstrates tumours filling the pituitary fossa and expanding into the suprasellar compartment, but **MRI** gives more anatomical detail, clearly delineating any suprasellar extension and the effect on adjacent structures.

Sagittal T1 weighted MRI with gadolinium showing a large pituitary tumour of mixed intensity with suprasellar extension.

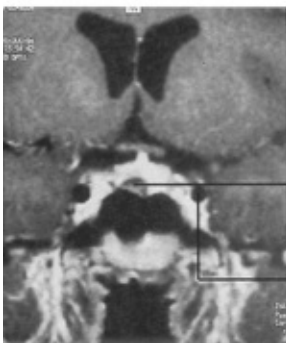
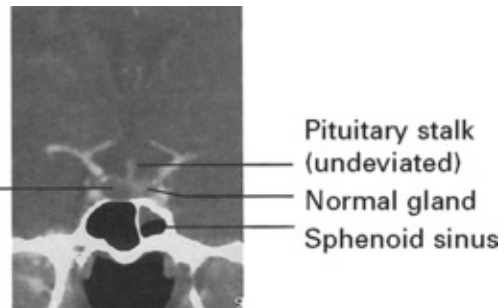


Coronal view of same patient showing relationships of the tumour to the carotid arteries and the cavernous sinus.



MICROADENOMAS

Coronal **CT Scanning** with contrast may demonstrate a low density region within the gland tissue (or may show deviation of the pituitary stalk from the midline). Tumours > 5 mm diameter produce these characteristic appearances. Tumours under this size are difficult to detect.



MRI is marginally better than CT scanning in the detection of microadenomas but both have false positives and false negatives.

CT angiography or MR angiography may be required before transphenoidal operation to exclude the presence of an incidental medially projecting aneurysm.

MANAGEMENT

A variety of different forms of treatment are available:

Drug therapy

Dopamine agonists lower abnormal hormone concentrations, especially prolactin. In prolactinoma, the prolactin levels usually fall and the tumour shrinks, but patients require long-term therapy as the source persists. Cessation of treatment can result in rapid tumour re-expansion. Agents used include bromocriptine and cabergoline, a long acting preparation.

Somatostatin analogues: e.g. octreotide, inhibit growth hormone production and cause some tumour shrinkage in a proportion of patients. No longer used for long-term therapy.

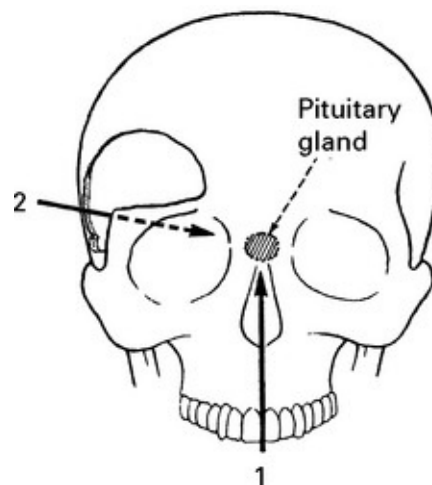
GH receptor antagonists: pegvisomant may be of value in GH secreting tumours with an inadequate response to surgery, radiation or octreotide.

Operative approach

From BELOW:

1. *Trans-sphenoidal*

Through an incision in the upper gum the nasal mucosa is stripped from the septum and the pituitary fossa approached through the sphenoid sinus. The microscope aids vision and either traditional intraoperative fluoroscopy, neuronavigation or real-time MRI ([page 386](#)) is used for guidance. Through this route the pituitary gland can be directly visualised and explored for microadenomas. Even large tumours with suprasellar extensions may be removed from below, avoiding the need for craniotomy.

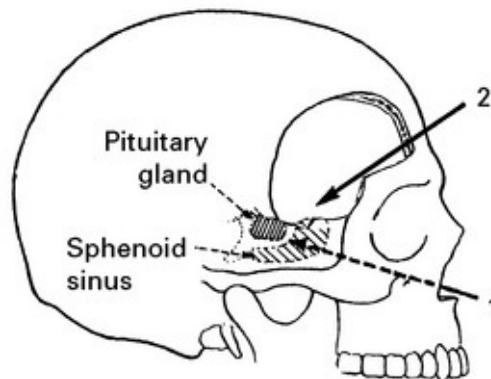


Many centres now use a *transnasal endoscopic approach* to remove the tumour. This avoids the sublabial incision and minimises septal retraction and postoperative discomfort. It greatly improves visualisation of the cavernous sinus and intrasellar structures.

From ABOVE

2. Transfrontal

Through a craniotomy flap the frontal lobe is retracted to provide direct access to the pituitary tumour. This approach is usually reserved for tumours with large frontal or lateral extensions.



N.B. Patients may require steroid cover before any anaesthetic or operative procedure.

Radiotherapy

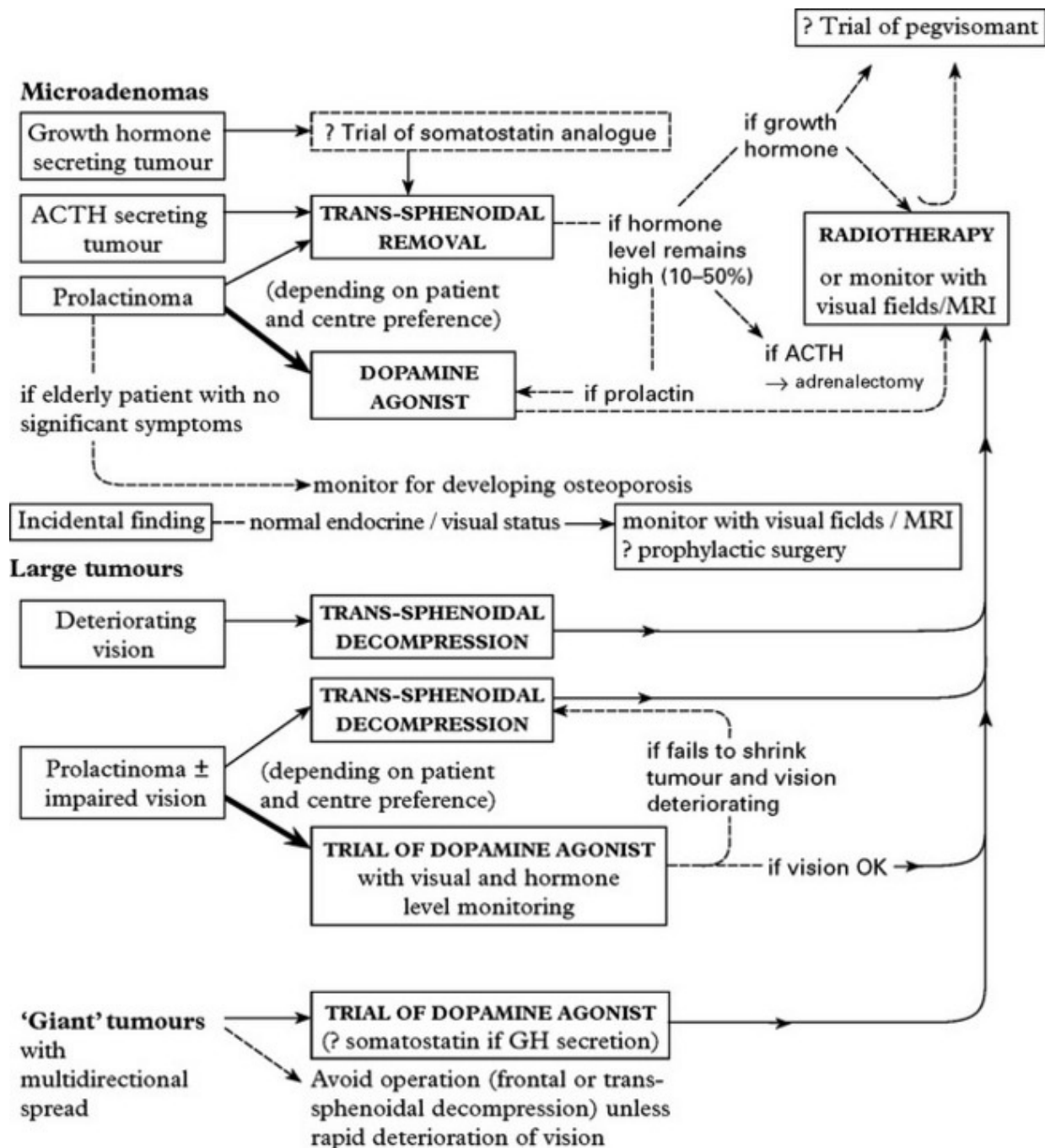
Pituitary adenomas are radiosensitive and external irradiation is commonly employed. Stereotactic radiosurgery is also used, but may not provide additional benefit. Occasionally, radioactive seeds of yttrium or gold are implanted into the pituitary fossa.

Several months elapse before hormone levels begin to fall. Pituitary function gradually declines over a 5–10 year period after treatment and most patients eventually require replacement hormone therapy to prevent symptoms of hypopituitarism developing.

Treatment selection

Treatment choice depends on:

- presenting problems and patient's requirements,
e.g. incidental finding, restoring fertility, halting visual deterioration.
- patient's age.
- preference and experience of the treatment centre.



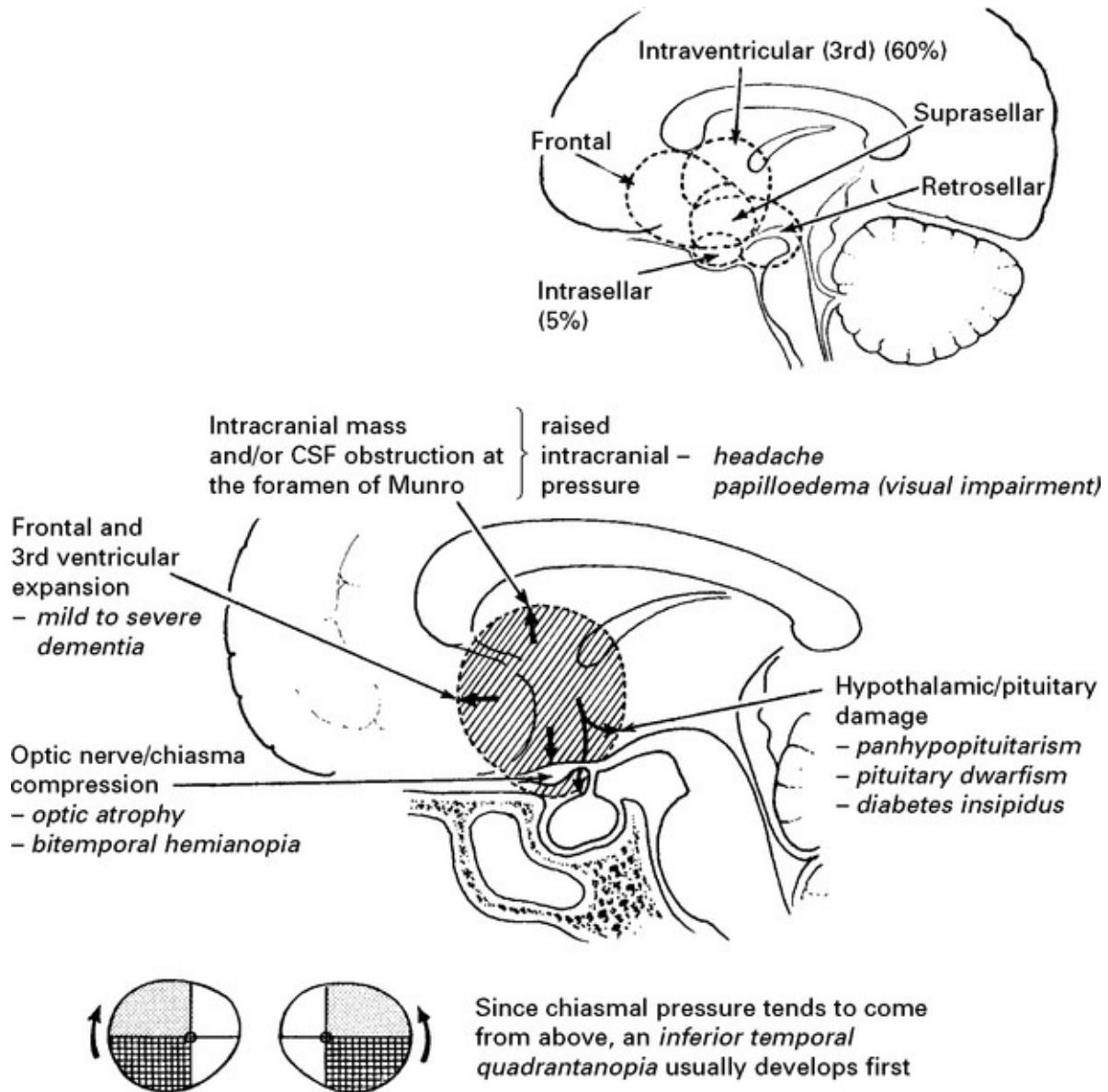
SELLAR/SUPRASELLAR TUMOURS

CRANIOPHARYNGIOMA

These tumours arise from remnants of Rathke's cleft and constitute about 3% of all primary intracranial tumours. They may present at any age, but occur predominantly in children from 5–14 years (adamantinomatous type) and in adults from 50–60 years (papillary type). Although benign, proximity to crucial structures poses complex problems of management. About 40% of craniopharyngiomas have solid components of squamous epithelium with calcified debris and one or more cystic regions containing greenish cholesteatomatous fluid. In 20% the tumour is solid throughout. Although the tumour capsule appears well defined, histological examination reveals finger-like projections extending into adjacent tissue with marked surrounding gliosis.

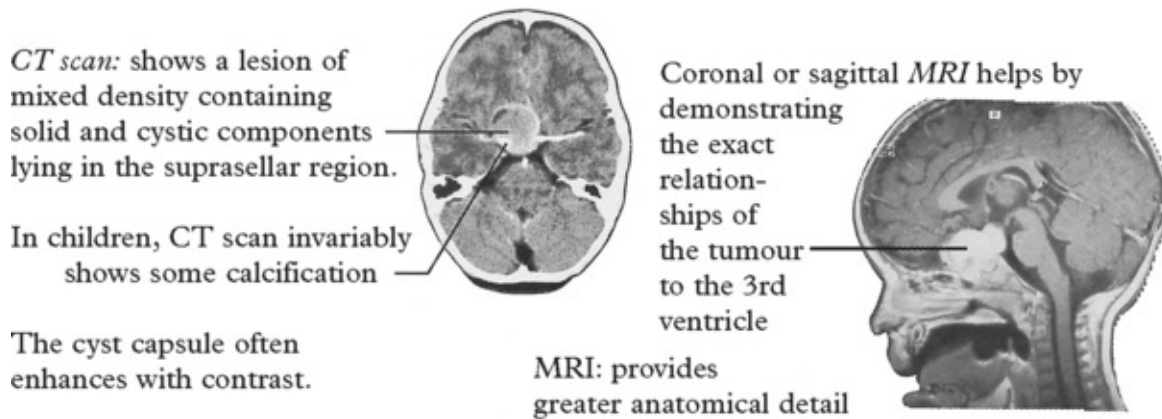
Sites: growth usually begins near the pituitary stalk, but may extend in any direction.

Clinical features: depend on the exact site and size of the tumour. Growth is slow and most signs and symptoms develop insidiously.



Investigations

Skull X-ray: shows calcification above or within the pituitary fossa in most children and in 25% of adults.



Pituitary function studies (page 342): often demonstrate the need for hormone replacement.

Management

Several options exist; the more aggressive the treatment, the higher the risks, but the lower the recurrence rate.

All patients require preoperative ophthalmological and endocrine assessment and steroid cover before any anaesthetic or operative procedure.

Operative removal usually involves a subfrontal or subtemporal craniotomy, perhaps combined with a transcallosal approach (i.e. splitting the anterior corpus callosum from above and approaching the tumour through the 3rd ventricle). The trans-sphenoidal route permits removal of purely intrasellar tumours.

Methods

1. Total tumour excision (+ radiotherapy if recurrence develops)
2. Subtotal tumour excision + radiotherapy (defer if <3 years)
3. Cyst drainage + (with an indwelling catheter and reservoir)
 - radiotherapy
 - or
 - implantation of yttrium-90 or chemotherapeutic agent (bleomycin)

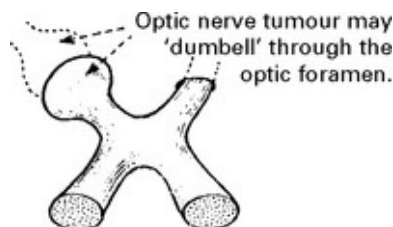
Although total excision avoids the immediate need and associated risks of radiotherapy to a developing brain, it carries a risk of life-threatening hypothalamic damage. Accepting a subtotal resection is often the safest option. Operative mortality lies between 0–10% and depends on the tumour site and the extent of the attempted removal. Some report a recurrence rate of up to 30% within 10 years of an apparent ‘total’ removal. This presumably results from residual tumour extensions lying beyond the capsule.

Within subtotal removal the recurrence rate approaches 90%, but with radiotherapy this falls to 30–50% after 5 years. The decision to aim for total or subtotal removal requires careful judgement. Preoperative investigations help but the final decision often awaits direct exploration.

Coronal or sagittal *MRI* helps by demonstrating the exact relationships of

OPTIC NERVE (GLIOMA) ASTROCYTOMA

This rare tumour usually presents in children under 10 years. Up to one-third are associated with neurofibromatosis (NF1) where the tumour may be bilateral. Tumour growth expands the nerve in a fusiform manner. Some extend anteriorly into the orbit, others posteriorly to involve the optic chiasma. All are of the pilocytic type and growth is slow. Spontaneous regression may occur, particularly in NF1 patients.



Clinical features

Visual field scotomas gradually progress to *complete visual loss*.

Orbital extension causes *proptosis*.

In some patients posterior expansion beyond the chiasma causes *hypothalamic damage* (*precocious puberty and other endocrine disturbance*) and/or hydrocephalus.

CT scans demonstrate an enhancing mixed attenuation mass within the orbit or lying in the suprasellar region. *MRI* is more sensitive for chiasmatic extensions.

Management		Prognosis
<i>Unilateral within orbit</i>	– Conservative approach but if imaging shows progression towards chiasma, complete excision (with orbital enucleation if necessary)	– Long-term survival expected.
<i>Lesion involving the optic chiasma</i>	– Conservative approach (the value of radiotherapy is not known and may risk vasculitis and intellectual deterioration).	– Patients may retain vision for many years; survival is often long-term. Those with hypothalamic damage have a poor prognosis.

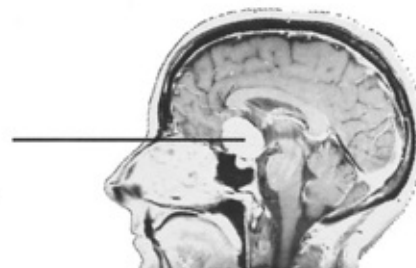
SUPRASellar MENINGIOMA

Meningiomas arising from the tuberculum sellae often present early as a result of chiasmal compression causing visual field defects – usually a *bitemporal hemianopia*.

CT scan shows a rounded, often partly calcified suprasellar mass homogeneously enhancing with contrast with or without *hyperostosis* of the tuberculum sellae or planum sphenoidale. *MRI* provides improved anatomical detail.

Unfortunately the visual defect often persists after operation, but attempted removal is essential to prevent further progression.

T1 weighted MRI with gadolinium



MENINGIOMA OF THE OPTIC NERVE SHEATH

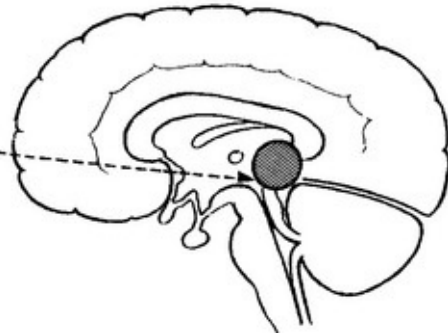
Rarely, meningiomas arise from the optic nerve sheath, usually extending in dumbbell fashion through the optic foramen. Some penetrate the orbital dura and invade the orbital contents. Total excision is impossible without sacrificing the adjacent optic nerve.

SUPRASellar EPIDERMoid/DERMoid (see [page 337](#)).

Note: large aneurysms or granulomas (TB, sarcoid) may simulate a sellar/suprasellar tumour on CT scan or MRI. If in doubt, perform CTA or MRA prior to operative exploration.

PINEAL REGION TUMOURS

Pineal region tumours are relatively uncommon. They consist of a variety of different pathological types and as a result of the direct anatomical relationship with the third ventricle include tumours arising at this site. Less than 20% actually originate from 'pineal' cells.



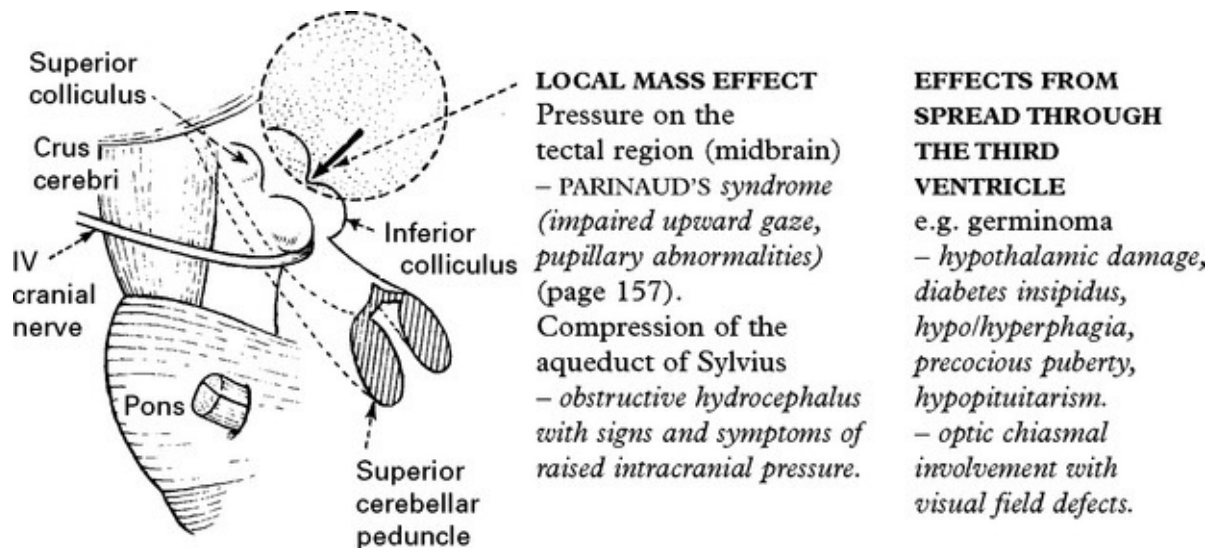
PATHOLOGICAL TYPES

Germ cell tumours: *Germinoma* is the commonest pineal region tumour of germ cell origin. It is malignant in nature and adheres firmly to surrounding tissues and cells may spread to the floor and anterior wall of the third ventricle. *Teratomas* are usually well differentiated, occurring predominantly in males, and formed from various cell types – muscle, bone, cartilage, dermis. Tumour consistency depends on the predominant cell type. In most the tumour margin is well defined, but malignant, poorly differentiated forms occasionally occur.

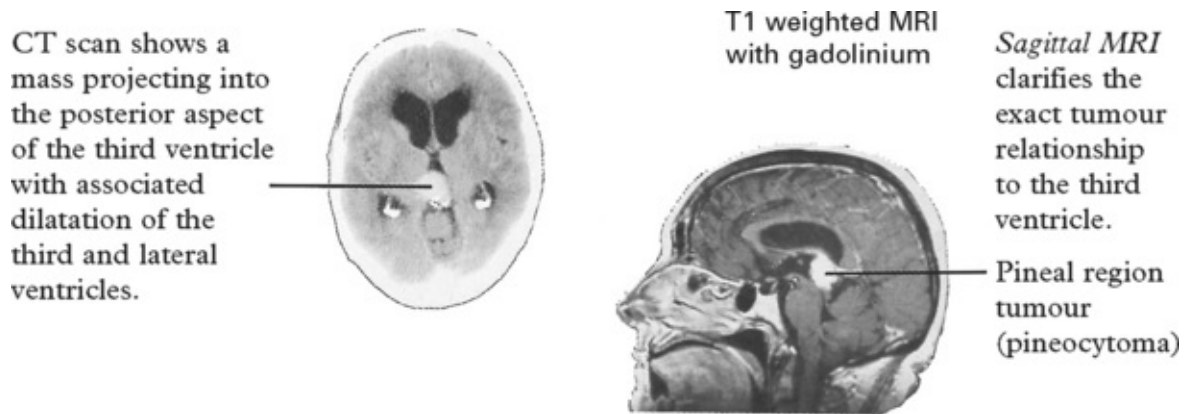
Other germ cell tumours include the highly malignant *yolk sac tumour*, *choriocarcinoma* and *embryonal carcinoma*.

Pineocytoma: well differentiated, slowly growing tumour
Pineoblastoma: poorly differentiated, highly malignant tumour } rare tumours of true 'pineal' origin.
Glial cell tumours – **astrocytoma** – arising from cells within the pineal gland, or from adjacent brain.
 – **ependymoma** – arising from cells lining the third ventricle.
Meningioma } rarely occur in the pineal region.
Dermoid
Epidermoid

CLINICAL FEATURES Develop due to:



INVESTIGATIONS



Pineocytomas – may appear calcified.

Teratomas – may contain mixed densities from fat to calcification.

Tumour markers – Serum/CSF human chorionic gonadotrophin – ↑ in choriocarcinoma (and slight ↑ in some germinomas)

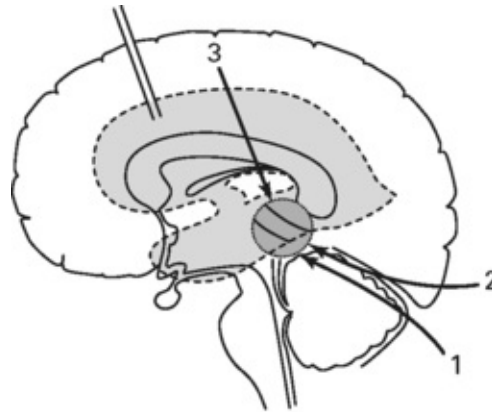
– Serum/CSF alpha fetoprotein – ↑ in yolk sac tumours

CSF cytology: Malignant pineal region tumours can metastasise through CSF and *cytology* is important in planning treatment.

MANAGEMENT

Hydrocephalus often requires urgent treatment with a *ventriculoperitoneal shunt* or 3rd ventriculostomy. Large tumours may obstruct the foramen of Munro, making *bilateral* ventricular drainage necessary.

If biopsy, either via an endoscope or by stereotaxy, confirms a germinoma, or if serum/CSF markers are significantly raised suggesting choriocarcinoma or a yolk sac tumour, then radiotherapy ± chemotherapy is the treatment of choice.



Routes of direct approach
 1 – Infratentorial supracerebellar
 2 – Suboccipital transtentorial
 [3 – Transventricular]

Teratomas, pineocytomas, dermoid or epidermoid cysts and meningiomas require *direct operative exploration* and excision, usually via either the supracerebellar or the transtentorial approach as shown. Pineoblastomas, certain pineocytomas and ependymomas require a combination of *excision + radiotherapy*. *Chemotherapy* may be added for the more malignant tumours.

When imaging shows disseminated tumour or when CSF cytology is positive, the entire craniospinal axis should be irradiated.

Outcome depends on tumour type. For germinomas and resectable tumours, the outlook is excellent and long-term survival is the rule.

TUMOURS OF THE VENTRICULAR SYSTEM

EPENDYMOMA

Intracranial ependymomas originate from cells lining the ventricular cavities. The majority arise in the 4th ventricle and in this site occur predominantly in children. Most are low grade (grade II), but an anaplastic form (grade III) exists and in about 10% tumour cells seed